The total number of degrees of freedom is 19 less the redundant information, which is as follows:

Redundant variables in interconnecting streams being eliminated:

Stream 1: (4 + 2) =	6
Stream 2: (4 + 2) =	6
Stream 3: (4 + 2) =	6
Stream 4: (4 + 2) =	6
	24

Redundant constraints being eliminated:

Stream 1:	
NH_3 concentration = 0	1
p = 100 atm	1
Stream 2:	
p = 100 atm	1
Stream 3:	
NH_3 concentration = 0	1
p = 100 atm	1
T = −50°C	1
Stream 4:	
NH_3 concentration = 0	1
T = −50°C	1
p = 100 atm	1
	9

Overall the number of degrees of freedom should be

$$N_d = 19 - 24 + 9 = 4$$

We can check the count for N_d by making a degrees-of-freedom analysis about the entire process as follows:

Examine Fig. E5.5b.

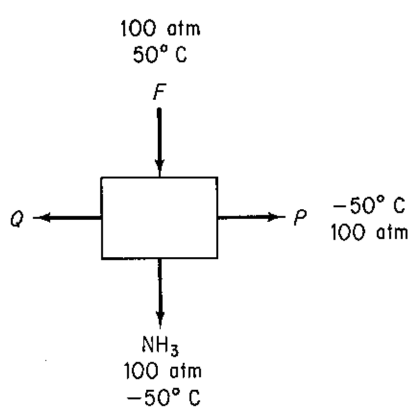


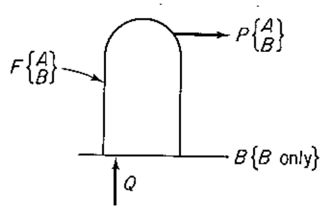
Figure E5.5b

$N_v = 3(4 + 2) + 1 =$		19
N_r:		
Material balances		
(H, N, A)	3	
Energy balance	1	
Specifications:		
Stream F (T = 50°C, p = 100 atm, NH_3 = 0)	3	
NH_3 stream (T = −50°C, p = 100 atm, three components have 0 concentration)	5	
Purge stream (T = −50°C, p = 100 atm, NH_3 = 0)	$\underline{3}$	$\underline{15}$
$N_d = 19 - 15 =$		$\underline{\underline{4}}$

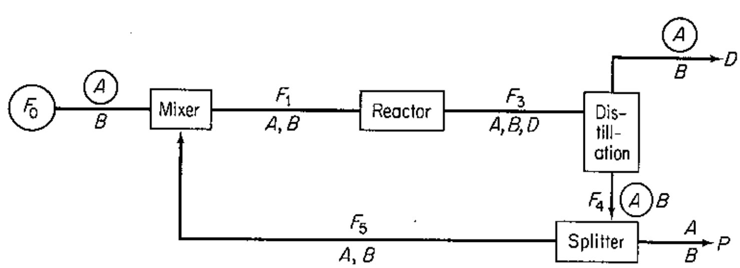
We do not have the space to illustrate additional combinations of simple units to form more complex units, but Kwauk[1] prepared several excellent tables summarizing the variables and degrees of freedom for distillation columns, absorbers, heat exchangers, and the like.

Self-Assessment Test

1. Is there any difference between the number of species present in a process and the number of components in the process?
2. Why are there $N_{sp} + 2$ variables associated with each stream?
3. Determine the number of degrees of freedom for a still.



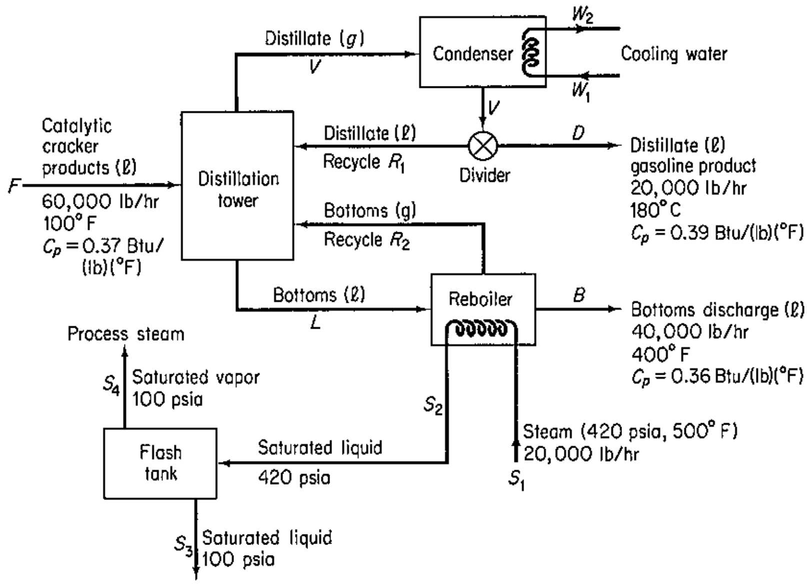
4. Determine the number of degrees of freedom in the following process:



[1] M. Kwauk, *AIChE J.*, v. 2, p. 240 (1956).

The encircled variables have known values. The reaction parameters in the reactor are known as is the fraction split at the splitter between F_4 and F_5. Each stream is a single phase.

5. The accompanying figure represents the schematic flowsheet of a distillation tower used to recover gasoline from the products of catalytic cracker. Is the problem completely specified, that is, is the number of degrees of freedom equal to zero for the purpose of calculating the heat transfer to the cooling water in the condenser?



5.2 SOLVING MATERIAL AND ENERGY BALANCES USING FLOWSHEETING CODES

In Sec. 2.4 we discussed combining units from the viewpoint of making material balances. As more and more units are connected together in a plant, you can understand that the degree of complexity requires that the solution of material and energy balances be carried out via a computer code. Such a program can also, at the same time, determine the size of equipment and piping, evaluate costs, and optimize performance.

Once the process flowsheet is specified, the solution of the appropriate steady-state process material and energy balances is referred to as **flowsheeting,** and the computer code used in the solution is known as a **flowsheeting package.** The essential problem in flowsheeting without associated optimization is to solve (satisfy) a large set of linear and nonlinear equations to an acceptable degree of precision, normally by an iterative procedure. In flowsheeting without optimization, you must make sufficient specifications to take up *all* the degrees of freedom. Table 2.5 lists some typical codes used to execute flowsheeting. Individual process units that make up the process flowsheet are represented in the form of modules (building blocks) or as equation sets.

Figure 5.2 illustrates the main features of a flowsheeting program. We will examine some of the details concerning the material and energy balance phase. Fundamental to all flowsheeting codes is the calculation of mass and energy balances for the entire process. Inputs to the material and energy balances phase of the calculations for the flowsheet must be defined in sufficient detail to determine all the intermediate and product streams and the unit performance variables for all units.

Frequently, process plants contain recycle streams and control loops, and the solution for the stream properties requires iterative calculations. Thus efficient numerical methods for convergence must be used. In addition, appropriate physical properties and thermodynamic data have to be retrieved from a data base. Finally, a master program must exist that links all the building blocks, physical property data, thermodynamic calculations, subroutines, and numerical subroutines, and that also supervises the information flow. You will find that optimization and economic analysis are really the ultimate goal in the use of flowsheet codes.

Two extremes are encountered in flowsheeting software. At one extreme the entire set of equations (and inequalities) representing the process is written down, including the material and energy balances, the stream connections, and the rela-

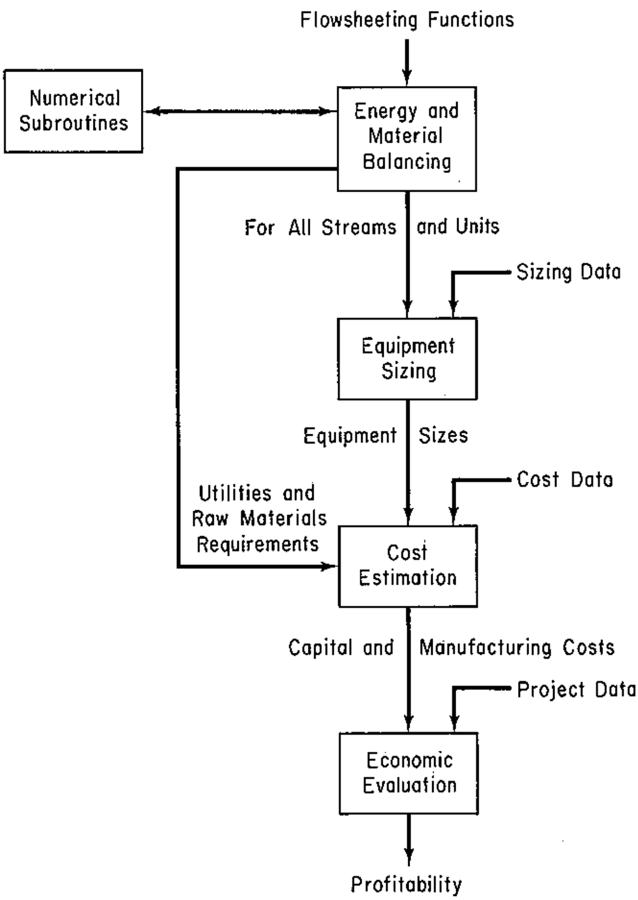


Figure 5.2 Information flow in a typical flowsheeting code.

tions representing the equipment functions. This representation is known as the **equation-oriented method** of flowsheeting. The equations can be solved in a sequential fashion analogous to the modular representation described below or simultaneously by Newton's method (or the equivalent), or by employing sparse matrix techniques to reduce the extent of matrix manipulations; references can be found at the end of this chapter.

At the other extreme, the process can be represented by a collection of modules (the **modular method**) in which the equations (and other information) representing each subsystem or piece of equipment are collected together and coded so that the module may be used in isolation from the rest of the flowsheet and hence is portable from one flowsheet to another. A module is a model of an individual element in a flowsheet (such as a reactor) that can be coded, analyzed, debugged, and interpreted by itself. Examine Fig. 5.3. Each module contains the equipment sizes, material and energy balance relations, the component flow rates, and the temperatures, pressures, and phase conditions of each stream that enters and leaves the physical equipment represented by the module. Values of certain of these parameters and variables determine the capital and operating costs for the units. Of course, the interconnections set up for the modules must be such that information can be transferred from module to module concerning the streams, compositions, flow rates, coefficients, and so on. In other words, the modules comprise a set of building blocks that can be arranged in general ways to represent any process.

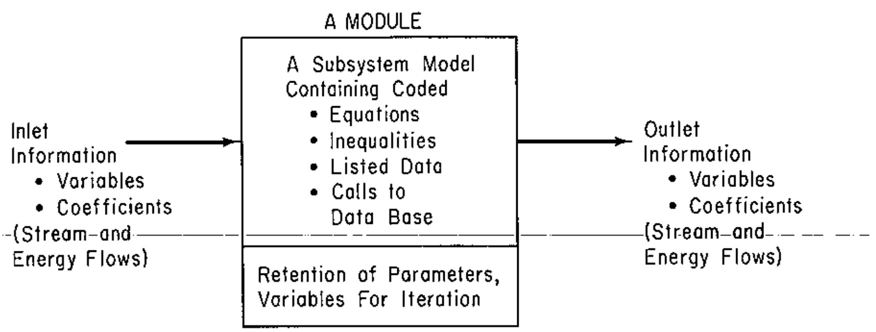


Figure 5.3 A typical process module showing the necessary interconnections of information.

Both sequential and simultaneous calculational sequences have been proposed for the modular approach as well as the equation-oriented approach. Either the program and/or the user must select the decision variables for recycle and provide estimates of certain stream values to make sure that covergence of the calculations occurs, especially in a process with many recycle streams. Reviews by Evans[2] and Rosen[3] point out many of the problems and practices pertaining to flowsheeting.

In addition to the two extremes, combinations of equations and modules can be

[2] L. B. Evans, "Flowsheeting: A State of the Art Review," *Proc. Chemcomp. 1982*, G. F. Froment, ed., p. 1, KVI, Antwerp, Belgium (1982).

[3] E. Rosen, "Steady State Chemical Simulation-State of the Art Review," *Computer Applications to Chemical Engineering*, R. G. Squires and G. V. Reklaitis, eds., ACS Symposuim Series, no. 124, p. 3, 1980.

used. Equations can be lumped into modules, whereas modules can be represented by their basic equations or by polynomials that fit the input–output information.

In current practice, modularly organized simulators seems to prevail because (1) the modules are easier to construct and understand; (2) addition and deletion of modules for a flowsheet is easily accomplished without changing the solution strategy; (3) modules are easier to program and debug than sets of equations, and diagnostics for them easier to analyze; and (4) the modules already exist and work, whereas equation blocks for equipment are not prevalent. Difficulties encountered in equation-based codes usually stem from sets of equations associated with particular units. Embedding these equations as subproblems with special routines often alleviates the difficulties. By permitting the user to mix equations and subroutines, use can be made of "macros" in the general network representing a process. We will review equation-based flowsheeting first because it is much closer to the techniques used up to this point in this book, and then turn to consideration of modular-based flowsheeting.

5.2-1 Equation-Based Flowsheeting

Sets of linear and/or nonlinear equations can be solved simultaneously using an appropriate computer code (see Table L.1) by one of the methods described in Appendix L. Equation-based flowsheeting codes pertaining to chemical engineering can be used for the same purpose. The latter have some advantages in that the physical property data needed for the coefficients in the equations are transparently transmitted from a data base at the proper time in the sequence of calculations.

Whatever the code used to solve material and energy balance problems, you must provide certain input information to the code in an acceptable format. All flowsheeting codes require that you convert the information in the flowsheet (see Fig. 5.4) to an information flowsheet as illustrated in Fig. 5.5, or something equivalent. In the information flowsheet, you use the name of the mathematical model (subroutine for modular-based flowsheeting) that will be used for the calculations instead of the name of the process unit.

Once the information flowsheet is set up, the determination of the process topology is easy, that is, you can immediately write down the stream interconnections between the modules (or subroutines) that have to be included in the input data set. For Fig. 5.5 the matrix of stream connections (the **process matrix**) is (a negative sign designates an exit stream)

Unit	Associated streams				
1	1	−2			
2	2	−3			
3	3	8	−4	−13	
4	4	7	11	−9	−5
5	5	−6			
6	6	−8	−7		
7	10	−11	−12		
8	9	−10			

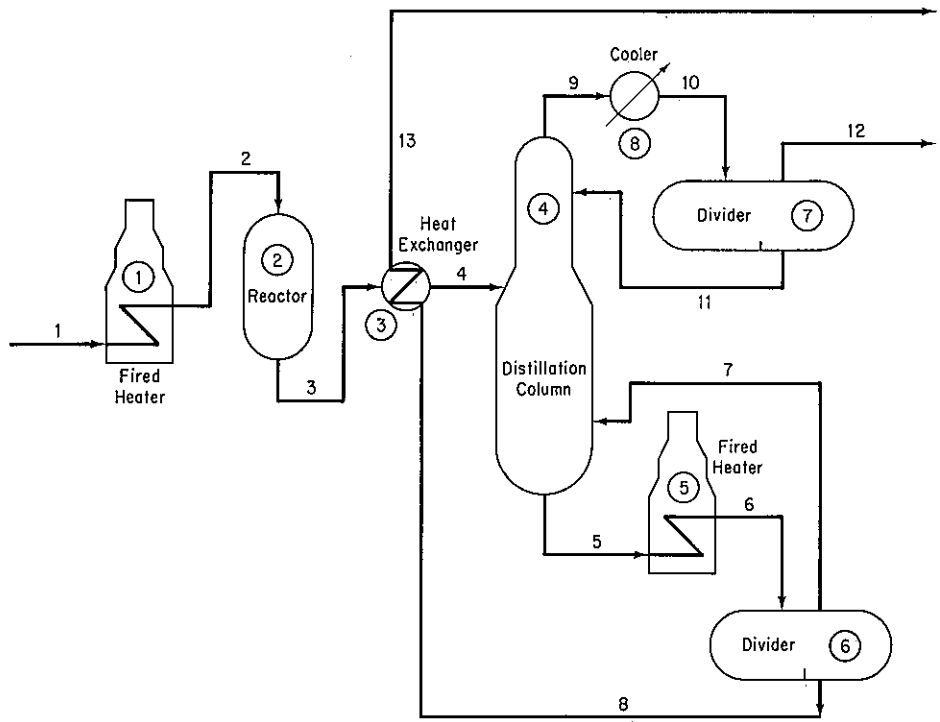


Figure 5.4 Hypothetical process flow sheet.

The interconnections between the unit modules may represent information flow as well as material and energy flow. In the mathematical representation of the plant, the interconnection equations are the material and energy balance flows between model subsystems. Equations for models such as mixing, reaction, heat exchange, and so on, must also be listed so that they can be entered into the computer code used to solve the equation. Table 5.1 lists the common type of equations that might be used for a single subsystem. In general, similar process units repeatedly occur in a plant and can be represented by the same set of equations, which differ only in the names of variables, the number of terms in the summations, and the values of any coefficients in the equations.

Equation-based codes can be formatted to include inequality constraints along with the equations. Such constraints might be of the form $a_1x_1 + a_2x_2 + \cdots \leqq b$, and might arise from such factors as

1. Conditions imposed in linearizing nonlinear equations
2. Process limits for temperature, pressure, concentration

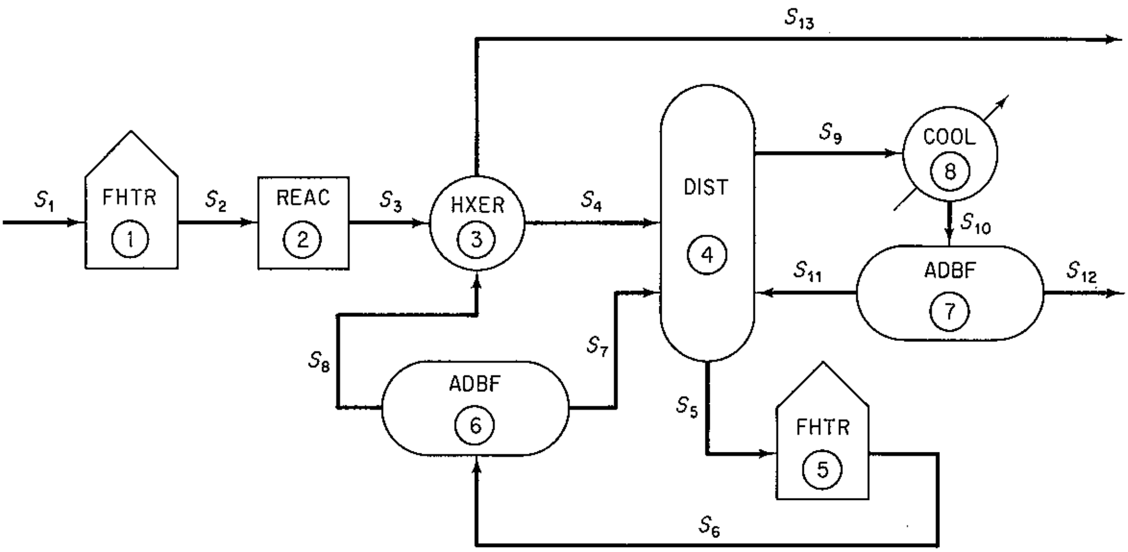
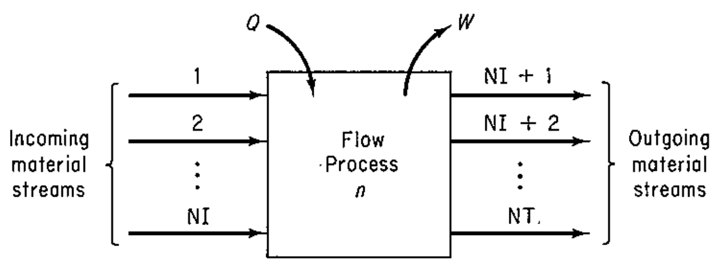


Figure 5.5 Information flow sheet for the hypothetical process (S stands for stream; module or computer code number is encircled).

TABLE 5.1 Generic Equations for a Steady-state Open System*

System diagram



Total mass balance (or mole balance without reaction)

$$\sum_{i=1}^{NI} F_i = \sum_{i=NI+1}^{NT} F_i$$

Component mass or mole balances (without reaction)

$$\sum_{i=1}^{NI} F_i w_{i,j} = \sum_{i=NI+1}^{NT} F_i w_{i,j}$$

$$\text{for } j = 1, 2, \ldots, NC$$

Energy balance

$$\sum_{i=1}^{NI} F_i H_i + Q_n - W_{s,n} = \sum_{i=NI+1}^{NI} F_i H_i$$

Summation of mole or mass fractions

$$\sum_{j=1}^{NC} w_{i,j} = 1.0 \quad \text{for } i = 1, 2, \ldots, NI$$

TABLE 5.1 *(cont.)*

Vapor-liquid equilibrium distribution

$$y_j = K_j x_j \quad \text{for} \quad j = 1, 2, \ldots, \text{NC}$$

Equilibrium vaporization coefficients

$$K_j = K(T_i, P_i, \overline{W}_i) \quad j = 1, 2, \text{for} \ldots, \text{NC}$$

Physical property functions

$$\begin{rcases} H_i = H_{\text{VL}}(T_i, P_i, \overline{W}_i) \\ S_i = S_{\text{VL}}(T_i, P_i, \overline{W}_i) \end{rcases} i = 1, 2, \ldots, \text{NI}$$

Total mole balance (with reaction)

$$\sum_{i=1}^{\text{NI}} F_i + \sum_{l=1}^{\text{NR}} R_l\left[\sum_{j=1}^{\text{NC}} V_{j,l}\right] = \sum_{i=\text{NI}+1}^{\text{NI}} F_i$$

Component mole balances (with reaction)

$$\sum_{i=1}^{\text{NI}} F_i w_{i,j} + \sum_{l=1}^{\text{NR}} V_{j,l} R_l = \sum_{i=\text{NI}+1}^{\text{NT}} F_i w_{i,j} \qquad \text{for } j = 1, 2, \ldots, \text{NC}$$

Molar atom balances

$$\sum_{i=1}^{\text{NI}} F_i\left[\sum_{j=1}^{\text{NC}} w_{i,j} a_{j,k}\right] = \sum_{i=\text{NI}+1}^{\text{NT}} F_i\left[\sum_{j=1}^{\text{NC}} w_{i,j} a_{j,k}\right] \qquad \text{for } k = 1, 2, \ldots, \text{NE}$$

Mechanical energy balance

$$\sum_{i=1}^{\text{NI}}(K_i + P_i) + \sum_{i=1}^{\text{NI}} \int_{P_{1,i}}^{P_{2,i}} V_i\, dp_i = \sum_{i=\text{NI}+1}^{\text{N}} (K_i + P_i) + \sum_{i=\text{NI}+1}^{\text{NT}} \int_{P_{1,i}}^{P_{2,i}} V_i\, dP_i + W_{s,n} + E_{v,n}$$

*Notation:

$a_{j,k}$	number of atoms of the kth chemical element in the jth component
F_i	total flow rate of the ith stream
H_i	relative enthalpy of the ith stream
K_j	vaporization coefficient of the jth component
NC	number of chemical components (compounds)
NE	number of chemical elements
NI	number of incoming material streams
NR	number of chemical reactions
NT	total number of material streams
p_i	pressure of the ith stream
Q_n	heat transfer for the nth process unit
R_l	reaction expression for the lth chemical reaction
T_i	temperature of the ith stream
$V_{j,l}$	stoichiometric coefficient of the jth component in the lth chemical reaction
$w_{i,j}$	fractional composition (mass of mole) of the jth component in the ith stream
$\overline{W}_i$	average composition in the ith stream
$W_{s,n}$	work for the nth process unit
x_j	mole fraction of component j in the liquid
y_j	mole fraction of component j in the vapor

3. Requirements that variables be in a certain order
4. Requirements that variables be positive or integer

A slack variable has to be included in such relations to make them into equations.

Four codes currently under development for equation-based flowsheeting are ASCEND,[4] Quasilin,[5] Speedup,[6] and TISFLO-II.[7] An equation-based code should include the following characteristics:

1. A method of easily entering the equations and inequality constraints in the code used to solve them. Slack variables can be used to transform the inequality constraints into equality constraints.
2. A possibility of using both continuous and discrete variables, the latter being particularly necessary to accommodate changes in phase or changes from one correlation to another.[8]
3. The option to use alternative forms of a function depending on the value of logical variables that outline the state of the process. A typical example is the calculation of p–V–T relations as the phase changes from gas to liquid. To fix on the appropriate form, the problem must be treated as an optimization problem in which feasibility is required on each cycle for each state with the logical variables treated as constraints for a state. Alternatively, the code can guess the state and if the logical constraints are violated, another guess can be made.
4. A method of solving submodels that defy normal equation-solving subroutines or are specially suited to ad hoc methods of solution.
5. The ability to build *macros*, that is, create complex models comprised of standard subelements. For example, distillation might be composed of trays, flash units, splitters, mixers, heat exchangers, and so on.
6. Provision for equivalence variables, that is, using the same name for two different variables if they are equal, such as the mass flow rate through a pump.
7. A method for solving individual models such as Newton or quasi-Newton methods combined with sparse matrix methods to convert the nonlinear alge-

[4] M. H. Locke, and A. W. Westerberg "The Ascend-II System—A Flowsheeting Application of Successive Quadratic Programming Methodology," *Comput. Chem. Eng.*, v. 7, p. 615 (1983).

[5] H. P. Hutchinson, D. J. Jackson, and W. Morton, "Equation Oriented Flowsheet Simulation, Design and Optimization," *Proc. Europ. Fed. Chem. Eng. Conf. Comput. Appl. Chem. Eng.*, Paris, April 1983; "The Development of an Equation-Oriented Flowsheet Simulation and Optimization Package," *Comput. Chem. Eng.*, v. 10, p. 19 (1986).

[6] R. W. H. Sargent, J. D. Perkins, and S. Thomas, "Speedup: Simulation Program for Economic Evaluation and Design of Unified Processes," in *Computer-Aided Process Plant Design*, M. E. Leesley, ed., Gulf Publishing Company, Houston, 1982.

[7] J. A. de Leeuw den Bouter, A. G. Swenber, and M. G. G. Van Meulebrouk, "Simulation and Optimization with TISFLO—II," *Proc. Chemplant 80*, Heviz, Hungary, September 1980.

[8] J. D. Perkins, "Equation Oriented Flowsheeting," in *Foundations of Computer-Aided Design*," A. W. Westerberg and J. J. Chien, eds., Cache Corp., Austin, Tex. 1983.

braic equations in the model to linearized approximates. Then the linearized equations can be solved iteratively taking advantage of their structure. The alternative is to use *tearing*. Tearing is discussed in detail in Sec. 5.2-2, treating modular-based flowsheeting, but briefly, by "tearing" we mean selecting certain output variables from a set of equations as known values so that the remaining variables can be solved by serial substitution. A residual set of equations equal to the number of tear variables will remain, and if these are not satisfied, new guesses are made for the values of the tear variables, and the sequence repeated. Examine Fig. 5.6c. Furthermore, the sensitivity of functions with respect to variables (the first partial derivatives) developed in Newton-like methods provides valuable information.

8. A method for *precedence ordering* so as to partition a model into a sequence of smaller models containing sets of irreducible equations (equations that have to be solved simultaneously as illustrated in Fig. 5.6).
9. A method to select initial guesses for the Newton-like solution procedure for the algebraic equations. Poor choices may lead to unsatisfactory results. You want the initial guesses to be as close to the correct answer as possible so that the procedure will converge. You can perhaps solve approximate models, and then pass to more complex ones.
10. Provision for *scaling* of the variables and equations. By scaling of variables we mean introducing transformations that make all the variables have ranges in the same order of magnitude. By scaling of equations we mean multiplying each equation by a factor that causes the value of the deviation of each equation from zero to be of the same order of magnitude.

An example of an equation-oriented flowsheeting code is ASCEND by Locke and Westerberg. Figure 5.7 shows the configuration of the various subroutines involved. Phasing of inputs, solving a problem, and generating outputs can be ordered by the user. Several independent programs operate on a file of the values of the variables and pointers that represent a flowsheet. A user supplies or makes use of existing subroutines containing equations that model a component in the flowsheet plus an input file that defines how these submodels should be assembled. ASCEND maintains a file that includes the user-supplied definition of the configuration plus definitions of variable packets (groups of associated variables), equation packets, types of variable and equation packets associated with an element in the flowsheet, a procedure to allow the user to create complex models (called "macros") from a simple set of inputs and default values so that the user has only to input the important parameters of interest or the exceptions to the default values.

Internal representation and storage in ASCEND-II of a flowsheet which may contain hundreds or even thousands of variables is carried out by the GEV (Generator, Equation packet, Variable packet) method proposed by Berna, et al. (1980). Figure 5.8 shows (1) a flowsheet element, (2) the internal formulation of the element, and (3) the matrix representation of the element. A generator is a subroutine. It calculates the necessary partial derivatives and equation residuals for the Newton-Raphson portion of a solution. Equation packets are groups of equations grouped together because they represent the equations of the process matrix reserved for a sin-

$$h_1:\ x_1^2 x_2 - 2x_3^{1.5} + 4 = 0$$
$$h_2:\ x_2 + 2x_5 - 8 = 0$$
$$h_3:\ x_1 x_4 x_5^2 - 2x_3 - 7 = 0$$
$$h_4:\ -2x_2 + x_5 + 5 = 0$$
$$h_5:\ x_2 x_4^2 x_5 + x_2 x_4 - 6 = 0$$

(a) The n independent equations involving n variables ($n = 5$).

	x_1	x_2	x_3	x_4	x_5
h_1	1	1	1		
h_2		1			1
h_3	1		1	1	1
h_4		1			1
h_5		1		1	1

(b) The occurrence matrix (the 1's represent the occurrence of a variable in an equation).

	x_2	x_5	x_4	x_1	x_3
h_2	1	1			
h_4	1	1			
h_5	1	1	1		
h_3		1	1	1	1
h_1	1			1	1

(c) The rearranged (partitioned) occurrence matrix with groups of equations (sets I, II, and III) that have to be solved simultaneously collected together in a precedence order for solution.

Figure 5.6 Partitioning and tearing. The equations are partitioned into blocks containing common variables, as in (c). Equations h_2 and h_4 (set I) are solved simultaneously for x_2 and x_5 first, then h_5 (set II) is solved for x_4, and lastly h_1 and h_3 are solved simultaneously. For example, assume a value for x_3; solve h_1 for x_1 then check to see if Equation h_3 is satisfied. If not, adjust x_3, solve h_1 for x_1, recheck h_3, and so on until both h_1 and h_3 are satisfied.

gle generator. In Fig. 5.8b, the corresponding equation packet would contain the equations for the material balance, equilibrium, and enthalpy balance for the flash unit. Variable packets are groups of associated variables grouped together for convenience. One example of a variable packet would be a stream variable packet containing the flowrate, temperature, pressure, enthalpy, and component mole fractions of

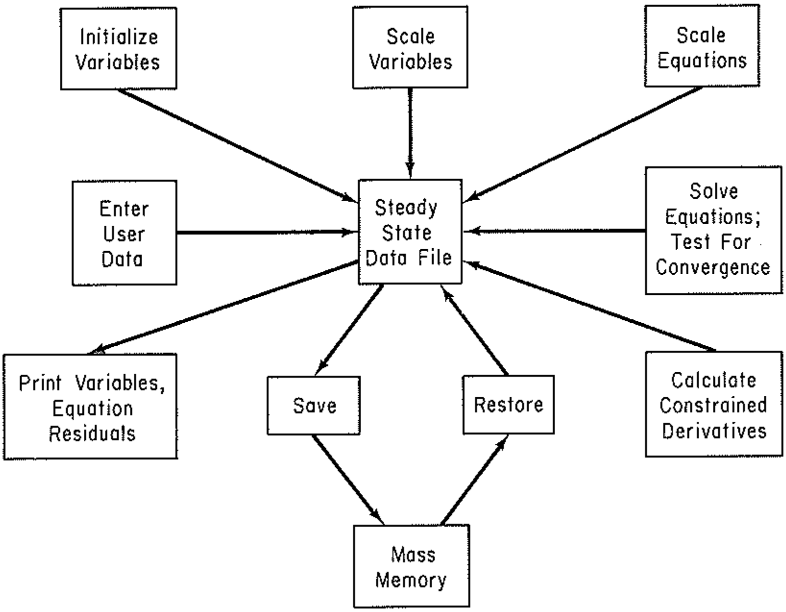


Figure 5.7 Information flow in ASCEND-II.

a stream. By specifying the elements in Fig. 5.8c, the user designates the generator to be used to calculate the partial derivatives and equation residuals for Newton or quasi-Newton methods of solution. Thus, an entire flowsheet can be characterized by a set of generators with associated variable packets and equation packets.

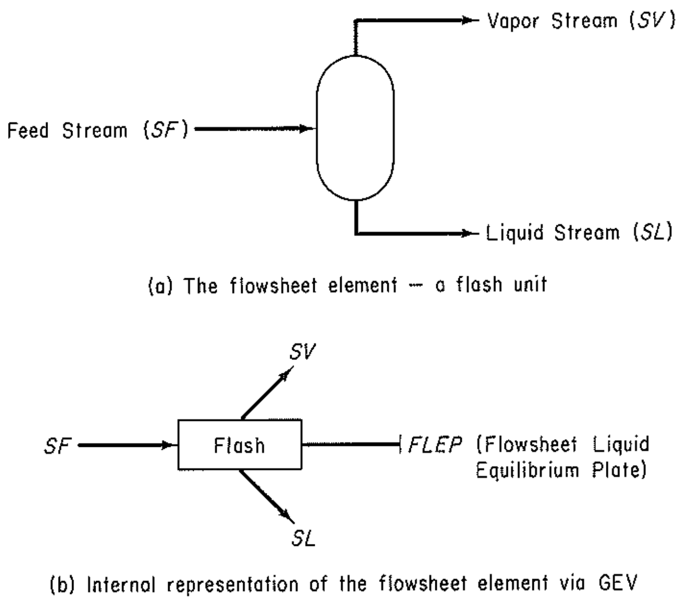


Figure 5.8 Three representations of an element in a flowsheet in ASCEND-II. (a) The flowsheet element—a flash unit. (b) Internal representation of the flowsheet element via GEV (c) Representation of the flowsheet element.

	SF	SV	SL
Material Balance	1	1	1
Equilibrium Relation	0	1	1
Enthalpy Balance	1	1	1

(c) Process matrix representation of the flowsheet element

Figure 5.8 *(cont.)*

EXAMPLE 5.6 Sequential Solution of Material Balances

Figure E5.6 shows 10 units that are part of a larger plant. The independent linear equations relating the flows of materials in and out of the units are listed in the figure. The incident matrix is shown below.

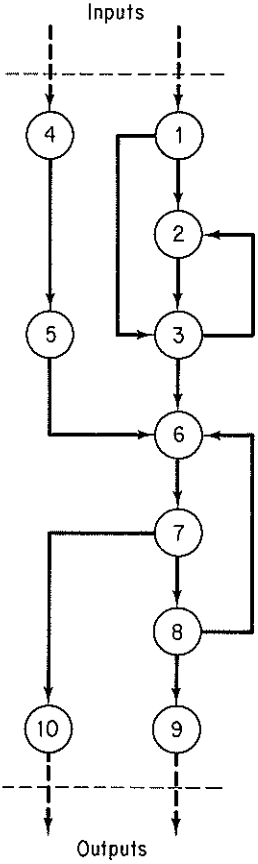
Equations	Variables												Interface variables			
	y_{12}	y_{13}	y_{23}	y_{32}	y_{36}	y_{45}	y_{56}	y_{67}	y_{86}	y_{78}	y_{710}	y_{89}	y_1	y_4	y_9	y_{10}
1	−1	−1	0	0	0	0	0	0	0	0	0	0	+1	0	0	0
2	+1	0	−1	+1	0	0	0	0	0	0	0	0	0	0	0	0
3	0	+1	+1	−1	−1	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	−1	0	0	0	0	0	0	0	+1	0	0
5	0	0	0	0	0	+1	−1	0	0	0	0	0	0	0	0	0
6	0	0	0	0	+1	0	+1	−1	+1	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	+1	0	−1	−1	0	0	0	0	0
8	0	0	0	0	0	0	0	0	−1	+1	0	−1	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0	0	+1	0	0	−1	0
10	0	0	0	0	0	0	0	0	0	0	+1	0	0	0	0	−1

Incidence matrix

Suppose that instead of a simultaneous solution of the equations, a sequential solution is wanted. In what order should the equations be solved? Partition the equations so that a sequential solution can be executed. Lump together blocks of equations that still have to be solved simultaneously.

Solution

From Fig. E5.6 you can see that it is not necessary to solve all 10 equations simultaneously. The system of equations can be broken up into lower-order subsystems, some of which can be comprised of individual equations or small groups of equations. These groups are associated with the so-called *irreducible* sets of equations. By inspection you can establish the following



Subsystem	Balance
1	$0 = y_1 - y_{12} - y_{13}$
2	$0 = y_{12} + y_{32} - y_{23}$
3	$0 = y_{23} - y_{32} - y_{36} + y_{13}$
4	$0 = y_4 - y_{45}$
5	$0 = y_{45} - y_{56}$
6	$0 = y_{36} + y_{56} + y_{86} - y_{67}$
7	$0 = y_{67} - y_{78} - y_{710}$
8	$0 = y_{78} - y_{86} - y_{89}$
9	$0 = y_{89} - y_9$
10	$0 = y_{710} - y_{10}$

Figure E5.6 An example system containing recycle and bypass streams.

precedence order:

Step	Subsystem equations(s)
1	1
2	4
3	5
4	2 and 3 simultaneously
5	6, 7, and 8 simultaneously
6	9
7	10

In the case of more complicated sets of equations that cannot easily be decomposed by inspection, refer to some of the supplementary references at the end of this chapter for suitable algorithms.

EXAMPLE 5.7 Simultaneous Material and Energy Balances

A distillation column separates 10,000 lb/hr of a 40% benzene–60% chlorobenzene liquid solution which is at 70°F. The liquid product from the top of the column is 99.5% benzene, while the bottoms (stream from the reboiler) contains 1% benezene. The total condenser uses water that enters at 60°F and leaves at 140°F, while the reboiler uses saturated steam at 280°F. The reflux ratio (the ratio of the liquid overhead returned to the column to the liquid overhead product removed) is 6 to 1. Assume that both the condenser and reboiler operate at 1 atm pressure, that the temperature calculated for the condenser is 178°F and for the reboiler 268°F, and that the calculated fraction benzene in the vapor from the reboiler is 3.9 wt% (5.5 mole %). Calculate the following:

(a) The pounds of overhead product (distillate) and bottoms per hour
(b) The pounds of reflux per hour
(c) The pounds of liquid entering the reboiler and the reboiler vapor per hour
(d) The pounds of steam and cooling water used per hour

Solution

Steps 1, 2, and 7 Figure E5.7a will help visualize the process and assist in pointing out what additional data have to be determined.

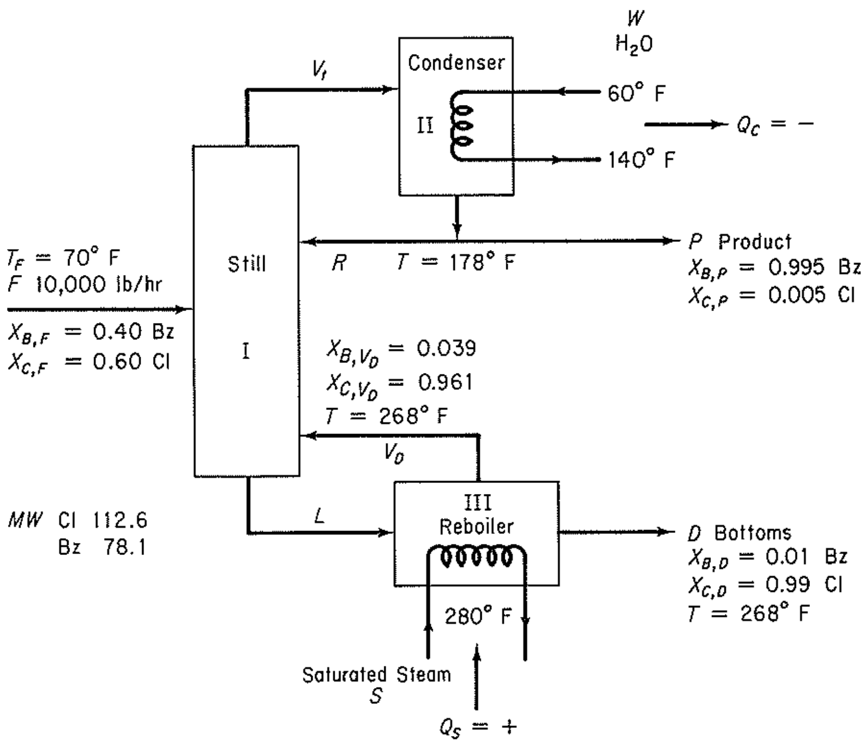


Figure E5.7a

Steps 5 and 6 An analysis of the degrees of freedom will reveal whether the process is properly specified or not. We assume that $\Delta\hat{H}_i$ is a function of temperature only so that $\Delta\hat{H}_i$ could be replaced as a variable by T_i. The water (W) and steam (S) can be calculated directly from Q_C and Q_S, respectively, hence will not be considered in the analysis of the degrees of freedom. Also, since the condenser is a total condenser, the liquid P and R have the same temperature and composition, and V_t must have the same composition as P. B is benzene and C is chlorobenzene.

Subsystem I (still)

7	unknown variables (R, V_D, V_t, L, $\omega_{B,L}$, T_{V_t}, T_L)
-3	equations (components B and C, and energy)
4	net degrees of freedom

Subsystem II (condenser)

5	unknown variables (V_t, R, P, Q_c, T_{V_t})
-2	equations (component B only independent, and energy)
3	net degrees of freedom

Subsystem III (reboiler)

6	unknown variables (L, D, V_D, Q_S, $\omega_{B,L}$, T_L)
-3	equations (components B and C, and energy)
3	net degrees of freedom

Whole system

10	subsystem degrees of freedom $(4 + 3 + 3)$
-7	interconnections (V_t, T_{V_t}, R, L, $\omega_{B,L}$, T_L, V_D)
-1	additional relations (reflex ratio specified: $R/P = 6$)
2	net degrees of freedom for the process

For the whole process, there are $7 + 5 + 6 = 18$ minus 7 equals 11 variables whose values are unknown, and 9 equations plus extra relations.

Table E5.7 lists the possible equations for review to help decide on the two additional specifications needed. Not all of the equations are independent, of course, and some are nonlinear. Let us assume that V_t is saturated vapor at $T_{V_t} = 178°F$ and L is subcooled liquid at $T_L = 248°F$.

Consequently, the process has the correct number of specifications and can be solved. You must always check to make sure that the equations are independent because the data given might in some instances (as shown in Chap. 2) lead to dependence among two or more equations.

Rather than solve nine equations simultaneously for the nine values of the variables which are unknown, some of which will be nonlinear, we will carry out a sequential procedure that leads to the final solution via solving small sets of linear equations simultaneously.

Step 4

Basis: 1 hr

TABLE E5.7

Balance	In	Out
Overall process		
(a) Total mass (lb)	10,000	$= P + D$
(b) Benzene (lb)	10,000 (0.40)	$= P(0.995) + D(0.01)$
(c) Energy (Btu)	$Q_C + Q_S + 10{,}000\Delta\hat{H}_{F,70°F}$	$= P\,\Delta\hat{H}_{P,178°F} + D\,\Delta\hat{H}_{d,268°F}$
Process I (still)		
(d) Total mass (lb)	$10{,}000 + R + V_D$	$= V_t + L$
(e) Benzene (lb)	$10{,}000(0.40) + R(0.995) + V_D(0.039)$	$= V_t(0.995) + L_{B,L}$
(*f*) *Energy* (*Btu*)	$10{,}000(\Delta\hat{H}_{F,70°F}) + R\,\Delta\hat{H}_{R,178°F} + V_D\,\Delta\hat{H}_{V_D,268°F}$	$= V_t\,\Delta\hat{H}_{V_t,T_{V_t}} + L\,\Delta\hat{H}_{L,T_L}$
Process II (condenser)		
(g) Total mass (lb)	V_t	$= R + P$
(h) Benzene (lb)	$V_t(0.995)$	$= R(0.995) + P(0.995)$
(i) Energy (Btu)	$Q_C + V_t\,\Delta\hat{H}_{V_t,T_{V_t}}$	$= R\,\Delta\hat{H}_{R,178°F} + P\,\Delta\hat{H}_{P,178°F}$
(j) Reflex ratio given	R	$= 6P$
Process III (reboiler)		
(k) Total (lb)	L	$= V_D + D$
(l) Benzene (lb)	$L_{B,L}$	$= V_D(0.039) + D(0.01)$
(m) Energy (Btu)	$Q_S + L\,\Delta\hat{H}_{L,T_L}$	$= V_d\Delta\hat{H}_{V_D,268°F} + D\Delta\hat{H}_{D,268°F}$

Step 3 First we have to get some pertinent enthalpy or heat capacity data so that the energy balances can be used. If you used a computer program, the data would be retrieved from a data base. (The exit streams have the same composition as the solutions in the condenser or reboiler, respectively.) The heat capacity data for liquid benzene (Bz) and chlorobenzene (Cl) will be assumed to be as follows (no enthalpy tables are available):

	C_p [Btu/(lb)(°F)]		$\Delta\hat{H}_{\text{vaporization}}$(Btu/lb)	
Temp. (°F)	Cl	Bz	Cl	Bz
70	0.31	0.405		
90	0.32	0.415		
120	0.335	0.43		
150	0.345	0.45		
180	0.36	0.47	140	170
210	0.375	0.485	135	166
240	0.39	0.50	130	160
270	0.40	0.52	126	154

SOURCE: Data estimated from tabulations in D. Q. Kern, *Process Heat Transfer*, McGraw-Hill, New York, 1950, Appendix.

Assume that the solutions are ideal so that each component can be treated independently.

Steps 7 and 8 Next we determine the sequence of solution of the system equations. If you used a computer code, you might be asked to provide such a sequence, or possibly the code would select a sequence for you if you wished. As indicated by Fig. 5.6, we first prepare an occurrence matrix and then rearrange the matrix (here by inspection since it is a small matrix) to give the precedence order for solution. Figure E5.7b is the occurrence matrix, and Fig. E5.7c is the precedence matrix with the encircled numbers and associated

Eq.	R	V_D	V_f	L	P	D	$\omega_{B,L}$	Q_C	Q_S
a					1	1			
b					1	1			
c					1	1		1	1
g	1		1		1				
i	1		1		1			1	
j	1				1				
k		1		1		1			
l		1		1		1	1		
m		1		1		1			1

Figure E5.7b Occurence Matrix

Eq.	D	P	R	V_f	Q_C	Q_S	L	V_D	$\omega_{B,L}$
a	1	1	(1)						
b	1	1							
j		1	1	(2)					
g		1	1	1	(3)				
i		1	1	1	1	(4)			
c	1	1			1	1	(5)		
m	1					1	1	1	(6)
k	1						1	1	
l	1						1	1	1 (7)

Figure E5.7c Precedence Ma

boxes indicating the equation to be solved, or set of equations to be solved, simultaneously. Note that we have substituted the overall balances for the process I (still) balances because common sense dictates making overall balances first.

Step 9

(1) Solution of Eqs. (a) and (b):

$$P = 3959 \text{ lb/hr} \longleftarrow (a_1)$$

$$D = 6041 \text{ lb/hr} \longleftarrow (a_2)$$

(The encircled letters indicate answers to the questions posed.)

(2) Solution of Eq. (j):

$$R = 6P = 6(3959) = 23{,}754 \text{ lb/hr} \longleftarrow (b)$$

(3) Solution of Eq. (g):

$$V_t = R + P = 23{,}754 + 3959 = 27{,}713 \text{ lb/hr}$$

(4) Solution of Eq. (i):

$$Q_C = -27{,}713\Delta\hat{H}_{V_t,178°F} + 23{,}754\Delta\hat{H}_{R,178°F} + 3959\Delta\hat{H}_{P,178°F}$$

Use a reference temperature of 178°F so that the sensible heat changes are zero. In a computer code, all the calculations might use the same common reference temperature such as 77°F or 32°F; hence all the terms in Eq. (i) would be involved, but we do not know the heat capacity data with sufficient accuracy here to integrate over large temperature ranges.

$$\Delta\hat{H}_{V_t,178°F} = 0.995(170) + 0.005(140) \cong 170 \text{ Btu/lb} \qquad \text{(vapor)}$$

$$\Delta\hat{H}_{R,178°F} = 0 \qquad \text{(liquid)}$$

$$\Delta\hat{H}_{P,178°F} = 0 \qquad \text{(liquid)}$$

$$Q_C = -4.711 \times 10^6 \text{ Btu/hr} \qquad \text{(heat removed)}$$

To calculate W, assume that $C_{p_{H_2O}} = 1$ Btu/(lb)(°F)

$$Q_C = WC_{p_{H_2O}}(140 - 60) = W(80)$$

$$W = 5.89 \times 10^4 \text{ lb } H_2O\text{/hr} \longleftarrow (d_1)$$

(5) Solution of Eq. (c):

$$-4.711 \times 10^6 + Q_S + 10{,}000\Delta H_{F,70°F} = 3959\Delta H_{P,178°F} + 6041\Delta H_{D,268°F}$$

Use as a reference temperature 70°F (all streams are liquid). To save space we make the assumption that the P stream is pure benzene and the B stream is pure chlorobenzene, and that average heat capacities are reasonable to use, quite satisfactory assumptions since the value of Q_C is one order of magnitude larger than the sensible heat terms.

$$\Delta\hat{H}_{F,70°F} = 0$$

$$\Delta\hat{H}_{P,178°F} = \int_{70}^{178} C_{p_P}\, dT = 0.435(178 - 70) = 47.0 \text{ Btu/lb}$$

$$\Delta\hat{H}_{D,268°F} = \int_{70}^{268} C_{p_D}\, dT = 0.355(268 - 70) = 70.3 \text{ Btu/lb}$$

$$Q_S = 5.32 \times 10^6 \text{ Btu/hr} \qquad \text{(heat introduced)}$$

To calculate S, assume that the saturated steam exits as saturated liquid at 280°F ($\Delta\hat{H}_{vap}$ = 923 Btu/lb from the steam tables)

$$Q_S = \Delta\hat{H}_{vap\,H_2O}\, S = 923(S)$$
$$\text{at } 280°\text{F}$$

$$S = \frac{5.32 \times 10^6 \text{ Btu/hr}}{923 \text{ Btu/lb}} = 5760 \text{ lb/hr} \longleftarrow \text{(d}_2\text{)}$$

(6) Solution of Eqs. (m) and (k):

$$L = V_D + 6041$$

$$5.32 \times 10^6 + L\,\Delta\hat{H}_{L,248°F} = V_D\,\Delta H_{V_D,268°F} + 6041\Delta\hat{H}_{D,268°F}$$

Use a reference temperature of 248°F, as we do not know the composition of stream L.

$$\Delta\hat{H}_{L,248°F} = 0 \qquad \text{(liquid)}$$

$$\Delta\hat{H}_{V_D,248°F} = \int_{248}^{268} C_{pV_D}\,dT + \Delta\hat{H}_{V_D,268°F}$$
$$= [0.039(0.515) + 0.961(0.395)](268 - 248)$$
$$+ [0.039(154) + 0.961(126) = 127.1 \text{ Btu/lb} \qquad \text{(vapor)}$$

$$\Delta\hat{H}_{D,268°F} = 70.3 \text{ Btu/lb} \qquad \text{(liquid)}$$

$$V_D = 38{,}520 \text{ lb/hr} \longleftarrow \text{(c}_2\text{)}$$

$$L = 44{,}560 \text{ lb/hr} \longleftarrow \text{(c)}$$

(7) Solution of Eq. (1):

$$44{,}560\ \omega_{B,L} = 38{,}520(0.039) + 6041(0.01)$$

$$\omega_{B,L} = 0.035 \quad \text{(not required)}$$

5.2-2 Modular-Based Flowsheeting

The *sequential modular* method of flowsheeting, as mentioned previously, is the one most commonly encountered in computer packages. A module exists for each process unit in the information flowsheet. Given the values of each input stream composition, flow rate, temperature, pressure, enthalpy, and the equipment parameters, the module calculates the properties of its outlet streams. The output stream for a module can become the input stream for another module for which the calculations proceed until the material and energy balances are resolved for the entire process.

The underlying concept of modularity in flowsheeting packages for design and analysis is to represent equipment or unit operations by portable computer subroutines. By portable we mean that such a subroutine can be assembled as an element of a large group of subroutines and successfully represent a certain type of equipment in any process. Figure 5.9(a and b) shows typical standardized unit operations modules together with their flowsheet symbols. Internally, a very simple module might

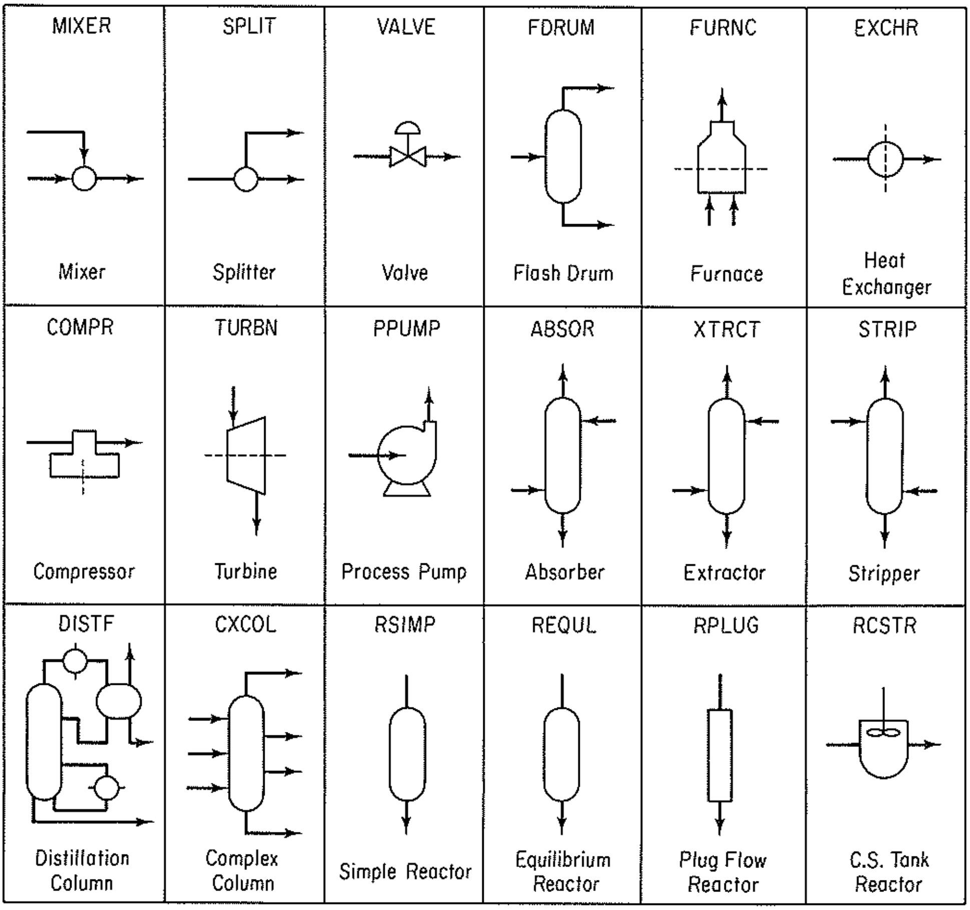


Figure 5.9a Typical process modules used in sequential modular-based flowsheeting codes with their subroutine names.

just be a table look-up program. However, most modules consist of Fortran subroutines that execute a sequence of calculations. Suroutines may consist of 100 to thousands of lines of code. Figure 5.10 illustrates the program used to execute a simple unit operation, the liquid pump subroutine PUMPX, in the FLOWTRAN package.

Other modules take care of equipment sizing and cost estimation, perform numerical calculations, handle recycle calculation (described in more detail below), optimize, and serve as controllers (executives) for the whole set of modules so that they function in the proper sequence. Figure 5.11 illustrates the information flow in a flowsheeting package composed of modules.

Modules representing equipment are written to be as general and as flexible as possible, so that they can be used for the analysis or design of a wide variety of plants. Then they can be applied to specific situations in the plant under consideration by introducing the parameters of the plant into the modules. Appropriate parameters are called upon as needed under the control of the executive program. These parameters may be physical dimensions (such as tube size in a heat ex-

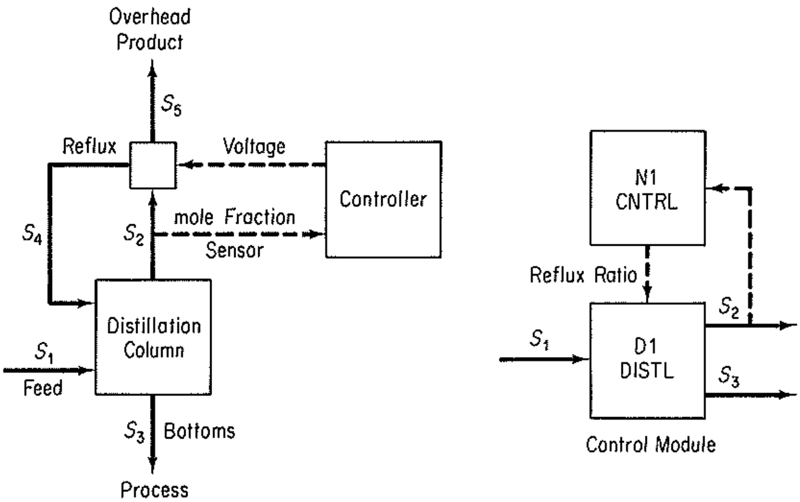


Figure 5.9b A feedback control module, N1: a graphical representation of a subroutine that adjusts equipment parameters to force a desired value of a stream variable. In the figure the feedback control block compares the stream variable value with the specification. When the convergence tolerance is satisfied, the next unit in the calculation order is computed. Otherwise, the control block adjusts the reflux ratio so as to achieve a specified distillate mole fraction.

```
      SUBROUTINE PUMPX (F, B, P, R)
C
C     Title        = Centrifugal pump size and power
C     Files        = FORTRAN 1, 6
C     Subprograms  = MOLVOL
C
C     B            Stream data for pump effluent
C     F            Stream data for pump feed
C     P(1)         Outlet pressure, psia
C     P(2)         Unit name
C     R(1)         Flowrate, gpm
C     R(2)         Pressure increase, psi
C     R(3)         Fluid horsepower, hp
C     R(4)         Pump efficiency
C     R(5)         Brake horsepower, hp
C     R(6)         Motor efficiency
C     R(7)         Electrical power, kilowatts
C
      IMPLICIT     DOUBLE PRECISION (A - H, O - Z)
      COMMON //    BHP, BHPL, DELP, EFFM, EFFP, EKW, FHP, GPM, GPML,
     1             VL, VV, X(20)
      COMMON /CONTRL/ NC, KOUT, KSKIP, KONV
      DIMENSION    B(*), F(*), P(2), R(7)
C
C                            Initialize
            UNAM   = P(2)
      IF    (KOUT .EQ. 2) GO TO 500
      DO 11 I      = 1, 7
   11       R(I)   = 0.
C                            Set outlet stream data from inlet stress
      DO 21 I      = 1, NC + 5
   21       B(I)   = F(I)
C                            Set outlet pressure
      IF    (P(1) .LE. 0.) GO TO 400
            B(NC + 3) = P(1)
C                            Check inlet flow and vapor fraction
      IF    (F(NC + 1) .LE. 0.) GO TO 402
      IF    (F(NC + 5) .GT. 0.) WRITE(1, 924) UNAM
```

```
C
C                               Compute pump flow and pressure increase
      DO 31 I      = 1, NC
   31       X(I)   = F(I)/F(NC + 1)
      IF    (MOLVOL(VL, VV, F(NC + 2), F(NC + 3), X, X, 2, 1) .EQ. 2)
     1            GO TO 406
            GPM    = F(NC + 1)*VL*.12468
            DELP   = B(NC + 3) - F(NC + 3)
C
C                               Compute fluid horsepower, pump efficiency,
C                               and brake horsepower
            FHP    = GPM*DELP*.5833E-3
            GPML   = LOG(MAX(20D0, MIN(GPM, 5000D0)))
            EFFP   = -.3161 + GPML*(.240123 - GPML*.0119889)
            BHP    = FHP/EFFP
C
C                               Compute motor efficiency and electrical
C                               power
            BHPL   = LOG(MAX(1D0, MIN(BHP, 5000D0)))
            EFFM   = .7982 + BHPL*(.033270 - BHPL*.0019334)
            EKW    = BHP/(1.34102*EFFM)
C
C                               Save results, print history report
            R(1)   = GPM
            R(2)   = DELP
            R(3)   = FHP
            R(4)   = EFFP
            R(5)   = BHP
            R(6)   = EFFM
            R(7)   = EKW
      WRITE (1, 902) UNAM, B(NC + 3), B(NC + 1)
      RETURN
C
C                               Print warning and error messages
  400 WRITE (1, 920) UNAM
      RETURN
  402 WRITE (1, 922) UNAM
      RETURN
  406 WRITE (1, 926) UNAM
      RETURN
C
C                               Print unit report
  500 WRITE (6, 950) UNAM, F(NC + 6), B(NC + 6)
      WRITE (6, 952) R(1), R(2), R(3), R(5), R(7), R(4), R(6)
      IF    (F(NC + 5) .GT. 0.) WRITE (6, 954)
      IF    (B(NC + 3) .LE. F(NC + 3)) WRITE (6, 956)
      RETURN
C
  902 FORMAT(1X, A6, 21H(PUMPX)  Outlet pres=, F7.1, 12H psia, Flow=,
     1      F10.1, 9H lbmol/hr)
C
  920 FORMAT(5H  ***, A6, '(PUMPX)  PARAM 1 is in error.  Feed',
     1                 ' pressure was used.')
  922 FORMAT(5H  ***, A6, '(PUMPX)  Feed flow is zero.')
  924 FORMAT(5H  ***, A6, '(PUMPX)  Feed is not all liquid.')
  926 FORMAT(5H  ***, A6, '(PUMPX)  Cannot compute fluid volume.')
C
  950 FORMAT(/ / 1X, A6, 17H(PUMPX)  Inlet = , A4, 12H , Outlet = , A4)
  952 FORMAT(5X, 11HFlow, gpm =, F9.2, 17H , Delta P, psi =, F8.2/
     1       5X, 11HFluid hp  =, F9.2, 17H , Brake hp      =, F8.2,
     2           12H , Elec kw =, F8.2/
     3       5X, 11HPump eff  =, F9.4, 17H , Driver eff    =, F8.4)
  954 FORMAT('     *Note: Inlet pressure is not less than outlet.')
  956 FORMAT('     *Note: Feed is not all liquid.  Outlet T AND FV ',
     1                 'may be wrong.')
      END
```

Figure 5.10 FLOWTRAN Subroutine for a Pump Block. The dummy arguments of the subroutine are: (1) F and B, the feed and effluent stream vectors containing stream flow information; (2) the equipment paramenter vector, R, which contains the outlet pressure; and (3) the retention vector, R, which contains the flowrate, pressure increase, fluid Hp, pump efficiency, brake Hp, motor efficiency, and electrical power in kilowatts, all of which are to be printed out. From Seader, I. D., W. D. Seider, and A. C. Pauls, *Flowtran Simulation—An Introduction* 3rd ed., CACHE, Austin, TX (1987).

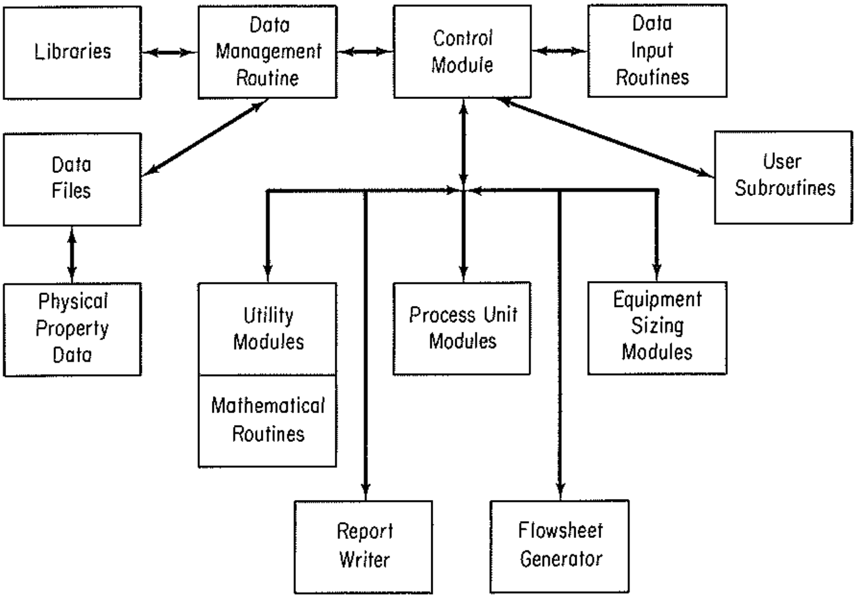


Figure 5.11 Information flow between modules in a typical sequential modular-based flowsheeting package.

changer), operating conditions (such as reflux ratio in a distillation column), reaction rate constants, convergence factors, and so one. Table 5.2 is a specification for a two-phase flash unit and Fig. 5.12 is the representation of a flash in FLOWTRAN.

Information flows between modules via the material streams. Associated with each stream is an ordered list of numbers that characterize the stream. Table 5.3 lists a typical set of parameters associated with a stream. As a user of a modular-based code, you have to provide

1. The process topology
2. Input stream information including physical properties, connections
3. Design parameters needed in the modules and equipment specifications
4. Convergence criteria

In addition, you may have to insert a preferred calculation order for the modules. When economic evaluation and optimization are being carried out, you must also provide cost data and optimization criteria.

Table 5.4 is a computer printout of the type of input data (for stream 11) needed for a process. The input data can be entered either in a batch mode or an interactive mode through a terminal, depending on the options available in the flowsheeting code, and its interactive flexibility. Most execute in a batch mode, but some allow preparation of input data in a conversational mode. In the initial processing of the input data, a main program is generated, compiled, loaded, and connected with the required modules and all the needed subprograms, and storage is allocated all prior to execution. Although you have to check the input data to make

TABLE 5.2 Specifications for a Two-phase Equilibrium Flash

Description

This flash capability solves the single-stage heat balance, material balance, and phase equilibrium relations for given inlet streams and user specifications. It applies to both the vapor–liquid and liquid–liquid cases. For liquid–liquid equilibria, replace "vapor" by "1st liquid" and "liquid" by "2nd liquid."

Inlet streams: Any number of vapor–liquid.

Outlet streams: One vapor and one liquid stream or one vapor–liquid stream.

Options for specifications

1. Temperature and pressure specified
2. Pressure and vapor fraction specified
3. Temperature and vapor fraction specified
4. Pressure and heat added specified
5. Temperature and heat added specified
6. Specified vapor flow rate

For all options, the vapor mole fraction and enthalpy, liquid mole fraction and enthalpy, and the unspecified values of temperature, pressure, or vapor fraction are calculated. In options 4 and 5, the inlet stream enthalpy must be known. In options 1, 2, and 3, the heat duty will be calculated if inlet stream enthalpy is known.

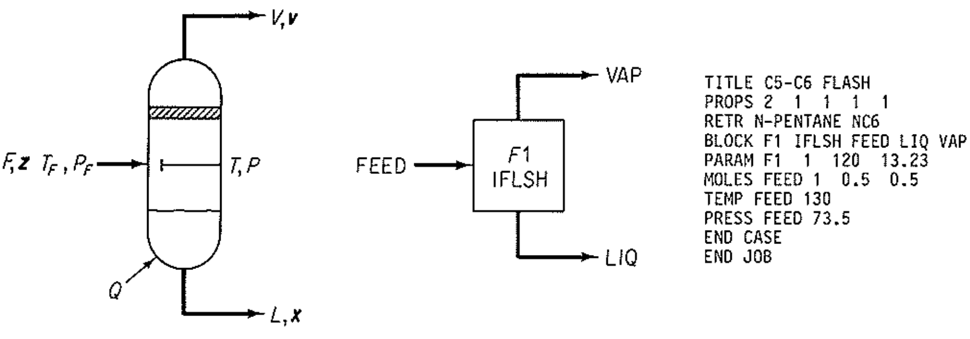
Under options 1, 4, and 5, it is possible for the result to be a subcooled liquid or a superheated vapor, in which case the vapor fraction is set equal to zero or 1 as appropriate, and the nonexistent phase composition has no meaning.

Additional results: Heat duty as discussed above

PPS requirements: K values and phase enthalpies

Outlet attributes: T, p, H, V

SOURCE: Adapted from *Functional Specifications for ASPEN,* Department of Chemical Engineering, Massachusetts Institute of Technology, December 1977.



(a) Flash Vessel (b) Simulation Flowsheet (c) FLOWTRAN Program

Figure 5.12 Simulation of a Flash Unit in FLOWTRAN. From Sender, J. D., W. D. Seider, and A. C. Pauls, *Flowtran Simulation-An Introduction,* CACHE, Austin, TX (1987).

TABLE 5.3 Stream Parameters

1.	Stream number*
2.	Stream flag (designates type of stream)
3.	Total flow, lb mol/hr
4.	Temperature, °F
5.	Pressure, psia
6.	Flow of component 1, lb mol/hr
7.	Flow of component 2, lb mol/hr
8.	Flow of component 3, lb mol/hr
9.	Molecular weight
10.	Vapor fraction
11.	Enthalpy
12.	Sensitivity

*Corresponds to an arbitrary numbering scheme used on the information flowsheet.

certain that they are correct, most packages have extensive input data capabilities as part of the preprocessing phase that give diagnostic error and warning messages.

You can regard a module as an information modifier. It receives information about feed streams, streams coming from other modules, together with parameters, carries out computations with the information, and produces output information which is forwarded to other modules and/or is reported. Under some circumstances output information may be fed back as input to the module.

Often, modules are prepared at several levels of sophistication for the same type of equipment. Approximate models in modules are used in the initial stages of setting up a study of the whole plant to get a rough idea of whether a design is worth pursuing, and to investigate the suitability of various convergence techniques. Subsequently, the approximate models can be replaced by more comprehensive models. The more comprehensive and accurate the model is, the greater the cost of program development and program execution. Also, you may find complex programs more difficult to understand and use, and it is easier to make mistakes in using them.

Modular-based flowsheeting exhibits several advantages in design. The flowsheet architecture is easily understood because it closely follows the process flowsheet. Individual modules can easily be added and removed from the computer package. Furthermore, new modules may be added to or removed from the flowsheet without any difficulty or affecting other modules. Modules at two different levels of accuracy can be substituted for one another as mentioned above.

Modular-based flowsheeting also has certain drawbacks.

1. The output of one module is the input to another. The input and output variables in a computer module are fixed so that you cannot arbitrarily introduce an output and generate an input as can be done with an equation-based code.
2. The modules make it costly to generate reasonable accurate derivatives or their substitutes, especially if a module contains tables, functions with discrete variables, discontinuities, and so on. Perturbation of the input to a module is the

TABLE 5.4 Typical Input Data for a Process

COMPONENT NO.	MOL. WT.	LBS.	LB MOLES	DENSITY, LB/CU. FT.	SPEC. HEAT, BTU/(LB)(F)	HT. VAPOR.
1	2	0	0		3.5	
2	60	0	0			
3	78	4644.9	59.551	54.7		
4	84	0	0	44.3	0.43	154
APPROX. TOTAL		4644.9	59.551			

NO. STREAM	NO. COMPON.	FLOW RATE	CONNECTIONS	CALC. ORDER
11	4	59.551	12 −4	6

TEMP., °F	PRESS, PSIG	AVG. MOL. WT.	CONVG. FRACT.	EQUIPMENT
100.0	285.	78.0	0.02	2

primary way in which a finite-difference substitute for a derivative can be generated.

3. The modules may require a fixed precedence order of solution, that is, the output of one module must become the input of another; hence convergence may be slower than in an equation-solving method, and the computational costs may be high.
4. To specify a parameter in a module as a design variable, you have to place a control block around the module and adjust the parameter such that design specifications are met. This arrangement creates a loop. If the values of many design variables are to be determined, you might end up with several nested loops of calculation (which do, however, enhance stability). A similar arrangement must be used if you want to impose constraints.
5. Conditions imposed on a process (or a set of equations for that matter) may cause the unit physical states to move from a two-phase to a single-phase operation, or the reverse. As the code shifts from one module to another to represent the process properly, a severe discontinuity occurs and physical property values may jump about.

To obtain a solution for the material and energy balances in a flowsheet by the sequential modular method, you must partition the flowsheet, select tear streams, nest the computations, and thus determine the computation sequence.

Partitioning of the modules in a block information diagram into minimum-sized subsets of modules that have to be solved simultaneously can be executed by many methods. As with solving sets of equations, you want to obtain the smallest block of modules in which the individual modules are tied together by the information flow of outputs and inputs. Between blocks, the information flow occurs serially. A simple algorithm is to trace a path of the flow of information (material usually but could be energy or a signal) from one module to the next through the module output streams. The tracing continues until (a) a module in the path is encountered again, in which case all the modules in the path up to the repeated module form a group together which is collapsed and treated as a single module in subsequent tracing, or (b) a module or group with no output is encountered, in which case the module or group of modules is deleted from the block diagram. The block diagram in Fig. 5.13 can be partitioned by the following steps.

Start with an arbitrary unit, say 4, and any sequence.

Path 1:	Start:	$4 \rightarrow 5 \rightarrow 6 \rightarrow 4$	collapse as (456)
	Continue:	$(456) \rightarrow 2 \rightarrow 4$	collapse as (4562)
	Continue:	$(4562) \rightarrow 1 \rightarrow 2$	collapse as (45621)
	Continue:	$(45621) \rightarrow 7 \rightarrow 8 \rightarrow 7$	collapse (78)
	Continue:	$(45621) \rightarrow (78) \rightarrow 9$	terminate (no output)

Put number 9 on the precedence order list and delete module 9 from the information block diagram.

Continue: $(45621) \rightarrow (78) \rightarrow 9$ Terminate (put 78 on the list before 9)

Delete (78) from the block diagram.

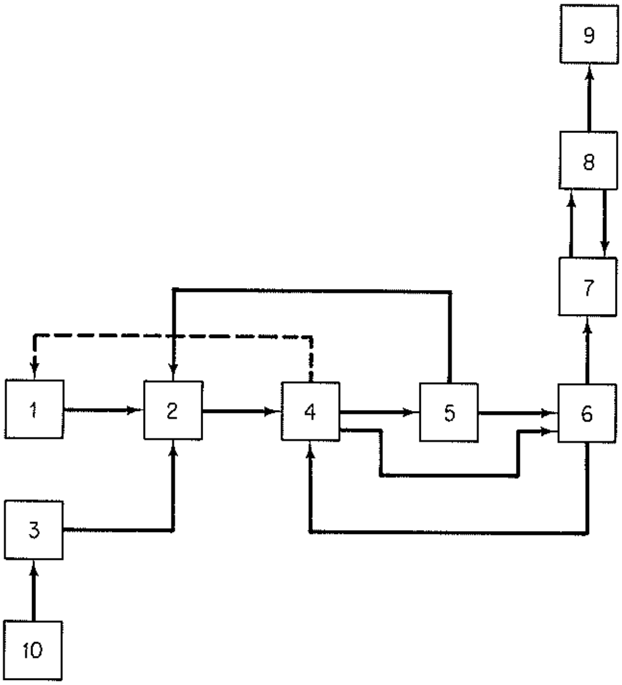


Figure 5.13 Block diagram to be partitioned.

Continue:	(45621)	Terminate (put 45621 before 78 in the list)

Delete (45621) from the block diagram

This completes the search in path 1, as no more modules exist.

Path 2:	Start: 10 → 3	Terminate (put 3 in list before 45621)
	Continue: 10	Terminate (put 10 in list before 3)

All of the modules have been deleted from the block diagram, and no more paths have to be searched. Computer techniques to partition complex sets of modules besides the one described above can be found in the supplementary references at the end of this chapter. Simple sets can be partitioned by inspection.

From a computational viewpoint, the presence of recycle streams is one of the major impedements in the sequential solution of a flowsheeting problem. Without recycle streams, the flow of information would proceed in a forward direction, and the calculational sequence for the modules could easily be determined from the precedence order analysis outlined above. With recycle streams present, large groups of modules have to be solved simultaneously, defeating the concept of a sequential solution module by module. For example, in Fig. 5.14, you cannot make a material balance on the reactor without knowing the information in stream S6, but you have to carry out the computations for the cooler module first to evaluate S6, which in turn depends on the separator module, which in turn depends on the reactor module. Partitioning will identify those collections of modules that have to be solved simultaneously (termed *maximal cyclical subsystems* or *irreducible nets*).

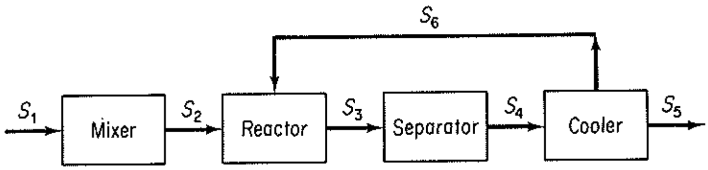


Figure 5.14 Modules in which recycle occurs; information (material) from the cooler module is fed back to the reactor.

To execute a sequential solution for a set of modules, you have to tear certain streams. *Tearing* in connection with modular flowsheeting involves decoupling the interconnections between the modules so that sequential information flow can take place. Tearing is required because of the loops of information created by recycle streams. What you do in tearing is to provide initial guesses for values of some of the unknowns (the tear variables), usually but not necessarily the recycle streams, and then calculate the values of the tear variables from the modules. These calculated values form new guesses, and so on, until the differences between the estimated and calculated values are sufficiently small. *Nesting* of the computations determines which tear streams are to be converged simultaneously and in which order collections of tear streams are to be converged.

Physical insight and experience in numerical analysis are important in selecting which variables to tear. For example, Fig. 5.15 illustrates an equilibrium vapor–liquid separator for which the combined material and equilibrium equations give the relation

$$\sum_{j=1}^{C} \frac{z_j(1 - K_j)}{1 - (V/F) + VK_j/F} = 0 \tag{5.4}$$

where z_j is the mole fraction of species j out of C components in the feed stream, $K_j = y_j/x_j$ is the vapor–liquid equilibrium coefficient, a function of temperature, and the stream flow rates are noted in the figure. For narrow-boiling systems, you can guess V/F, y_j, and x_j, and use Eq. (5.4) to calculate K_j and hence the temperature. This scheme works well because T lies within a narrow range. For wide-boiling ma-

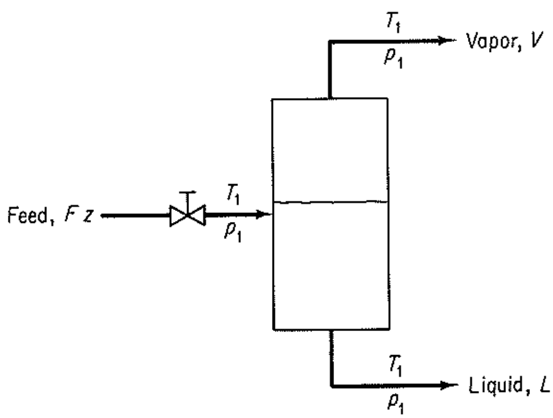


Figure 5.15 Vapor–liquid separator

terials, the scheme does not converge well. It is better to solve Eq. (5.4) for V/F by guessing T, y, and x, because V/F lies within a narrow range even for large changes in T. Usually, the convergence routines for the code comprise a separate module whose variables are connected to the other modules via the tear variables. Examine Fig. 5.16.

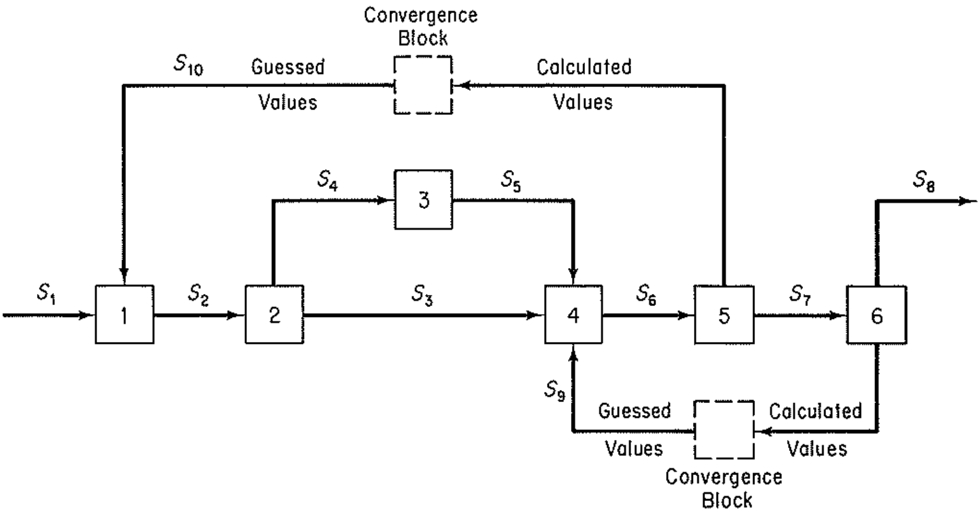


Figure 5.16 A computational sequence modular flowsheeting. Initial values of both recycles are guessed, then the modules are solved in the order 1, 2, 3, 4, 5, and 6. Calculated values for recycle streams S9 and S10 are compared to guessed values in a convergence block, and unless the difference is less than some prescribed tolerance, another iteration takes place with the calculated values, or estimates based on them, forming the new initial guessed values of the recycle streams.

An engineer can usually carry out the partitioning, tearing, nesting, and determine the computational sequence for a flowsheet by inspection if the flowsheet is not too complicated. In codes such as FLOWTRAN, the user supplies the computational sequence as input. Other codes determine the sequence automatically. In ASPEN, for example, the code is capable of determining the entire computational sequence, but the user can supply as many specifications as desired, up to and including the complete sequence. Consult one of the supplementary references at the end of this chapter for detailed information on optimal techniques of partitioning and tearing, techniques beyond our scope in this text.

Once the tear streams are identified and the sequence of calculations specified, everything is in order for the solution of material and energy balances. All that has to be done is to calculate the correct values for the stream flow rates and their properties. To execute the calculations, many computer codes use the method of successive substitution, which is described in Appendix L. The output(s) of each module on interation k is expressed as an explicit function of the input(s) calculated from the previous iteration, $k - 1$. For example, in Fig. 5.16 for module 1,

$$S2^{(k)} = f(S1^{(k-1)}, S10^{(k-1)}, \text{coefficients})$$

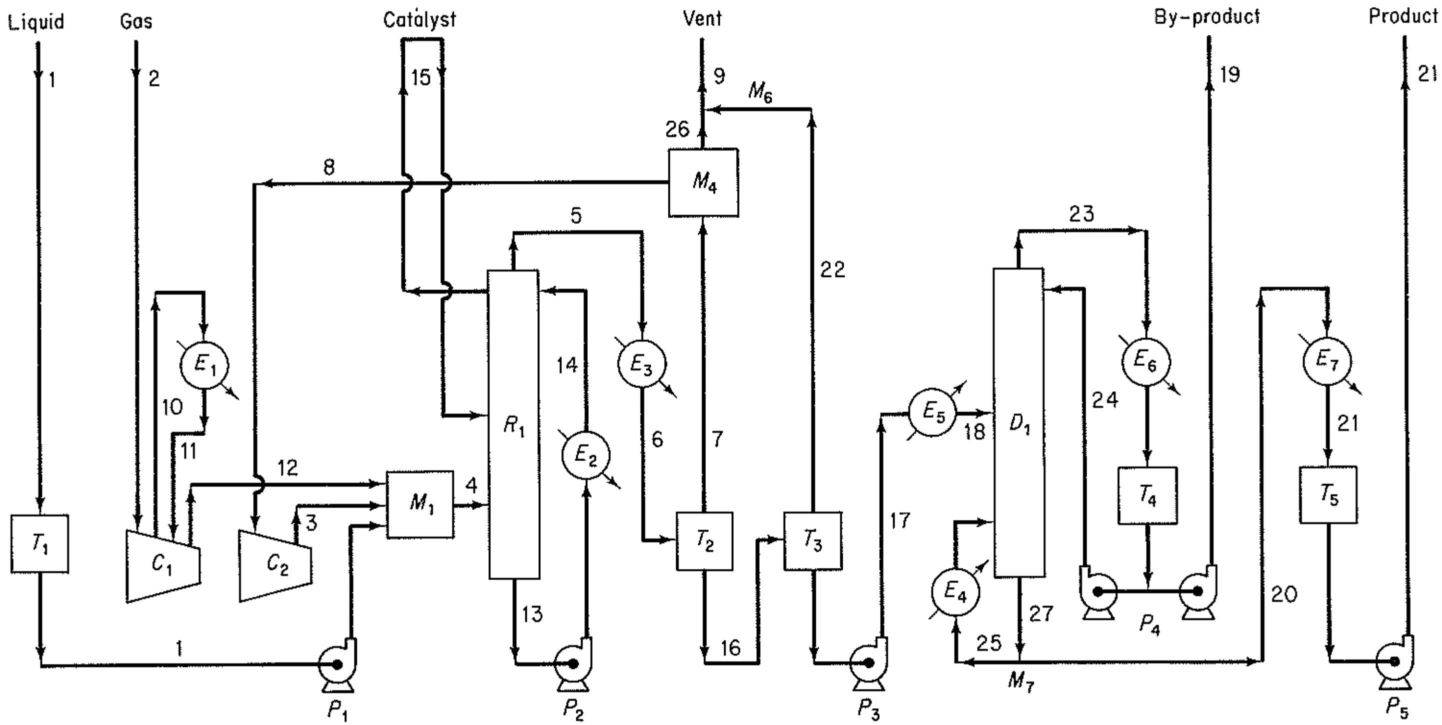


Figure E5.8 C = COMPRESSOR, D = DISTILLATION UNIT, E = HEAT EXCHANGER, P = PUMP, R = REACTOR, T = TANK, M = MIXER OR SPLITTER. Streams are identified by Arabic numbers.

To accelerate recycle stream convergence, the Wegstein method (refer to Appendix L) has been the mainstay of sequential modular flowsheeting for almost 25 years. Although this method neglects possible interaction between variables in tear streams, for most systems it works very well. A number of heuristics have been developed to improve convergence by delaying application of the Wegstein acceleration step until a specified number of direct substitution steps have been made and by setting bounds on the maximum acceleration allowed. With the use of Wegstein's method or direct substitution, it is necessary to control the convergence of recycle streams separately from the convergence of module specifications. Newton or quasi-Newton methods can also be used to solve for recycle streams.

EXAMPLE 5.8 Solution of Material and Energy Balances Using a Sequential Modular-Based Flowsheeting Code

Figure E5.8 is a hypothetical process used for demonstration by Diamond Shamrock Co. of their flowsheeting code PROVES. Makeup gas is compressed, combined with recycle gas, and fed, together with liquid raw material, into a three-phase, suspended bed catalytic reactor. The reactor is cooled by recirculating liquid through a heat reclamation steam generator. Reaction products are condensed and the pressure of the exit stream reduced in two stages. The gas from the first-stage separator is recirculated, whereas the liquid from the second-stage separator is fed into a distillation column. Pure product is withdrawn from the bottom of the column. The distillate is a by-product that is pumped to another plant.

Table E5.8a lists the input data for the flowsheeting code needed to solve the material and energy balances. Table E5.8b is a computer printout of the flow rates, temperatures, and pressures of the 21 major streams.

SUPPLEMENTARY REFERENCES

Flowsheeting Codes

BROYDEN, C. G., "A Class of Methods for Solving Nonlinear Simultaneous Equations," *Math Comput.*, v. 19, p. 577 (1965).

CHEN, H. S., and M. A. STADTHERR, "A Simultaneous-Modular Approach to Process Flowsheeting and Optimization: I. Theory and Implementation," *AIChE J.*, v. 30 (1984).

HANYAK, M. E., "Textual Expansion of Chemical Process Flowsheets into Algebraic Equation Sets," *Comput. Chem. Eng.*, v. 4, p. 223 (1980).

HILLESTAD, M., and T. HERTZBERG, "Dynamic Simulation of Chemical Engineering Systems by the Sequential Modular Approach," *Comput. Chem. Eng.*, v. 10, p. 377 (1986).

JOULIA, X., and B. KOEHRET, "Simultaneous Modular Approach to Process Flowsheeting," EFCE Pub. Ser. 1983, 27, Cong. Int. "Inf. Genie Chem," Vol. 2, C70/1-C70/7(1983).

MONTAGNA, J. M., and O. A. IRIBARREN, "Optimal Computation Sequence in the Simulation of Chemical Plants," *Comput. Chem. Eng.*, v. 12, p. 71 (1988).

NISHIDA, N., G. STEPHANOPOULOS, and A. W. WESTERBERG, "A Review of Process Synthesis," *AIChE J.*, v. 27, p. 321 (1981).

TABLE E5.8a Input Data

Number of streams	21	(first 21 only)
Number of components	4	
Number of operations	9	

Component [name, mol. wt., no., density (lb/ft^3), C_p (Btu/(lb) (F)), ΔH_{vap}]

Gas	2.	1	0	3.5	0
Catalyst	60.	2	0	0	0
Liquid A	78.	3	54.7	0.43	169.
Liquid B	84.	4	44.3	0.48	154.

Stream Data

No.	Flow Rate	Units*	Temp (°F)	Press (psig)
1	59.551	0	100.	0
2	183.	0	100.	0
3	0	0	140.	285.
4	0	0	100.	285.
5	0	0	400.	275.
6	0	0	120.	270.
7	0	0	120.	265.
8	0	0	120.	265.
9	0	0	120.	265.
10	183.	0	250.	95.
11	183.	0	120.	95.
12	183.	0	250.	285.
13	9000.	1	400.	285.
14	9000.	1	295.	285.
15	0.8	1	100.	0
16	0	0	120.	0
17	0	0	120.	0
18	0	0	200.	15.
19	0	0	200.	0
20	0	0	230.	5.
21	0	0	120.	0

Component Data

Stream No.	Gas	Catalyst	Liquid A	Liquid B
1	0	0	59.551	0
2	183.	0	0	0
3	0	0	0	0
4	0	0	0	0
5	0	0	0	0
6	0	0	0	0
7	0	0	0	0
8	0	0	0	0
9	9	9	9	9
10	183.	0	0	0
11	183.	0	0	0
12	183.	0	0	0

TABLE E5.8a *(Continued)*

13	0	760.	81.	89159.
14	0	760.	81.	89159.
15	0	0.8	0	0
16	0	0	0	0
17	0	0	0	0
18	0	0	0	0
19	0	0	0	0
20	0	0	0	0
21	0	0	0	0

Stream Connections

Module	Streams (−1 = out)				
T1	1	− 1			
C1	2	11	− 10	− 12		
E1	10	− 11				
C2	8	3				
M1	12	3	1	− 4		
R1	4	15	14	− 15	− 13	− 5
E2	13	− 14				
E3	5	− 6				
T2	6	− 7	− 16			
M4	7	− 8	− 26			
M6	26	22	− 9			
T3	16	− 22	− 17			
E4	25	− 25				
E5	17	− 18				
D1	18	25	24	− 23	− 27	
E6	23	− 23				
T4	23	− 23				
M7	27	− 20	− 25			
E7	20	− 21				
T5	21	− 21				
P1	1	− 1				
P2	13	− 13				
P3	17	− 17				
P4A	24	− 24				
P4B	19	− 19				
P5	21	− 21				

Suggested Calculation Order (1 = serial, 2 = in loop, 3 = end of loop)

M1	R1	T2	M4	E3	C2	E5	D1	E7
2	2	2	2	2	3	1	1	1

Convergence Tolerances

Total flow	0.05			
Component flow	0.02	0.02	0.02	0.02

TABLE E5.8a *(Continued)*

Designation of Operations (type, no. inlet streams, fractional split)

M1	MIXR	3	0
M4	MIXR	1	0.07479
E3	EXCH	1	0
C2	COMP	2	0
E5	EXCH	1	0
E7	EXCH	1	0

Separation
(type, fraction of inlet component that goes into first outlet stream)

T2	SEPN	1	0	0	0

Distillation
[mole fraction in outlet stream (+ = 1st stream; − = 2nd stream)]

D1	DIST	0	0	−0.01099	0.01958

Reactor
(type, fraction conversion of key component, key component number, stoichiometric coefficient of components 1, 2, . . .)

R1	0.9	3.0	3.01423	0	1	−1.00032

Cooling Water Temperatures [F (inlet, temperature difference)]

87	28

Generated Steam Pressure [psig, temperature, ΔH_{vap} (Btu/lb)]

40	287	920

Compressor
[stream suction, stream discharge, key component, C_p/C_v, efficiency, interstage temperature of cooling gas (°F)]

C1	2	12	1	1.41	0.72	120
C2	8	3	1	1.41	0.72	

Heat Exchanger (1 = liq., 2 = gas, 4 = liq. + solid, 6 = boiling liq.,
7 = condensing vapor, 8 = noncondensable gas + condensing vapor)

TABLE E5.8a (*Continued*)

	Stream No. at Cool End	Phase at Cool End	Stream No. at Warm End	Phase at Warm End	Heat Duty (Btu/hr)
E1	11	2	10	2	
E2	14	4	13	4	1.7×10^6
E3	6	8	5	8	
E4	25	1	25	1	
E5	17	1	18	6	
E6	23	7	23	1	
E7	21	1	20	1	

Pump

[stream no., key component no., equipment type, dynamic head (ft), efficiency]

P1	1	3	1	600.	0.45
P2	13	4	1	80.	0.45
P3	17	4	1	50.	0.45
P4	23	4	1	50.	0.45
P5	21	4	1	20.	0.45

Reactor–Fluidized Bed [inlet stream no., outlet stream no., key component of product, reaction temp. (F), avg. press (psig), productivity (lb/hr) (ft³), reactor/bed volume]

R1	4	5	4	395.	285.	16.	1.5

Tank	Stream No. Inlet	Outlet		Final Pressure (psig)	Phase†	Key Comp.	Retention Time (hr)
T1	1	− 1		0.	1	3	24.
T2	6	− 7	− 16	265.	1	4	0.1
T3	16	− 17	− 22		1	4	0.1
T4	23	− 19	− 24		1	4	0.1
T5	21	− 21			1	4	12.

*0, lbmol/hr; 1, lb/hr.

† liquid and/or solid; 2, gas; 3, gas + liquid (and solid).

TABLE E5.8b

COMPONENT			STREAM NO.			
			1		2	
NO.	NAME	MOL. WT.	LBS	LBMOLES	LBS	LBMOLES
1	GAS	2.0	0.0	0.000	366.0	183.000
2	CATALYST	60.0	0.0	0.000	0.0	0.000
3	LIQUID A	78.0	4644.9	59.551	0.0	0.000
4	LIQUID B	84.0	0.0	0.000	0.0	0.000
	APPROX. TOTAL		4644.9	59.551	366.0	183.000
	TEMPERATURE, F.		100.0		100.0	
	PRESSURE, PSIG.		0.0		0.0	
	AVG. MOL. WEIGHT		78.0		2.0	

			6		7	
NO.	NAME	MOL. WT.	LBS	LBMOLES	LBS	LBMOLES
1	GAS	2.0	477.6	238.813	477.6	238.813
2	CATALYST	60.0	0.0	0.000	0.0	0.000
3	LIQUID A	78.0	464.4	5.955	0.0	0.000
4	LIQUID B	84.0	4503.4	53.613	0.0	0.000
	APPROX. TOTAL		5445.6	298.381	477.6	238.813
	TEMPERATURE, F.		120.0		120.0	
	PRESSURE, PSIG.		270.0		265.0	
	AVG. MOL. WEIGHT		18.2		2.0	

			11		12	
NO.	NAME	MOL. WT.	LBS	LBMOLES	LBS	LBMOLES
1	GAS	2.0	366.0	183.000	366.0	183.000
2	CATALYST	60.0	0.0	0.000	0.0	0.000
3	LIQUID A	78.0	0.0	0.000	0.0	0.000
4	LIQUID B	84.0	0.0	0.000	0.0	0.000
	APPROX. TOTAL		366.0	183.000	366.0	183.000
	TEMPERATURE, F.		120.0		250.0	
	PRESSURE, PSIG.		95.0		285.0	
	AVG. MOL. WEIGHT		2.0		2.0	

S T R E A M   N O .

3		4		5	
LBS	LBMOLES	LBS	LBMOLES	LBS	LBMOLES
434.7	217.363	800.7	400.363	477.6	238.813
0.0	0.000	0.0	0.000	0.0	0.000
0.0	0.000	4644.9	59.551	464.4	5.955
0.0	0.000	0.0	0.000	4503.4	53.613
434.7	217.363	5445.7	459.914	5445.6	293.381
140.0		100.0		400.0	
285.0		285.0		275.0	
2.0		11.8		18.2	

8		9		10	
LBS	LBMOLES	LBS	LBMOLES	LBS	LBMOLES
441.9	220.952	35.7	17.860	366.0	183.000
0.0	0.000	0.0	0.000	0.0	0.000
0.0	0.000	0.0	0.000	0.0	0.000
0.0	0.000	0.0	0.000	0.0	0.000
441.9	220.952	35.7	17.860	366.0	183.000
120.0		120.0		250.0	
265.0		265.0		95.0	
2.0		2.0		2.0	

13		14		15	
LBS	LBMOLES	LBS	LBMOLES	LBS	LBMOLES
0.0	0.000	0.0	0.000	0.0	0.000
760.0	12.666	760.0	12.666	0.8	0.013
81.0	1.038	81.0	1.038	0.0	0.000
89159.0	1061.416	89159.0	1061.416	0.0	0.000
90000.0	1075.122	90000.0	1075.122	0.8	0.013
400.0		295.0		100.0	
285.0		285.0		0.0	
83.7		83.7		60.0	

TABLE E5.8b *(Continued)*

COMPONENT			STREAM NO. 16		STREAM NO. 17	
NO.	NAME	MOL. WT.	LBS	LBMOLES	LBS	LBMOLES
1	GAS	2.0	0.0	0.000	0.0	0.000
2	CATALYST	60.0	0.0	0.000	0.0	0.000
3	LIQUID A	78.0	464.4	5.955	464.4	5.955
4	LIQUID B	84.0	4503.4	53.613	4503.4	53.613
	APPROX. TOTAL		4967.9	59.568	4967.9	59.568
	TEMPERATURE, F.					
	PRESSURE, PSIG.		120.0		120.0	
	AVG. MOL. WEIGHT		0.0		0.0	
			83.4		83.4	

EQUIPMENT SPECIFICATION

ITEM NO.	NAME	EQUI. NO.	TYPE	REQUIRED NO.
1	COMPRESSOR	1	2-ST. DUPL. REC.	1
2	COMPRESSOR	2	1-STAGE RECIPR.	1
3	HEAT EXCHANGER	1	FIXED TUBE SHEET	1
4	HEAT EXCHANGER	2	FLOATING HEAD	1
5	HEAT EXCHANGER	3	FLOATING HEAD	1
6	PUMP	1	CENTRIFUGAL	1
7	PUMP	2	CENTRIFUGAL	1
8	REACTOR	1	FLUIDIZED BED	1
9	TANK	1	CYLINDRICAL	1
10	TANK	2	CYLINDRICAL	1
11	TANK	3	CYLINDRICAL	1
12	DISTILN. COLUMN	1	BUBBLE PLATE	1
13	BOILER	1	FIXED TUBES	1
14	CONDENSER	1	FLOATING HEAD	1
15	REFLUX PUMP	1	CENTRIFIGUAL	1
16	REFLUX TANK	1	CYLINDRICAL	1
17	HEAT EXCHANGER	1	FLOATING HEAD	3
18	HEAT EXCHANGER	2	FLOATING HEAD	1
19	PUMP	1	CENTRIFUGAL	2
20	PUMP	2	CENTRIFUGAL	1
21	PUMP	3	CENTRIFUGAL	1
22	TANK	1	CYLINDRICAL	2

STREAM NO.

18 LBS	18 LBMOLES	19 LBS	19 LBMOLES	20 LBS	20 LBMOLES	21 LBS	21 LBMOLES
0.0	0.000	0.0	0.000	0.0	0.000	0.0	0.000
0.0	0.000	0.0	0.000	0.0	0.000	0.0	0.000
464.4	5.955	418.1	5.360	46.3	0.594	46.3	0.594
4503.4	53.613	8.9	0.107	4494.5	53.505	4494.5	53.505
4967.9	59.568	427.1	5.467	4540.8	54.100	4540.8	54.100

18	19	20	21
200.0	200.0	230.0	120.0
285.0	0.0	5.0	0.0
83.4	78.1	83.9	83.9

TEMP. F	PRES. PSIG	PARAMETER 1		PARAMETER 2		MATERIAL
250.	285.	1094.	SCFPM	285.	PSI.DIF.	
140.	285.	1322.	SCFPM	20.	PSI.DIF.	
250.	95.	237.	SQ.FT.	166.	M BTU/HR	STEEL
400.	285.	571.	SQ.FT.	1700.	M BTU/HR	STEEL
400.	275.	212.	SQ.FT.	1862.	M BTU/HR	STEEL
100.	0.	10.	GPM	600.	FT.	CAST IRON
400.	285.	253.	GPM	80.	FT.	CAST IRON
395.	285.	422.	CU.FT.	0.		STEEL
100.	0.	19820.	GAL.	0.		STEEL
120.	265.	109.	GAL.	0.		STEEL
120.	0.	109.	GAL.	0.		STEEL
230.	10.	39.	PLATES	22.	IN.DIA.	STEEL
338.	100.	59.	SQ.FT.	701.	M BTU/HR	STEEL
200.	40.	47.	SQ.FT.	792.	M BTU/HR	STEEL
200.	10.	9.	GPM	70.	FT.	CAST IRON
200.	10.	78.	GAL.	0.		STEEL
338.	100.	13.	SQ.FT.	339.	M BTU/HR	STEEL
230.	40.	24.	SQ.FT.	239.	M BTU/HR	STEEL
120.	0.	13.	GPM	50.	FT.	CAST IRON
120.	0.	13.	GPM	50.	FT.	CAST IRON
120.	0.	699.	GPM	20.	FT.	CAST IRON
120.	0.	5981.	GAL.	0.		STEEL

TABLE E5.8b *(Continued)*

UTILITIES REQUIREMENTS

IT. NO.	EQUIP-MENT	EL. POWER 500 V KW	STEAM 100 PSIG LBS/HR	WATER COOLING GPM	G. STEAM* 50 PSIG LBS/HR	BOILER FEEDWAT. GPM
1	COMP 1	184.1	0.0	0.0	0.0	0.0
2	COMP 2	3.7	0.0	0.0	0.0	0.0
3	EXCH 1	0.0	0.0	13.0	0.0	0.0
4	EXCH 2	0.0	0.0	0.0	1663.0	3.6
5	EXCH 3	0.0	0.0	146.3	0.0	0.0
6	PUMP 1	2.3	0.0	0.0	0.0	0.0
7	PUMP 2	6.0	0.0	0.0	0.0	0.0
TOTAL		196.2	0.0	159.4	1663.0	3.6

* G. STEAM = GENERATED STEAM

Seader, J. D., W. D. Seider, and A. C. Pauls, *Flowtran Simulation—An Introduction,* CACHE Corp., Austin, Tex., 1987.

Shacham, M., S. Macchetto, L. F. Stutzman, and P. Babcock, "Equation-Oriented Approach to Flowsheeting," *Comput. Chem. Eng.*, v. 6, p. 79 (1982).

Simulation Sciences, Inc., *PROCESS Simulation Program Input Manual,* Simulation Sciences, Inc., Fullerton, Calif., 1984.

Stadtherr, M. A., H. S. Chen, and C. M. Hilton, "Development of a New Equation-Based Process Flowsheeting System: Numerical Studies," *Selected Topics on Computer-Aided Process Design and Analysis,* R. S. H. Mah and G. V. Reklaitis, eds., AIChE Symposium Series, v. 78, no. 214, p. 12 (1982).

Stephanopoulos, G., "Synthesis of Process Flowsheets: An Adventure in Heuristic Design or a Utopia of Mathematical Programming," in *Foundations of Computer-Aided Design,* Vol. 2, R. S. H. Mah and W. D. Seider, eds., Engineering Foundation, p. 439 (1981).

Westerberg, A. W., and H. H. Chien, eds., *Proc. 2d Int. Conf. Found. Comput.-Aided Process Des.,* CACHE Corp., Austin, Tex., 1984.

Westerberg, A. W., and H. H. Chien, eds., "Thoughts on a Future Equation-Oriented Flowsheeting System," *Comput. Chem. Eng.*, v. 9, p. 517 (1985).

Westerberg, A. W., H. P. Hutchinson, R. L. Motard, and P. Winter, *Process Flowsheeting,* Cambridge University Press, Cambridge, 1979.

PROBLEMS

An asterisk designates problems appropriate for solution using a computer. Refer also to the problems that require writing computer programs at the end of the chapter.

		EL. POWER 500 V	STEAM 100 PSIG	WATER COOLING	G. STEAM* 50 PSIG	BOILER FEEDWAT.
IT. NO.	EQUIP-MENT	KW	LBS/HR	GPM	LBS/HR	GPM
1	DIST 1	0.2	857.0	62.0	0.0	0.0
2	EXCH 5	0.0	415.3	0.0	0.0	0.0
3	EXCH 7	0.0	0.0	18.8	0.0	0.0
4	PUMP 3	0.1	0.0	0.0	0.0	0.0
5	PUMP 4	0.1	0.0	0.0	0.0	0.0
6	PUMP 5	0.0	0.0	0.0	0.0	0.0
TOTAL		0.6	1272.3	18.1	0.0	0.0

Section 5.1

5.1. Determine the number of degrees of freedom for the condenser shown in Fig. P5.1.

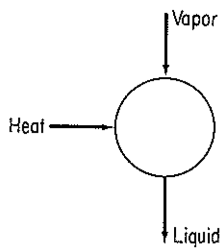


Figure P5.1

5.2. Determine the number of degrees of freedom for the reboiler shown in Fig. P5.2. What variables should be specified to make the solution of the material and energy balances determinate?

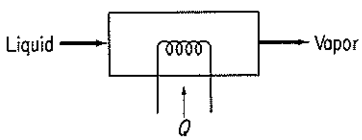


Figure P5.2

5.3. If to the equilibrium stage shown in Example 5.1 you add a feed stream, determine the number of degrees of freedom. See Fig. P5.3.

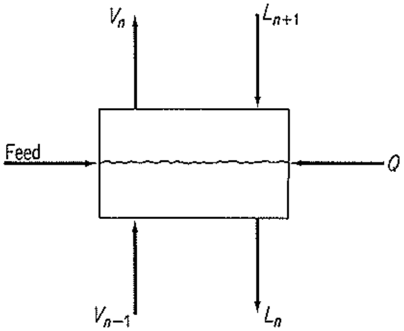


Figure P5.3

5.4. How many variables must be specified for the furnace shown in Fig. P5.4 to absorb all the degrees of freedom?

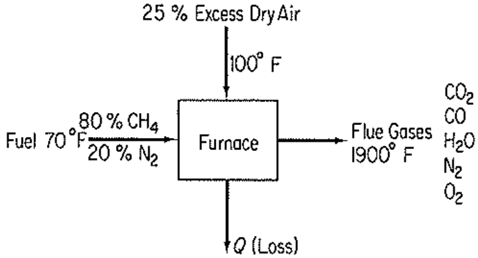


Figure P5.4

5.5. Figure P5.5 shows a simple absorber or extraction unit. S is the absorber oil (or fresh solvent), and F is the feed from which material is to be recovered. Each stage has a Q (not shown); the total number of equilibrium stages is N. What is the number of degrees of freedom for the column? What variables should be specified?

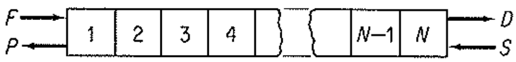


Figure P5.5

5.6. In a reactor model, rather than assume that the components exit from the reactor at equilibrium, an engineer will specify the r independent reactions that occur in the reactor, and the extent of each reaction, ξ_i. The reactor model must also provide for heating or cooling. How many degrees of freedom are associated with such a reactor model?

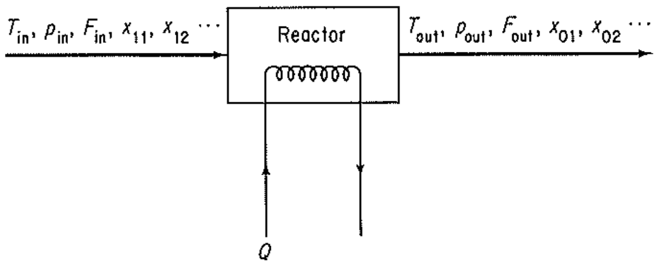


Figure P5.6

5.7. Set up decomposition schemes for the processes shown in Figure P5.7. What additional variables must be specified to make the system determinate if **(a)** the feed conditions are known; or **(b)** the product conditions are specified?

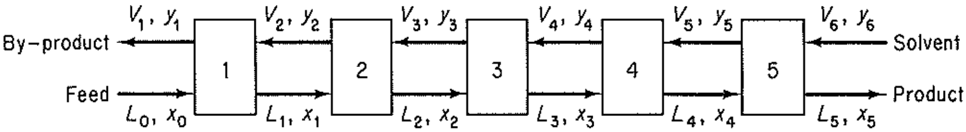


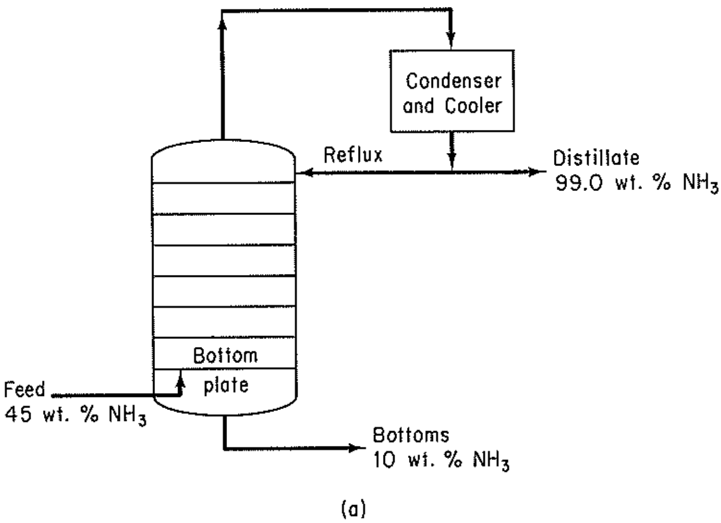
Figure P5.7

5.8. Determine whether or not the following problems are determinate in the sense that all the values of the material flows can be calculated.

(a) A vapor mixture containing 45 weight percent ammonia, balance water, and having an enthalpy of 1125 Btu per pound, is to be fractionated in a bubble-cap column operating at a pressure of 250 psia. The column is to be equipped with a total condenser. The distillate product is to contain 99.0 weight percent ammonia and the bottom product is to contain 10.0 weight percent ammonia. The distillate and the reflux leaving the condenser will have an enthalpy of 18 Btu/lb (Figure P5.8a).

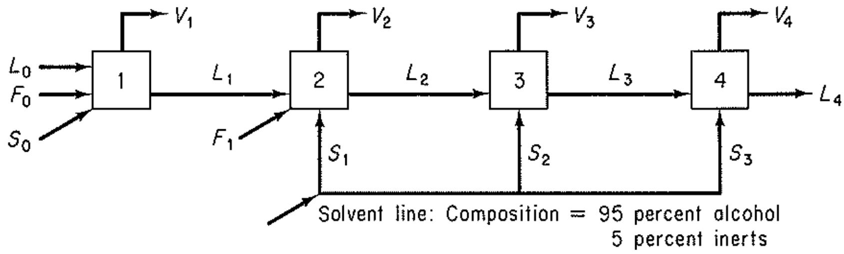
(b) An engineer designed an extraction unit (Fig. P5.8b) to recover oil from a pulp using alcohol as a solvent. The inerts refer to oil-free and solvent-free pulp. Several of the streams in the plant were also used for oil recovery. These streams are shown as F_0 and F_1. Notice that the extracts from the first two stages were not clear but contained some inerts. (Both V_1 and V_2 contain all three components: oil, solvent, and inerts.) Equal amounts of S_1 and L_1 were added to stage 2. There are 2 lb of L_2 for each lb of V_2 leaving the second stage.

The raffinate from stage 1, L_1, contains 32.5 percent alcohol and also in this same stream the weight ratio of inerts to solution is 60 lb inert/100 lb solution. The remaining raffinate streams, L_2, L_3, and L_4, contain 60 lb inert/100 lb alcohol. The L_2 stream contains 15 percent oil.



(a)

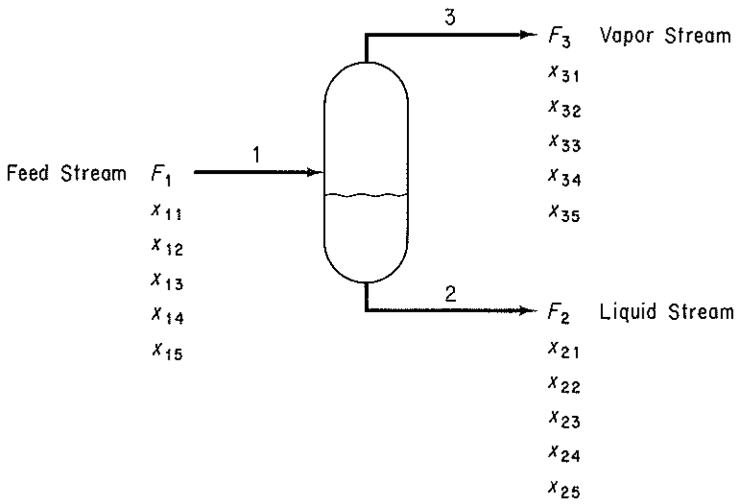
Figure P5.8a



(b)

Figure P5.8b

5.9. Examine Fig. P5.9. Values of F_1, x_{11}, x_{12}, x_{13}, x_{14}, and x_{15} are known. Streams F_2 and F_3 are in equilibrium, and the three streams all have the same (known) temperatures and pressures. Is the problem completely specified, underspecified, or overspecified? Assume that the values of K in the equilibrium relations can be calculated from the given temperatures and pressures.



Description of the variables:

F_i — Molar flow for i^{th} stream

x_{ij} — Mole fraction of j^{th} component in i^{th} stream

K_j — Phase-equilibrium coefficient for the j^{th} component

The design variables are:

The Feed Stream, F_1, x_{11}, x_{12}, x_{13}, x_{14}

The Phase-Equilibrium coefficients, K_1, K_2, K_3, K_4, K_5

Figure P5.9

5.10. Book[9] describes a mixer-heat exchanger section of a monoethylamine plant that is illustrated in Fig. P5.10 along with the notation. Trimethylamine recycle enters in stream 4, is cooled in the heat exchanger, and is mixed with water from stream 1 in mixer 1. The trimethylamine–water mixture is used as the cold-side fluid in the heat exchanger and is then mixed with the ammonia–methanol stream from the gas absorber in mixer 3. The mixture leaving mixer 3 is the reaction mixture which feeds into the preheater of the existing plant.

Table P5.10 lists the 31 equations that represent the process in Fig. P5.10. C_i

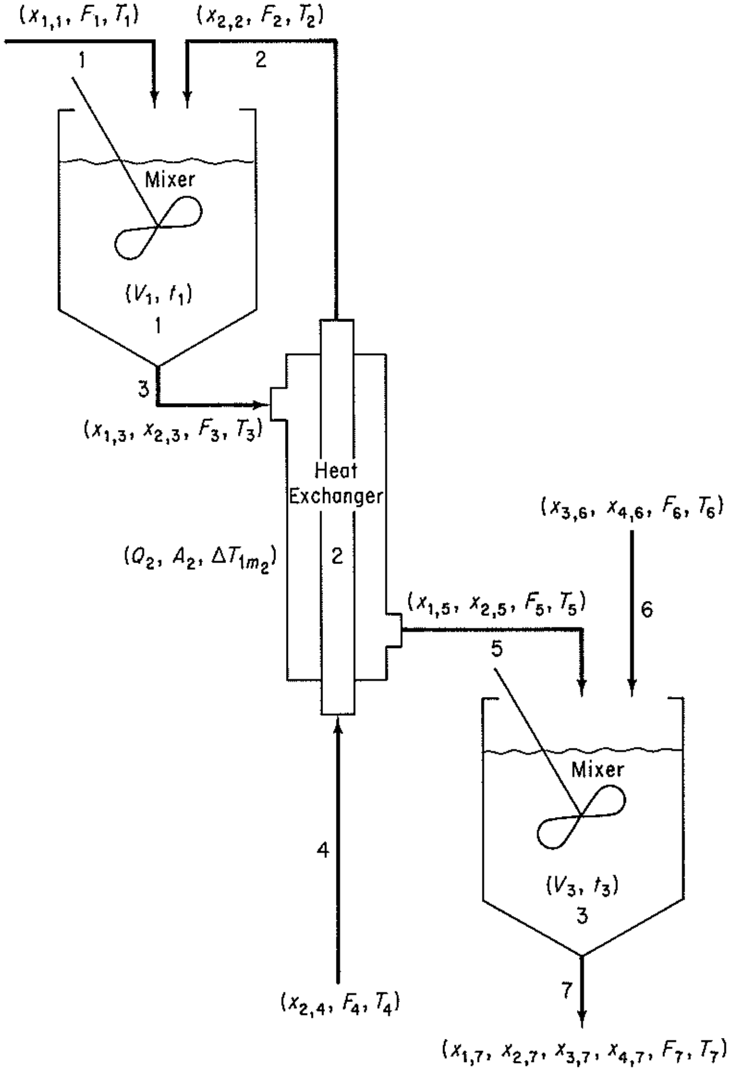


Figure P5.10

[9] N. L. Book, Structural Analysis and Solution of Systems of Algebraic Design Equations, Ph. D. dissertation, University of Colorado, 1976.

TABLE P5.10 List of Equations to Model the Process

Material Balance Equations

Mole fraction equations

1. $x_{1,1} - 1 = 0$
2. $x_{2,2} - 1 = 0$
3. $x_{1,3} + x_{2,3} - 1 = 0$
4. $x_{2,4} - 1 = 0$
5. $x_{1,5} + x_{2,5} - 1 = 0$
6. $x_{3,6} + x_{4,6} - 1 = 0$
7. $x_{1,7} + x_{2,7} + x_{3,7} + x_{4,7} - 1 = 0$

Flow balance equations

8. $F_1 + F_2 - F_3 = 0$
9. $F_2 - F_4 = 0$
10. $F_3 - F_5 = 0$
11. $F_5 + F_6 - F_7 = 0$

Component balance equations

12. $x_{1,1}F_1 - y_1 = 0$
13. $x_{1,3}F_3 - y_1 = 0$
14. $x_{1,5}F_5 - y_1 = 0$
15. $x_{1,7}F_7 - y_1 = 0$
16. $x_{2,2}F_2 - y_2 = 0$
17. $x_{2,3}F_3 - y_2 = 0$
18. $x_{2,4}F_4 - y_2 = 0$
19. $x_{2,5}F_5 - y_2 = 0$
20. $x_{2,7}F_7 - y_2 = 0$
21. $x_{3,6}F_6 - y_3 = 0$
22. $x_{3,7}F_7 - y_3 = 0$
23. $x_{4,6}F_6 - y_4 = 0$
24. $x_{4,7}F_7 - y_4 = 0$

Energy Balance Equations

25. $C_1F_1T_1 + C_2F_2T_2 - C_3F_3T_3 = 0$
26. $C_4F_4T_4 - Q_2 - C_2F_2T_2 = 0$
27. $C_3F_3T_3 + Q_2 - C_5F_5T_5 = 0$
28. $C_5F_5T_5 + C_6F_6T_6 - C_7F_7T_7 = 0$

Equipment Specification Equations

29. $V_1 - (F_3)\left(\dfrac{V_1}{F_1}\right)/\rho_3 = 0$
30. $Q_2 - U_2A_2\,\Delta T_{lm_2} = 0$
31. $V_3 - (F_7)\left(\dfrac{V_3}{F_3}\right)/\rho_7 = 0$

is a heat capacity (a constant), F_i a flow rate, A area (a constant), ΔT_{lm} a log mean temperature difference

$$\frac{(T_4 - T_5) - (T_2 - T_3)}{\ln [(T_4 - T_5)/(T_2 - T_3)]}$$

U is a heat transfer coefficient (a constant), V_i a volume, y_i the molar flow rate of component i, x_i the mole fraction of component i, Q the heat transfer rate, and ρ_i the molar density (a constant). The question is: How many degrees of freedom exist for the process?

5.11. Cavett proposed the following problem as a test problem for computer-aided design. Four flash drums are connected as shown in Fig. P5.11. The temperature in each flash drum is specified, and equilibrium is assumed to be independent of composition so that vapor-liquid equilibrium constants are truly constant. Is the problem properly specified, or do additional variables have to be given? If the latter, what should they be? The feed is as follows

Component	Feed
1. Nitrogen and helium	358.2
2. Carbon dioxide	4,965.6
3. Hydrogen sulfide	339.4
4. Methane	2,995.5
5. Ethane	2,395.5
6. Propane	2,291.0
7. Isobutane	604.1
8. *n*-Butane	1,539.9
9. Isopentane	790.4
10. *n*-Pentane	1,129.9
11. Hexane	1,764.7
12. Heptane	2,606.7
13. Octane	1,844.5
14. Nonane	1,669.0
15. Decane	831.7
16. Undecane plus	1,214.5
Total	27,340.6

5.12. The flowsheet (Fig. P5.12) has a high-pressure feed stream of gaseous component A contaminated with a small amount of B. It mixes first with a recycle stream consisting mostly of A and passes into a reactor, where an exothermic reaction to form C from A takes place. The stream is cooled to condense out component C and passed through a valve into a flash unit. Here most of the unreacted A and the contaminant B flash off, leaving a fairly pure C to be withdrawn as the liquid stream. Part of the recycle is bled off to keep the concentration of B from building up in the system. The rest is repressurized in a compressor and mixed, as stated earlier, with the feed stream. The number of parameters/variables for each unit are designated by the number within the symbol for the unit.

How many degrees of freedom exist for this process?

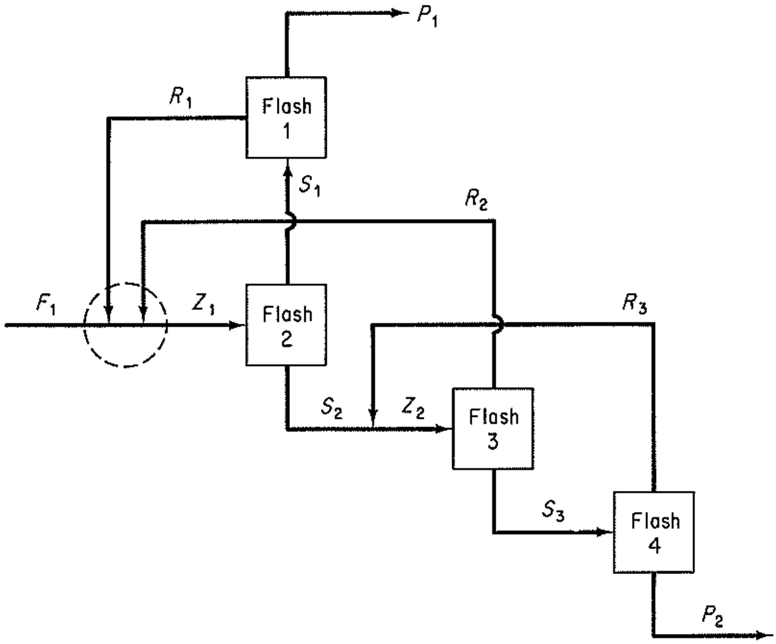


Figure P5.11

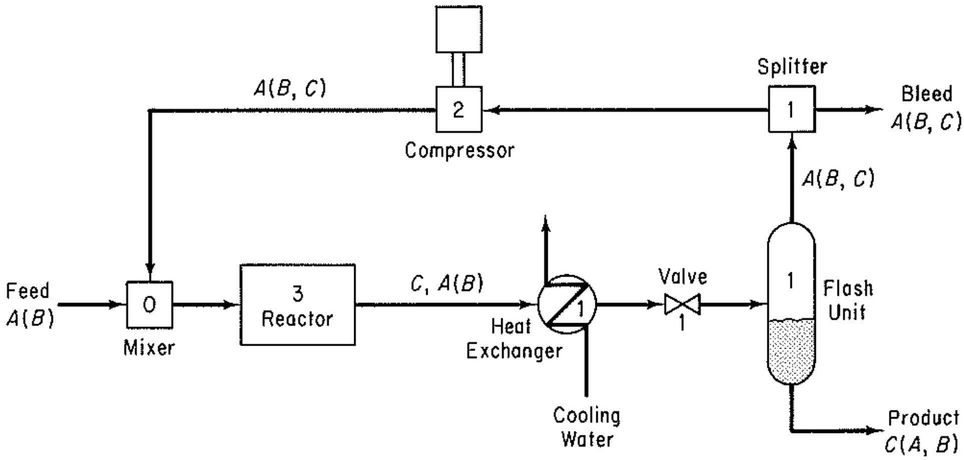


Figure P5.12

5.13. Determine a precedence order for solution of the set of seven equations and seven variables given by the following occurrence matrix:

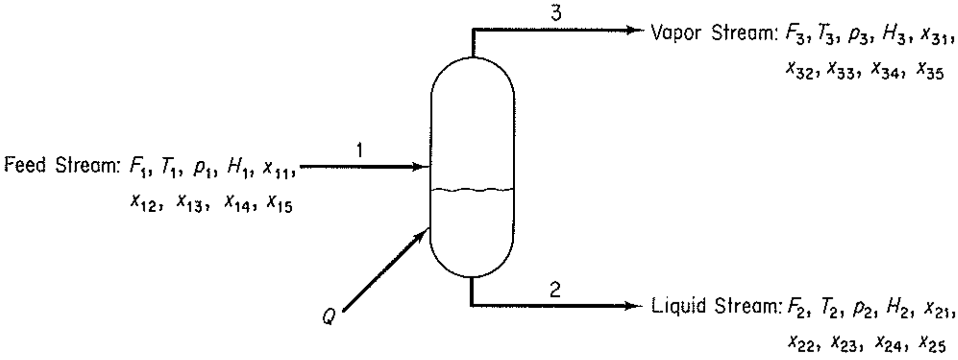
$$
\begin{array}{c}
\text{Variable} \\
\rightarrow
\end{array}
$$

$$
\begin{array}{c c}
\text{Equation} \downarrow & \begin{array}{ccccccc} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \end{array} \\
\begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{array} &
\begin{bmatrix}
1 & 0 & 1 & 0 & 1 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 1 & 1 & 1 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 1 & 0 & 1 & 0 & 1
\end{bmatrix}
\end{array}
$$

5.14. Suggest a solution order for the set of equations in Table P5.14 representing the flash separator in Fig. P5.14. The selection of the first seven of the eight design variables to be calculated (feed stream variables and the equilibrium pressure) is standard for flash separations. The last design variable is usually selected as either

TABLE P5.14 Equations for the Flash Separator in Figure P5.14.

Equation No.	Equation
	The overall material balance equation:
1	$F_2 + F_3 - F_1 = 0$
	The individual component balance equations:
2–5	$F_2x_{2j} + F_3x_{3j} - F_1x_{1j} = 0 \qquad j = 1, 2, 3, 4$
	The stream composition equations:
6	$x_{15} + \sum_{j=1}^{4} x_{1j} = 1$
7	$\sum_{j=1}^{5} x_{2j} = 1$
8	$\sum_{j=1}^{5} x_{3j} = 1$
	The phase equilibrium equations:
9–13	$K_j = K_j(T_2, p_2) \qquad j = 1, 2, 3, 4, 5$
14–18	$x_{3j} - K_jx_{2j} = 0 \qquad j = 1, 2, 3, 4, 5$
19	$T_3 - T_2 = 0$
20	$p_3 - p_2 = 0$
	The enthalpies and energy balance equations:
21	$H_1 = H_1(T_1, p_1, x_{11}, x_{12}, x_{13}, x_{14}, x_{15})$
22	$H_2 = H_2(T_2, p_2, x_{21}, x_{22}, x_{23}, x_{24}, x_{25})$
23	$H_3 = H_3(T_3, p_3, x_{31}, x_{32}, x_{33}, x_{34}, x_{35})$
24	$F_1H_1 - F_2H_2 - F_3H_3 + Q = 0$



Variables:

F_i — Molar flow rate for the i^{th} stream
T_i — Temperature of the i^{th} stream, in degrees Rankine
p_i — Pressure of the i^{th} stream, in atmospheres
H_i — Enthalpy of the i^{th} stream, in Btu/mole
x_{ij} — Mole fraction of the j^{th} component in the i^{th} stream
K_j — Phase equilibrium coefficient of the j^{th} component
Q — Heat transfer to the unit Btu/hour

The Design Variables are:

Feed Stream: $F_1, T_1, p_1, x_{11}, x_{12}, x_{13}, x_{14}$
Equilibrium pressure: p_2
Heat load to the unit: Q

Figure P5.14

the equilibrium temperature, T_2, or Q. The choice of T_2 uncouples the material and energy balances, hence here Q has been selected, requiring a more challenging solution.

5.15. For the units shown in Fig. P5.15, indicate which recycle streams should be cut first.

5.16. Prepare a block diagram for the material flows in a methane evaporation plant. The following blocks can be used in addition to those shown in Fig. 5.9a: evaporator, tank, boiler.

5.17. Prepare a diagram of the modules for sequential solution of the phenosolvan plant in Fig. P5.17. Use Fig. 5.9a for the modules.

5.18. Prepare a block diagram of the modules involved in solving the material and energy balances for the extraction process used in a refinery for the production of normal paraffins shown in Fig. P5.18. Which additional modules not in Fig. 5.9a would you need? Also indicate what streams might be cut for a sequential modular solution of the material and energy balances.

5.19. In petroleum refining, lubricating oil is treated with sulfuric acid to remove unsaturated compounds, and after settling, the oil and acid layers are separated. The acid layer is added to water and heated to separate the sulfuric acid from the sludge contained in it. The dilute sulfuric acid, now 20% H_2SO_4 at 82°C, is fed to a

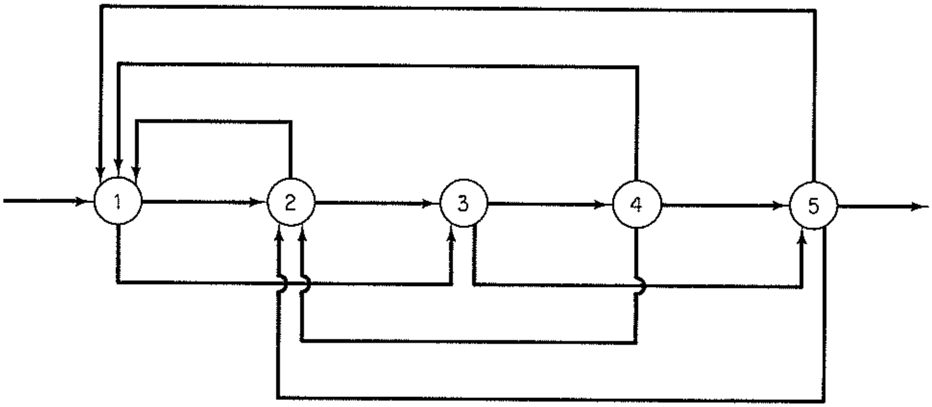


Figure P5.15

Simonson–Mantius evaporator, which is supplied with saturated steam at 400 kPa gauge to lead coils submerged in the acid, and the condensate leaves at the saturation temperature. A vacuum is maintained at 4.0 kPa by means of a barometric leg. The acid is concentrated to 80% H_2SO_4; the boiling point at 4.0 kPa is 121°C. How many kilograms of acid can be concentrated per 1000 kg of steam condensed?

5.20.* You are asked to perform a feasibility study on a continuous stirred tank reactor shown in Fig. P5.20 (which is presently idle) to determine if it can be used for the second-order reaction

$$2A \longrightarrow B + C$$

Since the reaction is exothermic, a cooling jacket will be used to control the reactor temperature. The total amount of heat transfer may be calculated from an overall heat transfer coefficient (U) by the equation

$$Q = UA\,\Delta T$$

where Q = total rate of heat transfer from the reactants to the water jacket in the steady state
U = empirical coefficient
A = area of transfer
ΔT = temperature difference (here $T_4 - T_2$)

Some of the energy released by the reaction will appear as sensible heat in stream $F2$, and some concern exists as to whether the fixed flow rates will be sufficient to keep the fluids from boiling while still obtaining good conversion. Feed data is as follows.

Component	Feed rate (lb mol/hr)	C_p [Btu/(lb mol)(°F)]	MW
A	214.58	41.4	46
B	23.0	68.4	76
C	0.0	4.4	6

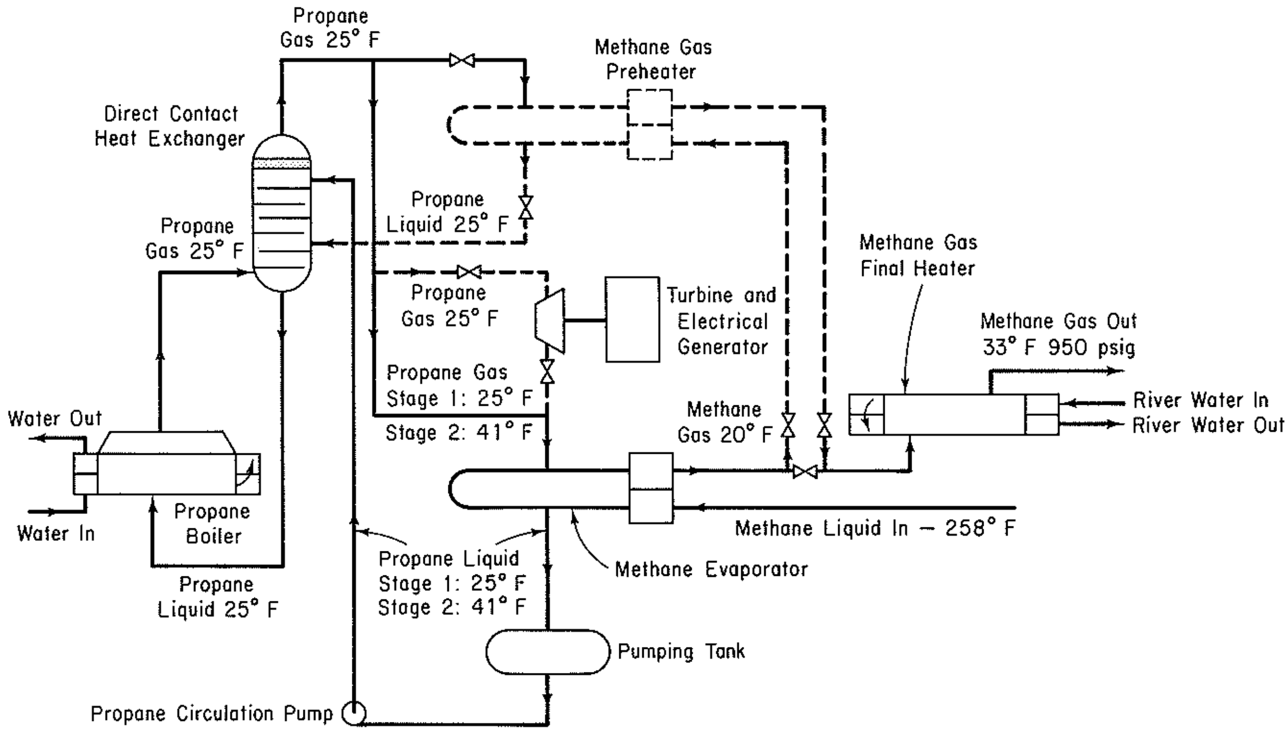

Propane
Gas 25° F
Methane Gas
Preheater
Direct Contact
Heat Exchanger
Propane
Liquid 25° F
Propane
Gas 25° F
Methane Gas
Final Heater
Propane
Gas 25° F
Turbine and
Electrical
Generator
Methane Gas Out
33° F 950 psig
Propane Gas
Stage 1: 25° F
Stage 2: 41° F
Methane
Gas 20° F
River Water In
River Water Out
Water Out
Water In
Propane
Boiler
Methane Liquid In − 258° F
Propane Liquid
Stage 1: 25° F
Stage 2: 41° F
Methane Evaporator
Propane
Liquid 25° F
Pumping Tank
Propane Circulation Pump

Figure P5.16

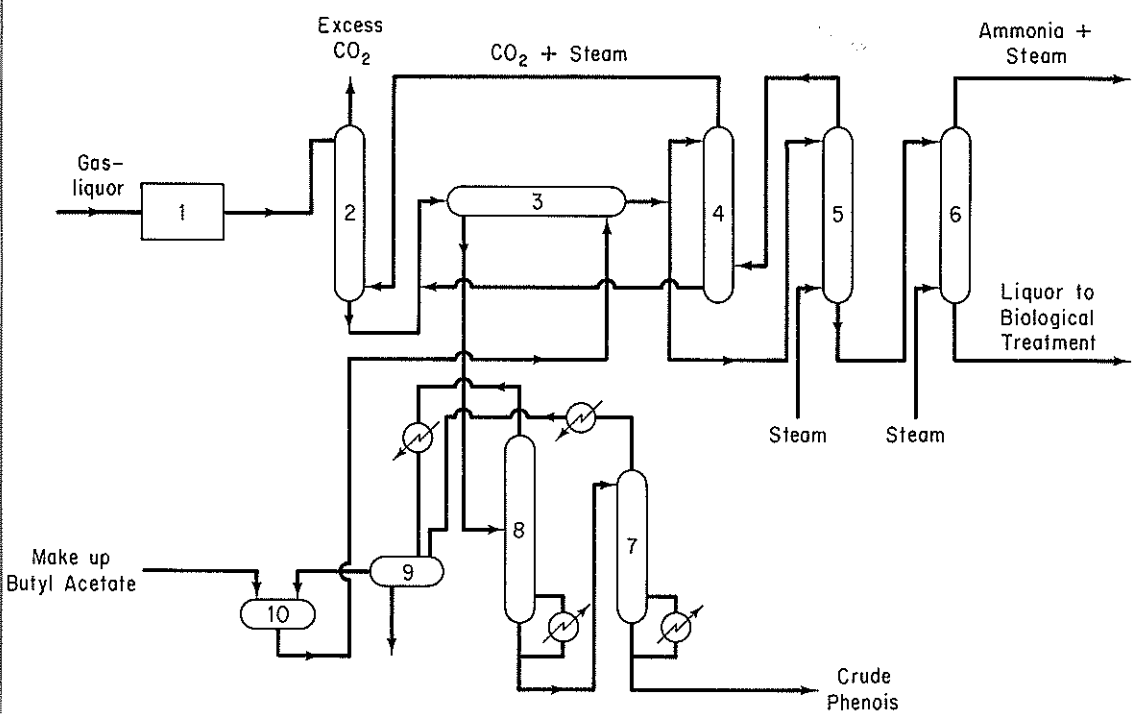


Figure P5.17 Phenosolvan plant—only one stream shown. Key: 1. Tar-oil separation (gravity); 2. Saturation tower; 3. Phenol extractor; 4. Butyl-acetate recovery tower; 5. Butyl-acetate stripping tower; 6. Ammonia stripping tower; 7. Atmospheric distillation tower; 8. Vacuum distillation tower; 9. Separator drum; 10. Butyl-acetate circulating drum.

The consumption rate of A may be expressed as

$$-2k(C_A)^2V_R$$

where

$$C_A = \frac{(F_{1,A})(\rho)}{\Sigma\,(F_{1,i})(MW_i)} = \text{concentration of } A,\ \text{lb mol/ft}^3$$

$$k = k_0 \exp\left(\frac{-E}{RT}\right)$$

k_0, E, R are constants and T is absolute temperature.

Write a computer program to solve for the temperatures of the exit streams and the product composition of the steady-state reactor using the following data:

Fixed parameters

$$\text{reactor volume} = V_R = 13.3\ \text{ft}^3$$

$$\text{heat transfer area} = A = 29.9\ \text{ft}^2$$

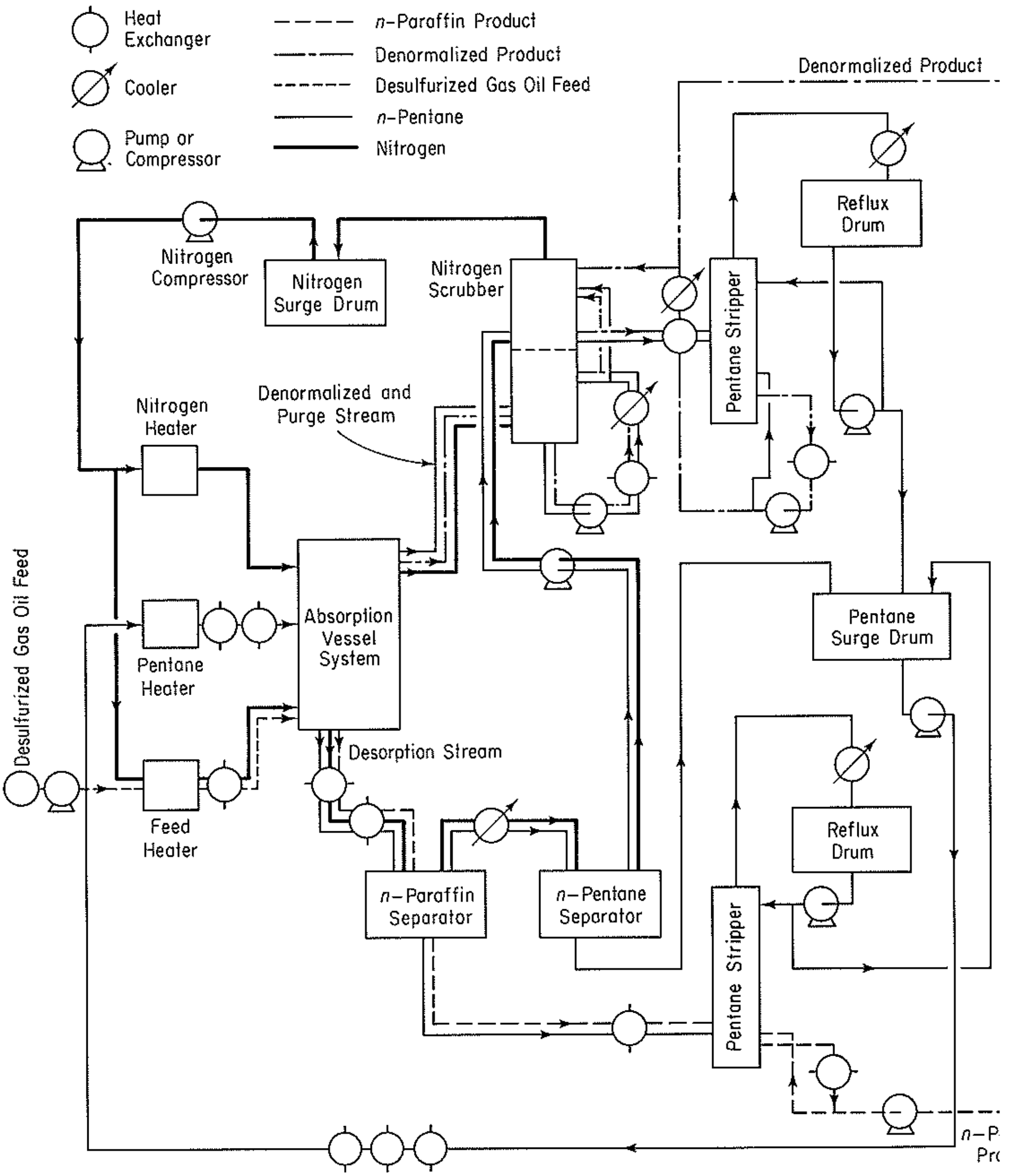


Figure P5.18 Flow chart for problem.

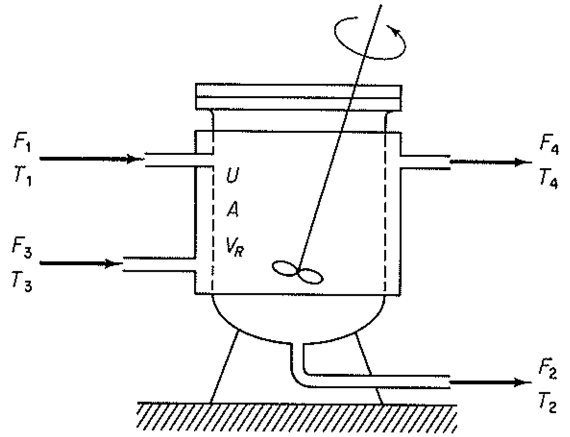


Figure P5.20

heat transfer coefficient = U = 74.5 Btu/(hr)(ft^2)(°F)

Variable input

reactant feed rate = F_i (see table above)

reactant feed temperature = T_1 = 80°F

water feed rate = F_3 = 247.7 lb mol/hr water

water feed temperature = T_3 = 75°F

Physical and thermodynamic data

reaction rate constant = k_0 = 34 lb mol B/(hr)(ft^2)

activation energy/gas constant = E/R = 1000°R

heat of reaction = ΔH = −5000 Btu/lb mol A

heat capacity of water = C_{p_w} = 18 Btu/(lb mol)(°F)

product component density = ρ = 55 lb/ft^3

The densities of each of the product components are essentially the same. Assume that the reactor contents are perfectly mixed as well as the water in the jacket, and that the respective exit stream temperatures are the same as reactor contents or jacket contents.

5.21. The steam flows for a plant are shown in Fig. P5.21. Write the material and energy balances for the system and calculate the unknown quantities in the diagram (A to F). There are two main levels of steam flow: 600 psig and 50 psig. Use the steam tables for the enthalpies.

5.22. Figure P5.22 shows a calciner and the process data. The fuel is natural gas. How can the energy efficiency of this process be improved by process modification? Suggest at least two ways based on the assumption that the supply conditions of the air and fuel remain fixed (but these streams can be possibly passed through heat exchangers). Show all calculations.

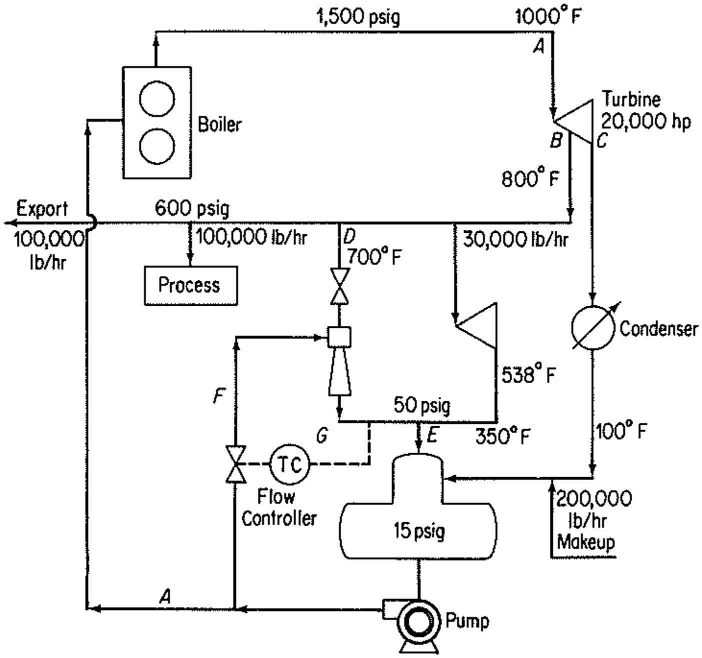


Figure P5.21

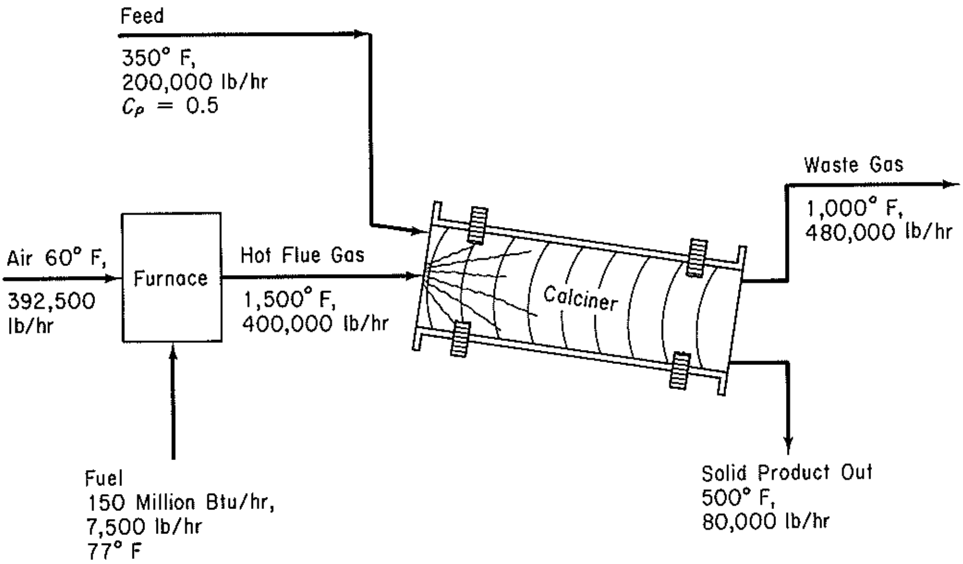


Figure P5.22

5.23. Limestone ($CaCO_3$) is converted into CaO in a continuous vertical kiln (see Fig. P5.23). Heat is supplied by combustion of natural gas (CH_4) in direct contact with the limestone using 50% excess air. Determine the kilograms of $CaCO_3$ that can be processed per kilograms of natural gas. Assume that the following average heat capacities apply:

$$C_p \text{ of } CaCO_3 = 234 \text{ J/(g mol)(°C)}$$

$$C_p \text{ of } CaO = 111 \text{ J/(g mol)(°C)}$$

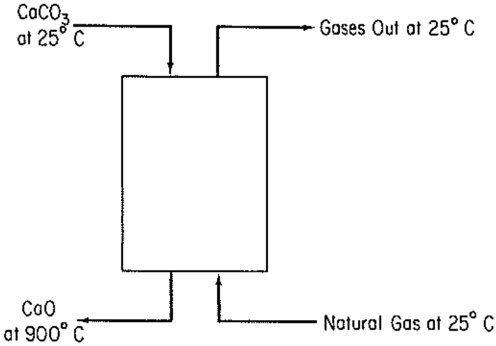


Figure P5.23

5.24. A feed stream of 16,000 lb/hr of 7% by weight NaCl solution is concentrated to a 40% by weight solution in an evaporator. The feed enters the evaporator, where it is heated to 180°F. The water vapor from the solution and the concentrated solution leave at 180°F. Steam at the rate of 15,000 lb/hr enters at 230°F and leaves as condensate at 230°F. See Fig. P5.24.

(a) What is the temperature of the feed as it enters the evaporator?

(b) What weight of 40% NaCl is produced per hour?

Assume that the following data apply:

Average C_p 7% NaCl soln:	0.92 Btu/(lb)(°F)
Average C_p 40% NaCl soln:	0.85 Btu/(lb)(°F)

$$\Delta \hat{H}_{vap} \; H_2O \text{ at } 180°F = 990 \text{ Btu/lb}$$

$$\Delta \hat{H}_{vap} \; H_2O \text{ at } 230°F = 959 \text{ Btu/lb}$$

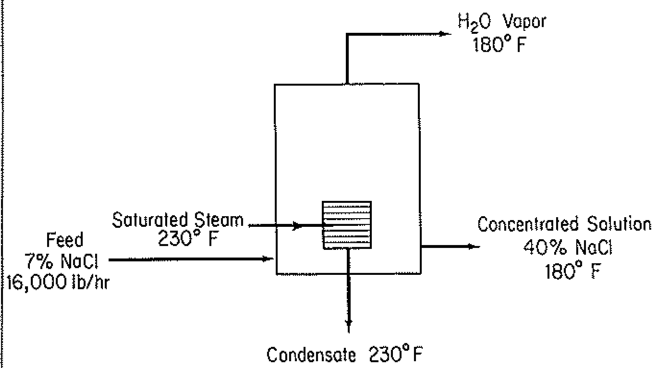


Figure P5.24

5.25. The Blue Ribbon Sour Mash Company plans to make commercial alcohol by a process shown in Fig. P5.25. Grain mash is fed through a heat exchanger where it is heated to 170°F. The alcohol is removed as 60% by weight alcohol from the first fractionating column; the bottoms contain no alcohol. The 60% alcohol is further fractionated to 95% alcohol and essentially pure water in the second column. Both stills operate at a 3 : 1 reflux ratio and heat is supplied to the bottom of the columns by steam. Condenser water is obtainable at 80°F. The operating data and physical properties of the streams have been accumulated and are listed for convenience:

Stream	State	Boiling point (°F)	C_p[Btu/(lb)(°F)] Liquid	Vapor	Heat of vaporization (Btu/lb)
Feed	Liquid	170	0.96	—	950
60% alcohol	Liquid or vapor	176	0.85	0.56	675
Bottoms I	Liquid	212	1.00	0.50	970
95% alcohol	Liquid or vapor	172	0.72	0.48	650
Bottoms II	Liquid	212	1.00	0.50	970

Prepare the material balances for the process, calculate the precedence order for solution, and

(a) Determine the weight of the following streams per hour:
  (1) Overhead product, column I
  (2) Reflux, column I
  (3) Bottoms, column I
  (4) Overhead product, column II
  (5) Reflux, column II
  (6) Bottoms, column II
(b) Calculate the temperature of the bottoms leaving heat exchanger III.
(c) Determine the total heat input to the system in Btu/hr.

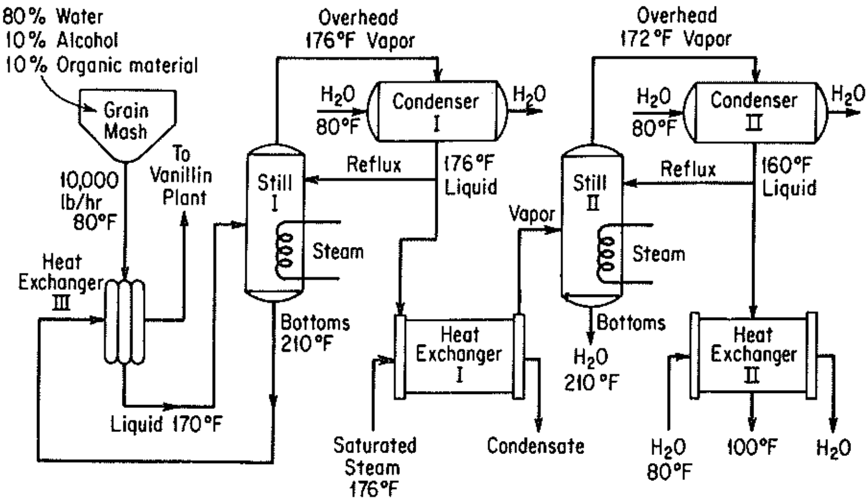
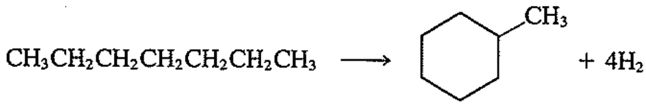


Figure P5.25

(d) Calculate the water requirements for each condenser and heat exchanger II in gal/hr if the maximum exit temperature of water from this equipment is 130°F.

5.26. Toluene, manufactured by the conversion of *n*-heptane with a Cr_2O_3-on-Al_2O_3 catalyst



by the method of hydroforming is recovered by use of a solvent. See Fig. P5.26 for the process and conditions.

The yield of toluene is 15 mole % based on the *n*-heptane charged to the reactor. Assume that 10 kg of solvent are used per kilogram of toluene in the extractors.

(a) Calculate how much heat has to be added or removed from the catalytic reactor to make it isothermal at 425°C.

(b) Find the temperature of the *n*-heptane and solvent stream leaving the mixer-settlers if both streams are at the same temperature.

(c) Find the temperature of the solvent stream after it leaves the heat exchanger.

(d) Calculate the heat duty of the fractionating column in kJ/kg of *n*-heptane feed to the process.

	$-\Delta H_f^\circ$* (kJ/g mol)	C_p[J/(g)(°C)] Liquid	Vapor	$\Delta H_{vaporization}$ (kJ/kg)	Boiling point (K)
Toulene†	12.00	2.22	2.30	364	383.8
n-Heptane	−224.4	2.13	1.88	318	371.6
Solvent	—	1.67	2.51	—	434

*As liquids.
† The heat of solution of toluene in the solvent is −23 J/g toluene.

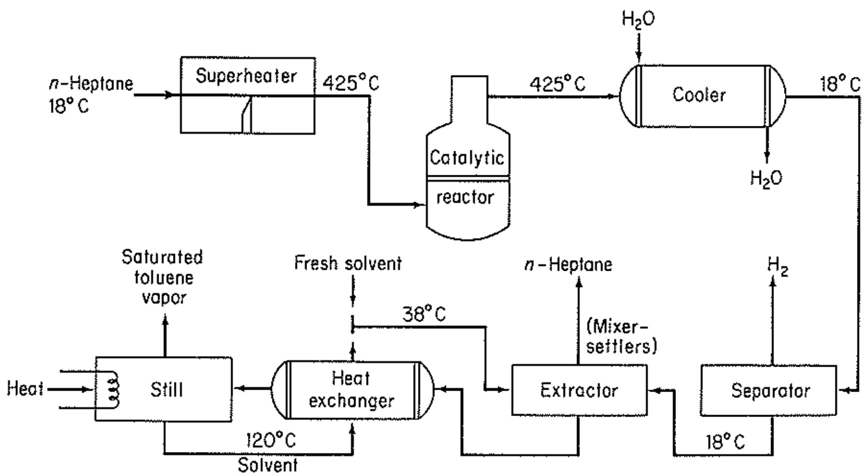


Figure P5.26

5.27. One hundred thousand pounds of a mixture of 50% benzene, 40% toluene, and 10% *o*-xylene is separated every day in a distillation-fractionation plant as shown on the flowsheet for Fig. P5.27.

	Boiling point (°C)	C_p Liquid [cal/(g)(°C)]	Latent heat of vap. (cal/g)	C_p vapor [cal/(g)(°C)]
Benzene	80	0.44	94.2	0.28
Toluene	109	0.48	86.5	0.30
o-Xylene	143	0.48	81.0	0.32
Charge	90	0.46	88.0	0.29
Overhead T_I	80	0.45	93.2	0.285
Residue T_I	120	0.48	83.0	0.31
Residue T_{II}	143	0.48	81.5	0.32

The reflux ratio for tower I is 6 : 1; the reflux ratio for tower II is 4 : 1; the charge to tower I is liquid; the charge to tower II is liquid. Compute:

(a) The temperature of the mixture at the outlet of the heat exchanger (marked as T^*)
(b) The Btu supplied by the steam reboiler in each column
(c) The quantity of cooling water required in gallons per day for the whole plant
(d) The energy balance around tower I

5.28. Sulfur dioxide emission from coal-burning power plants causes serious atmospheric pollution in the eastern and midwestern portions of the United States. Unfortunately, the supply of low-sulfur coal is insufficient to meet the demand. Processes presently under consideration to alleviate the situation include coal gasification followed by desulfurization and stack-gas cleaning. One of the more promising stack-gas-cleaning processes involves reacting SO_2 and O_2 in the stack gas with a solid metal oxide sorbent to give the metal sulfate and then thermally regenerating the sorbent and absorbing the resulting SO_3 to produce sulfuric acid. Recent laboratory experiments indicate that sorption and regeneration can be carried out with several metal oxides, but no pilot or full-scale processes have yet been put into operation.

You are asked to provide a preliminary design for a process that will remove 95% of the SO_2 from the stack gas of a 1000-MW power plant. Some data are given below and in the flow diagram of the process (Fig. P5.28). The sorbent consists of fine particles of a dispersion of 30% by weight CuO in a matrix of inert porous Al_2O_3. This solid reacts in the fluidized-bed sorber at 315°C. Exit solid is sent to the regenerator, where SO_3 is evolved at 700°C, converting all the $CuSO_4$ present back to CuO. The fractional conversion of CuO to $CuSO_4$ that occurs in the sorber is called α and is an important design variable. You are asked to carry out your calculations for $\alpha = 0.2$, 0.5, and 0.8. The SO_3 produced in the regenerator is swept out by recirculating air. The SO_3-laden air is sent to the acid tower, where the SO_3 is absorbed in recirculating sulfuric acid and oleum, part of which is withdrawn as salable by-products. You will notice that the sorber, regenerator, and perhaps the acid tower are adiabatic; their temperatures are adjusted by heat exchange with incoming streams. Some of the heat exchangers (nos. 1 and 3) recover heat by countercurrent exchange between the feed and exit streams. Additional heat is provided

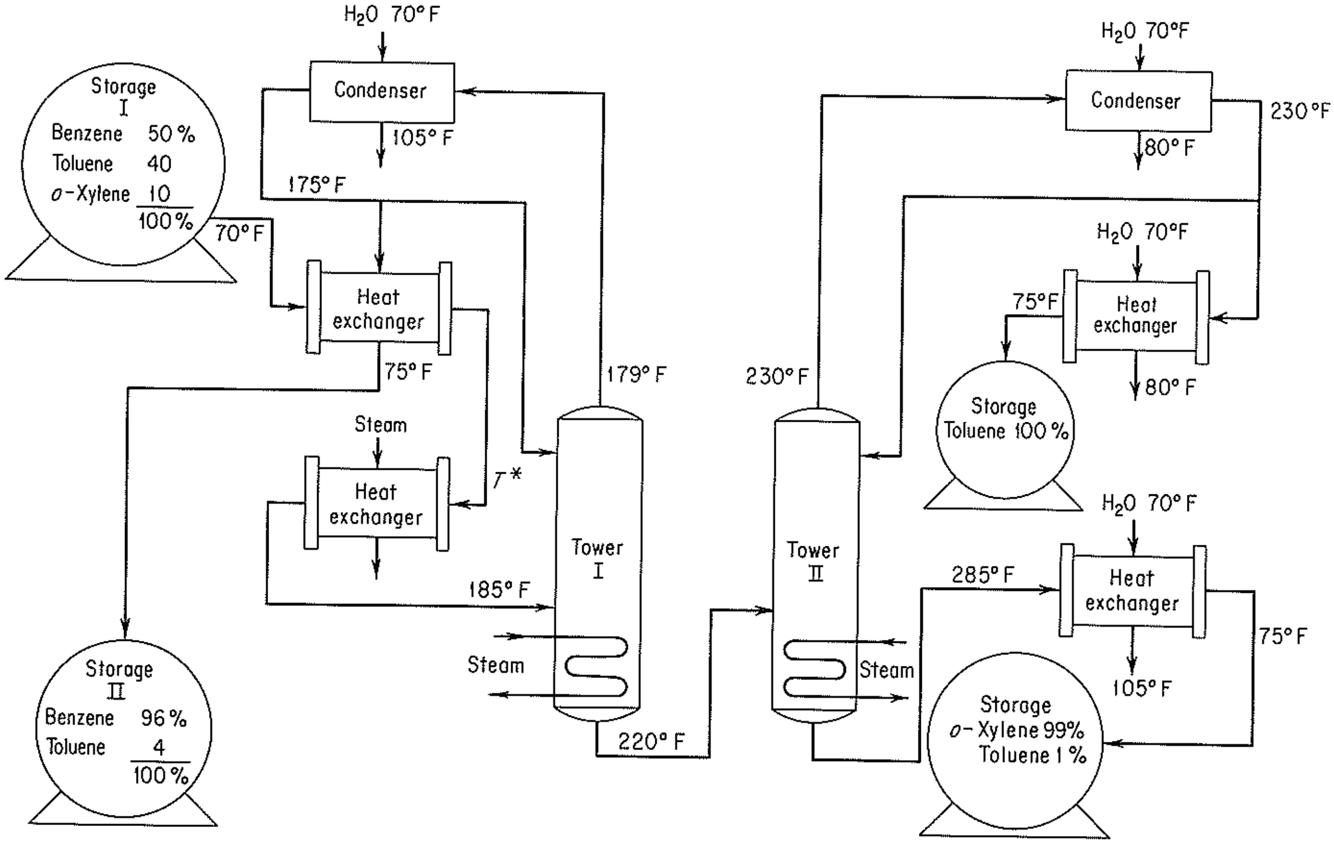
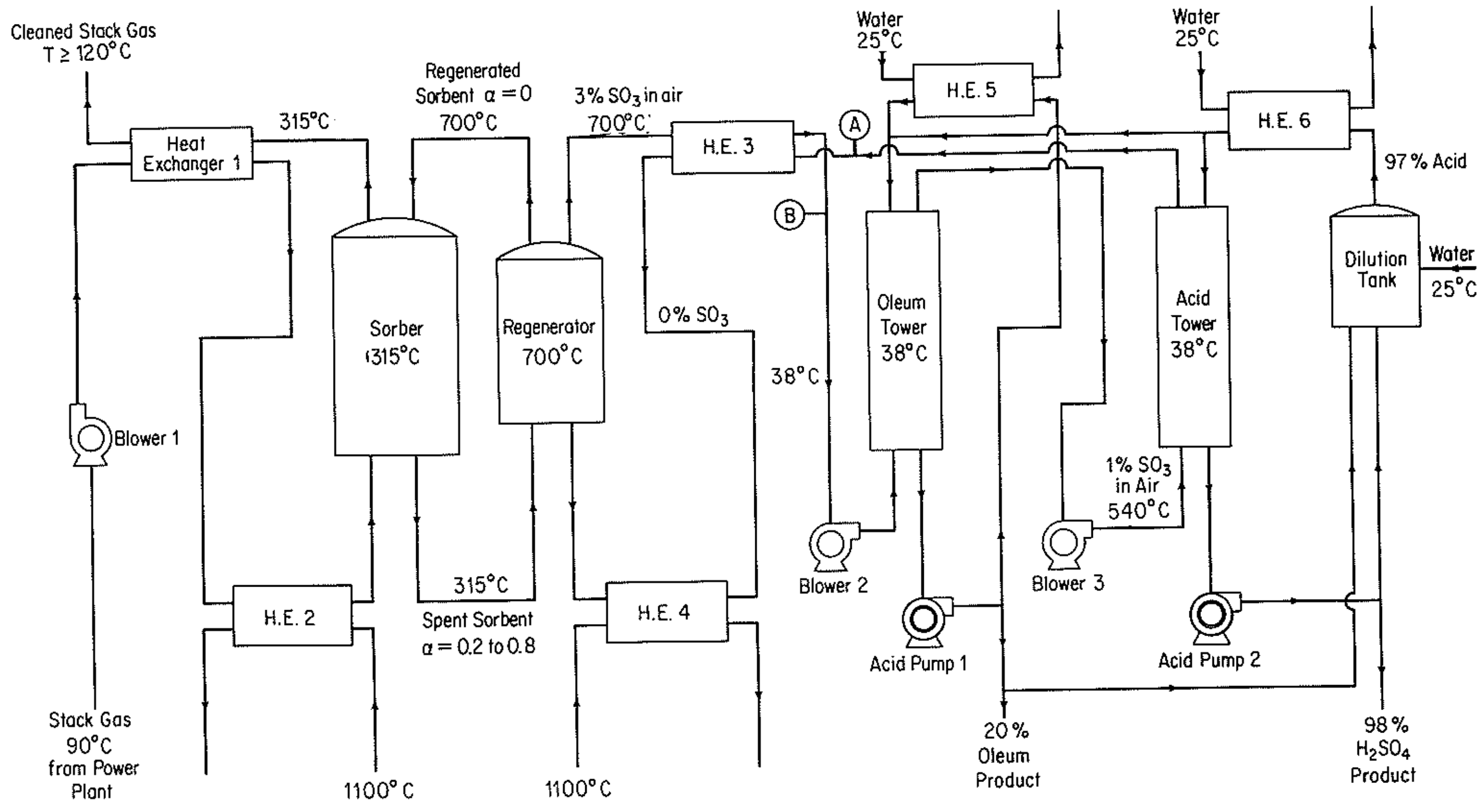

Storage
I
Benzene 50 %
Toluene 40
o-Xylene 10
100 %
70° F
H2O 70° F
Condenser
105° F
175° F
Heat exchanger
75° F
Steam
Heat exchanger
T*
185° F
179° F
Tower I
Steam
220° F
Storage
II
Benzene 96 %
Toluene 4
100 %
230° F
Tower II
Steam
H2O 70°F
Condenser
230 °F
80° F
H2O 70°F
Heat exchanger
75°F
80° F
Storage
Toluene 100 %
H2O 70° F
285° F
Heat exchanger
75° F
105° F
Storage
o-Xylene 99%
Toluene 1 %

Figure P5.27



Cleaned Stack Gas
T ≥ 120°C
Heat Exchanger 1
315°C
Blower 1
Stack Gas
90°C
from Power
Plant
H.E. 2
1100° C
Sorber
315°C
Regenerated
Sorbent α = 0
700°C
Regenerator
700°C
315°C
Spent Sorbent
α = 0.2 to 0.8
3% SO3 in air
700°C
H.E. 3
0% SO3
H.E. 4
1100° C
A
B
38°C
Water
25°C
H.E. 5
Oleum
Tower
38°C
Blower 2
Acid Pump 1
20%
Oleum
Product
Blower 3
1% SO3
in Air
540°C
Water
25°C
H.E. 6
Acid
Tower
38°C
Acid Pump 2
97% Acid
Dilution
Tank
Water
25°C
98%
H2SO4
Product

Figure P5.28

by withdrawing flue gas from the power plant at any desired high temperature up to 1100°C and then returning it at a lower temperature. Cooling is provided by water at 25°C. As a general rule, the temperature difference across heat-exchanger walls separating the two streams should average about 28°C. The nominal operating pressure of the whole process is 100 kPa. The three blowers provide 6 kPa additional head for the pressure losses in the equipment, and the acid pumps have a discharge pressure of 90 kPa gauge. You are asked to write the material and energy balances and some equipment specifications as follows:

(a) Sorber, regenerator, and acid tower. Determine the flow rate, composition, and temperature of all streams entering and leaving.
(b) Heat exchangers. Determine the heat load, and flow rate, temperature, and enthalpy of all streams.
(c) Blowers. Determine the flow rate and theoretical horsepower.
(d) Acid pump. Determine the flow rate and theoretical horsepower.

Use SI units. Also, use a basis of 100 kg of coal burned for all your calculations; then convert to the operating basis at the end of the calculations.

Power plant operation. The power plant burns 340 metric tons/hr of coal having the analysis given below. The coal is burned with 18% excess air, based on complete combustion to CO_2, H_2O, and SO_2. In the combustion only the ash and nitrogen is left unburned; all the ash has been removed from the stack gas.

Element	Wt. %
C	76.6
H	5.2
O	6.2
S	2.3
N	1.6
Ash	8.1

Data on Solids*

	Al_2O_3		CuO		$CuSO_4$	
T (K)	C_p	$H_T - H_{298}$	C_p	$H_T - H_{298}$	C_p	$H_T - H_{298}$
298	79.04	0.00	42.13	0.00	98.9	0.00
400	96.19	9.00	47.03	4.56	114.9	10.92
500	106.10	19.16	50.04	9.41	127.2	23.05
600	112.5	30.08	52.30	14.56	136.3	36.23
700	117.0	41.59	54.31	19.87	142.9	50.25
800	120.3	53.47	56.19	25.40	147.7	64.77
900	122.8	65.65	58.03	31.13	151.0	79.71
1000	124.7	77.99	59.87	37.03	153.8	94.98

*Units of C_p are J/(g mol)(K); units of H are kJ/g mol.

5.29. When coal is distilled by heating without contact with air, a wide variety of solid, liquid, and gaseous products of commercial importance is produced, as well as some significant air pollutants. The nature and amounts of the products produced depend on the temperature used in the decomposition and the type of coal. At low temperatures (400 to 750°C) the yield of synthetic gas is small relative to the yield of liquid products, whereas at high temperatures (above 900°C) the reverse is true. For the typical process flow sheet, shown in Fig. P5.29:

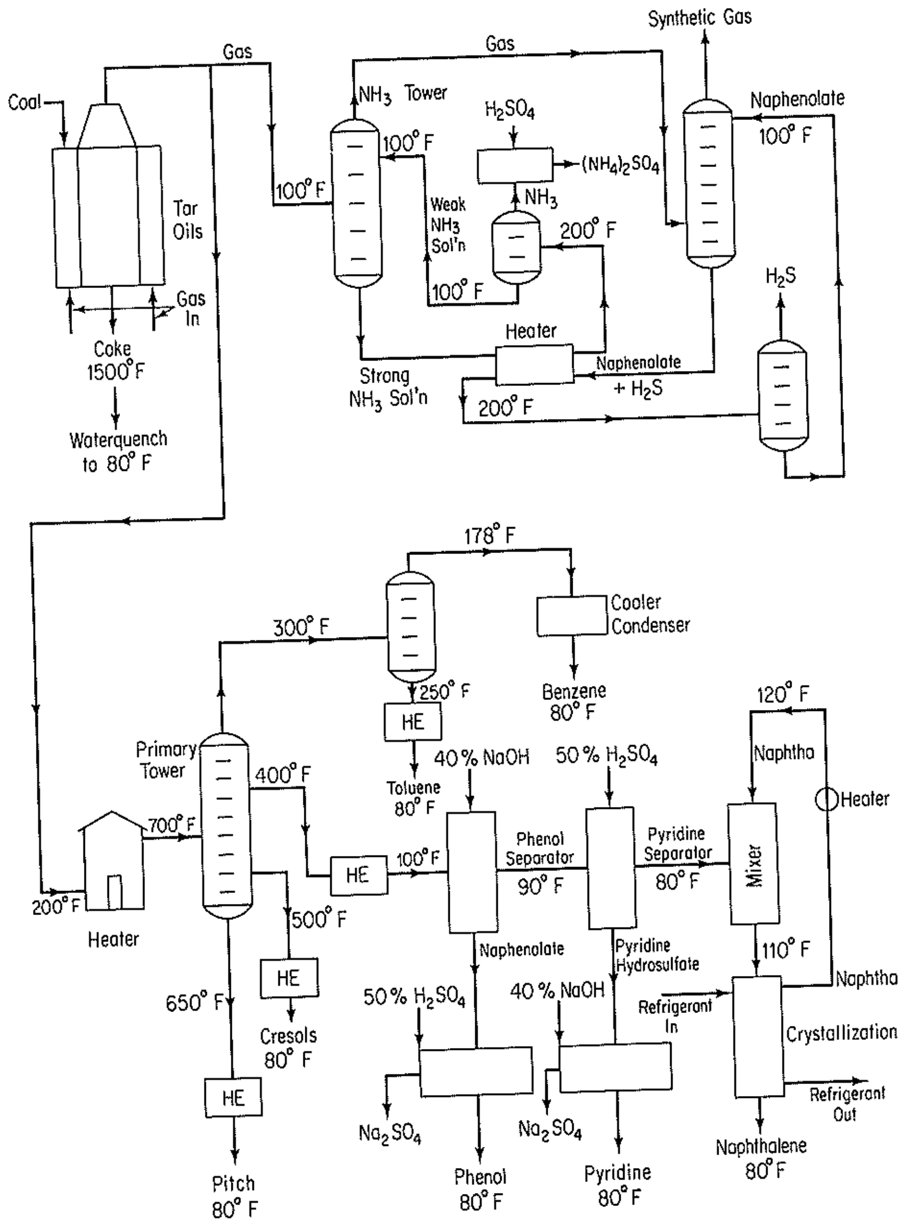


Figure P5.29

(a) How many tons of the various products are being produced?
(b) Make an energy balance around the primary distillation tower and benzol tower.
(c) How much (in pounds) of 40% NaOH solution is used per day for the purification of the phenol?
(d) How much 50% H_2SO_4 is used per day in the pyridine purification?
(e) What weight of Na_2SO_4 is produced per day by the plant?
(f) How many cubic feet of gas per day is produced? What percent of the gas (volume) is needed for the ovens?

Products produced per ton of coal charged	Mean C_p liquid (cal/g)	Mean C_p vapor (cal/g)	Mean C_p solid (cal/g)	Melting point (°C)	Boiling point (°C)
Synthetic gas—10,000 ft^3 (555 Btu/ft^3)					
$(NH_4)_2SO_4$, 22 lb					
Benzol, 15 lb	0.50	0.30	—	—	60
Toluol, 5 lb	0.53	0.35	—	—	109.6
Pyridine, 3 lb	0.41	0.28	—	—	114.1
Phenol, 5 lb	0.56	0.45	—	—	182.2
Naphthalene, 7 lb	0.40	0.35	0.281 $+0.00111T_{°F}$	80.2	218
Cresols, 20 lb	0.55	0.50	—	—	202
Pitch, 40 lb	0.65	0.60	—	—	400
Coke, 1500 lb	—	—	0.35		—

	ΔH_{vap} (cal/g)	ΔH_{fusion} (cal/g)
Benzol	97.5	—
Toluol	86.53	—
Pyridine	107.36	—
Phenol	90.0	—
Naphthalene	75.5	35.6
Cresols	100.6	—
Pitch	120	—

5.30.* A gas consisting of 95 mol % hydrogen and 5 mol % methane at 100°F and 30 psia is to be compressed to 569 psia at a rate of 440 lb mol/hr. A two-stage compresser system has been proposed with intermediate cooling of the gas to 100°F via a heat exchanger. See Fig. P5.30. The pressure drop in the heat exchanger from the inlet stream (S1) to the exit stream (S2) is 2.0 psia. Using a process simulator program, analyze all of the steam parameters subject to the following constraints; the exit stream from the first stage is 100 psia; both compressers are positive-displacement type and have a mechanical efficency of 0.8, a polytropic efficiency of 1.2, and a clearance fraction of 0.05. Flowtran is organized to solve this problem with minimual effort, but other flowsheeting packages can be used.

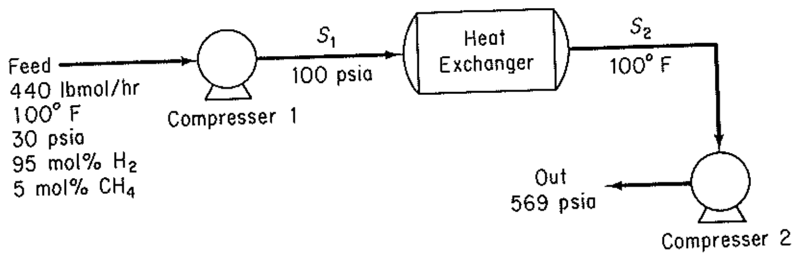


Figure P5.30

5.31. A gas feed mixture at 85°C and 100 psia having the composition shown in Fig. P5.31 is flashed to separate the majority of the light components from the heavy. The flash chamber operates at 5°C and 25 psia. To improve the separation process, it has been suggested to introduce a recycle as shown in Fig. P5.31. Will a significant improvement be made by adding a 25% recycle of the bottoms? 50%? With the aid of a computer process simulator, determine the molar flow rates of the streams for each of the three cases. This problem may readily be solved using Flowtran; however, other flowsheeting packages can be used.

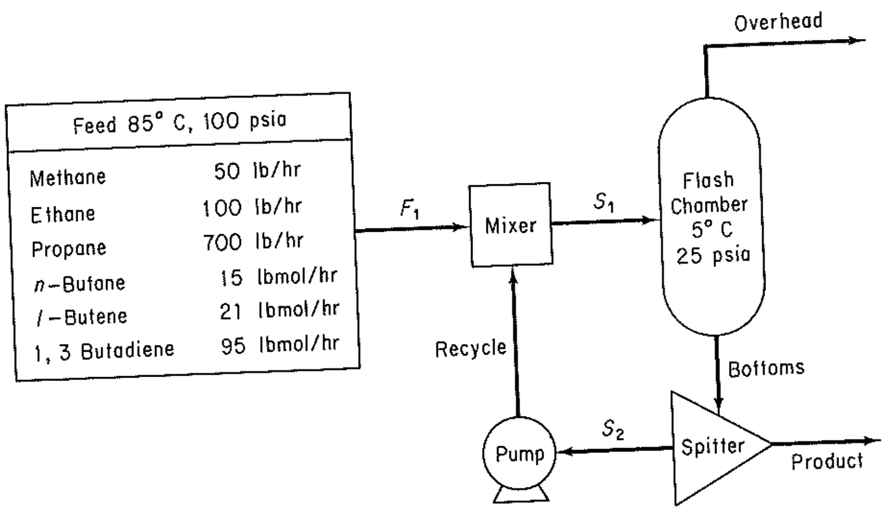


Figure P5.31

5.32. Conventional refrigeration cycles require vapor compression to high pressures. This requires a substantial amount of work, and typically it is the most expensive step of the process. However, a thermal compressor allows the mixture of a high-pressure vapor to be mixed with a low-pressure vapor to form an intermediate-pressure vapor. Using this device, it is possible to create a refrigeration cycle in which only vapor is pumped to a higher pressure, thus allowing substantial savings in energy cost. Solve the material and energy balances for the process shown in Fig. P5.32 subject to the following specifications: 100 lb mol/hr of isopropyl alcohol refrigerant flow through the condenser, and there is a high/low pressure recycle ratio of 9 to 1 at th

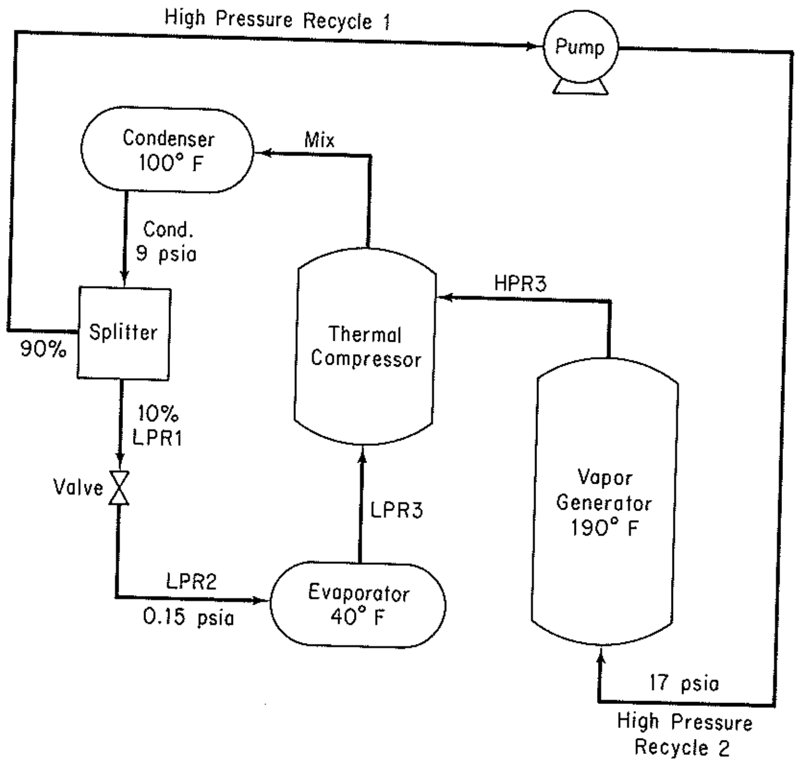


Figure P5.32

stream split. Calculate the energy exchange of each process unit. This problem may be readily solved using Flowtran; however, other flowsheeting packages can be used.

5.33. A mixture of three petroleum fractions containing lightweight hydrocarbons is to be purified and recycled back to a process. Each of the fractions is denoted by its normal boiling point: BP135, BP260, and BP500. The gases separated from this feed are to be compressed as shown in Fig. P5.33. The inlet feed stream (1) is at 45°C and 450 kPa, and has the composition shown. The exit gas (10) is compressed to 6200 kPa by a three-stage compressor process with intercooling of the vapor streams to 60°C by passing through a heat exchanger. The exit pressure for compressor 1 is 1100 kPa and 2600 kPa for compressor 2. The efficiencies for compressors 1, 2, and 3, with reference to an adiabatic compression, are 78, 75, and 72%, respectively. Any liquid fraction drawn off from a separator is recycled to the previous stage. Estimate the heat duty (in kJ/hr) of the heat exchangers and the various stream compositions (in kg mol/hr) for the system. Note that the separators may be considered as an adiabatic flash tanks in which the pressure decrease is zero. This problem has been formulated from *Application Briefs of Process,* the user manual for the computer simulation software package of Simulation Science, Inc., but other flowsheeting packages can be used to solve it.

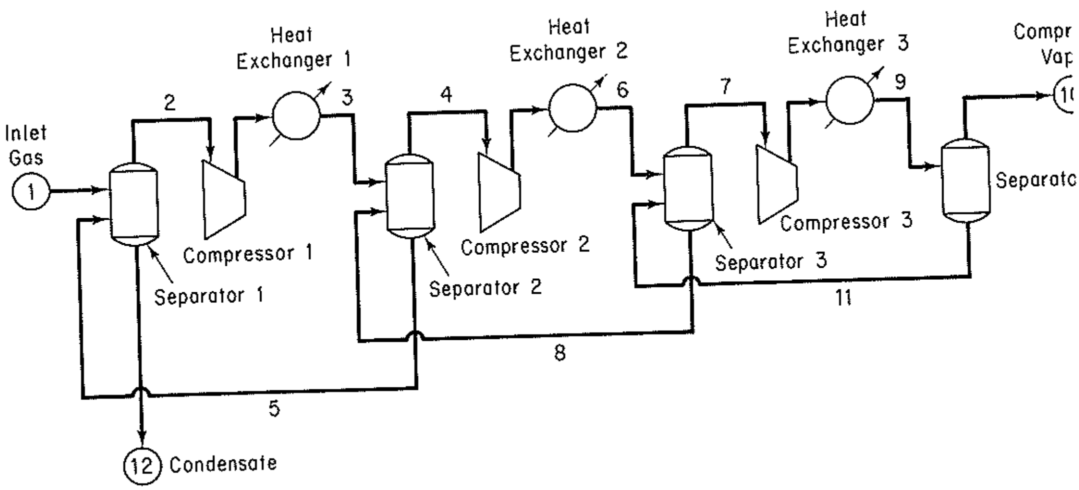


Figure P5.33

Component	kg mol/hr	M.W.	sp gr	Normal boiling point (°C)
Nitrogen	181			
Carbon dioxide	1,920			
Methane	14,515			
Ethane	9,072			
Propane	7,260			
Isobutane	770			
n-Butane	2,810			
Isopentane	953			
n-Pentane	1,633			
Hexane	1,542			
BP135	11,975	120	0.757	135
BP260	9,072	200	0.836	260
BP500	9,072	500	0.950	500

5.34. A demethanizer tower is used in a refinery to separate natural gas from a light hydrocarbon gas mixture stream (1) having the composition listed below. However, initial calculations show that there is a considerable energy wastage in the process. A proposed improved system is outlined in Fig. P5.34. Calculate the temperature (°F), pressure (psig), and composition (lb mol/hr) of all the process streams in the proposed system.

Inlet gas at 120°F and 588 psig stream (1) is cooled in the tube side of a gas–gas heat exchanger by passing the tower overhead stream (8) through the shell side. The temperature difference between the exit streams (2) and (10) of the heat exchanger is to be 10°F. Note that the pressure drop through the tube side is 10 psia and 5 psia on the shell side. The feed stream (2) is then passed through a chiller in which the temperature drops to -84°F and a pressure loss of 5 psi results. An adiabatic flash separater is used to separate the partially condensed vapor from the remaining gas. The vapor then passes through an expander turbine and is fed to the first tray of the tower at 125 psig. The liquid stream (5) is passed through a valve

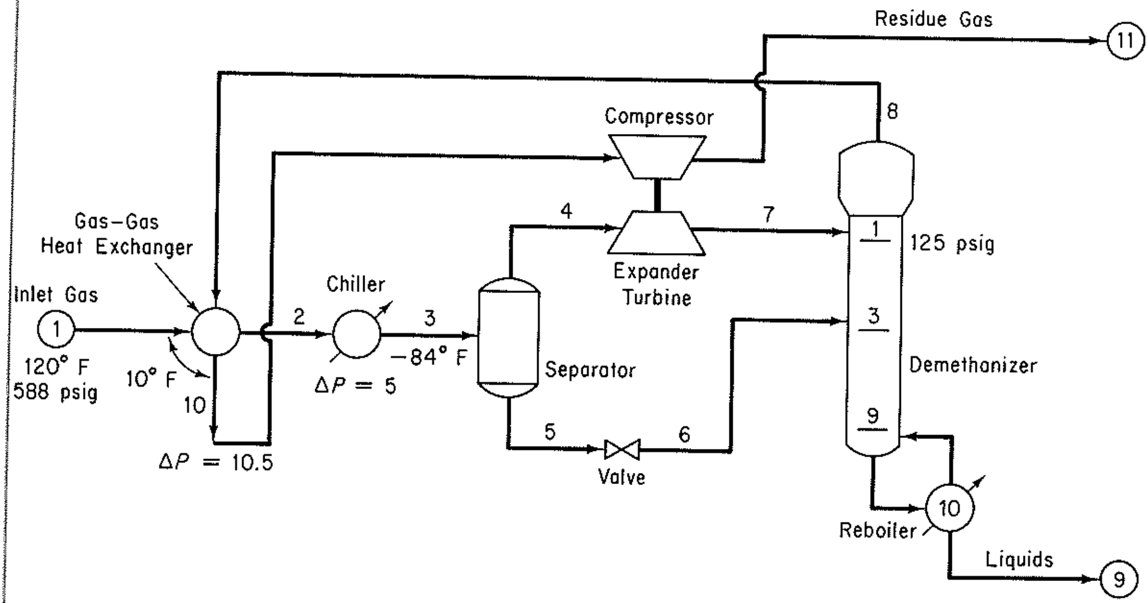


Figure P5.34

reducing the pressure to that of the third tray on the tower. The expander transfers 90% of its energy output to the compressor. The efficiency with respect to an adiabatic compression is 80% for the expander and 75% for the compressor. The process requirements are such that the methane-to-ethane ratio in the demethanizer liquids in stream 9 is to be 0.015 by volume; the heat duty on the reboiler is variable to achieve this ratio. A process rate of 23.06×10^6 standard cubic feet per day of feed stream 1 is required.

Component	Mol %
nitrogen	7.91
methane	73.05
ethane	7.68
propane	5.69
isopropane	0.99
n-butane	2.44
isopentane	0.69
n-pentane	0.82
C_6	0.42
C_7	0.31
Total	100.00

The tower has 10 trays, including the reboiler. *Note:* To reduce the number of trials, the composition of stream 3 may be referenced to stream 1 and if the exit stream of the chiller is given a dummy symbol, the calculations sequence can begin at the separator, thus eliminating the recycle loop.

Carry out the solution of the material and energy balances for the flowsheet in P5.34, determine the component and total mole flows, and determine the enthalpy flows for each stream. Also find the heat duty of each heat exchanger.

This problem has been formulated from *Application Briefs of Process*, the user manual for the computer simulation software package of Simulation Sciences, Inc., and is conveniently solved using Process, but other flowsheeting packages can be used.

5.35. A crude oil stream (1) at 1000 psia and 150°F having the composition shown in Table P5.35a is to be processed for shipping. The objectives are to (1) separate the hydrocarbons from the water and (2) reduce the pressure of the separated oil to atmospheric. Figure A in Fig. P5.35 illustrates a simplified version of a three-stage letdown process to carry out the objectives. The exit pressure of stream 3 from the first flash chamber, flash 1, is to be determined such that the product oil, stream 5, has a vapor pressure close to 14.7 psia. This specification is made directly on the process controller, and the exit pressure of the flash chamber 2 is the controlled parameter. To save computation time, each of the two compressor–heat exchanger–separator series shown in Figure B in Fig. P5.35 may first be modeled as isothermal flash units 4 and 5 inside the recycle loop. However, once the recycle is solved, the series should be substituted for these isothermal flash units in figure A in Fig. P5.35 and a rigorous calculation performed. Finally, a bubble point flash at 100°F should be performed on stream 5 as a check on the vapor pressure.

Other restrictions that are made on the system by the available equipment and the process are as follows. The water stream (W1) separated from flash 1 is split and 40% of the mass is delivered to the flash unit 2. All other water streams are discarded. The heat exchangers use utility water at 65°F and expel it at 75°F to cool the

TABLE P5.35a Feed Composition

	lb mol/hr
Water	3000.0
Carbon dioxide	35.0
Nitrogen	30.0
Methane	890.0
Ethane	300.0
Propane	520.0
Isobutane	105.0
n-Butane	283.0
Isopentane	100.0
n-Pentane	133.0
Cut 1	165.0
Cut 2	303.0
Cut 3	560.0
Cut 4	930.0
Cut 5	300.0
Cut 6	300.0
Cut 7	300.0
Cut 8	280.0
Cut 9	260.0
Cut 10	0.0
Total	8794.0

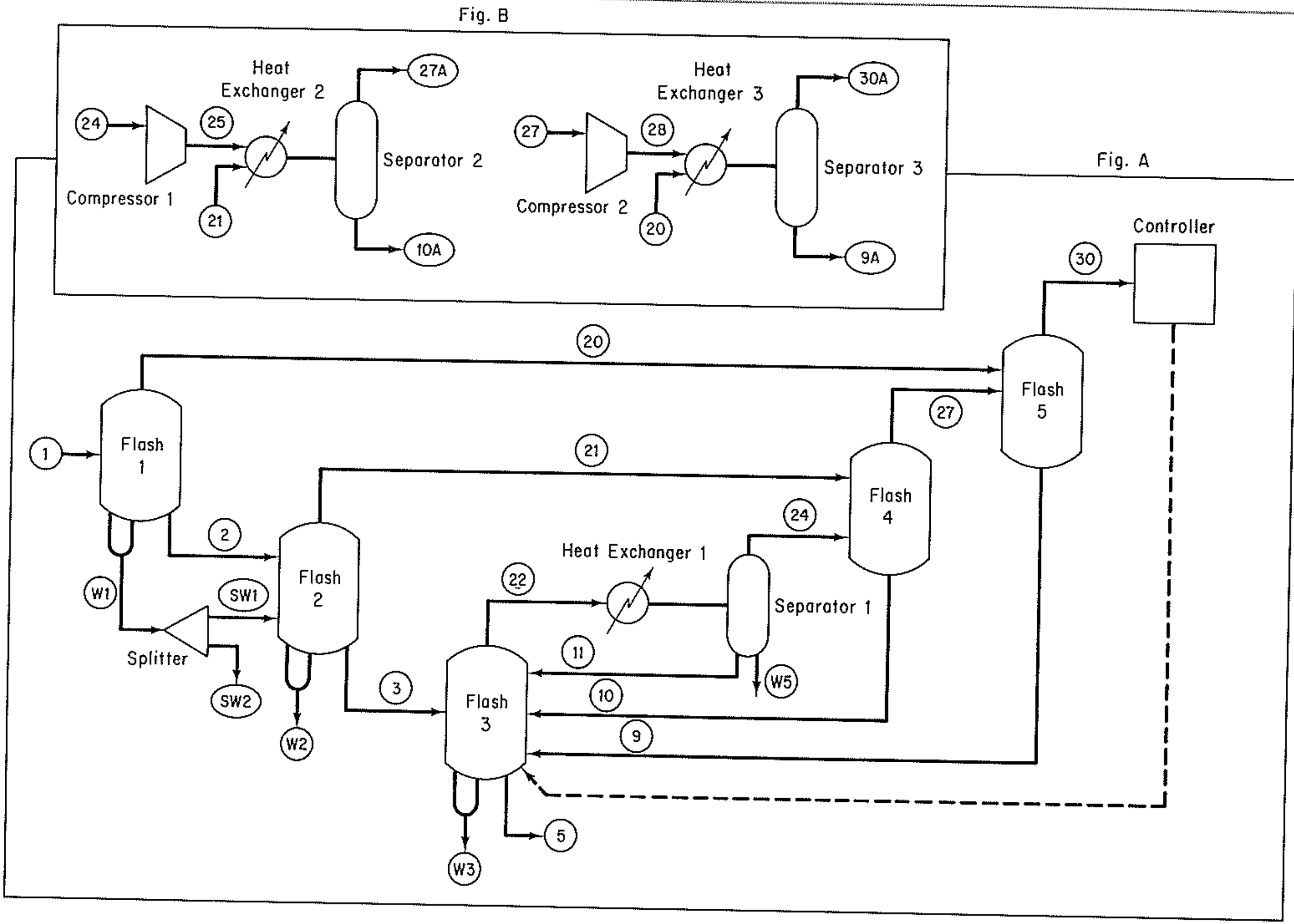


Fig. B
Heat Exchanger 2
Compressor 1
Separator 2
24
25
21
27A
10A
Heat Exchanger 3
Compressor 2
Separator 3
27
28
20
30A
9A
Fig. A
Controller
30
20
Flash 5
27
21
Flash 1
1
Flash 4
24
2
Flash 2
W1
SW1
Splitter
SW2
W2
Heat Exchanger 1
22
Separator 1
11
W5
3
Flash 3
10
9
5
W3

Figure P5.35

inlet streams to an exit temperature of 120°F. The pass through each exchanger results in a pressure drop of 3 psia in the hydrocarbon stream. Each compressor is 70% efficient with respect to an adiabatic compression and the outlet of pressure for compressors 1 and 2 are 100 psia and 300 psia, respectively. The operating conditions of the flash units are given in Table P5.35b. Additional data on the petroleum cuts are given in Table P5.35c.

Find the temperature, pressure, and liquid fraction of the exit product, and heat duty (if any) of each of the units in P5.35A.

This problem has been formulated from *Application Briefs of Process,* the user manual for the computer simulation software package of Simulation Sciences, Inc., but other flowsheeting packages can be used.

TABLE P5.35b

Flash unit	T (°F)	P (psia)	Operating condition
1	200	300	Isothermal
2	—	100	Adiabatic
3	—	14.7	Adiabatic
4	120	97	Isothermal
5	120	297	Isothermal

TABLE P5.35c Petroleum Cut Data

	MW	API Gravity	NBP °F
Cut 1	91	64	180
Cut 2	100	61	210
Cut 3	120	55	280
Cut 4	150	48	370
Cut 5	200	40	495
Cut 6	245	35	590
Cut 7	300	30	687
Cut 8	360	26	770
Cut 9	430	22	865
Cut 10	500	19	940

PROBLEMS THAT REQUIRE WRITING COMPUTER PROGRAMS

5.1. Determine the values of the unknown quantities in Fig. CP5.1 by solving the following set of linear material and energy balances that represent the steam balance:

(a) $181.60 - x_3 - 132.57 - x_4 - x_5 = -y_1 - y_2 + y_5 + y_4 = 5.1$

(b) $1.17x_3 - x_6 = 0$

(c) $132.57 - 0.745x_7 = 61.2$

(d) $x_5 + x_7 - x_8 - x_9 - x_{10} + x_{15} = y_7 + y_8 - y_3 = 99.1$

(e) $x_8 + x_9 + x_{10} + x_{11} - x_{12} - x_{13} = -y_7 = -8.4$

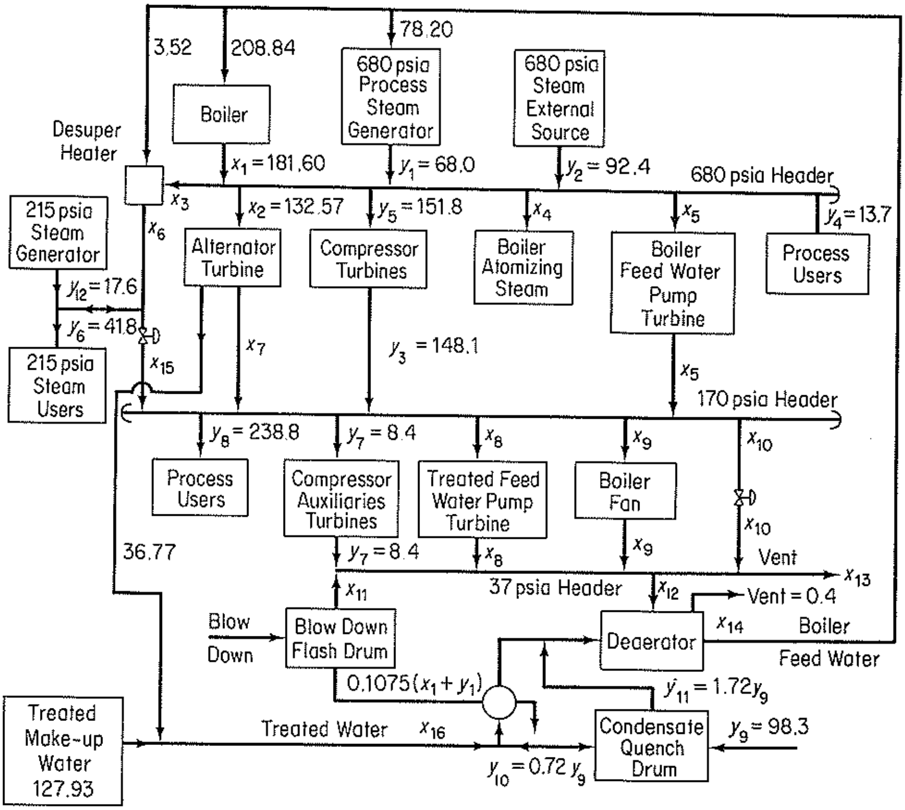


Figure CP5.1

(f) $x_6 - x_{15} = y_{12}^{*} - y_5 = 24.2$

(g) $-1.15(181.60) + x_3 - x_6 + x_{12} + x_{16} = 1.15y_1 - y_9 + 0.4 = -19.7$

(h) $181.60 - 4.594x_{12} - 0.11x_{16} = -y_1 + 1.0235y_9 + 2.45 = 35.05$

(i) $-0.0423(181.60) + x_{11} = 0.0423y_1 = 2.88$

(j) $-0.016(181.60) + x_4 = 0$

(k) $x_8 - 0.0147x_{16} = 0$

(l) $x_5 - 0.07x_{14} = 0$

(m) $-0.0805(181.60) + x_9 = 0$

(n) $x_{12} - x_{14} + x_{16} = 0.4 - y_9 = -97.9$

There are four levels of steam: 680, 215, 170, and 37 psia. The 14 x_i, $i = 3, \ldots,$ 16, are the unknowns and the y_i are given parameters for the system. Both x_i and y_i have the units of 10^3 lb/hr.

5.2. Wilson and Belkin (D. B. Wilson and H. M. Belkin, in *Stoichiometry*, E. Henley, ed., Cache Corp., Austin, Tex., 1972, p. 180) described a simplified refinery shown in Fig. CP5.2a that used three different crude oils and could produce eight products. Figure CP5.2b shows the boiling-point elevation curves that characterize each type of crude. Table CP5.2a lists the respective products and uses. Table CP5.2b lists the desired production rates.

You are asked to specify the proper operating temperatures (10 of them) indi-

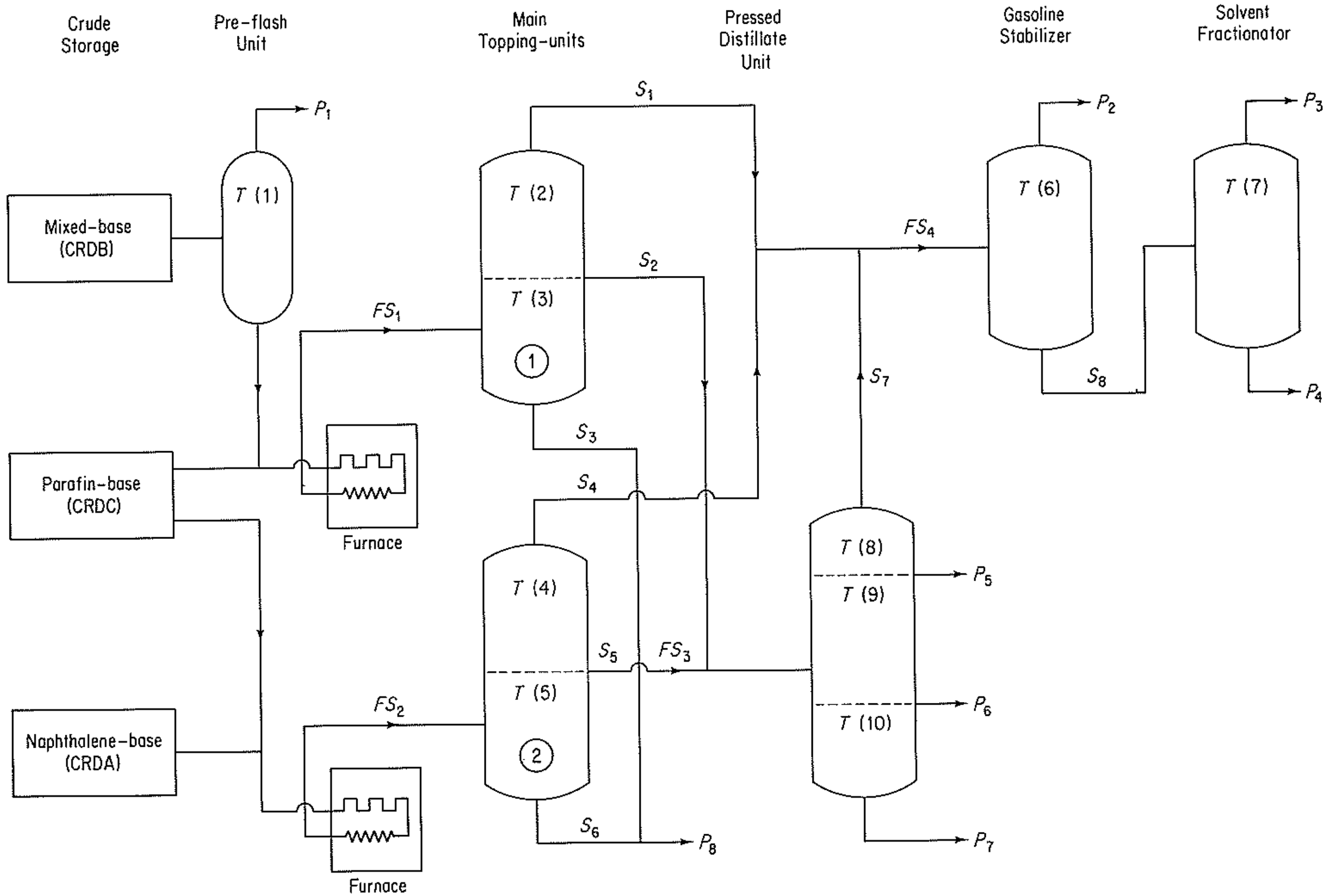


Figure CP5.2a

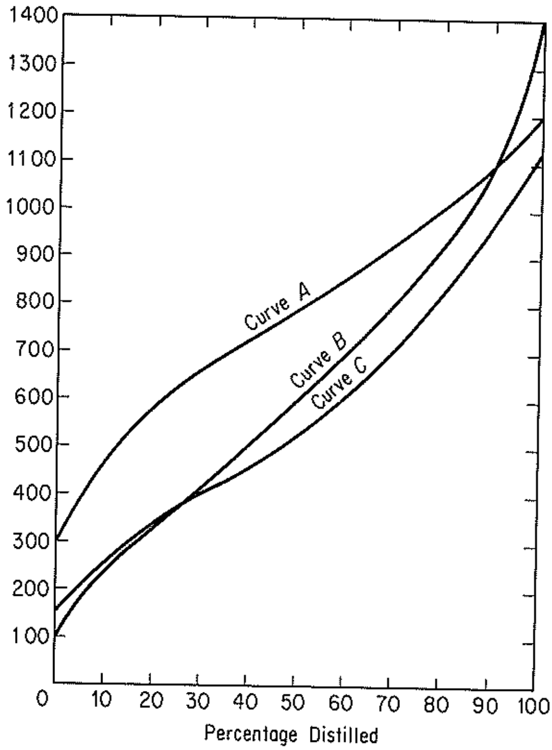


Figure CP5.2b

cated in Fig. CP5.2a by T(1) to T(10), given that the refinery will be supplied with 30,000 lb/hr of mixed-base crude and 43,000 lb/hr of paraffin-based crude.

Show that the process has 17 streams and 51 variables plus the fraction split between the two main topping units to give 52 variables in total and 34 degrees of freedom. Also show that if 10 values of the temperature are specified for three crude streams, 30 degrees of freedom are used. The other 4 degrees of freedom are taken up by specifying values for the three crude stream flows and the split between the topping units.

Table CP5.2c lists some of the restrictions placed on the temperatures selected and the possible range of values for each temperature. Functional relations for the boiling-point curves in terms of weight % distilled (FA, FB, and FC) versus temperature (°F) are

$$FA(T) = (27.50241 - 0.2061958T + 0.4441638 * 10^{-3}T^2 - 0.1853541 * 10^{-6}T^3)/100$$

$$FB(T) = (-0.2065866 + 0.7854635 * 10^{-1}T + 0.6124981 * 10^{-4}T^2 - 0.4395732 * 10^{-7}T^3)/100$$

$$FC(T) = (0.2424332 - 0.1151307T + 0.7828064 * 10^{-3}T^2 - 0.9227250 * 10^{-6}T^3 + 0.3437206 * 10^{-9}T^4)/100$$

Prepare a computer program in Fortran that determines the 10 temperatures in the refinery given the crude flow rates cited above and the fraction of crude C that is sent to topping unit 1. To adjust the temperatures, use any iterative technique that you wish.

TABLE CP5.2a Hydroskimming Refinery Products

Symbol	Name	True boiling-point range (°F)	Description of uses
P1	Fuel gas	150	Volatile products used by refinery for fuel
P2	Gasoline	125–425	Fuel for internal combustion engines
P3	Naphtha	355–510	Cleaner's solvents, paint thinners, chemical solvents, and stocks for blending of motor fuels
P4	Kerosene	422–555	Tractor fuel oil, stove oil, jet fuel
P5	Gas oil	440–700	Diesel engine fuel, fuels for industrial and household furnaces
P6	Cylinder stock	615–830	Unfinished heavy-oil stocks used directly as lubricants for steam engine cylinders; usually filtered but not dewaxed
P7	Fuel oil	750–1000	Industrial fuel
P8	Tar and asphalt	950–1400	Asphalt, road oil, roofing materials, and protective coating

TABLE CP5.2b

Product	Desired production rates (lb/hr)
Gasoline	As much as possible
Naphtha	Any amount
Kerosene	7400
Gas oil	Any amount
Cylinder stock	13,400
Fuel oil	11,000
Tar and asphalt	21,000
Fuel gas (to run equipment)	380

TABLE CP5.2c Temperature Ranges

Temperature point	Range of Values (°F)	Restrictions
T(1)	100–150	T(1) < T(2)
T(2)	225–375	T(2) < T(3)
T(3)	300–1000	
T(4)	225–375	T(4) < T(5)
T(5)	300–1050	
T(6)	125–350	T(6) < T(7)
T(7)	325–500	
T(8)	350–650	T(8) < T(9) and T(8) < T(10)
T(9)	440–700	T(9) < T(10)
T(10)	610–950	

UNSTEADY-STATE MATERIAL AND ENERGY BALANCES

6

In previous chapters all the material and all the energy balances you have encountered, except for the batch system balances, were *steady-state* balances, that is, balances for processes in which the accumulation term was zero. Now it is time for us to focus our attention briefly on **unsteady-state** processes. These are processes in which quantities or operating conditions within the system *change with time*. Sometimes you will hear the word *transient state* applied to such processes. The unsteady state is somewhat more complicated than the steady state and in general problems involving unsteady-state processes are somewhat more difficult to formulate and solve than those involving steady-state processes. However, a wide variety of important industrial problems fall into this category, such as the startup of equipment, batch heating or reactions, the change from one set of operating conditions to another, and the perturbations that develop as process conditions fluctuate. In this chapter we consider only one category of unsteady-state processes, but it is the one that is the most widely used, the lumped or macroscopic balances.

6.1 UNSTEADY-STATE MATERIAL AND ENERGY BALANCES

Your objectives in studying this section are to be able to:

1. Write down the macroscopic unsteady-state material and energy balances in words and in symbols.
2. Solve single linear ordinary differential material or energy balance equations given the initial conditions.
3. Take a word problem and translate it into a differential equation(s).

The basic expression in words, for either the material or the energy balance should by now be well known to you and is repeated below as Eq. (6.1). However, you should also realize that this equation can be applied at various levels, or strata, of description. In other words, an engineer can portray the operation of a real process by writing balances on a number of physical scales. A typical illustration of this concept might be in meterology, where the following different degrees of detail can be used on a descending scale of magnitude in the real world:

Global weather pattern
Local weather pattern
Individual clouds
Convective flow in clouds
Molecular transport
The molecules themselves

Similarly, in chemical engineering we write material and energy balances from the viewpoint of various scales of information:

(a) Molecular and atomic balances
(b) Microscopic balances
(c) Dispersion balances
(d) Plug flow balances
(e) Macroscopic balances (overall balances, lumped balances)

in decreasing order of degree of detail about a process.[1] In this chapter, the type of balance to be described and applied is the simplest one, the macroscopic balance [(e) in the list above].

The macroscopic balance ignores all the detail within a system and consequently results in a balance about the entire system. Only time remains as an independent variable in the balance. The dependent variables, such as concentration and temperature, are not functions of position but represent overall averages throughout the entire volume of the system. In effect, the system is assumed to be sufficiently well mixed so that the output concentrations and temperatures are equivalent to the concentrations and temperatures inside the system.

To assist in the translation of Eq. (6.1)

$$\left\{\begin{matrix}\text{accumulation}\\ \text{or depletion}\\ \text{within the}\\ \text{system}\end{matrix}\right\} = \left\{\begin{matrix}\text{transport into}\\ \text{system through}\\ \text{system}\\ \text{boundary}\end{matrix}\right\} - \left\{\begin{matrix}\text{transport out}\\ \text{of system}\\ \text{through system}\\ \text{boundary}\end{matrix}\right\} + \left\{\begin{matrix}\text{generation}\\ \text{within}\\ \text{system}\end{matrix}\right\} - \left\{\begin{matrix}\text{consumption}\\ \text{within}\\ \text{system}\end{matrix}\right\} \tag{6.1}$$

[1] Additional information together with applications concerning these various types of balances can be found in D. M. Himmelblau and K. B. Bischoff, *Process Analysis and Simulation*, Swift Publishing Co., Austin, Tex., 1980.

into mathematical symbols, you should refer to Fig. 6.1. Equation (6.1) can be applied to the mass of a single species or to the total amount of material or energy in the system. Let us write each of the terms in Eq. (6.1) in mathematical symbols for a very small time interval Δt. Let the accumulation be positive in the direction in which time is positive, that is, as time increases from t to $t + \Delta t$. Then, using the component mass balance as an example, the accumulation will be the mass of A in the system at time $t + \Delta t$ minus the mass of A in the system at time t,

$$\text{accumulation} = \rho_A V\big|_{t+\Delta t} - \rho_A V\big|_t$$

where[2] ρ_A = mass of component A per unit volume
V = volume of the system

Note that the net dimensions of the accumulation term are the mass of A.

We shall split the mass transport across the system boundary into two parts, transport through defined surfaces S_1 and S_2, whose areas are known, and transport across the system boundary through other (undefined) surfaces. The net transport of A into (through S_1) and out of (through S_2) the system through defined surfaces can be written as

$$\text{net flow across boundary via } S_1 \text{ and } S_2 = \rho_A v S\,\Delta t\big|_{S_1} - \rho_A v S\,\Delta t\big|_{S_2}$$

where v = fluid velocity in a duct of cross section S
S = defined cross-sectional area perpendicular to material flow

Again note that the net dimensions of the transport term are the mass of A. Other types of transport across the system boundary can be represented by

$$\text{net residual flow across boundary} = \dot{w}_A\,\Delta t$$

where $\dot{w}_A$ is the rate of mass flow of component A through the system boundaries other than the defined surfaces S_1 and S_2.

Finally, the net generation-consumption term will be assumed to be due to a chemical reaction r_A:

$$\text{net generation-consumption} = \dot{r}_A\,\Delta t$$

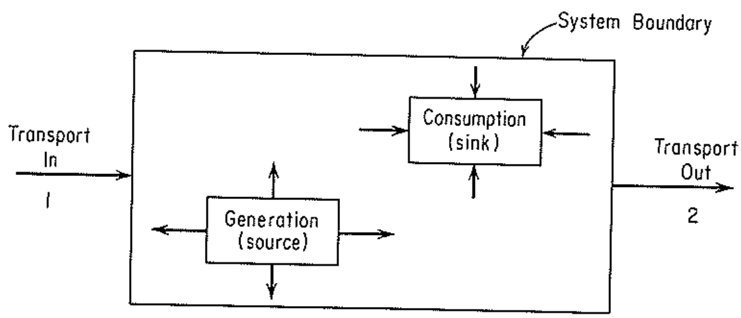


Figure 6.1 General unsteady-state process with transport in and out and internal generation and consumption.

[2] The symbol $|_t$ means that the quantities preceding the vertical line are evaluated at time t, or time $t + \Delta t$, or at surface S_1, or at surface S_2, as the case may be as denoted by the subscript.

where $\dot{r}_A$ is the net rate of generation-consumption of component A by chemical reaction.

Introduction of all these terms into Eq. (6.1) gives Eq. (6.2). Equations (6.3) and (6.4) can be developed from exactly the same type of analysis. Try to formulate Eqs. (6.3) and (6.4) yourself.

Species material balances:

$$\underbrace{\rho_A V|_{t+\Delta t} - \rho_A V|_t}_{\text{accumulation}} = \underbrace{\rho_A vS\ \Delta t|_{S_1} - \rho_A vS\ \Delta t|_{S_2}}_{\text{transport through defined boundaries}} + \underbrace{\dot{w}_A\ \Delta t}_{\substack{\text{transport}\\\text{through}\\\text{other}\\\text{boundaries}}} + \underbrace{\dot{r}_A\ \Delta t}_{\substack{\text{genera-}\\\text{tion or}\\\text{consump-}\\\text{tion}}} \tag{6.2}$$

Total material balance:

$$\underbrace{\rho V|_{t+\Delta t} - \rho V|_t}_{\text{accumulation}} = \underbrace{\rho vS\ \Delta t|_{S_1} + \rho vS\ \Delta t|_{S_2}}_{\text{transport through defined boundaries}} + \underbrace{\dot{w}\ \Delta t}_{\substack{\text{transport}\\\text{through}\\\text{other}\\\text{boundaries}}} \tag{6.3}$$

Energy balance:

$$\underbrace{E|_{t+\Delta t} - E|_t}_{\text{accumulation}} = \underbrace{\left|\left(\hat{H} + \frac{v^2}{2} + gh\right)\dot{m}\ \Delta t\right|_{S_1} - \left|\left(\hat{H} + \frac{v^2}{2} + gh\right)\dot{m}\ \Delta t\right|_{S_2}}_{\text{transport through defined boundaries}}$$

$$+ \underbrace{\dot{Q}\ \Delta t}_{\text{heat}} - \underbrace{\dot{W}\ \Delta t}_{\text{work}} + \underbrace{\dot{B}\ \Delta t}_{\substack{\text{transport}\\\text{through}\\\text{other}\\\text{boundaries}}} \tag{6.4}$$

where $\dot{B}$ = rate of energy transfer accompanying $\dot{w}$
$\dot{m}$ = rate of mass transfer through defined surfaces
$\dot{Q}$ = rate of heat transfer to system
$\dot{W}$ = rate of work done by the system
$\dot{w}$ = rate of total mass flow through the system boundaries other than through the defined surfaces S_1 and S_2
ρ = total mass per unit volume

The other notation for the material and energy balances is identical to that of Chaps. 2 and 4; note that the work, heat, generation, and mass transport are now all expressed as rate terms (mass or energy per unit time).

If each side of Eq. (6.2) is divided by Δt, we obtain

$$\frac{\rho_A V|_{t+\Delta t} - \rho_A V|_t}{\Delta t} = \rho_A vS|_{S_1} - \rho_A vS|_{S_2} + \dot{w}_A + \dot{r}_A \tag{6.5}$$

Similar relations can be obtained from Eqs. (6.3) and (6.4). Next, if we take the limit of each side of Eq. (6.5) as $\Delta t \to 0$, we get

$$\frac{d(\rho_A V)}{dt} = -\Delta(\rho_A vS) + \dot{w}_A + \dot{r}_A \tag{6.6}$$

Similar treatment of the total mass balance and the energy balance yields the following two equations:

$$\frac{d(\rho V)}{dt} = -\Delta(\rho v S) + \dot{w} \tag{6.7}$$

$$\frac{d(E)}{dt} = -\Delta\left[\left(\hat{H} + \frac{v^2}{2} + gh\right)\dot{m}\right] + \dot{Q} - \dot{W} + \dot{B} \tag{6.8}$$

Can you get Eqs. (6.7) and (6.8) from (6.3) and (6.4), respectively? Try it.

The relation between the energy balance given by Eq. (6.8), which has the units of energy per unit time, and the energy balance given by Eq. (4.24), which has the units of energy, should now be fairly clear. Equation (4.24) represents the integration of Eq. (6.8) with respect to time, expressed formally as follows:

$$E_{t_2} - E_{t_1} = \int_{t_1}^{t_2} \{-\Delta[(\hat{H} + \hat{K} + \hat{P})\dot{m}] + \dot{Q} + \dot{B} - \dot{W}\}dt \tag{6.9}$$

The quantities designated in Eqs. (4.24) or (4.24a) without the superscript dot are the respective integrated values from Eq. (6.9).

To solve one or a combination of the very general equations (6.6), (6.7), or (6.8) analytically may be quite difficult, and in the following examples we shall have to restrict our analyses to simple cases. If we make enough (reasonable) assumptions and work with simple problems, we can consolidate or eliminate enough terms of the equations to be able to integrate them and develop some analytical answers. Numerical solutions are also possible for nonlinear equations.

In the solution of unsteady-state problems you have to apply the usual procedures of problem solving discussed in Chaps. 1, 2, and 4. Two major tasks exist:

(a) To set up the unsteady-state equation(s)
(b) To solve it (them) once the equation(s) is (are) established

After you draw a diagram of the system and set down all the information available, you should try to recognize the important variables and represent them by letters. Next, decide which variable is the independent one, and label it. The independent variable is the one you select to have one or a series of values, and then the other variables are all dependent ones—they are determined by the first variable(s) and the initial state of the process you selected. Which is chosen as an independent and which as a dependent variable is usually fixed by the problem but may be arbitrary. Although there are no general rules applicable to all cases, the quantities that appear prominently in the statement of the problem are usually the best choices. You will need as many independent equations as you have dependent variables.

As mentioned before, in the macroscopic balance the independent variable is time. In mathematics, when the quantity x varies with time, we consider dx to be the change in x that occurs during time dt if the process continues through the interval beginning at time t. Let us say that t is the independent variable and x is the dependent one. In solving problems you can use one or a combination of Eqs. (6.6)–(6.8) directly, or alternatively you can proceed, as shown in some of the examples, to set up the differential equations from scratch exactly in the same fashion as Eqs. (6.2)–(6.4) were formulated. For either approach the objective is to translate the problem statement in words into one or more simultaneous differential equations having the form

$$\frac{dx}{dt} = f(x, t) \tag{6.10}$$

Then, assuming this differential equation can be solved, x can be found as a function of t. Of course, we have to know some initial condition(s) or at one (or more) given time(s) know the value(s) of x.

If dt and dx are always considered positive when increasing, you can use Eq. (6.1) without having any difficulties with signs. However, if you for some reason transfer an output to the left-hand side of the equation, or the equation is written in some other form, you should take great care in the use of signs (see Example 6.2 below). We are now going to examine some very simple unsteady-state problems which are susceptible to reasonably elementary mathematical analysis. You can (and will) find more complicated examples in texts dealing with all phases of mass transfer, heat transfer, and fluid dynamics.

EXAMPLE 6.1 Unsteady-State Material Balance without Generation

A tank holds 100 gal of a water-salt solution in which 4.0 lb of salt is dissolved. Water runs into the tank at the rate of 5 gal/min and salt solution overflows at the same rate. If the mixing in the tank is adequate to keep the concentration of salt in the tank uniform at all times, how much salt is in the tank at the end of 50 min? Assume that the density of the salt solution is essentially the same as that of water.

Solution

We shall set up from scratch the differential equations that describe the process.

Step 1 Draw a picture, and put down the known data. See Fig. E6.1.

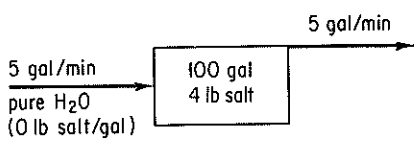


Figure E6.1

Step 2 Choose the independent and dependent variables. Time, of course, is the independent variable, and either the salt quantity or concentration in the tank can be the dependent variable. Suppose that we make the mass (quantity) of salt the dependent variable. Let x = lb of salt in the tank at time t.

Step 3 Write the known value of x at a given value of t. This is the initial condition:

$$\text{at } t = 0 \qquad x = 4.0 \text{ lb}$$

Step 4 It is easiest to make a total material balance and a component material balance on the salt. (No energy balance is needed because the system can be assumed to be isothermal.)

Total balance:

$$\textbf{accumulation} = \textbf{in}$$

$$[m_{tot}\ \text{lb}]_{t+\Delta t} - [m_{tot}\ \text{lb}]_t = \frac{5\ \text{gal}}{\text{min}}\left|\frac{1\ \text{ft}^3}{7.48\ \text{gal}}\right|\frac{\rho_{H_2O}\ \text{lb}}{\text{ft}^3}\left|\frac{\Delta t\ \text{min}}{}\right.$$

$$\textbf{out}$$

$$-\frac{5\ \text{gal}}{\text{min}}\left|\frac{1\ \text{ft}^3}{7.48\ \text{gal}}\right|\frac{\rho_{soln}\ \text{lb}}{\text{ft}^3}\left|\frac{\Delta t\ \text{min}}{}\right. = 0$$

This equation tells us that the flow of water into the tank equals the flow of solution out of the tank if $\rho_{H_2O} = \rho_{soln}$ as assumed. Otherwise, there is an accumulation term.

Salt balance:

$$\textbf{accumulation} = \textbf{in} - \textbf{out}$$

$$[x\ \text{lb}]_{t+\Delta t} - [x\ \text{lb}]_t = 0 - \frac{5\ \text{gal}}{\text{min}}\left|\frac{x\ \text{lb}}{100\ \text{gal}}\right|\frac{\Delta t\ \text{min}}{}$$

Dividing by Δt and taking the limit as Δt approaches zero,

$$\lim_{\Delta t \to 0} \frac{[x]_{t+\Delta t} - [x]_t}{\Delta t} = -0.05x$$

or

$$\frac{dx}{dt} = -0.05x \tag{a}$$

Notice how we have kept track of the units in the normal fashion in setting up these equations. Because of our assumption of uniform concentration of salt in the tank, the concentration of salt leaving the tank is the same as that in the tank, or x lb/100 gal of solution.

Step 5 Solve the unsteady-state material balance on the salt. By separating the independent and dependent variables, we get

$$\frac{dx}{x} = -0.05\ dt$$

This equation is easily integrated between the definite limits of

$$t = 0 \qquad x = 4.0$$

$$t = 50 \qquad X = \text{the unknown value of } x \text{ lb}$$

$$\int_{4.0}^{X} \frac{dx}{x} = -0.05 \int_0^{50} dt$$

$$\ln\frac{X}{4.0} = -2.5 \qquad \ln\frac{4.0}{X} = 2.5$$

$$\frac{4.0}{X} = 12.2 \qquad X = \frac{4.0}{12.2} = 0.328 \text{ lb salt}$$

An equivalent differential equation to Eq. (a) can be obtained directly from the component mass balance in the form of Eq. (6.6) if we let ρ_A = concentration of salt in the tank at any time t in terms of lb/gal:

$$\frac{d(\rho_A V)}{dt} = -\left(\frac{5\ \text{gal}}{\text{min}}\left|\frac{\rho_A\ \text{lb}}{\text{gal}}\right.\right) - 0$$

If the tank holds 100 gal of solution at all times, V is a constant and equal to 100, so that

$$\frac{d\rho_A}{dt} = -\frac{5\rho_A}{100} \tag{b}$$

The initial conditions are

$$\text{at } t = 0 \qquad \rho_A = 0.04$$

The solution of Eq. (b) would be carried out exactly as the solution of Eq. (a).

EXAMPLE 6.2 Unsteady-State Material Balance without Generation

A square tank 4 m on a side and 10 m high is filled to the brim with water. Find the time required for it to empty through a hole in the bottom 5 cm² in area.

Solution

Step 1 Draw a diagram of the process, and put down the data. See Fig. E6.2a.

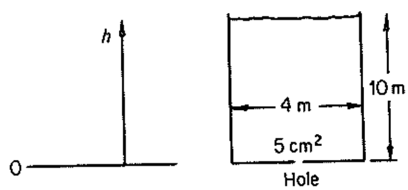


Figure E6.2a

Step 2 Select the independent and dependent variables. Again, time will be the independent variable. We could select the quantity of water in the tank as the dependent variable, but since the cross section of the tank is constant, let us choose h, the height of the water in the tank, as the dependent variable.

Step 3 Write the known value of h at a given value of t:

$$\text{at } t = 0 \qquad h = 10 \text{ m}$$

Step 4 Develop the unsteady-state balance(s) for the process. In an elemental time Δt, the height of the water in the tank drops Δh. The mass of water leaving the tank is in the form of a cylinder 5 cm² in area, and we can calculate the quantity as

$$\frac{5 \text{ cm}^2}{} \Bigg| \left(\frac{1 \text{ m}}{100 \text{ cm}}\right)^2 \Bigg| \frac{v^* \text{m}}{\text{s}} \Bigg| \frac{\rho \text{ kg}}{\text{m}^3} \Bigg| \frac{\Delta t \text{ s}}{} = 5 \times 10^{-4} v^* \rho\, \Delta t \qquad \text{kg}$$

where ρ = density of water
v^* = average velocity of the water leaving the tank

The depletion of water inside the tank in terms of the variable h, expressed in pounds, is

$$\left.\frac{16 \text{ m}^2}{} \Bigg| \frac{\rho \text{ kg}}{\text{m}^3} \Bigg| \frac{h \text{ m}}{}\right|_{t+\Delta t} - \left.\frac{16 \text{ m}^2}{} \Bigg| \frac{\rho \text{ kg}}{\text{m}^3} \Bigg| \frac{h \text{ m}}{}\right|_{t} = 16\rho\Delta h \qquad \text{kg}$$

An overall material balance indicates that

$$\textbf{accumulation} = \textbf{in} - \textbf{out}$$

$$16\rho\Delta h = 0 - 5 \times 10^{-4} \rho v^* \Delta t \tag{a}$$

Although Δh is a negative value, our equation takes account of this automatically. The accumulation is positive if the right-hand side is positive, and the accumulation is negative if the right-hand side is negative. Here the accumulation is really a depletion. You can see that the term ρ, the density of water, cancels out, and we could just as well have made our material balance on a volume of water.

Equation (a) becomes

$$\frac{\Delta h}{\Delta t} = -\frac{5 \times 10^{-4} v^*}{16}$$

Taking the limit as Δh and Δt approach zero, we get

$$\frac{dh}{dt} = -\frac{5 \times 10^{-4} v^*}{16} \tag{b}$$

Step 5 Unfortunately, this is an equation with one independent variable, t, and two dependent variables, h and v^*. We must find another equation to eliminate either h or v^* if we want to obtain a solution. Since we want our final equation to be expressed in terms of h, the next step is to find some function that relates v^* to h and t, and then we can substitute for v^* in the unsteady-state equation.

We shall employ the steady-state mechanical energy balance for an incompressible fluid, discussed in Chap. 4, to relate v^* and h. See Fig. E6.2b. Recall from Eq. (4.30) with $W = 0$ and $E_v = 0$ that

$$\Delta\left(\frac{v^2}{2} + gh\right) = 0 \tag{c}$$

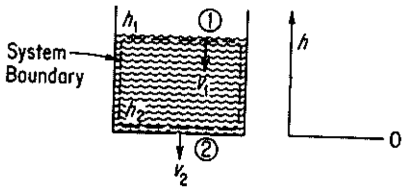


Figure E6.2b

We assume that the pressures are the same at section ①—the water surface—and section ②—the hole—for the system consisting of the water in the tank. Equation (c) reduces to

$$\frac{v_2^2 - v_1^2}{2} + g(h_2 - h_1) = 0 \tag{d}$$

where v_2 = exit velocity through the 5 cm² hole at boundary ②
v_1 = velocity of the water in the tank at boundary ①

If $v_1 \simeq 0$, a reasonable assumption for the water in the large tank at any time, at least compared to v_2, we have

$$\begin{aligned} v_2^2 &= -2g(0 - h_1) = 2gh \\ v_2 &= \sqrt{2gh} \end{aligned} \tag{e}$$

Because the exit-stream flow is not frictionless and because of turbulence and orifice effects in the exit hole, we must correct the value of v given by Eq. (e) for frictionless flow by an empirical adjustment factor as follows:

$$v_2 = c\sqrt{2gh} = v^* \tag{f}$$

where c is an orifice correction which we could find (from a text discussing fluid dynamics) has a value of 0.62 for this case. Thus

$$v^* = 0.62\sqrt{2(9.80)h} = 2.74\sqrt{h} \qquad \text{m/s}$$

Let us substitute this relation into Eq. (b) in place of v^*. Then we obtain

$$\frac{dh}{dt} = \frac{(5.4 \times 10^{-4})(2.74)(h)^{1/2}}{16}$$

or

$$-1.08 \times 10^4 \frac{dh}{h^{1/2}} = dt \tag{g}$$

Equation (g) can be integrated between

$$h = 10 \text{ m} \qquad \text{at } t = 0$$

and

$$h = 0 \text{ m} \qquad \text{at } t = \theta, \quad \text{the unknown time}$$

$$-1.08 \times 10^4 \int_{10}^{0} \frac{dh}{h^{1/2}} = \int_0^{\theta} dt$$

to yield θ,

$$\theta = 1.08 \times 10^4 \int_0^{10} \frac{dh}{h^{1/2}} = 1.08 \times 10^4 [2\sqrt{h}]_0^{10} = 6.83 \times 10^4 \text{ s}$$

EXAMPLE 6.3 Material Balance in Batch Distillation

A small still is separating propane and butane at 135°C and initially contains 10 kg moles of a mixture whose composition is $x = 0.30$ (x = mole fraction butane). Additional mixture ($x_F = 0.30$) is fed at the rate of 5 kg mol/hr. If the total volume of the liquid in the still is constant, and the concentration of the vapor from the still (x_D) is related to x_S as follows:

$$x_D = \frac{x_S}{1 + x_S}$$

how long will it take for the value of x_S to change from 0.30 to 0.40? What is the steady-state ("equilibrium") value of x_S in the still (i.e., when x_S becomes constant)? See Fig. E6.3.

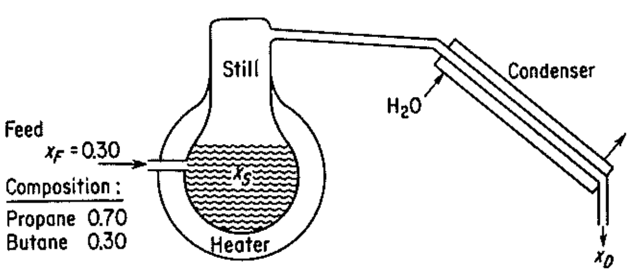


Figure E6.3

Solution

Since butane and propane form ideal solutions, we do not have to worry about volume changes on mixing or separation. Only the material balance is needed to answer the questions posed. If t is the independent variable and x_S the dependent variable, we can say that

Butane Balance (C_4): The input to the still is

$$\frac{5 \text{ mol feed}}{\text{hr}}\left|\frac{0.30 \text{ mol } C_4}{\text{mol feed}}\right|\frac{\Delta t \text{ hr}}{}$$

The output from the still is equal to the amount condensed,

$$\frac{5 \text{ mol cond.}}{\text{hr}}\left|\frac{x_D \text{ mol } C_4}{\text{mol cond.}}\right|\frac{\Delta t \text{ hr}}{}$$

The accumulation is

$$\frac{10 \text{ mol in still}}{}\left|\frac{x_S \text{ mol } C_4}{\text{mol in still}}\right|_{t+\Delta t} - \frac{10 \text{ mol in still}}{}\left|\frac{x_S \text{ mol } CH_4}{\text{mol in still}}\right|_{t} = 10\Delta x_S$$

Our unsteady-state material balance is then

$$\textbf{accumulation} = \textbf{in} \quad - \textbf{out}$$

$$10\,\Delta x_S = 1.5\,\Delta t - 5x_D\,\Delta t$$

or, dividing by Δt and taking the limit as Δt approaches zero,

$$\frac{dx_S}{dt} = 0.15 - 0.5x_D$$

As in Example 6.2, it is necessary to reduce the equation to one dependent variable by substituting for x_D:

$$x_D = \frac{x_S}{1 + x_S}$$

Then

$$\frac{dx_S}{dt} = 0.15 - \frac{x_S}{1 + x_S}(0.5)$$

$$\frac{dx_S}{0.15 - \dfrac{0.5x_S}{1 + x_S}} = dt$$

The integration limits are

$$\text{at } t = 0 \qquad x_S = 0.30$$

$$t = \theta \qquad x_S = 0.40$$

$$\int_{0.30}^{0.40} \frac{dx_S}{0.15 - [0.5x_S/(1 + x_S)]} = \int_0^\theta dt = \theta$$

$$\int_{0.30}^{0.04} \frac{(1 + x_S)\,dx_S}{0.15 - 0.35x_S} = \theta = \left[-\frac{x_S}{0.35} - \frac{1}{(0.35)^2}\ln(0.15 - 0.35x_S)\right]_{0.30}^{0.40}$$

$$\theta = 5.85 \text{ hr}$$

If you did not know how to integrate the equation analytically, or if you only had experimental data for x_D as a function of x_S instead of the given equation, you could always integrate the equation numerically on a digital computer.

The steady-state value of x_S is established at infinite time or, alternatively, when the accumulation is zero. At that time,

$$0.15 = \frac{0.5x_S}{1 + x_S} \qquad \text{or} \qquad x_S = 0.428$$

The value of x_S could never be greater than 0.428 for the given conditions.

EXAMPLE 6.4 Unsteady-State Chemical Reaction

A compound dissolves in water at a rate proportional to the product of the amount undissolved and the difference between the concentration in a saturated solution and the concentration in the actual solution at any time. A saturated solution of compound contains 40 g/100 g H_2O. In a test run starting with 20 kg of undissolved compound in 100 kg of pure compound is found that 5 kg is dissolved in 3 hr. If the test continues, how many kilograms of compound will remain undissolved after 7 hr? Assume that the system is isothermal.

Solution

Step 1 See Fig. E6.4.

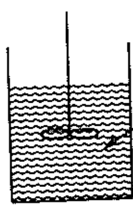


Figure E6.4

Step 2 Let the dependent variable x = the kilograms of *undissolved* compound a time t (the independent variable).

Step 3 At any time t the concentration of dissolved compound is

$$\frac{20 - x}{100}\left(\frac{\text{kg compound}}{\text{lb } H_2O}\right)$$

The rate of dissolution of compound according to the problem statement is

$$\text{rate } \frac{\text{kg}}{\text{hr}} = k(x \text{ kg})\left[\frac{40 \text{ kg compound}}{100 \text{ kg } H_2O} - \frac{(20 - x) \text{ kg compound}}{100 \text{ kg } H_2O}\right]$$

where k is the proportionality constant. We could have utilized a concentration measured on volume basis rather than one measured on a weight basis, but this would just change the valu and units of the constant k.

Next we make an unsteady-state balance for the compound in the solution (the system

$$\textbf{accumulation} = \textbf{in} - \textbf{out} + \textbf{generation}$$

$$\frac{dx}{dt} = 0 - 0 + kx\left(\frac{40}{100} - \frac{20 - x}{100}\right)$$

or

$$\frac{dx}{dt} = \frac{kx}{100}(20 + x)$$

Step 4 Solution of the unsteady-state equation:

$$\frac{dx}{x(20 + x)} = \frac{k}{100}\,dt$$

This equation can be split into two parts and integrated,

$$\frac{dx}{x} - \frac{dx}{x + 20} = \frac{k}{5}\,dt$$

or integrated directly for the conditions

$$\begin{array}{ll} t_0 = 0 & x_0 = 20 \\ t_1 = 3 & x_1 = 15 \\ t_2 = 7 & x_2 = ? \end{array}$$

Since k is an unknown, the first pair of conditions is given in order to evaluate k.
To find k, we integrate between the limits stated in the test run:

$$\int_{20}^{15} \frac{dx}{x} - \int_{20}^{15} \frac{dx}{x + 20} = \frac{k}{5}\int_0^3 dt$$

$$\left[\ln \frac{x}{x + 20}\right]_{20}^{15} = \frac{k}{5}(3 - 0)$$

$$k = -0.257$$

(Note that the negative value of k corresponds to the actual physical situation in which undissolved material is consumed.) To find the unknown amount of undissolved compound at the end of 7 hr, we have to integrate again:

$$\int_{20}^{x_2} \frac{dx}{x} - \int_{20}^{x_2} \frac{dx}{x + 20} = \frac{-0.257}{5}\int_0^7 dt$$

$$\left[\ln \frac{x}{x + 20}\right]_{20}^{x_2} = \frac{-0.0257}{5}(7 - 0)$$

$$x_2 = 10.7 \text{ kg}$$

EXAMPLE 6.5 Overall Unsteady-State Process

A 15% Na_2SO_4 solution is fed at the rate of 12 lb/min into a mixer that initially holds 100 lb of a 50-50 mixture of Na_2SO_4 and water. The exit solution leaves at the rate of 10 lb/min. Assuming uniform mixing, what is the concentration of Na_2SO_4 in the mixer at the end of 10 min? Ignore any volume changes on mixing.

Solution

In contrast to Example 6.1, both the water and the salt concentrations, and the total material in the vessel, change with time in this problem.

Step 1 See Fig. E6.5.

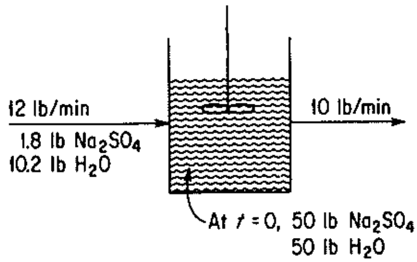


Figure E6.5

Step 2

$$\text{Let } x = \text{fraction } Na_2SO_4 \text{ in the tank at time } t, \left(\frac{\text{lb } Na_2SO_4}{\text{lb total}}\right)$$

$$y = \text{total pounds of material in the tank at time } t$$

Step 3 At time $t = 0$, $x = 0.50$, $y = 100$.
Step 4 Material balances between time t and $t + \Delta t$:
(a) Total balance:

$$\begin{array}{ccccc} \textbf{accumulation} & = & \textbf{in} & - & \textbf{out} \end{array}$$

$$[y]_{t+\Delta t} - [y]_t = \Delta y = \frac{12 \text{ lb}}{\text{min}}\Big|\,\Delta t \text{ min} - \frac{10 \text{ lb}}{\text{min}}\Big|\,\Delta t \text{ min}$$

or

$$\frac{dy}{dt} = 2 \tag{a}$$

(b) Na_2SO_4 balance:

$$\textbf{accumulation}$$

$$\left[\frac{x \text{ lb } Na_2SO_4}{\text{lb total}}\Big|\,y \text{ lb total}\right]_{t+\Delta t} - \left[\frac{x \text{ lb } Na_2SO_4}{\text{lb total}}\Big|\,y \text{ lb total}\right]_t$$

$$= \quad \textbf{in} \quad - \quad \textbf{out}$$

$$= \frac{1.8 \text{ lb } Na_2SO_4}{\text{min}}\Big|\,\Delta t \text{ min} - \frac{10 \text{ lb total}}{\text{min}}\Big|\,\frac{x \text{ lb } Na_2SO_4}{\text{lb total}}\Big|\,\Delta t \text{ min}$$

or

$$\frac{d(xy)}{dt} = 1.8 - 10x \tag{b}$$

Step 5 There are a number of ways to handle the solution of these two differential equations with two unknowns. Perhaps the easiest is to differentiate the xy product in Eq. (b) and substitute Eq. (a) into the appropriate term containing dy/dt:

$$\frac{d(xy)}{dt} = x\frac{dy}{dt} + y\frac{dx}{dt}$$

To get y, we have to integrate Eq. (a):

$$\int_{100}^{y} dy' = \int_{0}^{t} 2\,dt'$$

$$y - 100 = 2t$$

$$y = 100 + 2t$$

Then

$$\frac{d(xy)}{dt} = x(2) + (100 + 2t)\frac{dx}{dt} = 1.8 - 10x$$

$$\frac{dx}{dt}(100 + 2t) = 1.8 - 12x$$

$$\int_{0.50}^{x} \frac{dx'}{1.8 - 12x} = \int_{0}^{10} \frac{dt'}{100 + 2t}$$

$$-\frac{1}{12}\ln\frac{1.8 - 12x}{1.8 - 6.0} = \frac{1}{2}\ln\frac{100 + 20}{100}$$

$$x = 0.267\ \frac{\text{lb } Na_2SO_4}{\text{lb total solution}}$$

An alternative method of setting up the dependent variable would have been to let x = lb of Na_2SO_4 in the tank at the time t. Then the Na_2SO_4 balance would be

$$\frac{dx}{dt}\frac{\text{lb } Na_2SO_4}{\text{min}} = \frac{1.8 \text{ lb } Na_2SO_4}{\text{min}} - \frac{10 \text{ lb total}}{\text{min}}\left|\frac{x \text{ lb } Na_2SO_4}{y \text{ lb total}}\right.$$

$$\frac{dx}{dt} = 1.8 - \frac{10x}{100 + 2t}$$

or

$$\frac{dx}{dt} + \frac{10}{100 + 2t}x = 1.8$$

The solution to this linear differential equation gives the pounds of Na_2SO_4 after 10 min; the concentration then can be calculated as x/y. To integrate the linear differential equation, we introduce the integrating factor $e^{\int (dt/10+0.2t)}$:

$$xe^{\int (dt/10+0.2t)} = \int (e^{\int (dt/10+0.2t)})(1.8)\,dt + C$$

$$x(10 + 0.2t)^5 = 1.8\int (10 + 0.2t)^5\,dt + C$$

at $x = 50$, $t = 0$, and $C = 35 + 10^5$;

$$x = 1.5(10 + 0.2t) + \frac{35 \times 10^5}{(10 + 0.2t)^5}$$

at $t = 10$ min;

$$x = 1.5(12) + \frac{35 \times 10^5}{(12)^5} = 32.1$$

The concentration is

$$\frac{x}{y} = \frac{32.1}{100 + 20} = 0.267 \frac{\text{lb } Na_2SO_4}{\text{lb total solution}}$$

EXAMPLE 6.6 Unsteady-State Energy Balance

Five thousand pounds of oil initially at 60°F is being heated in a stirred (perfectly mixed) tank by saturated steam which is condensing in the steam coils at 40 psia. If the rate of heat transfer is given by Newton's heating law,

$$\dot{Q} = \frac{dQ}{dt} = h(T_{\text{steam}} - T_{\text{oil}})$$

where Q is the heat transferred in Btu and h is the heat transfer coefficient in the proper units, how long does it take for the discharge from the tank to rise from 60°F to 90°F? What is the maximum temperature that can be achieved in the tank?

Additional Data:

$$\text{motor horsepower} = 1 \text{ hp; efficiency is } 0.75$$

$$\text{entering oil flow rate} = 1018 \text{ lb/hr at a temperature of } 60°\text{F}$$

$$\text{discharge oil flow rate} = 1018 \text{ lb/hr at a temperature of } T$$

$$h = 291 \text{ Btu/(hr)(°F)}$$

$$C_{P_{\text{oil}}} = 0.5 \text{ Btu/(lb)(°F)}$$

Solution

The process is shown in Fig. E6.6. The independent variable will be t, the time; the dependent variable will be the temperature of the oil in the tank, which is the same as the temperature of the oil discharged. The material balance is not needed because the process, insofar as the quantity of oil is concerned, is assumed to be in the steady state.

Our next step is to set up the unsteady-state energy balance for the interval Δt. Let T_S = the steam temperature and T = the oil temperature:

$$\text{accumulation} = \text{input} - \text{output}$$

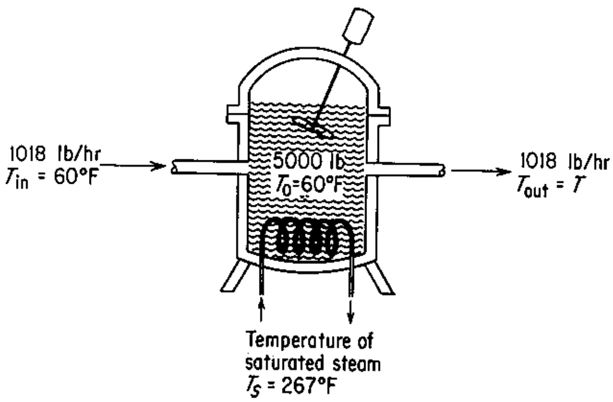


Figure E6.6

A good choice for a reference temperature for the enthalpies is 60°F because this choice makes the input enthalpy zero.

$$\text{enthalpy of the input steam} \quad \frac{1018 \text{ lb}}{\text{hr}}\left|\frac{0.5 \text{ Btu}}{(\text{lb})(^\circ\text{F})}\right|\overbrace{(T_{\text{in}} - T_{\text{ref}})}^{60^\circ\text{F} - 60^\circ\text{F}}\Big|\,\Delta t \text{ hr} = 0$$

$$\text{heat transfer} \quad h(T_S - T) = \frac{291 \text{ Btu}}{(\text{hr})(^\circ\text{F})}\Big|\,(267 - T)^\circ\text{F}\,\Big|\,\Delta t \text{ hr}$$

input (Btu)

$$\text{enthalpy of the output stream} \quad \frac{1018 \text{ lb}}{\text{hr}}\left|\frac{0.5 \text{ Btu}}{(\text{lb})(^\circ\text{F})}\right|(T - 60)\text{F}\,\Big|\,\Delta t \text{ hr}$$

output (Btu)

$$\text{enthalpy change inside tank} \quad \left[5000 \text{ lb}\left|\frac{0.5 \text{ Btu}}{(\text{lb})(^\circ\text{F})}\right|(T - 60)^\circ\text{F}\right]_{t+\Delta t} - \left[5000 \text{ lb}\left|\frac{0.5 \text{ Btu}}{(\text{lb})(^\circ\text{F})}\right|(T - 60)^\circ\text{F}\right]_{t}$$

accumulation (Btu)

The rate of change of energy inside the tank is nothing more than the rate of change of internal energy, which, in turn, is essentially the same as the rate of change of enthalpy (i.e., $dE/dt = dU/dt = dH/dt$) because $d(pV)/dt \simeq 0$.

We use Eq. (6.8) with $\dot{B} = 0$. The energy introduced by the motor enters the tank as $\dot{W}$ and is:

$$\dot{W} = \frac{3 \text{ hp}}{4}\left|\frac{0.7608 \text{ Btu}}{(\text{sec})(\text{hp})}\right|\frac{3600 \text{ sec}}{1 \text{ hr}} = \frac{1910 \text{ Btu}}{\text{hr}}$$

Then

$$2500\frac{dT}{dt} = 291(267 - T) - 509(T - 60) + 1910$$

$$\frac{dT}{dt} = 44.1 - 0.32T$$

$$\int_{60}^{90} \frac{dT}{44.1 - 0.32T} = \int_0^{\theta} dt = \theta$$

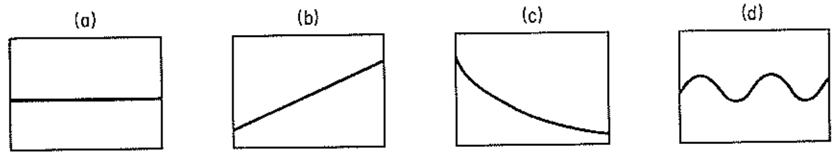
$\theta = 1.52$ hr.

A numerical solution using an appropriate computer program can be carried out for those differential equations for which analytical solutions are not possible.

Self-Assessment Test

1. In a batch type of chemical reactor, do we have to have the starting conditions to predict the yield?
2. Is time the independent or dependent variable in macroscopic unsteady-state equations?

3. How can you obtain the steady-state balances from Eqs. (6.6)–(6.8)?



4. Which of the following plots of the response vs. time are not transient state processes?
5. Group the following words in sets of synonyms: (1) response, (2) input variable, (3) parameter, (4) state variable, (5) system parameter, (6) initial condition, (7) output, (8) independent variable, (9) dependent variable, (10) coefficient, (11) output variable, (12) constant.
6. A chemical inhibitor must be added to the water in a boiler to avoid corrosion and scale. The inhibitor concentration must be maintained between 4 and 30 ppm. The boiler system always contains 100,000 kg of water and the blowdown (purge) rate is 15,000 kg/hr. The makeup water contains no inhibitor. An initial 2.8 kg of inhibitor is added to the water and thereafter 2.1 kg is added periodically. What is the maximum time interval until the first addition of inhibitor?
7. In a reactor a small amount of undesirable by-product is continuously formed at the rate of 0.5 lb/hr when the reactor contains a steady inventory of 10,000 lb of material. The reactor inlet stream contains traces of the by-product (40 ppm), and the effluent stream is 1400 lb/hr. If on startup, the reactor contains 5000 pm of the by-product, what will be the concentration in ppm in 10 hr?

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PROBLEMS

6.1. A tank containing 100 kg of a 60% brine (60% salt) is filled with a 10% salt solution at the rate of 10 kg/min. Solution is removed from the tank at the rate of 15 kg/min. Assuming complete mixing, find the kilograms of salt in the tank after 10 min.

6.2. A defective tank of 1500 ft^3 volume containing 100% propane gas (at 1 atm) is to be flushed with air (at 1 atm) until the propane concentration reduced to less than 1%. At that concentration of propane the defect can be repaired by welding. If th , flow rate of air into the tank is 30 ft^3/min, for how many minutes must the tank be flushed out? Assume that the flushing operation is conducted so that the gas in the tank is well mixed.

6.3. A 2% uranium oxide slurry (2 lb UO_2/100 lb H_2O) flows into a 100-gal tank at the rate of 2 gal/min. The tank initially contains 500 lb of H_2O and no UO_2. The slurry is well mixed and flows out at the same rate at which it enters. Find the concentration of slurry in the tank at the end of 1 hr.

6.4. The catalyst in a fluidized-bed reactor of 200-m^3 volume is to be regenerated by contact with a hydrogen stream. Before the hydrogen can be introduced into the reactor, the O_2 content of the air in the reactor must be reduced to 0.1%. If pure N_2 can be fed into the reactor at the rate of 20 m^3/min, for how long should the reactor be purged with N_2? Assume that the catalyst solids occupy 6% of the reactor volume and that the gases are well mixed.

6.5. An advertising firm wants to get a spherical inflated sign out of a warehouse. The sign is 20 ft in diameter and is filled with H_2 at 15 psig. Unfortunately, the door to the warehouse is only 19 ft high by 20 ft wide. The maximum rate of H_2 that can be safely vented from the balloon is 5 ft^3/min (measured at room conditions). How long will it take to get the sign small enough to just pass through the door?

(a) First assume that the pressure inside the balloon is constant so that the flow rate is constant.

(b) Then assume the amount of H_2 escaping is proportional to the volume of the balloon, and initially is 5 ft^3/min.

(c) Could a solution to this problem be obtained if the amount of escaping H_2 were proportional to the pressure difference inside and outside the balloon?

6.6. A plant at Canso, Nova Scotia, makes fish-protein concentrate (FPC). It takes 6.6 kg of whole fish to make 1 kg of FPC, and therein is the problem—to make money, the plant must operate most of the year. One of the operating problems is the drying of the FPC. It dries in the fluidized dryer at a rate approximately proportional to its moisture content. If a given batch of FPV loses one-half of its initial moisture in the first 15 min, how long will it take to remove 90% of the water in the batch of FPC?

6.7. Water flows from a conical tank at the rate of $0.020(2 + h^2)$ m^3/min, as shown in Fig. P6.7. If the tank is initially full, how long will it take for 75% of the water to flow out of the tank? What is the flow rate at that time?

6.8. A sewage disposal plant has a big concrete holding tank of 100,000 gal capacity. It is three-fourths full of liquid to start with and contains 60,000 lb of organic material in suspension. Water runs into the holding tank at the rate of 20,000 gal/hr and the

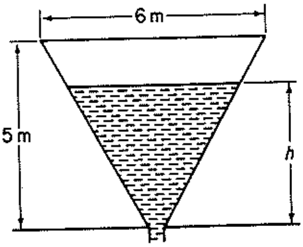


Figure P6.7

solution leaves at the rate of 15,000 gal/hr. How much organic material is in the tank at the end of 3 hr?

6.9. Suppose that in Problem 6.8 the bottom of the tank is covered with sludge (precipitated organic material) and that the stirring of the tank causes the sludge to go into suspension at a rate proportional to the difference between the concentration of sludge in the tank at any time and 10 lb of sludge/gal. If no organic material were present, the sludge would go into suspension at the rate of 0.05 lb/(min) (gal solution) when 75,000 gal of solution are in the tank. How much organic material is in the tank at the end of 3 hr?

6.10. In a chemical reaction the products X and Y are formed according to the equation

$$C \longrightarrow X + Y$$

The rate at which each of these products is being formed is proportional to the amount of C present. Initially: $C = 1, X = 0, Y = 0$. Find the time for the amount of X to equal the amount of C.

6.11. A tank is filled with water. At a given instant two orifices in the side of the tank are opened to discharge the water. The water at the start is 3 m deep and one orifice is 2 m below the top while the other one is 2.5 m below the top. The coefficient of discharge of each orifice is known to be 0.61. The tank is a vertical right circular cylinder 2 m in diameter. The upper and lower orifices are 5 and 10 cm in diameter, respectively. How long will be required for the tank to be drained so that the water level is at a depth of 1.5 m?

6.12. Suppose that you have two tanks in series, as diagrammed in Fig. P6.12. The volume of liquid in each tank remains constant because of the design of the overflow lines. Assume that each tank is filled with a solution containing 10 lb of A, and that the tanks contain 100 gal of aqueous solution each. If fresh water enters at the rate of 10 gal/hr, what is the concentration of A in each tank at the end of 3 hr? Assume complete mixing in each tank and ignore any change of volume with concentration.

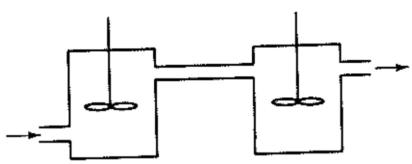
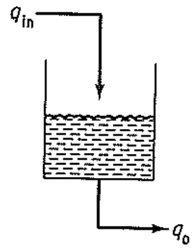


Figure P6.12

6.13. A well-mixed tank has a maximum capacity of 100 gal and it is initially half full. The discharge pipe at the bottom is very long and thus it offers resistance to the flow of water through it. The force that causes the water to flow is the height of the water in the tank, and in fact that flow is just proportional to this height. Since the height is proportional to the total volume of water in the tank, the volumetric flow rate of water out, q_o, is

$$q_o = kV$$

The flow rate of water into the tank, q_i, is constant. Use the information given in Fig. P6.13 to decide whether the amount of water in the tank increases, decreases, or remains the same. If it changes, how much time is required to completely empty or fill the tank, as the case may be?



Volume of Tank = 100 gal
Initial Amount of H_2O = 50 gal
q_i = 2 gal/min
k = 0.01 min^{-1}

Figure P6.13

6.14. A stream containing a radioactive fission product with a decay constant of 0.01 hr^{-1} (i.e., $dn/dt = 0.01\ n$), is run into a holding tank at the rate of 100 gal/hr for 24 hr. Then the stream is shut off for 24 hr. If the initial concentration of the fission product was 10 mg/liter and the tank volume is constant at 10,000 gal of solution (owing to an overflow line), what is the concentration of fission product:
(a) At the end of the first 24-hr period?
(b) At the end of the second 24-hr period?
What is the maximum concentration of fission product? Assume complete mixing in the tank.

6.15. A radioactive waste that contains 1500 ppm of ^{92}Sr is pumped into a holding tank that contains 0.40 m^3 at the rate of 1.5×10^{-3} m^3/min. ^{92}Sr decays as follows:

$$^{92}\text{Sr} \longrightarrow {}^{92}\text{Y} \longrightarrow {}^{92}\text{Zr}$$

half-life: 2.7 hr 3.5 hr

If the tank contains clear water initially and the solution runs out at the rate of 1.5×10^{-3} m^3/min, assuming perfect mixing:
(a) What is the concentration of Sr, Y, and Zr after 1 day?
(b) What is the equilibrium concentration of Sr and Y in the tank?
The rate of decay of such isotopes is $dN/dt = -\lambda N$, where $\lambda = 0.693/t_{1/2}$ and the half-life is $t_{1/2}$. N = moles.

6.16. A cylinder contains 3 m^3 of pure oxygen at atmospheric pressure. Air is slowly pumped into the tank and mixes uniformly with the contents, an equal volume of which is forced out of the tank. What is the concentration of oxygen in the tank after 9 m^3 of air has been admitted?

6.17. Suppose that an organic compound decomposes as follows:

$$C_6H_{12} \longrightarrow C_4H_8 + C_2H_4$$

If 1 mole of C_6H_{12} exists at $t = 0$ but no C_4H_8 and C_2H_4, set up equations showing the moles of C_4H_8 and C_2H_4 as a function of time. The rates of formation of C_4H_8 and C_2H_4 are each proportional to the number of moles of C_6H_{12} present.

6.18. A large tank is connected to a smaller tank by means of a valve. The large tank contains N_2 at 690 kPa while the small tank is evacuated. If the valve leaks between the two tanks and the rate of leakage of gas is proportional to the pressure difference between the two tanks $(p_1 - p_2)$, how long does it take for the pressure in the small tank to be one-half its final value? The instantaneous initial flow rate with the small tank evacuated is 0.091 kg mol/hr.

	Tank 1	Tank 2
Initial pressure (kPa)	700	0
Volume (m^3)	30	15

Assume that the temperature in both tanks is constant and is 20°C.

6.19. The following chain reactions take place in a constant-volume batch tank:

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

Each reaction is first order and irreversible. If the initial concentration of A is C_{A_0} and if only A is present initially, find an expression for C_B as a function of time. Under what conditions will the concentration of B be dependent primarily on the rate of reaction of A?

6.20. Consider the following chemical reaction in a constant-volume batch tank:

$$\begin{tikzcd} A \arrow[r, shift left, "k_1"] \arrow[d, "k_3"'] & B \arrow[l, shift left, "k_2"] \\ C & \end{tikzcd}$$

All the indicated reactions are first order. The initial concentration of A is C_{A_0}, and nothing else is present at that time. Determine the concentrations of A, B, and C as functions of time.

6.21. Tanks A, B, and C are each filled with 1000 gal of water. See Fig. P6.21. Workers have instructions to dissolve 2000 lb of salt in each tank. By mistake, 3000 lb is dissolved in each of tanks A and C and none in B. You wish to bring all the compositions to within 5% of the specified 2 lb/gal. If the units are connected A–B–C–A by three 50-gpm pumps,

(a) Express the concentrations C_A, C_B, and C_C in terms of t (time).

(b) Find the shortest time at which all concentrations are within the specified range. Assume the tanks are all well mixed.

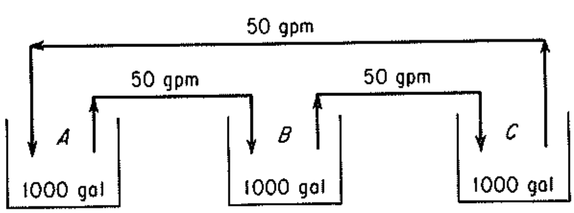


Figure P6.21

6.22. Determine the time required to heat a 10,000-lb batch of liquid from 60°F to 120°F using an external, counterflow heat exchanger having an area of 300 ft^2. Water a

180°F is used as the heating medium and flows at a rate of 6000 lb/hr. An overall heat transfer coefficient of 40 Btu/(hr)(ft^2)(°F) may be assumed; use Newton's law of heating. The liquid is circulated at a rate of 6000 lb/hr, and the specific heat of the liquid is the same as that of water (1.0). Assume that the residence time of the liquid in the external heat exchanger is very small and that there is essentially no holdup of liquid in this circuit.

6.23. A ground material is to be suspended in water and heated in preparation for a chemical reaction. It is desired to carry out the mixing and heating simultaneously in a tank equipped with an agitator and a steam coil. The cold liquid and solid are to be added continuously and the heated suspension will be withdrawn at the same rate. One method of operation for starting up is to (1) fill the tank initially with water and solid in the proper proportions, (2) start the agitator, (3) introduce fresh water and solid in the proper proportions and simultaneously begin to withdraw the suspension for reaction, and (4) turn on the steam. An estimate is needed of the time required, after the steam is turned on, for the temperature of the effluent suspension to reach a certain elevated temperature.

(a) Using the nomenclature given below, formulate a differential equation for this process. Integrate the equation to obtain n as a function of B and ϕ (see nomenclature).

(b) Calculate the time required for the effluent temperature to reach 180°F if the initial contents of the tank and the inflow are both at 120°F and the steam temperature is 220°F. The surface area for heat transfer is 23.9 ft^2, and the heat transfer coefficient is 100 Btu/(hr)(ft^2)(°F). The tank contains 6000 lb, and the rate of flow of both streams is 1200 lb/hr. In the proportions used, the specific heat of the suspension may be assumed to be 1.00.

If the area available heat transfer is doubled, how will the time required be affected? Why is the time with the larger area less than half that obtained previously? The heat transferred is $Q = UA(T_{\text{tank}} - T_{\text{steam}})$.

Nomenclature

W = weight of tank contents, lb

G = rate of flow of suspension, lb/hr

T_s = temperature of steam, °F

T = temperature in tank at any instant, perfect mixing assumed, °F

T_0 = temperature of suspension introduced into tank; also initial temperature of tank contents, °F

U = heat-transfer coefficient, Btu/(hr)(ft^2)(°F)

A = area of heat-transfer surface, ft^2

C_p = specific heat of suspension, Btu/(lb)(°F)

t = time elapsed from the instant the steam is turned on, hr

n = dimensionless time, Gt/W

B = dimensionless ratio, UA/GC_p

ϕ = dimensionless temperature (relative approach to the steam temperature) $(T - T_0)/(T_s - T_0)$

6.24. Consider a well-agitated cylindrical tank in which the heat transfer surface is in the form of a coil that is distributed uniformly from the bottom of the tank to the top of the tank. The tank itself is completely insulated. Liquid is introduced into the tank at a uniform rate, starting with no liquid in the tank, and the steam is turned on at the instant that liquid flows into the tank.

(a) Using the nomenclature of Problem 6.23, formulate a differential equation for this process. Integrate this expression to obtain an equation for ϕ as a function of B and f, where f = fraction filled = W/W_{filled}.

(b) If the heat transfer surface consists of a coil of 10 turns of 1-in.-OD tubing 4 ft in diameter, the feed rate is 1200 lb/hr, the heat capacity of the liquid is 1.0 Btu/(lb)(°F), the heat transfer coefficient is 100 Btu/(hr)(°F)(ft^2) of covered area, the steam temperature is 200°F, and the temperature of the liquid introduced into the tank is 70°F, what is the temperature in the tank when it is completely full? What is the temperature when the tank is half full? The heat transfer is given by $Q = UA(T_{\text{tank}} - T_{\text{steam}})$.

6.25. A cylindrical tank 5 ft in diameter and 5 ft high is full of water at 70°F. The water is to be heated by means of a steam jacket around the sides only. The steam temperature is 230°F, and the overall coefficient of heat transfer is constant at 40 Btu/(hr)(ft^2)(°F). Use Newton's law of cooling (heating) to estimate the heat transfer. Neglecting the heat losses from the top and the bottom, calculate the time necessary to raise the temperature of the tank contents to 170°F. Repeat, taking the heat losses from the top and the bottom into account. The air temperature around the tank is 70°F, and the overall coefficient of heat transfer for both the top and the bottom is constant at 10 Btu/(hr)(ft^2)(°F).

PROBLEM THAT REQUIRES WRITING A COMPUTER PROGRAM

6.1. Write a Fortran computer program for batch distillation. Calculate the mass fraction composition in the pot and the mass retrieved from the condenser given the initial mass and composition in the pot, and the final mass in the pot. Use Example 6.3 for the vapor–liquid equilibrium data and the general setup.

Appendix A

ANSWERS TO SELF-ASSESSMENT TESTS

Section 1.1

1. 2.10×10^{-5} m^3/s
2. **(a)** 252 lb_m; **(b)** 29.6 lb_f
3. Examine the conversion factors inside the cover.
4. c is dimensionless.
5. 32.174 (ft)(lb_m)/(s^2)(lb_f)
6. (a)
7. A has the same units as k; B has the units of T.
8. **(a)** 28.349, 0.454; **(b)** 1.609; **(c)** 37.85

Section 1.2

1. 60.05
2. A kilogram mole contains 6.02×10^{26} molecules, whereas a pound mole contains $(6.02 \times 10^{26})(0.454)$ molecules.
3. 0.123 kg mol NaCl/kg mol H_2O
4. 1.177 lb mol
5. 0.121 kg/s

Section 1.3

1. **(a)** F; **(b)** T; **(c)** F; **(d)** T
2. 13.6 g/cm^3
3. 62.4 lb_m/ft^3 (10^3 kg/m^3)
4. This means that the density at 10°C of liquid HCN is 1.2675 times the density of water at 4°C.
5. 0.79314 g/cm^3 (Note that you need the density of water at 60°F.)
6. 9
7. **(a)** 63%; **(b)** 54.3; **(c)** 13.8

8. 8.11 ft^3
9. (a) 0.33; (b) 18.7
10. (a) C_4 – 0.50, C_5 – 0.30, C_6 – 0.20; (b) C_4 – 0.57, C_5 – 0.28, C_6 – 0.15; (c) C_4 – 57, C_5 – 28, C_6 – 15; (d) 62.2 kg/kg mol

Section 1.4

1. See p. 32.
2. (a) 100 kg or 100 lb
   (b) 1 ton of water
   (c) The stated mass for each case
   (d) 100 cm^3 mixture

Section 1.5

1. (a) 0°C and 100°C; (b) 32°F and 212°F
2. $\Delta°F(1.8) = \Delta°C$.
3. Yes. Yes.
4. $92.76 + 0.198T_{°F}$.
5.

°C	°F	K	°R
−40.0	−40.0	233	420
25.0	77.0	298	537
425	796	698	1256
−234	−390	38.8	69.8

6. Immerse in ice-water bath and mark 0°C. Immerse in boiling water at 1 atm pressure and mark 100°C. Interpolate between 0°C and 100°C in desired intervals.
7. (a) 1°C; (b) 1°C; (c) 1 Δ°C

Section 1.6

1. Gauge pressure + barometric pressure = absolute pressure.
2. See p. 49.
3. Barometric pressure − vacuum pressure = absolute pressure.
4. (a) 15.5; (b) 106.6 kPa; (c) 1.052; (d) 35.6
5. (a) Gauge pressure; (b) barometric pressure, absolute pressure; (c) 50 in. Hg
6. In the absence of the barometric pressure, assume 101.3 kPa; then absolute pressure is 61.3 kPa.
7. (a) T; (b) T

Section 1.7

1. See p. 57.
2. (a) *Chemical Engineers' Handbook* and/or *Properties of Gases and Liquids*
   (b) *Technical Data Book—Petroleum Refining*
   (c) *Handbook of Physics and Chemistry* and others
   (d) *Chemical Engineers' Handbook*

Section 1.8

1. A serial sequence is adequate; some steps can be in parallel. Feedback may be inserted at various stages.

3. (a) Find a reason to remember; talk to others who are interested to ascertain what motivated them. Write down the reason for periodic pep talks with yourself.
   (b) Get a "bird's-eye view" of the whole section. Use the figure numbered 0 to see the relation to other sections. Look at a similar textbook. Ask for help; do not wait.
   (c) After scanning the serious reading count.
   (d) Take a few minutes as you read to mentally test yourself on what you have just read to reinforce learning. Continue to test at periodic intervals.
   (e) Practice using the information for an assumed quiz; predict quiz questions; use text examples or problems as samples.

Section 1.9

1. (a) $C_9H_{18} + \frac{27}{2}O_2 \longrightarrow 9CO_2 + 9H_2O$
   (b) $FeS_2 + 11O_2 \longrightarrow 2Fe_2O_3 + 8SO_2$
2. 3.08
3. 323
4. (a) $CaCO_3$ – 43.4%; CaO – 56.6%; (b) 0.308
5. (a) H_2O; (b) NaCl; (c) NaCl, H_2O, NaOH (assuming that the gas escapes)

Section 2.1

1, 2. The systems are somewhat arbitrary, as are the time intervals selected, but (a) and (c) can be closed systems (ignoring evaporation) and (b) open.
3. See Eqs. (2.1), (2.2), (2.3), and (2.3a).
4. (a) batch; (b) flow; (c) flow; (d) flow.
5. Accumulation is zero.
6. Figure 2.3—No, because many streams have no values and some streams, such as gases, are not included in the diagram. Figure 2.4—Yes.
7. Equation (2.1) or simplifications thereof.
8. (a) T; (b) T if *C* is regarded as an element; F if the exit compound is CO_2 and the entering reactant is *C*; (c) F
9. No chemical reaction takes place, or with reaction, the moles of reactants may equal the moles of product by accident.

Section 2.2

1. A set of unique values for the variables in the equations representing the problem.
2. (a) one; (b) three; (c) three
3. (a) two; (b) two of these three: acetic acid, water, total; (c) two; (d) feed of the 10% solution and mass fraction ω of the acetic acid in *P*; (e) 14% acetic acid, 86% water
4. Not for a unique solution. Only two are independent.
5. Substitution; Gauss–Jordan elimination; use of determinants.
6. Select the most accurate equations that will provide a unique solution. (Or use least squares if all must be used.)

7. The sum of the mass or mole fractions in one stream is unity.
8. Collect additional data so as to specify or evaluate the values of the excess unknown variables.
9. $F, D, P, \omega_{D2}, \omega_{P1}$.
10. Only two independent material balances can be written. The sum of the mass fractions for streams D and P yields two additional independent equations. One value of F, D, or P must be specified to obtain a solution. Note that specifying W_{D2} or W_{P1} will not help.
11. One

Section 2.3

1. Orsat (dry basis) does not include the water vapor in the analysis.
2. SO_2 is not included in the gas analysis.
3. See Eqs. (2.9)-(2.11).
4. Yes. See Eq. (2.9).
5. More
6. 4.5%
7. moles: 5.95 CO_2, 3.747 O_2, 44.0 N_2
8. 178 kg/hr
9. **(a)** 28% Na_2SO_4; **(b)** 33.3

Section 2.4

1. Three components = three independent equations. Any three of the four (1, 2, 3, total) will do.
2. A = 35 kg/hr; B = 65 kg/hr
3. 3.7321, 0.2680
4. $x = 0.2268$; $y = 0.3696$
5. **(a)** 252; **(b)** 1.063; **(c)** 2.31; **(d)** 33.8%
6. 33.3 kg
7. 186 kg
8. **(a)** 23.34; **(b)** 66.5%
9. 0.994
10. 0.81

Section 2.5

1.

	lb	fr.
tol	396	0.644
bz	19.68	0.032
xyl	200.00	0.325

2. 863 lb air/lb S

	Converter	Burner
SO_2	0.5%	9.5%
SO_3	9.4	—
O_2	7.4	11.5
N_2	82.7	79.0

Section 2.6

1. Recycle is feedback from downstream to upstream in the materials flow bypassing one or more units. Bypassing is feedforward bypassing one or more units.
2. Purge is a side stream usually small in quantity relative to main stream that is bled from the main stream to remove impurities from the system.
3. The stoichiometric ratio
4. **(a)** R = 3000 kg/hr; **(b)** air = 85,100 kg/hr
5. **(a)** 908 recycled and 3.2 purged ; **(b)** 9.05% conversion
6. **(a)** 1200.6 kg/hr; **(b)** 194.9 kg/hr

Section 3.1-1

1. $pV = nRT$
2. T: absolute temperature in degrees; p: absolute pressure in mass/(length)(time)^2; V: (length)^3/mole; n: mole; R: (mass)(length)^2/(time)^2(mole)(degree)
3. See p. 240.
4. 1883 ft^3
5. 2.98 kg
6. 1.32
7. 28.3 m^3/hr

Section 3.1-2

1. 0.0493 kg/m^3
2. 1.02 (lb CH_4/ft^3 CH_4 at 70°F and 2 atm)/(lb air/ft^3 at S.C.)

Section 3.1-3

1. See Eqs. (3.3) and (3.6).
2. **(a)** N_2, 0.28 psia; CH_4, 10.9 psia; C_2H_6, 2.62 psia
   (b) N_2, 0.04 ft^3; CH_4, 1.58 ft^3; C_2H_6, 0.38 ft^3
   (c) Same as the mole fractions
3. **(a)** 11.18 psia at 2 ft^3 and 120°F; **(b)** 0.28 psia at 2 ft^3 and 120°F

Section 3.1-4

1. **(a)** 2,735 ft^3/hr; **(b)** 5,034 ft^3/hr, **(c)** 22,429 ft^3/hr, **(d)** 30,975 ft^3/hr
2. 118,400 ft^3/hr

Section 3.2-1

1. See the last paragraph in Sec. 3.2-1. Also, they can be used for interpolation and extrapolation.
2. **(a)** Least squares and theory; **(b)** least squares would give the least error.
3. Comparison with experimental data; reduction in limit of ideal gas to $pV = nRT$; go through the critical point; holding one variable constant, the residual expression should be correct; derivatives give proper functions.
4. The volume occupied by the moles of the gas—by using $(V - nb)$, and the attractive forces existing between the molecules of the gas—by adding $[p + (n^2a/V^2)]$.
5. b is m^3; a is (K)$^{1/2}$(m)^6(kPa).
6. $V = 0.60$ ft^3
7. 314 K
8. **(a)** 50.7 atm; **(b)** 34.0 atm
9. 0.316 m^3/kg mol

Section 3.2-2

1. $\hat{V}_{c_i} = RT_c/P_c$. It can be used to calculate $\hat{V}_{ri}$, which is a parameter on the Nelson and Obert charts.

2. (a) No; (b) 5.08 ft^3; (c) 1.35 ft^3
3. 1.65 kg
4. 14.9 atm

Section 3.2-4

1. (a) 289 atm
   (b) 300 atm
   (c) 262 atm

Section 3.3

1. See Fig. 3.9.
2. Ice at its vapor pressure changes to liquid water at 32°F (at 0.0886 psia) and the pressure increases as shown in Fig. 3.12 with vapor and liquid in equilibrium. At 250°F, the pressure is 29.82 psia.
3. (a) 75 psia (5.112 atm); (b) it sublimes.
4. Experimental $p^* = 219.9$ mm Hg; predicted 220.9 mm Hg
5. 80.1°C

Section 3.4

1. The partial pressure of the vapor equals the vapor pressure of the gas. Liquid and vapor are in equilibrium.
2. Yes; yes
3. 21°C; benzene
4. 0.0373
5. 4.00 lb
6. (a) Both gas; (b) some liquid water, residual is gas; (c) both gas; (d) some liquid water, residual is gas.
7. 190 psia; $C_2H_6 = 0.0677$, $C_3H_8 = 0.660$, i-$C_4H_{10} = 0.2415$, n-$C_4H_{10} = 0.0308$

Section 3.5

1. 0.063 2. 57.3% 3. 86% $\mathcal{RH}$

Section 3.6

1. 53°C (126°F) 2. 71.2% H_2O 3. 94.0%; 100%

Section 3.7

1. No, gas and liquid in equilibrium. The triple point in the p–T projection is actually a line on the pVT surface. The pressure and temperature are fixed but the volume is not fixed.
2. Nothing. The points are not realizable equilibrium values.
3. Yes for a liquid–vapor if the process goes above the critical point. Probably not for a solid–vapor unless a solid–liquid region can be formed with a smooth property transition.

4. See Fig. 3.23.
5. **(a)** $C = 3$, $\mathcal{P} = 1$, $F = 4$; **(b)** $C = 2$, $\mathcal{P} = 2$, $F = 2$
6. No. $C = 2$, $\mathcal{P} = 6$, F is negative, which is not possible.
7. Two
8. One

Section 4.1

1. Intensive—independent of quantity of material; extensive—dependent of quantity of material. Measurable—temperature, pressure; unmeasurable—internal energy, enthalpy. State variable—difference in value between two states depends only on the states; path variable—difference in two states depends on trajectory reaching in the final state.
2. Heat: energy transfer across a system boundary due to a temperature difference. Work: energy transfer across a system boundary by means of a vector force acting through a vector deplacement on the boundary.
3. **(a)** Open; **(b)** open; **(c)** open; **(d)** closed; **(e)** closed
4. (a)
6. Decrease
7. **(a)** $K_0 = 0$, $P_0 = 4900$ J; **(b)** $K_f = 0$; $P_f = 0$; **(c)** $\Delta K = 0$, $\Delta P = -4900$ J (a decrease); **(d)** 1171 cal, 4.64 Btu
8. Assume $P_{atm} = 1$ atm is constant; 1.26×10^5 J

Section 4.2

1. $C_p = 4.25 + 0.002T$
2. $\frac{7}{2}R$
3. **(a)** cal/(g mol)(ΔK); **(b)** cal/(g mol)(ΔK)(K); **(c)** cal/(g mol)(ΔK)(K^2); **(d)** cal/(g mol)(ΔK)(K^3)
4. 7.11 cal/(g mol)(K)
5. **(a)** 0.58 cal/(g)(°C); **(b)** 0.49 cal/(g)(°C); experimental = 0.535 cal/(g)(°C)
6. $27{,}115 + 6.55T - 9.98 \times 10^{-4}T^2$

Section 4.3

1. 32,970 J (7880 cal)
2. 192 Btu for 2 lb
3. **(a)** Liquid; **(b)** two phase; **(c)** two phase; **(d)** vapor; **(e)** vapor
4. 13,900 Btu
5. 19,013 J/kg

Section 4.4

1. −334 Btu/lb
2. 6567 Btu

Section 4.5-1

1. **(a)** $T = 90.19$ K; **(b)** $m = 1.48 \times 10^7$ g; **(c)** $x = 3.02 \times 10^{-4}$; **(d)** 49.6 hr; **(e)** from a handbook, $T = 100$ K approx (for process at constant volume)
2. **(a)** Down; **(b)** probably up, depending on room temperature; **(c)** down
3. **(a)** $T_1 = 166.2°C$, $T_2 = 99.63°C$; **(b)** $Q = -6.40 \times 10^4$ J
4. 594 Btu
5. **(a)** 0; **(b)** 0; **(c)** 0; **(d)** if ideal gas $\Delta T = 0$, T_2 = room temperature; **(e)** 0.26 atm
6. 7m

Section 4.5-2

1. 12,200 lb/hr
2. 99°C
3. 1847 watts (2.48 hp)
4. −19.7 Btu/s

Section 4.6

1. (2)
2. Thermal—presumably internal energy or heat—not freely convertible to mechanical energy. The latter types of energy are freely convertible one to the other.
3. Process in which total of freely convertible forms of energy are not reduced.
4. $W = 27.6$ kJ; $Q = -27.6$ kJ; $\Delta E = \Delta H = 0$
5. 38.8 hp

Section 4.7-1

1. −74.83 kJ/g mol CH_4 (−17.88 kcal)
2. 597.32 kJ/g mol C_6H_6 (142.76 kcal)

Section 4.7-2

1. −148.53 k cal/g mol
2. No
3. 230 Btu/ft³ at 60°F and 30.0 in. Hg

Section 4.7-3

1. No. The result depends on the values in Eq. (4.38) and may be positive or negative.
2. **(a)** $p \simeq 2$ atm; **(b)** $Q_v = -3.55$ kcal/g mol CO (evolved)

Section 4.7-4

1. 125 Btu/ft³ at SC.

Section 4.7-5

1. −248,100 J (−59,299 kcal)
2. 6020 Btu/lb methane
3. 0.52 lb

Section 4.7-6

1. 975K

Section 4.8-1

1. Almost always
2. (a) HNO_3, HCl, or H_2SO_4 in water; (b) NaCl, KCl, NH_4NO_3 in water
3. (a) 25°C and 1 atm; (b) 0
4. −4,655 cal/g mol soln (heat transfer is from solution to surroundings)
5. $Q = -1.61 \times 10^6$ Btu

Section 4.8-2

1. $\omega_V = 0.85$; $\omega_L = 0.15$
2. 500 Btu/lb
4. Two phase; for the liquid $\omega_{H_2O} = 0.50$ and $\hat{H} = 8$ Btu/lb; for the vapor $\omega_{H_2O} = 1.00$ and $\hat{H} = 1174$ Btu/lb

Section 4.9

1. t_{DB} = gas temperature; t_{WB} = temperature registered at equilibrium between evaporating water around temperature sensor and surroundings at t_{DB}
2. No
3. See Eqs. (5.21)–(5.26).
4. (a) 0.03 kg/kg dry air; (b) 1.02 m^3/kg dry air; (c) 38°C_{in}; (d) 151 kJ/kg dry air; (e) 31.5°C
5. (a) $\mathcal{H} = 0.0808$ lb H_2O/lb dry air; (b) $H = 118.9$ Btu/lb dry air; (c) $V = 16.7$ ft^3/lb dry air; (d) $\mathcal{H} = 0.0710$ lb H_2O/lb dry air
6. 49,700 Btu
7. (a) 1.94×10^6 ft^3/hr; (b) 3.00×10^4 ft^3/hr; (c) 2.44×10^5 Btu/hr

Section 5.1

1. Yes. See Appendix L.
2. For one phase, $N_{sp} - 1 + 2$ intensive variable (from the phase rule) plus one extensive variable are needed to completely specify a stream.
3. 9
4. Any number from 7 to 11 may be acceptable, depending on assumptions concerning the relations among the pressures in the reactor and mixer streams.
5. Yes

Chapter 6

1. The solution of the differential equation requires the initial conditions to be able to predict the response at other times.
2. Independent
3. Let the derivatives vanish (delete them from the equations).
4. (a)
5. (1), (4), (7), (9), (11); (2), (8); (3), (5), (10), (12); (6)

Appendix B

ATOMIC WEIGHTS AND NUMBERS

TABLE B.1 Relative Atomic Weights, 1965 (Based on the Atomic Mass of $^{12}C = 12$)

The values for atomic weights given in the table apply to elements as they exist in nature, without artificial alteration of their isotopic composition, and, further, to natural mixtures that do not include isotopes of radiogenic origin.

Name	Symbol	Atomic Number	Atomic Weight	Name	Symbol	Atomic Number	Atomic Weight
Actinium	Ac	89	—	Mercury	Hg	80	200.59
Aluminum	Al	13	26.9815	Molybdenum	Mo	42	95.94
Americium	Am	95	—	Neodymium	Nd	60	144.24
Antimony	Sb	51	121.75	Neon	Ne	10	20.183
Argon	Ar	18	39.948	Neptunium	Np	93	—
Arsenic	As	33	74.9216	Nickel	Ni	28	58.71
Astatine	At	85	—	Niobium	Nb	41	92.906
Barium	Ba	56	137.34	Nitrogen	N	7	14.0067
Berkelium	Bk	97	—	Nobelium	No	102	—
Beryllium	Be	4	9.0122	Osmium	Os	75	190.2
Bismuth	Bi	83	208.980	Oxygen	O	8	15.9994
Boron	B	5	10.811	Palladium	Pd	46	106.4
Bromine	Br	35	79.904	Phosphorus	P	15	30.9738
Cadmium	Cd	48	112.40	Platinum	Pt	78	195.09
Caesium	Cs	55	132.905	Plutonium	Pu	94	—
Calcium	Ca	20	40.08	Polonium	Po	84	—
Californium	Cf	98	—	Potassium	K	19	39.102
Carbon	C	6	12.01115	Praseodym	Pr	59	140.907
Cerium	Ce	58	140.12	Promethium	Pm	61	—
Chlorine	Cl	17	35.453[b]	Protactinium	Pa	91	—
Chromium	Cr	24	51.996[b]	Radium	Ra	88	—
Cobalt	Co	27	58.9332	Radon	Rn	86	—
Copper	Cu	29	63.546[b]	Rhenium	Re	75	186.2
Curium	Cm	96	—	Rhodium	Rh	45	102.905
Dysprosium	Dy	66	162.50	Rubidium	Rb	37	84.57
Einsteinium	Es	99	—	Ruthenium	Ru	44	101.07
Erbium	Er	68	167.26	Samarium	Sm	62	150.35
Europium	Eu	63	151.96	Scandium	Sc	21	44.956
Fermium	Fm	100	—	Selenium	Se	34	78.96
Fluorine	F	9	18.9984	Silicon	Si	14	28.086
Francium	Fr	87	—	Silver	Ag	47	107.868
Gadolinium	Gd	64	157.25	Sodium	Na	11	22.9898
Gallium	Ga	31	69.72	Strontium	Sr	38	87.62
Germanium	Ge	32	72.59	Sulfur	S	16	32.064
Gold	Au	79	196.967	Tantalum	Ta	73	180.948
Hafnium	Hf	72	178.49	Technetium	Tc	43	—
Helium	He	2	4.0026	Tellurium	Te	52	127.60
Holmium	Ho	67	164.930	Terbium	Tb	65	158.924
Hydrogen	H	1	1.00797	Thallium	Tl	81	204.37
Indium	In	49	114.82	Thorium	Th	90	232.038
Iodine	I	53	126.9044	Thulium	Tm	59	168.934
Iridium	Ir	77	192.2	Tin	Sn	50	118.69
Iron	Fe	26	55.847	Titanium	Ti	22	47.90
Krypton	Kr	36	83.80	Tungsten	W	74	183.85
Lanthanum	La	57	138.91	Uranium	U	92	238.03
Lawrencium	Lr	103	—	Vanadium	V	23	50.942
Lead	Pb	82	207.19	Xenon	Xe	54	131.30
Lithium	Li	3	6.939	Ytterbium	Yb	70	173.04
Lutetium	Lu	71	174.97	Yttrium	Y	39	88.905
Magnesium	Mg	12	24.312	Zinc	Zn	30	65.37
Manganese	Mn	25	54.9380	Zirconium	Zr	40	91.22
Mendelevium	Md	101	—				

SOURCE: *Comptes Rendus*, 23rd IUPAC Conference, 1965, Butterworth's, London, 1965, pp. 177–178.

Appendix C

STEAM TABLES

***Note:* Tables in SI and American Engineering units for superheated steam are on the chart in the pocket in the back of the book.**

Absolute pressure = atmospheric pressure − vacuum.

Barometer and vacuum columns may be corrected to mercury at 32°F by subtracting 0.00009 × (t − 32) × column height, where t is the column temperature in °F.

One inch of mercury at 32°F = 0.4912 lb/in.^{2}

Example:

Barometer reads 30.17 in. at 70°F. Vacuum column reads 28.26 in. at 80°F. Pounds pressure = (30.17 − 0.00009 × 38 × 30.17) − 28.26 − 0.00009 × 48 × 28.26) = 1.93 in. of mercury at 32°F.

Saturation temperature (from table) = 100°F.

TABLE C.1 Saturated Steam: Temperature Table

	Absolute pressure			*Specific volume*			*Enthalpy*	
Temp. Fahr. t	Lb/in.^{2} p	In. Hg 32°F	Sat. Liquid v_f	Evap. v_{fg}	Sat. Vapor v_g	Sat. Liquid h_f	Evap. h_{fg}	Sat. Vapor h_g
32	0.0886	0.1806	0.01602	3305.7	3305.7	0	1075.1	1075.1
34	0.0961	0.1957	0.01602	3060.4	3060.4	2.01	1074.9	1076.0
36	0.1041	0.2120	0.01602	2836.6	2836.6	4.03	1072.9	1076.9
38	0.1126	0.2292	0.01602	2632.2	2632.2	6.04	1071.7	1077.7
40	0.1217	0.2478	0.01602	2445.1	2445.1	8.05	1070.5	1078.6
42	0.1315	0.2677	0.01602	2271.8	2271.8	10.06	1069.3	1079.4
44	0.1420	0.2891	0.01602	2112.2	2112.2	12.06	1068.2	1080.3
46	0.1532	0.3119	0.01602	1965.5	1965.5	14.07	1067.1	1081.2
48	0.1652	0.3364	0.01602	1829.9	1829.9	16.07	1065.9	1082.0
50	0.1780	0.3624	0.01602	1704.9	1704.9	18.07	1064.8	1082.9
52	0.1918	0.3905	0.01603	1588.4	1588.4	20.07	1063.6	1083.7
54	0.2063	0.4200	0.01603	1482.4	1482.4	22.07	1062.5	1084.6
56	0.2219	0.4518	0.01603	1383.5	1383.5	24.07	1061.4	1085.5
58	0.2384	0.4854	0.01603	1292.7	1292.7	26.07	1060.2	1086.3
60	0.2561	0.5214	0.01603	1208.1	1208.1	28.07	1059.1	1087.2
62	0.2749	0.5597	0.01604	1129.7	1129.7	30.06	1057.9	1088.0
64	0.2949	0.6004	0.01604	1057.1	1057.1	32.06	1056.8	1088.9

TABLE C.1 (cont.)

Temp. Fahr. t	Absolute pressure Lb/in.² p	In. Hg 32°F	Specific volume Sat. Liquid v_f	Evap. v_{fg}	Sat. Vapor v_g	Enthalpy Sat. Liquid h_f	Evap. h_{fg}	Sat. Vapor h_g
68	0.3388	0.6898	0.01605	927.0	927.0	36.05	1054.5	1090.6
70	0.3628	0.7387	0.01605	868.9	868.9	38.05	1053.4	1091.5
72	0.3883	0.7906	0.01606	814.9	814.9	40.04	1052.3	1092.3
74	0.4153	0.8456	0.01606	764.7	764.7	42.04	1051.2	1093.2
76	0.4440	0.9040	0.01607	718.0	718.0	44.03	1050.1	1094.1
78	0.4744	0.9659	0.01607	674.4	674.4	46.03	1048.9	1094.9
80	0.5067	1.032	0.01607	633.7	633.7	48.02	1047.8	1095.8
82	0.5409	1.101	0.01608	595.8	595.8	50.02	1046.6	1096.6
84	0.5772	1.175	0.01608	560.4	560.4	52.01	1045.5	1097.5
86	0.6153	1.253	0.01609	527.6	527.6	54.01	1044.4	1098.4
88	0.6555	1.335	0.01609	497.0	497.0	56.00	1043.2	1099.2
90	0.6980	1.421	0.01610	468.4	468.4	58.00	1042.1	1100.1
92	0.7429	1.513	0.01611	441.7	441.7	59.99	1040.9	1100.9
94	0.7902	1.609	0.01611	416.7	416.7	61.98	1039.8	1101.8
96	0.8403	1.711	0.01612	393.2	393.2	63.98	1038.7	1102.7
98	0.8930	1.818	0.01613	371.3	371.3	65.98	1037.5	1103.5
100	0.9487	1.932	0.01613	350.8	350.8	67.97	1036.4	1104.4
102	1.0072	2.051	0.01614	331.5	331.5	69.96	1035.2	1105.2
104	1.0689	2.176	0.01614	313.5	313.5	71.96	1034.1	1106.1
106	1.1338	2.308	0.01615	296.5	296.5	73.95	1033.0	1107.0
108	1.2020	2.447	0.01616	280.7	280.7	75.94	1032.0	1107.9
110	1.274	2.594	0.01617	265.7	265.7	77.94	1030.9	1108.8
112	1.350	2.749	0.01617	251.6	251.6	79.93	1029.7	1109.6
114	1.429	2.909	0.01618	238.5	238.5	81.93	1028.6	1110.5
116	1.512	3.078	0.01619	226.2	226.2	83.92	1027.5	1111.4
118	1.600	3.258	0.01620	214.5	214.5	85.92	1026.4	1112.3
120	1.692	3.445	0.01620	203.45	203.47	87.91	1025.3	1113.2
122	1.788	3.640	0.01621	193.16	193.18	89.91	1024.1	1114.0
124	1.889	3.846	0.01622	183.44	183.46	91.90	1023.0	1114.9
126	1.995	4.062	0.01623	174.26	174.28	93.90	1021.8	1115.7
128	2.105	4.286	0.01624	165.70	165.72	95.90	1020.7	1116.6
130	2.221	4.522	0.01625	157.55	157.57	97.89	1019.5	1117.4
132	2.343	4.770	0.01626	149.83	149.85	99.89	1018.3	1118.2
134	2.470	5.029	0.01626	142.59	142.61	101.89	1017.2	1119.1
136	2.603	5.300	0.01627	135.73	135.75	103.88	1016.0	1119.9
138	2.742	5.583	0.01628	129.26	129.28	105.88	1014.9	1120.8
140	2.887	5.878	0.01629	123.16	123.18	107.88	1013.7	1121.6
142	3.039	6.187	0.01630	117.37	117.39	109.88	1012.5	1122.4
144	3.198	6.511	0.01631	111.88	111.90	111.88	1011.3	1123.2
146	3.363	6.847	0.01632	106.72	106.74	113.88	1010.2	1124.1
148	3.536	7.199	0.01633	101.82	101.84	115.87	1009.0	1124.9
150	3.716	7.566	0.01634	97.18	97.20	117.87	1007.8	1125.7
152	3.904	7.948	0.01635	92.79	92.81	119.87	1006.7	1126.6

TABLE C.1 *(cont.)*

	Absolute pressure			*Specific volume*			*Enthalpy*	
Temp. Fahr. t	Lb/in.² p	In. Hg 32°F	Sat. Liquid v_f	Evap. v_{fg}	Sat. Vapor v_g	Sat. Liquid h_f	Evap. h_{fg}	Sat. Vapor h_g
154	4.100	8.348	0.01636	88.62	88.64	121.87	1005.5	1127.4
156	4.305	8.765	0.01637	84.66	84.68	123.87	1004.4	1128.3
158	4.518	9.199	0.01638	80.90	80.92	125.87	1003.2	1129.1
160	4.739	9.649	0.01639	77.37	77.39	127.87	1002.0	1129.9
162	4.970	10.12	0.01640	74.00	74.02	129.88	1000.8	1130.7
164	5.210	10.61	0.01642	70.79	70.81	131.88	999.7	1131.6
166	5.460	11.12	0.01643	67.76	67.78	133.88	998.5	1132.4
168	5.720	11.65	0.01644	64.87	64.89	135.88	997.3	1133.2
170	5.990	12.20	0.01645	62.12	62.14	137.89	996.1	1134.0
172	6.272	12.77	0.01646	59.50	59.52	139.89	995.0	1134.9
174	6.565	13.37	0.01647	57.01	57.03	141.89	993.8	1135.7
176	6.869	13.99	0.01648	56.64	54.66	143.90	992.6	1136.5
178	7.184	14.63	0.01650	52.39	52.41	145.90	991.4	1137.3
180	7.510	15.29	0.01651	50.26	50.28	147.91	990.2	1138.1
182	7.849	15.98	0.01652	48.22	48.24	149.92	989.0	1138.9
184	8.201	16.70	0.01653	46.28	46.30	151.92	987.8	1139.7
186	8.566	17.44	0.01654	44.43	44.45	153.93	986.6	1140.5
188	8.944	18.21	0.01656	42.67	42.69	155.94	985.3	1141.3
190	9.336	19.01	0.01657	40.99	41.01	157.95	984.1	1142.1
192	9.744	19.84	0.01658	39.38	39.40	159.95	982.8	1142.8
194	10.168	20.70	0.01659	37.84	37.86	161.96	981.5	1143.5
196	10.605	21.59	0.01661	36.38	36.40	163.97	980.3	1144.3
198	11.057	22.51	0.01662	34.98	35.00	165.98	979.0	1145.0
200	11.525	23.46	0.01663	33.65	33.67	167.99	977.8	1145.8
202	12.010	24.45	0.01665	32.37	32.39	170.01	976.6	1146.6
204	12.512	25.47	0.01666	31.15	31.17	172.02	975.3	1147.3
206	13.031	26.53	0.01667	29.99	30.01	174.03	974.1	1148.1
208	13.568	27.62	0.01669	28.88	28.90	176.04	972.8	1148.8
210	14.123	28.75	0.01670	27.81	27.83	178.06	971.5	1149.6
212	14.696	29.92	0.01672	26.81	26.83	180.07	970.3	1150.4
215	15.591		0.01674	25.35	25.37	186.10	968.3	1151.4
220	17.188		0.01677	23.14	23.16	188.14	965.2	1153.3
225	18.915		0.01681	21.15	21.17	193.18	961.9	1155.1
230	20.78		0.01684	19.371	19.388	198.22	958.7	1156.9
235	22.80		0.01688	17.761	17.778	203.28	955.3	1158.6
240	24.97		0.01692	16.307	16.324	208.34	952.1	1160.4
245	27.31		0.01696	15.010	15.027	213.41	948.7	1162.1
250	29.82		0.01700	13.824	13.841	218.48	945.3	1163.8
255	32.53		0.01704	12.735	12.752	223.56	942.0	1165.6
260	35.43		0.01708	11.754	11.771	228.65	938.6	1167.3

TABLE C.1 *(cont.)*

	Absolute pressure			*Specific volume*			*Enthalpy*	
Temp. Fahr. t	Lb/in.^{2} p	In. Hg 32°F	Sat. Liquid v_f	Evap. v_{fg}	Sat. Vapor v_g	Sat. Liquid h_f	Evap. h_{fg}	Sat. Vapor h_g
270	41.85		0.01717	10.053	10.070	238.84	931.8	1170.6
275	45.40		0.01721	9.313	9.330	243.94	928.2	1172.1
280	49.20		0.01726	8.634	8.651	249.06	924.6	1173.7
285	53.25		0.01731	8.015	8.032	254.18	921.0	1175.2
290	57.55		0.01735	7.448	7.465	259.31	917.4	1176.7
295	62.13		0.01740	6.931	6.948	264.45	913.7	1178.2
300	67.01		0.01745	6.454	6.471	269.60	910.1	1179.7
305	72.18		0.01750	6.014	6.032	274.76	906.3	1181.1
310	77.68		0.01755	5.610	5.628	279.92	902.6	1182.5
315	83.50		0.01760	5.239	5.257	285.10	898.8	1183.9
320	89.65		0.01765	4.897	4.915	290.29	895.0	1185.3
325	96.16		0.01771	4.583	4.601	295.49	891.1	1186.6
330	103.03		0.01776	4.292	4.310	300.69	887.1	1187.8
335	110.31		0.01782	4.021	4.039	305.91	883.2	1189.1
340	117.99		0.01788	3.771	3.789	311.14	879.2	1190.3
345	126.10		0.01793	3.539	3.557	316.38	875.1	1191.5
350	134.62		0.01799	3.324	3.342	321.64	871.0	1192.6
355	143.58		0.01805	3.126	3.144	326.91	866.8	1193.7
360	153.01		0.01811	2.940	2.958	332.19	862.5	1194.7
365	162.93		0.01817	2.768	2.786	337.48	858.2	1195.7
370	173.33		0.01823	2.607	2.625	342.79	853.8	1196.6
375	184.23		0.01830	2.458	2.476	348.11	849.4	1197.5
380	195.70		0.01836	2.318	2.336	353.45	844.9	1198.4
385	207.71		0.01843	2.189	2.207	358.80	840.4	1199.2
390	220.29		0.01850	2.064	2.083	364.17	835.7	1199.9
395	233.47		0.01857	1.9512	1.9698	369.56	831.0	1200.6
400	247.25		0.01864	1.8446	1.8632	374.97	826.2	1201.2
405	261.67		0.01871	1.7445	1.7632	380.40	821.4	1201.8
410	276.72		0.01878	1.6508	1.6696	385.83	816.6	1202.4
415	292.44		0.01886	1.5630	1.5819	391.30	811.7	1203.0
420	308.82		0.01894	1.4806	1.4995	396.78	806.7	1203.5
425	325.91		0.01902	1.4031	1.4221	402.28	801.6	1203.9
430	343.71		0.01910	1.3303	1.3494	407.80	796.5	1204.3
435	362.27		0.01918	1.2617	1.2809	413.35	791.2	1204.6
440	381.59		0.01926	1.1973	1.2166	418.91	785.9	1204.8
445	401.70		0.01934	1.1367	1.1560	424.49	780.4	1204.9
450	422.61		0.01943	1.0796	1.0990	430.11	774.9	1205.0
455	444.35		0.0195	1.0256	1.0451	435.74	769.3	1205.0
460	466.97		0.0196	0.9745	0.9941	441.42	763.6	1205.0
465	490.43		0.0197	0.9262	0.9459	447.10	757.8	1204.9
470	514.70		0.0198	0.8808	0.9006	452.84	751.9	1204.7
475	539.90		0.0199	0.8379	0.8578	458.59	745.9	1204.5
480	566.12		0.0200	0.7972	0.8172	464.37	739.8	1204.2

TABLE C.1 *(cont.)*

Temp. Fahr. t	*Absolute pressure* Lb/in.^{2} p	In. Hg 32°F	*Specific volume* Sat. Liquid v_f	Evap. v_{fg}	Sat. Vapor v_g	*Enthalpy* Sat. Liquid h_f	Evap. h_{fg}	Sat. Vapor h_g
485	593.28		0.0201	0.7585	0.7786	470.18	733.6	1203.8
490	621.44		0.0202	0.7219	0.7421	476.01	727.3	1203.3
495	650.59		0.0203	0.6872	0.7075	481.90	720.8	1202.7
500	680.80		0.0204	0.6544	0.6748	487.80	714.2	1202.0
505	712.19		0.0206	0.6230	0.6436	493.8	707.5	1201.3
510	744.55		0.0207	0.5932	0.6139	499.8	700.6	1200.4
515	777.96		0.0208	0.5651	0.5859	505.8	693.6	1199.4
520	812.68		0.0209	0.5382	0.5591	511.9	686.5	1198.4
525	848.37		0.0210	0.5128	0.5338	518.0	679.2	1197.2
530	885.20		0.0212	0.4885	0.5097	524.2	671.9	1196.1
535	923.45		0.0213	0.4654	0.4867	530.4	664.4	1194.8
540	962.80		0.0214	0.4433	0.4647	536.6	656.7	1193.3
545	1003.6		0.0216	0.4222	0.4438	542.9	648.9	1191.8
550	1045.6		0.0218	0.4021	0.4239	549.3	640.9	1190.2
555	1088.8		0.0219	0.3830	0.4049	555.7	632.6	1188.3
560	1133.4		0.0221	0.3648	0.3869	562.2	624.1	1186.3
565	1179.3		0.0222	0.3472	0.3694	568.8	615.4	1184.2
570	1226.7		0.0224	0.3304	0.3528	575.4	606.5	1181.9
575	1275.7		0.0226	0.3143	0.3369	582.1	597.4	1179.5
580	1326.1		0.0228	0.2989	0.3217	588.9	588.1	1177.0
585	1378.1		0.0230	0.2840	0.3070	595.7	578.6	1174.3
590	1431.5		0.0232	0.2699	0.2931	602.6	568.8	1171.4
595	1486.5		0.0234	0.2563	0.2797	609.7	558.7	1168.4
600	1543.2		0.0236	0.2432	0.2668	616.8	548.4	1165.2
605	1601.5		0.0239	0.2306	0.2545	624.1	537.7	1161.8
610	1661.6		0.0241	0.2185	0.2426	631.5	526.6	1158.1
615	1723.4		0.0244	0.2068	0.2312	638.9	515.3	1154.2
620	1787.0		0.0247	0.1955	0.2202	646.5	503.7	1150.2
625	1852.4		0.0250	0.1845	0.2095	654.3	491.5	1145.8
630	1919.8		0.0253	0.1740	0.1993	662.2	478.8	1141.0
635	1989.0		0.0256	0.1638	0.1894	670.4	465.5	1135.9
640	2060.3		0.0260	0.1539	0.1799	678.7	452.0	1130.7
645	2133.5		0.0264	0.1441	0.1705	687.3	437.6	1124.9
650	2208.8		0.0268	0.1348	0.1616	696.0	422.7	1118.7
655	2286.4		0.0273	0.1256	0.1529	705.2	407.0	1112.2
660	2366.2		0.0278	0.1167	0.1445	714.4	390.5	1104.9
665	2448.0		0.0283	0.1079	0.1362	724.5	372.1	1096.6
670	2532.4		0.0290	0.0991	0.1281	734.6	353.3	1087.9
675	2619.2		0.0297	0.0904	0.1201	745.5	332.8	1078.3
680	2708.4		0.0305	0.0810	0.1115	757.2	310.0	1067.2
685	2800.4		0.0316	0.0716	0.1032	770.1	284.5	1054.6

TABLE C.1 *(cont.)*

Temp. Fahr. t	*Absolute pressure* Lb/in.^{2} p	In. Hg 32°F	*Specific volume* Sat. Liquid v_f	Evap. v_{fg}	Sat. Vapor v_g	*Enthalpy* Sat. Liquid h_f	Evap. h_{fg}	Sat. Vapor h_g
690	2895.0		0.0328	0.0617	0.0945	784.2	254.9	1039.1
695	2992.7		0.0345	0.0511	0.0856	801.3	219.1	1020.4
700	3094.1		0.0369	0.0389	0.0758	823.9	171.7	995.6
705	3199.1		0.0440	0.0157	0.0597	870.2	77.6	947.8
705.34*	3206.2		0.0541	0	0.0541	910.3	0	910.3

*Critical temperature. v = specific volume, ft^3/lb. h = enthalpy, Btu/lb.

Source: Combustion Engineering, Inc.

Appendix D

PHYSICAL PROPERTIES OF VARIOUS ORGANIC AND INORGANIC SUBSTANCES

General Sources of Data for Tables on the Physical Properties, Heat Capacities, and Thermodynamic Properties in Appendices D, E, and F

1. Kobe, Kenneth A., and Associates, "Thermochemistry of Petrochemicals," Reprint from *Petroleum Refiner*, Gulf Publishing Company, Houston, Jan. 1949–July 1958. (Enthalpy tables D.2–D. and heat capacities of several gases in Table E.1, Appendix E.)
2. Lange, N. A., *Handbook of Chemistry*, 12th ed., McGraw-Hill, New York, 1979.
3. Maxwell, J. B., *Data Book on Hydrocarbons*, Van Nostrand Reinhold, New York, 1950.
4. Perry, J. H., and C. H. Chilton, eds., *Chemical Engineers' Handbook*, 5th ed., McGraw-Hill, New York, 1973.
5. Rossini, Frederick D., et al., "Selected Values of Chemical Thermodynamic Properties," from *National Bureau of Standards Circular 500*, U.S. Government Printing Office, Washington, D.C., 1952.
6. Rossini, Frederick D., et al., "Selected Values of Physical and Thermodynamic Properties of Hydrocarbons and Related Compounds," American Petroleum Institute Research Project 44, 1953 and subsequent years.
7. Weast, Robert C., *Handbook of Chemistry and Physics*, 59th ed., CRC Press, Boca Raton, Florida, 1979.

TABLE D.1 Physical Properties of Various Organic and Inorganic Substances*

To convert to kcal/g mol multiply by 0.2390; to Btu/lb mol multiply by 430.2.

Compound	Formula	Formula Wt	Sp Gr	Melting Temp. (K)	$\Delta\hat{H}$ Fusion (kJ/g mol)	Normal b.p.(K)	$\Delta\hat{H}$ Vap. at b.p. (kJ/g mol)	T_c (°K)	p_c (atm)	$\hat{V}_c$ (cm³/g mol)	z_c
Acetaldehyde	C_2H_4O	44.05	$0.783^{18°/4°}$	149.5		293.2		461.0			
Acetic acid	CH_3CHO_2	60.05	1.049	289.9	12.09	390.4	24.4	594.8	57.1	171	0.200
Acetone	C_3H_6O	58.08	0.791	178.2	5.69	329.2	30.2	508.0	47.0	213	0.238
Acetylene	C_2H_2	26.04	0.9061(A)	191.7	3.7	191.7	17.5	309.5	61.6	113	0.274
Air			1.000					132.5	37.2		
Ammonia	NH_3	17.03	$0.817^{-79°}$ 0.597(A)	195.40	5.653	239.73	23.35	405.5	111.3	72.5	0.243
Ammonium carbonate	$(NH_4)_2CO_3 \cdot H_2O$	114.11			(decomposes at 331 K)						
Ammonium chloride	NH_4Cl	53.50	$2.53^{17°}$		(decomposes at 623 K)						
Ammonium nitrate	NH_4NO_3	80.05	$1.725^{25°}$	442.8	5.4	(decomposes at 483.2 K)					
Ammonium sulfate	$(NH_4)_2SO_4$	132.14	1.769	786		(decomposes at 786 K after melting)					
Aniline	C_6H_7N	93.12	1.022	266.9		457.4		699	52.4		
Benzaldehyde	C_6H_5CHO	106.12	1.046	247.16		452.16	38.40				
Benzene	C_6H_6	78.11	0.879	278.693	9.837	353.26	30.76	562.6	48.6	260	0.274
Benzoic acid	$C_7H_6O_2$	122.12	$1.316^{28°/4°}$	395.4		523.0					
Benzyl alcohol	C_7H_8O	108.13	1.045	257.8		478.4					
Boron oxide	B_2O_3	69.64	1.85	723	22.0						
Bromine	Br_2	159.83	$3.119^{20°}$ 5.87(A)	265.8	10.8	331.78	31.0	584	102	144	0.306
1, 2-Butadiene	C_4H_6	54.09	$0.652^{20°}$	136.7		283.3		446			
1, 3-Butadiene	C_4H_6	54.09	0.621	164.1		268.6		425	42.7	221	0.271
Butane	n-C_4H_{10}	58.12	0.579	134.83	4.661	272.66	22.31	425.17	37.47	255	0.374
iso-Butane	iso-C_4H_{10}	58.12	0.557	113.56	4.540	261.43	21.29	408.1	36.0	263	0.283

phthalate	see Dibutyl phthalate										
tyric acid	n-$C_4H_8O_2$	88.10	0.958	267		437.1		628	52.0	290	0.293
utyric d	iso-$C_4H_8O_2$	88.10	0.949	226		427.7		609			
um senate	$Ca_3(AsO_4)_2$	398.06		1723							
um rbide	Ca_2C_2	64.10	$2.22^{18°}$	2573							
um rbonate	$CaCO_3$	100.09	2.93	(decomposes at 1098 K)							
um oride	$CaCl_2$	110.99	$2.152^{15°}$	1055	28.4						
	$CaCl_2{\cdot}H_2O$	129.01									
	$CaCl_2{\cdot}2H_2O$	147.03									
	$CaCl_2{\cdot}6H_2O$	219.09	$1.78^{17°}$	303.4	37.3	($-6H_2O$ at 473 K)					
um anamide	$CaCN_2$	80.11	2.29								
um anide	$Ca(CN)_2$	92.12									
um droxide	$Ca(OH)_2$	74.10	2.24	($-H_2O$ at 853 K)							
um oxide	CaO	56.08	2.62	2873	50	3123					
um osphate	$Ca_3(PO_4)_2$	310.19	3.14	1943							
um silicate	$CaSiO_3$	117.17	2.915	1803	48.62						
um sulfate ypsum)	$CaSO_4{\cdot}2H_2O$	172.18	2.32	($-1\frac{1}{2}H_2O$ at 301°K)							
on	C	12.010	2.26	3873	46.0	4473					
on dioxide	CO_2	44.01	1.53(A) 1.229 satd. liq. at 5162 kPa	$217.0^{5.2\ atm}$	8.32	(sublimes at 195 K)		304.2	72.9	94	0.275
on ulfide	CS_2	76.14	$1.261^{22°/20°}$ 2.63(A)	161.1	4.39	319.41	26.8	552.0	78.0	170	0.293
on noxide	CO	28.01	0.968(A)	68.10	0.837	81.66	6.042	133.0	34.5	93	0.294
on rachloride	CCl_4	153.84	1.595	250.3	2.5	349.9	30.0	556.4	45.0	276	0.272
ine	Cl_2	70.91	2.49(A)	172.16	6.406	239.10	20.41	417.0	76.1	124	0.276

ces of data are listed at the beginning of Appendix D.
= 20°C/4°C unless specified. Sp gr for gas referred to air (A).

TABLE D.1 *(cont.)*

Compound	Formula	Formula Wt	Sp Gr	Melting Temp. (K)	$\Delta\hat{H}$ Fusion (kJ/g mol)	Normal b.p.(K)	$\Delta\hat{H}$ Vap. at b.p. (kJ/g mol)	T_c (K)	p_c (atm)	$\hat{V}_c$ (cm^3/g mol)	z_c
Chlorobenzene	C_6H_5Cl	112.56	1.107	228		405.26	36.5	632.4	44.6	308	0.265
Chloroform	$CHCl_3$	119.39	$1.489^{20°}$	209.5		334.2		536.0	54.0	240	0.294
Chromium	Cr	52.01	7.1								
Copper	Cu	63.54	8.92	1356.2	13.0	2855	305				
Cumene	C_9H_{12}	120.19	0.862	177.125	7.1	425.56	37.5	636	31.0	440	0.260
Cupric sulfate	$CuSO_4$	159.61	$3.605^{15°}$	(decomposes at 873 K)							
Cyclohexane	C_6H_{12}	84.16	0.779	279.83	2.677	353.90	30.1	553.7	40.4	308	0.274
Cyclopentane	C_5H_{10}	70.13	0.745	179.71	0.6088	322.42	27.30	511.8	44.55	260	0.27
Decane	$C_{10}H_{22}$	142.28	$0.730^{20°}$	243.3		447.0		619.0	20.8	602	0.2476
Dibutyl phthalate	$C_8H_{22}O_4$	278.34	$1.045^{21°}$			613					
Diethyl ether	$(C_2H_5)_2O$	74.12	$0.708^{25°}$	156.86	7.301	307.76	26.05	467	35.6	281	0.261
Ethane	C_2H_6	30.07	1.049(A)	89.89	2.860	184.53	14.72	305.4	48.2	148	0.285
Ethanol	C_2H_6O	46.07	0.789	158.6	5.021	351.7	38.6	516.3	63.0	167	0.248
Ethyl acetate	$C_4H_8O_2$	88.10	0.901	189.4		350.2		523.1	37.8	286	0.252
Ethyl benzene	C_8H_{10}	106.16	0.867	178.185	9.163	409.35	36.0	619.7	37.0	360	0.260
Ethyl bromide	C_2H_5Br	108.98	1.460	154.1		311.4		504	61.5	215	0.320
Ethyl chloride	CH_3CH_2Cl	64.52	$0.903^{10°}$	134.83	4.452	285.43	25	460.4	52.0	199	0.274
3-Ethyl hexane	C_8H_{18}	114.22	0.7169			391.69	34.3	567.0	26.4	466	0.264
Ethylene	C_2H_4	28.05	0.975(A)	103.97	3.351	169.45	13.54	283.1	50.5	124	0.270
Ethylene glycol	$C_2H_6O_2$	62.07	$1.113^{19°}$	260	11.23	470.4	56.9				
Ferric oxide	Fe_2O_3	159.70	5.12	1833		(decomposes at 1833 K)					
Ferric sulfide	Fe_2S_3	207.90	4.3	(decomposes)							
Ferrous sulfide	FeS	87.92	4.84	1466		(decomposes)					
Formaldehyde	H_2CO	30.03	$0.815^{-20°}$	154.9		253.9	24.5				
Formic acid	CH_2O_2	46.03	1.220	281.46	12.7	373.7	22.3				
Glycerol	$C_3H_8O_3$	92.09	$1.260^{50°}$	291.36	18.30	563.2					
Helium	He	4.00	0.1368(A)	3.5	0.02	4.216	0.084	5.26	2.26	58	0.304
Heptane	C_7H_{16}	100.20	0.684	182.57	14.03	371.59	31.69	540.2	27.0	426	0.260

ogen	H_2	2.016	0.06948(A)	13.96	0.12	20.39	0.904	33.3	12.8	65	0.304
ogen loride	HCl	36.47	1.268(A)	158.94	1.99	188.11	16.15	324.6	81.5	87	0.266
ogen oride	HF	20.01	1.15	238		293		503.2			
ogen fide	H_2S	34.08	1.1895(A)	187.63	2.38	212.82	18.67	373.6	88.9	98	0.284
e	I_2	253.8	4.93^{20°	386.5		457.4		826.0			
	Fe	55.85	7.7	1808	15	3073	353				
oxide	Fe_3O_4	231.55	5.2	1867	138	(decomposes at 1867 K after melting)					
	Pb	207.21	11.337^{20°	600.6	5.10	2023	180				
oxide	PbO	223.21	9.5	1159	11.7	1745	213				
nesium	Mg	24.32	1.74	923	9.2	1393	132				
nesium oride	$MgCl_2$	95.23	2.325^{25°	987	43.1	1691	137				
nesium droxide	$Mg(OH)_2$	58.34	2.4	(decomposes at 623 K)							
nesium oxide	MgO	40.32	3.65	3173	77.4	3873					
ury	Hg	200.61	13.546^{20°								
ane	CH_4	16.04	0.554(A)	90.68	0.941	111.67	8.180	190.7	45.8	99	0.290
anol	CH_3OH	32.04	0.792	175.26	3.17	337.9	35.3	513.2	78.5	118	0.222
yl acetate	$C_3H_6O_2$	74.08	0.933	174.3		330.3		506.7	46.3	228	0.254
yl amine	CH_5N	31.06	0.699^{-11°	180.5		266.3^{758mm}		429.9	73.6		
yl oride	CH_3Cl	50.49	1.785(A)	175.3		249		416.1	65.8	143	0.276
yl ethyl one	C_4H_8O	72.10	0.805	186.1		352.6					
yl lohexane	C_7H_{14}	98.18	0.769	146.58	6.751	374.10	31.7	572.2	34.32	344	0.251
bdenum	Mo	95.95	10.2								
nalene	$C_{10}H_8$	128.16	1.145	353.2		491.0					
l	Ni	58.69	8.90^{20°	1725		3173					
acid	HNO_3	63.02	1.502	231.56	10.47	359	30.30				
benzene	$C_6H_5O_2N$	123.11	1.203	278.7		483.9					
gen	N_2	28.02	12.5(D)	63.15	0.720	77.34	5.577	126.2	33.5	90	0.291
gen xide	NO_2	46.01	1.448	263.86	7.334	294.46	14.73	431.0	100.0	82	0.232

ces of data are listed at the beginning of Appendix D.
= 20°C/4°C unless specified. Sp gr for gas referred to air (A).

TABLE D.1 (cont.)

Compound	Formula	Formula Wt	Sp Gr	Melting Temp. (K)	$\Delta\hat{H}$ Fusion (kJ/g mol)	Normal b.p.(K)	$\Delta\hat{H}$ Vap. at b.p. (kJ/g mol)	T_c (K)	p_c (atm)	$\hat{V}_c$ (cm^3/g mol)	z_c
Nitrogen oxide	NO	30.01	1.0367(A)	109.51	2.301	121.39	13.78	19.2	65.0	58	0.256
Nitrogen pentoxide	N_2O_5	108.02	$1.63^{18°}$	303		320					
Nitrogen tetraoxide	N_2O_4	92	$1.448^{20°}$	263.7		294.3		431.0	99.0		
Nitrogen trioxide	N_2O_3	76.02	1.447^{2o}	171		276.5					
Nitrous oxide	N_2O	44.02	$1.226^{-89°}$ 1.530(A)	182.1		184.4		309.5	71.7	96.3	0.272
n-Nonane	C_9H_{20}	128.25	0.718	219.4		423.8		595	23		
n-Octane	C_8H_{18}	114.22	0.703	216.2		398.7		595.0	22.5	543	0.250
Oxalic acid	$C_2H_2O_4$	90.04	1.90	(decomposes at 459 K)							
Oxygen	O_2	32.00	1.1053(A)	54.40	0.443	90.19	6.820	154.4	49.7	74	0.290
n-Pentane	C_5H_{12}	72.15	$0.630^{18°}$	143.49	8 .393	309.23	25.77	469.8	33.3	311	0.269
iso-Pentane	iso-C_5H_{12}	72.15	$0.621^{19°}$	113.1		300.9		461.0	32.9	308	0.268
1-Pentane	C_5H_{10}	70.13	0.641	107.96	4.937	303.13		474	39.9		
Phenol	C_6H_5OH	94.11	$1.071^{25°}$	315.66	11.43	454.56		692.1	60.5		
Phenyl hydrazine	$C_6H_8N_2$	108.14	$1.097^{23°}$	292.76	16 .43	51.66					
Phosphoric acid	H_3PO_4	98.00	$1.834^{18°}$	315.51	10.5	$(-\frac{1}{2}H_2O$ at 486 K)					
Phosphorus (red)	P_4	123.90	2.20	863	81.17	863	41.84				
Phosphorus (white)	P_4	123.90	1.82	317.4	2.5	553	49.71				
Phosphorus pentoxide	P_2O_5	141.95	2.387	(sublimes at 523 K)							
Propane	C_3H_8	44.09	1.562(A)	85.47	3.524	231.09	18 .77	369.9	42.0	200	0.277
Propene	C_3H_6	42.08	1.498(A)	87.91	3.002	225.46	18 .42	365.1	45.4	181	0.274
Propionic acid	$C_3H_6O_2$	74.08	0.993	252.2		414.4		612.5	53.0		

ohol											
opyl ohol	C_3H_8O	60.09	0.785	183.5		355.4		508.8	53.0	219	0.278
pyl benzene	C_9H_{12}	120.19	0.862	173.660	8.54	432.38	38.2	638.7	31.3	429	0.257
n dioxide	SiO_2	60.09	2.25	1883	8.54	2503					
m bisulfate	$NaHSO_4$	120.07	2.742	455							
m carbonate soda)	$Na_2CO_3 \cdot 10H_2O$	286.15	1.46	306.5		($-H_2O$ at 306.5 K)					
m carbonate la ash)	Na_2CO_3	105.99	2.533	1127	33.4	(decomposes)					
m chloride	NaCl	58.45	2.163	1081	28.5	1738	171				
m cyanide	NaCN	49.01		835	16.7	1770	155				
m hydorxide	NaOH	40.00	2.130	592	8.4	1663					
m nitrate	$NaNO_3$	85.00	2.257	583	15.9	(decomposes at 653 K)					
m nitrite	$NaNO_2$	69.00	$2.168^{0°}$	544		(decomposes at 593 K)					
m sulfate	Na_2SO_4	142.05	2.698	1163	24.3						
m sulfide	Na_2S	78.05	1.856	1223	6.7						
m sulfite	Na_2SO_3	126.05	$2.633^{15°}$	(decomposes)							
m sulfate	$Na_2S_2O_3$	158.11	1.667								
ombic)	S_8	256.53	2.07	386	10.0	717.76	84				
noclinic)	S_8	256.53	1.96	392	14.17	717.76	84				
chloride no)	S_2CL_2	135.05	1.687	193.0		411.2	36.0				
dioxide	SO_2	64.07	2.264(A)	197.68	7.402	263.14	24.92	430.7	77.8	122	0.269
trioxide	SO_3	80.07	2.75(A)	290.0	24.5	316.5	41.8	491.4	83.8	126	0.262
ic acid	H_2SO_4	98.08	$1.834^{18°}$	283.51	9.87				(decomposes at 613 K)		
ne	$C_6H_5CH_3$	92.13	0.866	178.169	6.619	383.78	33.5	593.9	40.3	318	0.263
	H_2O	18.016	$1.00^{4°}$	273.16	6.009	373.16	40.65	647.4	218.3	56	0.230
ene	C_8H_{10}	106.16	0.864	225.288	11.57	412.26	34.4	619	34.6	390	0.27
ne	C_8H_{10}	106.16	0.880	247.978	13.60	417.58	36.8	631.5	35.7	380	0.26
ne	C_8H_{10}	106.16	0.861	286.423	17.11	411.51	36.1	618	33.9	370	0.25
	Zn	65.38	7.140	692.7	6.673	1180	114.8				
ulfate	$ZnSO_4$	161.44	$3.74^{15°}$	(decomposes at 1013 K)							

ces of data are listed at the beginning of Appendix D.
= 20°C/4°C unless specified. Sp gr for gas referred to air (A).

TABLE D.2 Enthalpies of Paraffinic Hydrocarbons, C_1–C_6 (J/G MOL)

To convert to Btu/lb mol, multiply by 0.4306.

K	C_1	C_2	C_3	n-C_4	i-C_4	n-C_5	n-C_6
273	0						
291	630	912	1,264	1,709	1,658	2,125	2,545
298	879	1,277	1,771	2,394	2,328	2,976	3,563
300	950	1,383	1,919	2,592	2,522	3,222	3,858
400	4,740	7,305	10,292	13,773	13,623	17,108	20,463
500	9,100	14,476	20,685	27,442	27,325	34,020	40,622
600	14,054	22,869	32,777	43,362	43,312	53,638	64,011
700	19,585	32,342	46,417	61,186	61,220	75,604	90,123
800	25,652	42,718	61,337	80,600	80,767	99,495	118,532
900	32,204	53,931	77,404	101,378	101,754	125,101	148,866
1,000	39,204	65,814	94,432	123,428	123,971	152,213	181,041
1,100	46,567	78,408	112,340	146,607	147,234	180,665	214,764
1,200	54,308	91,504	131,042	170,707	171,418	210,246	249,868
1,300	62,383	105,143	150,331	195,727	196,480	240,872	286,143
1,400	70,709	119,202	170,205	221,375	222,212	272,378	323,465
1,500	79,244	133,678	190,581	247,650	248,571	304,595	361,539
1,600	88,031						
1,800	106,064						
2,000	124,725						
2,200	143,804						
2,500	173,050						

TABLE D.3 Enthalpies of Monoolefinic Hydrocarbons, C_2–C_4 (J/G MOL)

To convert to Btu/lb mol, multiply by 0.4306.

K	Ethylene	Propylene	1-Butene	iso-Butene	*cis*-2-Butene	*trans*-2-Butene
273	0	0	0	0	0	0
291	753	1,104	1,536	1,538	1,354	1,520
298	1,054	1,548	2,154	2,154	1,895	2,125
300	1,125	1,665	2,313	2,322	2,020	2,263
400	6,008	8,882	12,455	12,367	11,070	12,112
500	11,890	17,572	24,765	24,468	22,346	23,995
600	18,648	27,719	38,911	38,425	35,614	37,668
700	26,158	39,049	54,643	53,889	50,542	53,011
800	34,329	51,379	71,755	70,793	66,985	69,705
900	43,053	64,642	89,997	88,826	84,684	87,654
1,000	52,258	78,742	109,286	107,947	103,470	106,692
1,100	61,923	93,470	129,494	123,846	123,260	126,649
1,200	71,964	108,825	150,456	148,866	143,845	147,402
1,300	82,341	124,683	172,087	170,414	165,184	168,824
1,400	92,968	141,000	194,304	188,363	187,150	190,874
1,500	103,888	157,736	217,065	215,183	209,702	213,467

TABLE D.4 Enthalpies of Nitrogen and Some of its Oxides (J/G MOL)

To convert to Btu/lb mol, multiply by 0.4306.

K	N_2	NO	N_2O	NO_2	N_2O_4
273	0	0	0	0	0
291	524	537	681	658	1,384
298	728	746	951	917	1,937
300	786	801	9,660	985	2,083
400	3,695	3,785	13,740	4,865	10,543
500	6,644	6,811	18,179	9,070	19,915
600	9,627	9,895	22,919	13,564	30,124
700	12,652	13,054	27,924	18,305	
800	15,756	16,292	33,154	23,242	
900	18,961	19,597	38,601	28,334	
1,000	22,171	22,970	44,258	33,551	
1,100	25,472	26,392	50,115	38,869	
1,200	28,819	29,861	56,170	44,266	
1,300	32,216	33,371	62,425	49,731	
1,400	35,639	36,915	68,868	55,258	
1,500	39,145	40,488	75,504	60,826	
1,750	47,940	49,505			
2,000	56,902	58,634			
2,250	65,981	67,856			
2,500	75,060	77,127			

TABLE D.5 Enthalpies of Sulfur Compounds (J/G MOL)

To convert to Btu/lb mol, multiply by 0.4306.

K	$S_2(g)$	SO_2	SO_3	H_2S	CS_2
273	0	0	0	0	0
291	579	706	899	607	807
298	805	984	1,255	845	1,125
300	869	1,064	1,338	909	1,217
400	4,196	5,234	6,861	4,372	5,995
500	7,652	9,744	13,033	7,978	11,108
600	11,192	14,514	19,832	11,752	16,455
700	14,790	19,501	27,154	15,706	21,974
800	18,426	24,647	34,748	19,840	27,631
900	22,087	29,915	42,676	24,145	33,388
1,000	25,769	35,275	50,835	28,610	39,220
1,100	29,463	40,706	59,203	33,216	45,103
1,200	33,174	46,191	67,738	37,953	51,044
1,300	36,898	51,714	76,399	42,802	57,027
1,400	40,630	57,320		47,739	63,052
1,500	44,371	62,927		52,802	69,119
1,600	48,116	68,533		57,906	75,186
1,700	51,881	74,182		63,094	81,295
1,800	55,605	79,872		68,324	87,361
1,900	59,370	85,520			

Appendix E

HEAT CAPACITY EQUATIONS

LE E.1 Heat Capacity Equations for Organic and Inorganic Compounds (at Low Pressures)*

s: (1) $C_p^\circ = a + b(T) + c(T)^2 + d(T)^3$;

(2) $C_p^\circ = a + b(T) + c(T)^{-2}$.

Units of C_p° are J/(g mol)(K or °C).

To convert to cal/(g mol)(K or °C) = Btu/(lb mol)(°R or °F), multiply by 0.2390.

Compound	Formula	Mol. Wt.	State	Form	T	a	$b \cdot 10^2$	$c \cdot 10^5$	$d \cdot 10^9$	Temp. Range (in T)
one	CH_3COCH_3	58.08	g	1	°C	71.96	20.10	−12.78	34.76	0–1200
ylene	C_2H_2	26.04	g	1	°C	42.43	6.053	−5.033	18.20	0–1200
		29.0	g	1	°C	28.94	0.4147	0.3191	−1.965	0–1500
			g	1	K	28.09	0.1965	0.4799	−1.965	273–1800
onia	NH_3	17.03	g	1	°C	35.15	2.954	0.4421	−6.686	0–1200
onium sulfate	$(NH_4)_2SO_4$	132.15	c	1	K	215.9				275–328
ene	C_6H_6	78.11	l	1	K	−7.27329	77.054	−164.82	1,897.9	279–350
			g	1	°C	74.06	32.95	−25.20	77.57	0–1200
tane	C_4H_{10}	58.12	g	1	°C	89.46	30.13	−18.91	49.87	0–1200
ane	C_4H_{10}	58.12	g	1	°C	92.30	27.88	−15.47	34.98	0–1200
tene	C_4H_8	56.10	g	1	°C	82.88	25.64	− 17.27	50.50	0–1200
um carbide	CaC_2	64.10	c	2	K	68.62	1.19	-8.66×10^{10}	—	298–720
um carbonate	$CaCO_3$	100.09	c	2	K	82.34	4.975	-12.87×10^{10}	—	273–1033
um hydroxide	$Ca(OH)_2$	74.10	c	1	K	89.5				276–373
um oxide	CaO	56.08	c	2	K	41.84	2.03	-4.52×10^{10}		10273–1173
on	C	12.01	c†	2	K	11.18	1.095	-4.891×10^{10}		273–1373
on dioxide	CO_2	44.01	g	1	°C	36.11	4.233	−2.887	7.464	0–1500
on monoxide	CO	28.01	g	1	°C	28.95	0.4110	0.3548	−2.220	0–1500
on tetrachloride	CCl_4	153.84	l	1	K	12.285	0.01095	−318.26	3,425.2	273–343
ine	Cl_2	70.91	g	1	°C	33.60	1.367	−1.607	6.473	0–1200
er	Cu	63.54	c	1	K	22.76	0.06117			273–1357
ene (opropyl benzene)	C_9H_{12}	120.19	g	1	°C	139.2	53.76	−39.79	120.5	0–1200
exane	C_6H_{12}	84.16	g	1	°C	94.140	49.62	−31.90	80.63	0–1200

hite.

TABLE E.1 (*cont.*)

Compound	Formula	Mol. Wt.	State	Form	T	a	$b \cdot 10^2$	$c \cdot 10^5$	$d \cdot 10^9$	Temp. Range (in T)
Cylopentane	C_5H_{10}	70.13	g	1	°C	73.39	39.28	−25.54	68.66	0–1200
Ethane	C_2H_6	30.07	g	1	°C	49.37	13.92	−5.816	7.280	0–1200
Ethyl alcohol	C_2H_6O	46.07	l	1	K	−325.137	0.041379	−1,403.1	1.7035 × 10⁴	250–400
			g	1	°C	61.34	15.72	−8.749	19.83	0–1200
Ethylene	C_2H_4	28.05	g	1	°C	40.75	11.47	−6.891	17.66	0–1200
Ferric oxide	Fe_2O_3	159.70	c	2	K	103.4	6.711	−17.72 × 10¹⁰	—	273–1097
Formaldehyde	CH_2O	30.03	g	1	°C	34.28	4.268	0.0000	−8.694	0–1200
Helium	He	4.00	g	1	°C	20.8				All
n-Hexane	C_6H_{14}	86.17	l	1	K	31.421	0.97606	−235.37	3,092.7	273–400
			g	1	°C	137.44	40.85	−23.92	57.66	0–1200
Hydrogen	H_2	2.016	g	1	°C	28.84	0.00765	0.3288	−0.8698	0–1500
Hydrogen bromide	HBr	80.92	g	1	°C	29.10	−0.0227	0.9887	−4.858	0–1200
Hydrogen chloride	HCl	36.47	g	1	°C	29.13	−0.1341	0.9715	−4.335	0–1200
Hydrogen cyanide	HCN	27.03	g	1	°C	35.3	2.908	1.092		0–1200
Hydrogen sulfide	H_2S	34.08	g	1	°C	33.51	1.547	0.3012	−3.292	0–1500
Magnesium chloride	$MgCl_2$	95.23	c	1	K	72.4	1.58			273–991
Magnesium oxide	MgO	40.32	c	2	K	45.44	0.5008	−8.732 × 10¹⁰		273–2073
Methane	CH_4	16.04	g	1	°C	34.31	5.469	0.3661	−11.00	0–1200
			g	1	K	19.87	5.021	1.268	−11.00	273–1500
Methyl alcohol	CH_3OH	32.04	l	1	K	−259.25	0.03358	−1,1639	1.4052 × 10⁴	273–400
			g	1	°C	42.93	8.301	−1.87	−8.03	0–700

l cyclohexane	C_7H_{14}	98.18	g	1	°C	121.3	56.53	−37.72	100.8	0–1200
l cyclopentane	C_6H_{12}	84.16	g	1	°C	98.83	45.857	−30.44	83.81	0–1200
acid	HNO_3	63.02	l	1	°C	110.0				25
oxide	NO	30.01	g	1	°C	29.50	0.8188	−0.2925	0.3652	0–3500
en	N_2	28.02	g	1	°C	29.00	0.2199	0.5723	−2.871	0–1500
en dioxide	NO_2	46.01	g	1	°C	36.07	3.97	−2.88	7.87	0–1200
en tetraoxide	N_2O_4	92.02	g	1	°C	75.7	12.5	−11.3		0–300
s oxide	N_2O	44.02	g	1	°C	37.66	4.151	−2.694	10.57	0–1200
n	O_2	32.00	g	1	°C	29.10	1.158	−0.6076	1.311	0–1500
tane	C_5H_{12}	72.15	l	1	K	33.24	192.41	−236.87	17,944	270–350
			g	1	°C	114.8	34.09	−18.99	42.26	0–1200
ne	C_3H_8	44.09	g	1	°C	68.032	22.59	−13.11	31.71	0–1200
ene	C_3H_6	42.08	g	1	°C	59.580	17.71	−10.17	24.60	0–1200
n carbonate	Na_2CO_3	105.99	c	1	K	121				288–371
n carbonate	$Na_2CO_3 \cdot 10H_2O$	286.15	c	1	K	535.6				298
	S	32.07	c‡	1	K	15.2	2.68			273–368
			c§	1	K	18.5	1.84			368–392
ic acid	H_2SO_4	98.08	l	1	°C	139.1	15.59			10–45
dioxide	SO_2	64.07	g	1	°C	38.91	3.904	−3.104	8.606	0–1500
trioxide	SO_3	80.07	g	1	°C	48.50	9.188	−8.540	32.40	0–1000
e	C_7H_8	92.13	l	1	K	1.8083	81.222	−151.27	1,630	270–370
			g	1	°C	94.18	38.00	−27.86	80.33	0–1200
	H_2O	18.016	l	1	K	18.2964	47.212	−133.88	1,314.2	273–373
			g	1	°C	33.46	0.6880	0.7604	−3.593	0–1500

bic. §Monoclinic.

TABLE E.2 Heat Capacity Equations for Organic and Inorganic Compounds (at Low Pressures)*

Form: (1) $C_p^\circ = a + b(T) + c(T)^2 + d(T)^3$;

The following relations have the units of cal/(g mol)(°C or K) = Btu/(lb mol)(°F or °R), and T is in °F or °R.

Compound	Formula	Mol. Wt.	State	Form	T	a	$b \cdot 10^2$	$c \cdot 10^5$	$d \cdot 10^9$	Temp. Range (in T)
Acetylene	C_2H_2	26.04								
			g	1	°F	9.89	0.8273	−0.3783	0.7457	32–2200
Air			g	1	°F	6.900	0.02884	0.02429	−0.08052	32–2700
			g	1	°R	6.713	0.02609	0.03540	−0.08052	492–3200
Ammonia	NH_3	17.03								
			g	1	°F	8.2765	0.39006	0.035245	−0.2740	32–2200
Nitrogen	N_2	28.02	g	1	°F	6.895	0.07624	−0.007009		
Oxygen	O_2	32.00	g	1	°F	7.104	0.07851	−0.005528		
Carbon dioxide	CO_2	44.01	g	1	°F	8.448	0.5757	−0.2159	0.3059	0–3500
Carbon monoxide	CO	28.01	g	1	°F	6.865	0.08024	−0.007367		
Cyclohexane	C_6H_{12}	84.16	l	1	°F	37.05				50
					°F	41.26				150
			g		°F	30.84				50
Cyclopentane	C_5H_{10}	70.13	l	1	°F	32.95				100
Toluene	$C_6H_5{\cdot}CH_3$			1	°F	20.869	5.293	−2 .086	3.929	32–2200

*Sources of data are listed at the beginning of Appendix D.

Appendix F

HEATS OF FORMATION AND COMBUSTION

TABLE F.1 Heats of Formation and Heats of Combustion of Compounds at 25°C*†

Standard states of products for $\Delta\hat{H}_c^\circ$ are $CO_2(g)$, $H_2O(l)$, $N_2(g)$, $SO_2(g)$, and $HCl(aq)$. To convert to Btu/lb mol, multiply by 430.6.

Compound	Formula	Mol. wt.	State	$-\Delta\hat{H}_f^\circ$ (kJ/g mol)	$-\Delta\hat{H}_c^\circ$ (kJ/g mol)
Acetic acid	CH_3COOH	60.05	l	486.2	871.69
			g		919.73
Acetaldehyde	CH_3CHO	40.052	g	166.4	1192.36
Acetone	C_3H_6O	58.08	aq, 200	410.03	
			g	216.69	1821.38
Acetylene	C_2H_2	26.04	g	−226.75	1299.61
Ammonia	NH_3	17.032	l	67.20	
			g	46.191	382.58
Ammonium carbonate	$(NH_4)_2CO_3$	96.09	c		
			aq	941.86	
Ammonium chloride	NH_4Cl	53.50	c	315.4	
Ammonium hydroxide	NH_4OH	35.05	aq	366.5	
Ammonium nitrate	NH_4NO_3	80.05	c	366.1	
			aq	339.4	
Ammonium sulfate	$(NH_4)SO_4$	132.15	c	1179.3	
			aq	1173.1	
Benzaldehyde	C_6H_5CHO	106.12	l	88.83	
			g	40.0	
Benzene	C_6H_6	78.11	l	−48.66	3267.6

TABLE F.1 *(cont.)*

Compound	Formula	Mol. wt.	State	$-\Delta\hat{H}_f^\circ$ (kJ/g mol)	$-\Delta\hat{H}_c^\circ$ (kJ/g mol)
Bromine	Br_2	159.832	l	0	
			g	−30.7	
n-Butane	C_4H_{10}	58.12	l	147.6	2855.6
			g	124.73	2878.52
Isobutane	C_4H_{10}	58.12	l	158.5	2849.0
			g	134.5	2868.8
1-Butene	C_4H_8	56.104	g	−1.172	2718.58
Calcium arsenate	$Ca_3(AsO_4)_2$	398.06	c	3330.5	
Calcium carbide	CaC_2	64.10	c	62.7	
Calcium carbonate	$CaCO_3$	100.09	c	1206.9	
Calcium chloride	$CaCl_2$	110.99	c	794.9	
Calcium cyanamide	$CaCN_2$	80.11	c	352	
Calcium hydroxide	$Ca(OH)_2$	74.10	c	986.59	
Calcium oxide	CaO	56.08	c	635.6	
Calcium phosphate	$Ca_3(PO_4)_2$	310.19	c	4137.6	
Calcium silicate	$CaSiO_3$	116.17	c	1584	
Calcium sulfate	$CaSO_4$	136.15	c	1432.7	
			aq	1450.5	
Calcium sulfate (gypsum)	$CaSO_4{\cdot}2H_2O$	172.18	c	2021.1	
Carbon	C	12.01	c	0	393.51
			Graphite (β)		
Carbon dioxide	CO_2	44.01	g	393.51	
			l	412.92	
Carbon disulfide	CS_2	76.14	l	−87.86	1075.2
			g	−115.3	1102.6
Carbon monoxide	CO	28.01	g	110.52	282.99
Carbon tetrachloride	CCl_4	153.838	l	139.5	352.2
			g	106.69	384.9
Chloroethane	C_2H_5Cl	64.52	g	105.0	1421.1
			l	41.20	5215.44
Cumene (isopropylbenzene)	$C_6H_5CH(CH_3)_2$	120.19	g	−3.93	5260.59
			c	769.86	
Cupric sulfate	$CuSO_4$	159.61	aq	843.12	
			c	751.4	
Cyclohexane	C_6H_{12}	84.16	g	123.1	3953.0
Cyclopentane	C_5H_{10}	70.130	l	105.8	3290.9
			g	77.23	3319.5
Ethane	C_2H_6	30.07	g	84.667	1559.9
Ethyl acetate	$CH_3CO_2C_2H_5$	88.10	l	442.92	2274.48
Ethyl alcohol	C_2H_5OH	46.068	l	277.63	1366.91
			g	235.31	1409.25
Ethyl benzene	$C_6H_5{\cdot}C_2H_5$	106.16	l	12.46	4564.87
			g	−29.79	4607.13
Ethyl chloride	C_2H_5Cl	64.52	g	105	
Ethylene	C_2H_4	28.052	g	−52.283	1410.99
Ethylene chloride	C_2H_3Cl	62.50	g	−31.38	1271.5

TABLE F.1 *(cont.)*

Compound	Formula	Mol. wt.	State	$-\Delta\hat{H}_f^\circ$ (kJ/g mol)	$-\Delta\hat{H}_c^\circ$ (kJ/g mol)
3-Ethyl hexane	C_8H_{18}	114.22	l	250.5	5470.12
			g	210.9	5509.78
Ferric chloride	$FeCl_3$		c	403.34	
Ferric oxide	Fe_2O_3	159.70	c	822.156	
Ferric sulfide	FeS_2	*see* Iron sulfide	*see* Iron sulfide		
Ferrosoferric oxide	Fe_3O_4	231.55	c	1116.7	
Ferrous chloride	$FeCl_2$		c	342.67	303.76
Ferrous oxide	FeO	71.85	c	267	
Ferrous sulfide	FeS	87.92	c	95.06	
Formaldehyde	H_2CO	30.026	g	115.89	563.46
n-Heptane	C_7H_{16}	100.20	l	224.4	4816.91
			g	187.8	4853.48
n-Hexane	C_6H_{14}	86.17	l	198.8	4163.1
			g	167.2	4194.753
Hydrogen	H_2	2.016	g	0	285.84
Hydrogen bromide	HBr	80.924	g	36.23	
Hydrogen chloride	HCl	36.465	g	92.311	
Hydrogen cyanide	HCN	27.026	g	−130.54	
Hydrogen sulfide	H_2S	34.082	g	20.15	562.589
Iron sulfide	FeS_2	119.98	c	177.9	
Lead oxide	PbO	223.21	c	219.2	
Magnesium chloride	$MgCl_2$	95.23	c	641.83	
Magnesium hydroxide	$Mg(OH)_2$	58.34	c	924.66	
Magnesium oxide	MgO	40.32	c	601.83	
Methane	CH_4	16.041	g	74.84	890.4
Methyl alcohol	CH_3OH	32.042	l	238.64	726.55
			g	201.25	763.96
Methyl chloride	CH_3Cl	50.49	g	81.923	766.63†
Methyl cyclohexane	C_7H_{14}	98.182	l	190.2	4565.29
			g	154.8	4600.68
Methyl cyclopentane	C_6H_{12}	84.156	l	138.4	3937.7
			g	106.7	3969.4
Nitric acid	HNO_3	63.02	l	173.23	
			aq	206.57	
Nitric oxide	NO	30.01	g	−90.374	
Nitrogen dioxide	NO_2	46.01	g	−33.85	
Nitrous oxide	N_2O	44.02	g	−81.55	
n-Pentane	C_5H_{12}	72.15	l	173.1	3509.5
			g	146.4	3536.15
Phosphoric acid	H_3PO_4	98.00	c	1281	
			aq ($1H_2O$)	1278	
Phosphorus	P_4	123.90	c	0	
Phosphorus pentoxide	P_2O_5	141.95	c	1506	

TABLE F.1 *(cont.)*

Compound	Formula	Mol. wt.	State	$-\Delta\hat{H}_f^\circ$ (kJ/g mol)	$-\Delta\hat{H}_c^\circ$ (kJ/g mol)
n-Propyl alcohol	C_3H_8O	60.09	g	255	2068.6
n-Propylbenzene	$C_6H_5{\cdot}CH_2{\cdot}C_2H_5$	120.19	l	38.40	5218.2
			g	−7.824	5264.5
Silicon dioxide	SiO_2	60.09	c	851.0	
Sodium bicarbonate	$NaHCO_3$	84.01	c	945.6	
Sodium bisulfate	$NaHSO_4$	120.07	c	1126	
Sodium carbonate	Na_2CO_3	105.99	c	1130	
Sodium chloride	NaCl	58.45	c	411.00	
Sodium cyanide	NaCN	49.01	c	89.79	
Sodium nitrate	$NaNO_3$	85.00	c	466.68	
Sodium nitrite	$NaNO_2$	69.00	c	359	
Sodium sulfate	Na_2SO_4	142.05	c	1384.5	
Sodium sulfide	Na_2S	78.05	c	373	
Sodium sulfite	Na_2SO_3	126.05	c	1090	
Sodium thiosulfate	$Na_2S_2O_3$	158.11	c	1117	
Sulfur	S	32.07	c (rhombic)	0	
			c (monoclinic)	−0.297	
Sulfur chloride	S_2Cl_2	135.05	l	60.3	
Sulfur dioxide	SO_2	64.066	g	296.90	
Sulfur trioxide	SO_3	80.066	g	395.18	
Sulfuric acid	H_2SO_4	98.08	l	811.32	
			aq	907.51	
Toluene	$C_6H_5{\cdot}CH_3$	92.13	l	−11.99	3909.9
			g	−50.000	3947.9
Water	H_2O	18.016	l	285.840	
			g	241.826	
m-Xylene	$C_6H_4(CH_3)_2$	106.16	l	25.42	4551.86
			g	−17.24	4594.53
o-Xylene	$C_6H_4(CH_3)_2$	106.16	l	24.44	4552.86
			g	−19.00	4596.29
p-Xylene	$C_6H_4(CH_3)_2$	106.16	l	24.43	4552.86
			g	−17.95	4595.25
Zinc sulfate	$ZnSO_4$	161.45	c	978.55	
			aq	1059.93	

*Sources of data are given at the beginning of Appendix D, References 1, 4, and 5.
†Standard state HCl(g).

Appendix G

VAPOR PRESSURES

TABLE G.1 Vapor Pressures of Various Substances

Antoine equation:

$$\ln (p^*) = A - \frac{B}{C + T}$$

where p^* = vapor pressure, mm Hg
T = temperature, K
A, B, C = constants

Name	Formula	Range (K)	A	B	C
Acetic acid	$C_2H_4O_2$	290–430	16.8080	3405.57	−56.34
Acetone	C_3H_6O	241–350	16.6513	2940.46	−35.93
Ammonia	NH_3	179–261	16.9481	2132.50	−32.98
Benzene	C_6H_6	280–377	15.9008	2788.51	−52.36
Carbon disulfide	CS_2	288–342	15.9844	2690.85	−31.62
Carbon tetrachloride	CCl_4	253–374	15.8742	2808.19	−45.99
Chloroform	$CHCl_3$	260–370	15.9732	2696.79	−46.16
Cyclohexane	C_6H_{12}	280–380	15.7527	2766.63	−50.50
Ethyl acetate	$C_4H_8O_2$	260–385	16.1516	2790.50	−57.15
Ethyl alcohol	C_2H_6O	270–369	18.5242	3578.91	−50.50
Ethyl bromide	$C_2H_5B_r$	226–333	15.9338	2511.68	−41.44
n-Heptane	C_7H_{16}	270–400	15.8737	2911.32	−56.51
n-Hexane	C_6H_{14}	245–370	15.8366	2697.55	−48.78
Methyl alcohol	CH_4O	257–364	18.5875	3626.55	−34.29
n-Pentane	C_5H_{12}	220–330	15.8333	2477.07	−39.94
Sulfur dioxide	SO_2	195–280	16.7680	2302.35	−35.97
Toluene	$C_6H_5CH_3$	280–410	16.0137	3096.52	−53.67
Water	H_2O	284–441	18.3036	3816.44	−46.13

Appendix H

HEATS OF SOLUTION AND DILUTION

TABLE H.1 Integral Heats of Solution and Dilution at 25°c

Formula	Description	State	$-\Delta\hat{H}_f^\circ$ (kJ/g mol)	$-\Delta\hat{H}_{soln}^\circ$ (kJ/g mol)	$-\Delta\hat{H}_{dil}^\circ$ (kJ/g mol)
NaOH	crystalline II		426.726		
	in 3 H_2O	aq	455.612	28.869	28.869
	4 H_2O	aq	461.156	34.434	5.564
	5 H_2O	aq	464.486	37.739	3.305
	10 H_2O	aq	469.227	42.509	4.769
	20 H_2O	aq	469.591	42.844	.334
	30 H_2O	aq	469.457	42.718	.125
	40 H_2O	aq	469.340	42.593	.125
	50 H_2O	aq	469.252	42.509	.083
	100 H_2O	aq	469.059	42.342	.167
	200 H_2O	aq	469.026	42.258	.083
	300 H_2O	aq	469.047	42.300	.041
	500 H_2O	aq	469.097	42.383	.083
	1,000 H_2O	aq	469.189	42.467	.083
	2,000 H_2O	aq	469.285	42.551	.083
	10,000 H_2O	aq	469.448	42.718	.041
	50,000 H_2O	aq	469.528	42.802	.083
	∞ H_2O	aq	469.595	42.886	.083
H_2SO_4		liq	811.319		
	in 0.5 H_2O	aq	827.051	15.731	15.731
	1.0 H_2O	aq	839.394	28.074	12.343
	1.5 H_2O	aq	848.222	36.902	8.823
	2 H_2O	aq	853.243	41.923	5.021
	3 H_2O	aq	860.314	48.994	7.071
	4 H_2O	aq	865.376	54.057	5.063
	5 H_2O	aq	869.351	58.032	3.975
	10 H_2O	aq	878.347	67.027	8.995
	25 H_2O	aq	883.618	72.299	5.272
	50 H_2O	aq	884.664	73.345	1.046
	100 H_2O	aq	885.292	73.973	0.628
	500 H_2O	aq	888.054	76.734	2.761
	1,000 H_2O	aq	889.894	78.575	1.841
	10,000 H_2O	aq	898.388	87.069	2.636
	100,000 H_2O	aq	904.957	93.637	6.568
	500,000 H_2O	aq	906.630	95.311	1.674
	∞ H_2O	aq	907.509	96.190	0.879

SOURCE: F. D. Rossini et al., "Selected Values of Chem. Thermo. Properties," *Natl. Bur. Std. Circ. 500*, U.S. Government Printing Office, Washington, D.C., 1952.

Appendix I

ENTHALPY-CONCENTRATION DATA

TABLE I.1 Enthalpy-concentration Data for the Single-phase Liquid Region and Also the Saturated Vapor of the Acetic Acid-water System at 1 Atmosphere

Reference state: Liquid water at 32°F and 1 atm; solid acid at 32°F and 1 atm.

Liquid or Vapor		Enthalpy—Btu/lb Liquid Solution						Enthalpy Saturated Vapor Btu/lb
Mole Fraction Water	Weight Fraction Water	20°C 68°F	40°C 104°F	60°C 140°F	80°C 176°F	100°C 212°F	Saturated Liquid	
0.00	0.00	93.54	111.4	129.9	149.1	169.0	187.5	361.8
0.05	0.01555	93.96	112.2	130.9	150.4	170.6	186.9	
0.10	0.03225	93.82	112.3	131.5	151.3	172.0	186.5	374.6
0.20	0.0698	92.61	111.9	131.7	152.2	173.8	185.2	395.3
0.30	0.1140	90.60	110.7	131.3	152.6	175.0	183.8	423.7
0.40	0.1667	87.84	108.9	130.6	152.9	176.1	182.9	461.4
0.50	0.231	83.96	106.3	129.1	152.7	177.1	182.4	510.5
0.55	0.268	81.48	104.5	128.1	152.5	177.5	182.3	
0.60	0.3105	78.53	102.5	126.9	152.1	178.0	182.0	573.4
0.65	0.358	75.36	100.2	125.5	151.6	178.3	181.9	
0.70	0.412	71.72	97.71	123.9	151.1	178.8	181.7	656.0
0.75	0.474	67.59	94.73	122.2	149.9	179.3	182.1	
0.80	0.545	62.88	91.43	120.2	149.7	179.9	181.6	767.3
0.85	0.630	57.44	87.56	117.8	148.9	180.5	181.6	
0.90	0.730	51.03	83.02	115.1	147.8	180.8	181.7	921.6
0.95	0.851	43.74	77.65	111.7	146.4	181.2	181.5	
1.00	1.00	36.06	71.91	107.7	143.9	180.1	180.1	1150.4

SOURCE: Data calculated from miscellaneous literature sources and smoothed.

TABLE I.2 Vapor-liquid Equilibrium Data for the Acetic Acid-water System; Pressure = 1 Atmosphere

x Mole Fraction Water in the Liquid	y Mole Fraction Water in the Vapor
0.020	0.035
0.040	0.069
0.060	0.103
0.080	0.135
0.100	0.165
0.200	0.303
0.300	0.425
0.400	0.531
0.500	0.627
0.600	0.715
0.700	0.796
0.800	0.865
0.900	0.929
0.940	0.957
0.980	0.985

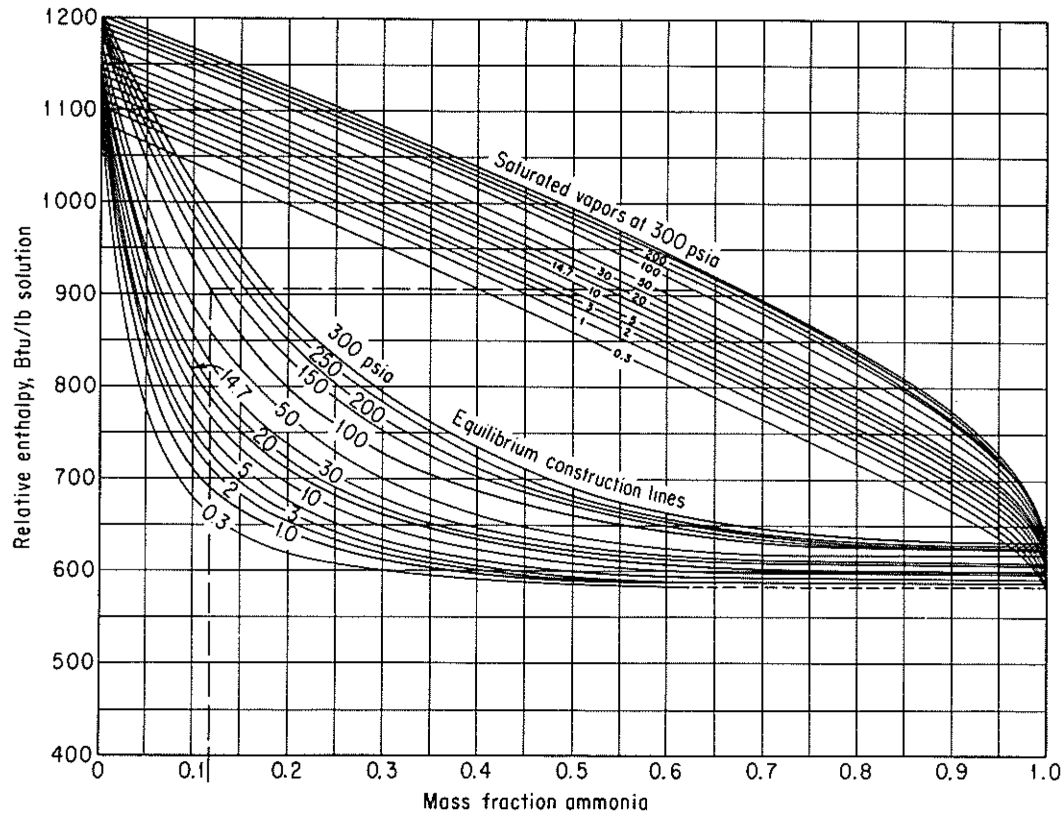

Saturated vapors at 300 psia
200
100
50
30
20
14.7
10
5
3
2
1
0.5
300 psia
250
200
150
100
50
30
20
10
5
3
2
1.0
0.3
Equilibrium construction lines
Relative enthalpy, Btu/lb solution
1200
1100
1000
900
800
700
600
500
400
Mass fraction ammonia
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0

Figure I.1

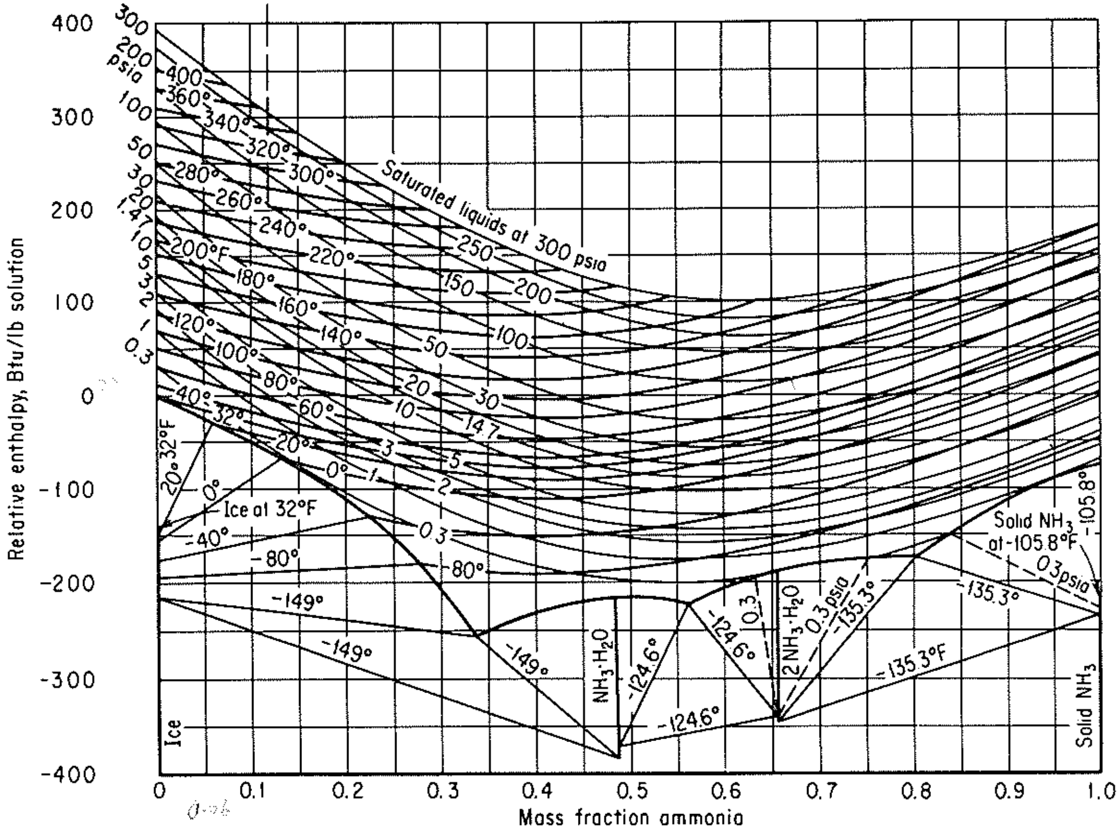


Figure I.1 Enthalpy-concentration chart for NH_3-H_2O.

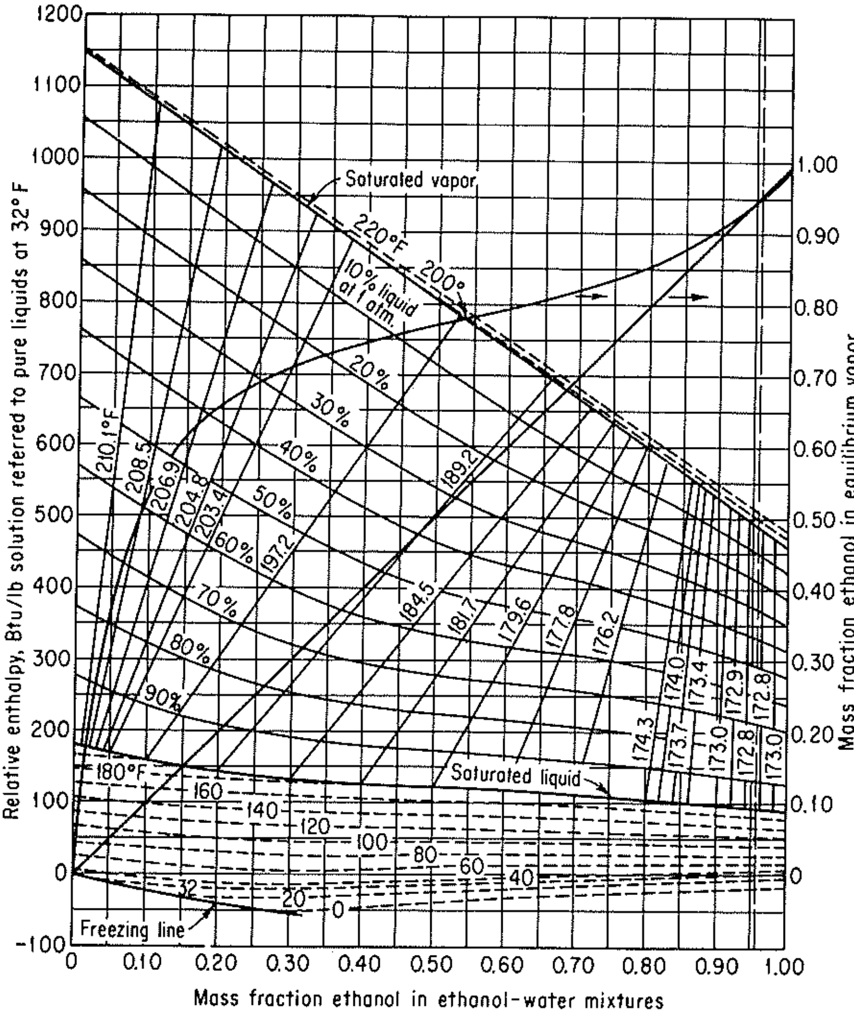


Figure I.2 Enthalpy-composition diagram for the ethanol-water system, showing liquid and vapor phases in equilibrium at 1 atm.

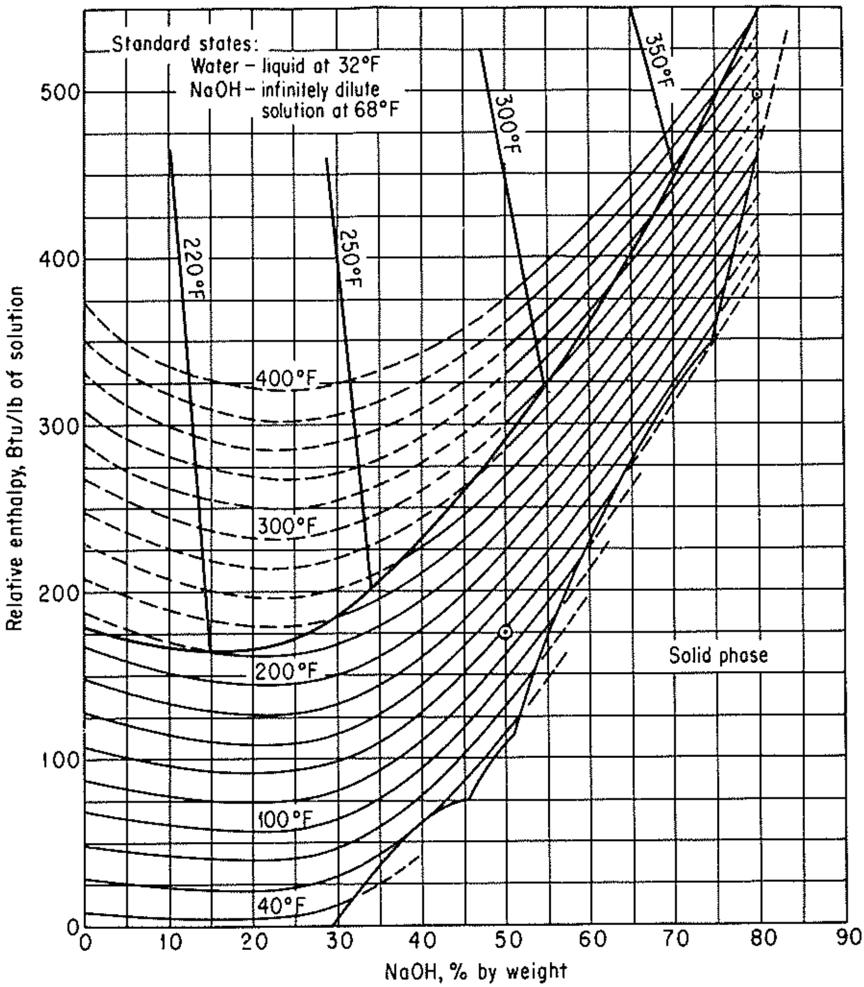


Figure I.3 Enthalpy-concentration chart for sodium hydroxide-water.

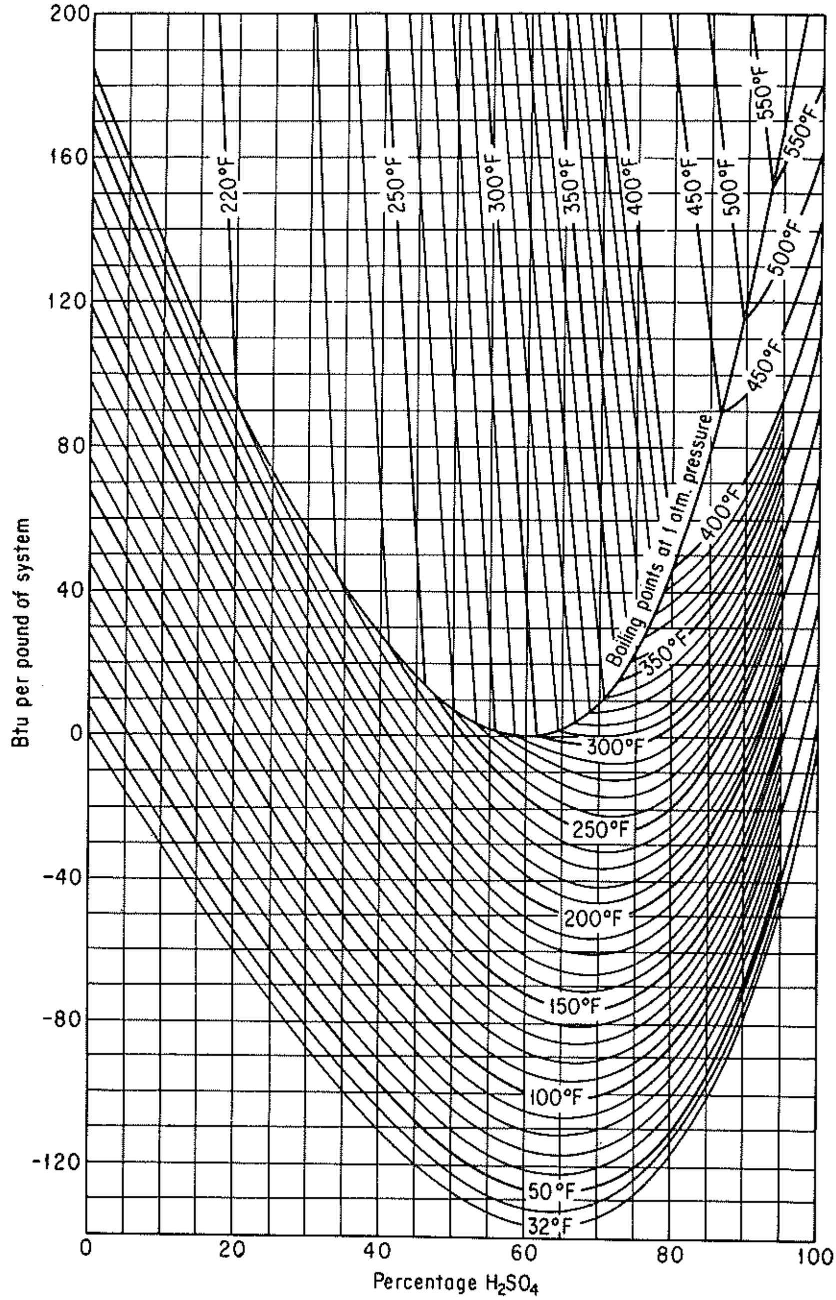


Figure I.4 Enthalpy-concentration of sulfuric acid-water system relative to pure components. (Water and H_2SO_4 at 32°F and own vapor pressure.) (Data from International Critical Tables, © 1943 O. A. Hougen and K. M. Watson.)

Appendix J

THERMODYNAMIC CHARTS

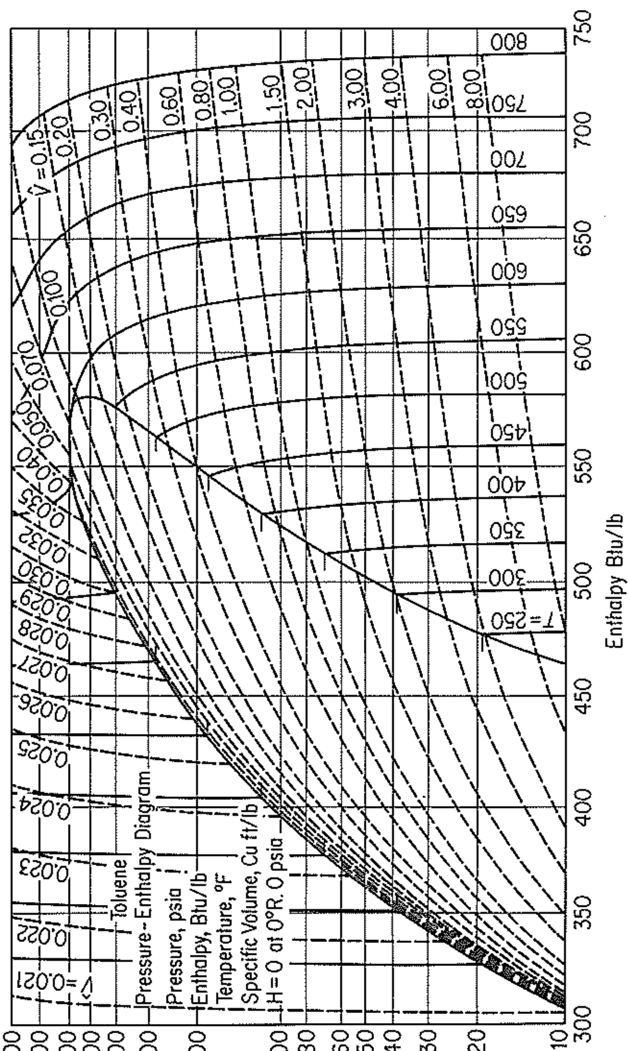


Figure J.1 Enthalpy-pressure chart for toluene.

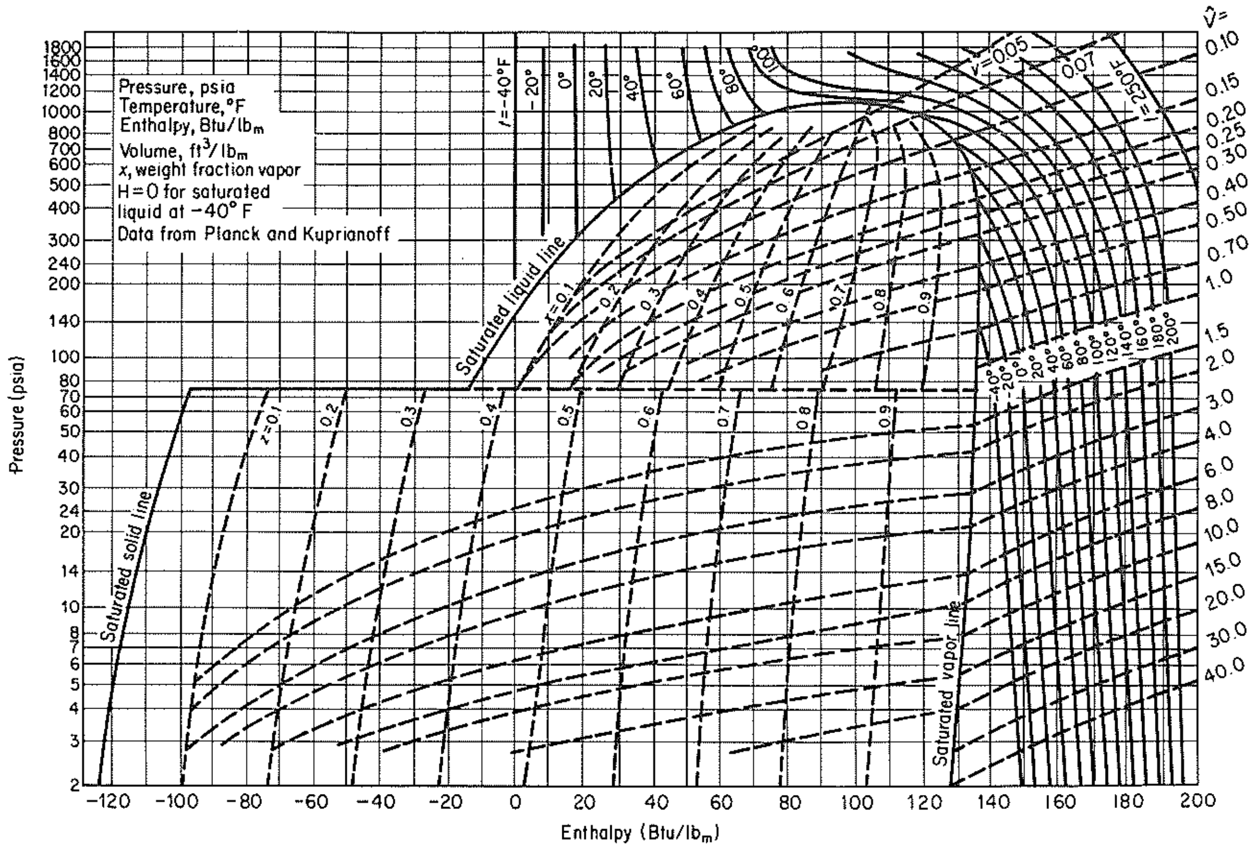


Figure J.2 Pressure-enthalpy chart for carbon dioxide.

Appendix K

PHYSICAL PROPERTIES OF PETROLEUM FRACTIONS

In the early 1930s, tests were developed which characterized petroleum oils and petroleum fractions, so that various physical characteristics of petroleum products could be related to these tests. Details of the tests can be found in *Petroleum Products and Lubricants,* an annual publication of the Committee D-2 of the American Society for Testing Materials.[1] These tests are not scientifically exact, and hence the procedure used in the tests must be followed faithfully if reliable results are to be obtained. However, the tests have been adopted because they are quite easy to perform in the ordinary laboratory and because the properties of petroleum fractions can be predicted from the results. The specifications for fuels, oils, and so on, are set out in terms of these tests plus many other properties, such as the flashpoint, the percent sulfur, and the viscosity.

Over the years various phases of the initial work have been extended and development of a new characterization scheme using the pseudocompound approach is evolving. Daubert[2] summarizes the traditional and new methods insofar as predicting molecular weights, pseudocritical temperature and pressure, acentric factor, and characterization factors.

In this appendix we present the results of the work of Smith and Watson and associates,[3–5] who related petroleum properties to a factor known as the *characterization factor* (sometimes called the *UOP characterization factor*). It is defined as

$$K = \frac{(T_B)^{1/3}}{S}$$

where K = *UOP* characterization factor
T = cubic average boiling point, °R
S = specific gravity at 60°F/60°F

[1] Report of Committee D-2, ASTM, Philadelphia, annually.

[2] T. E. Daubert, "Property Predictions," *Hydrocarbon Proc.*, pp. 107–110 (March 1980).

[3] R. L. Smith and K. M. Watson, *Ind. Eng. Chem.*, v. 29, p. 1408 (1937).

[4] K. M. Watson and E. F. Nelson, *Ind. Eng. Chem.*, v. 25, p. 880 (1933).

Other averages for boiling points are used in evaluating K and the other physical properties in this Appendix. (Refer to Danbert or Maxwell[6] for details.) This factor has been related to many of the other simple tests and properties of petroleum fractions, such as viscosity, molecular weight, critical temperature, and percentage of hydrogen, so that it is quite easy to estimate the factor for any particular sample. Furthermore, tables of the UOP characterization factor are available for a wide variety of common types of petroleum fractions as shown in Table K.1 for typical liquids.

Van Winkle[7] discusses the relationships among the volumetric average boiling point, the molal average boiling point, the cubic average boiling point, the weight average boiling point, and the mean average boiling point, and illustrates how the K and other properties of petroleum fractions can be evaluated from experimental data. In Table K.2 are shown the source, boiling-point basis, and any special limitations of the various charts in this appendix.

TABLE K.1 Typical UOP Characterization Factors

Type of stock	K	Type of stock	K
Pennsylvania crude	12.2–12.5	Propane	14.7
Mid-Continent crude	11.8–12.0	Hexane	12.8
Gulf Coast crude	11.0–11.8	Octane	12.7
East Texas crude	11.9	Natural gasoline	12.7–12.8
California crude	10.98–11.9	Light gas oil	10.5
Benzene	9.5	Kerosene	10.5–11.5

Riazi[8] has proposed an equation to predict the basic properties of crude and products based on the equation

$$\text{value of property} = aT_B^b S^c$$

where T_B is the normal boiling point in °R and S is the specific gravity at 60°F; a, b, and c are empirical constants.

For use in computer-aided calculations, the following empirical correlations can be of assistance. Refer to the original references and Daubert[9] for an assessment of the accuracy of each relation with respect to any material.

Heat capacity of hydrogen vapor[10]

$$C_p = (0.0450K - 0.233) + (0.440 + 0.0177K)(10^{-3}t) - 0.1520(10^{-6}t^2)$$

where C_p = specific heat of vapor, Btu/(lb$_m$)(°F)
t = temperature, °F
K = characterization factor

[6] J. B. Maxwell, *Data Book on Hydrocarbons*, New York; Van Nostrand, 1950.

[7] M. Van Winkle, *Pet. Refiner*, v. 34, pp. 136–138 (June 1955).

[8] M. R. Riazi, Ph.D. dissertation, Pennsylvania State University, 1980.

[9] Daubert, *Hydrocarbon Process.*, p. 107.

[10] J. F. Fallon and K. M. Watson, "Thermal Properties of Hydrocarbons," *Nat. Pet. News (Tech. Sec.)*, p. R-372 (June 7, 1944).

TABLE K.2 Information Concerning Charts in Appendix K

1. *Specific heats of hydrocarbon liquids*
   Source: J. B. Maxwell, *Data Book on Hydrocarbons,* Van Nostrand Reinhold, New York, 1950, p. 93 (original from M. W. Kellogg Co.).
   Description: A chart of C_p (0.4 to 0.8) vs. t (0 to 1000°F) for petroleum fractions from 0 to 120° API.
   Boiling-point basis: Volumetric average boiling point, which is equal to graphical integration of the differential ASTM distillation curve (Van Winkle's "exact method").
   Limitations: This chart is not valid at temperatures within 50°F of the pseudocritical temperatures.
2. *Vapor pressure of hydrocarbons*
   Source: Maxwell, *Data Book on Hydrocarbons,* p. 42.
   Description: Vapor pressure (0.002 to 100 atm) vs. temperature (50 to 1200°F) for hydrocarbons with normal boiling point of 100 to 1200°F (C_4H_{10} and C_5H_{12} lines shown).
   Boiling-point basis: Normal boiling points (pure hydrocarbons).
   Limitations: These charts apply well to all hydrocarbon series except the lowest-boiling members of each series.
3. *Heat of combustion of fuel oils and petroleum fractions*
   Source: Maxwell, *Data Book on Hydrocarbons,* p. 180.
   Description: Heats of combustion above 60°F (17,000 to 25,000 Btu/lb) vs. gravity (0 to 60°API) with correction for sulfur and inerts included (as shown on chart).
4. *Properties of petroleum fractions*
   Source: O. A. Hougen and K. M. Watson, *Chemical Process Principles Charts,* Wiley, New York, 1946, Chart 3.
   Description: °API(−10 to 90°API) vs. boiling point (100 to 1000°F) with molecular weight, critical temperature, and K factors as parameters.
   Boiling-point basis: Use cubic average boiling point when using the K values; use mean average boiling point when using the molecular weights.
5. *Heats of vaporization of hydrocarbons and petroleum fractions at* 1.0 *atm pressure*
   Source: Hougen and Watson, *Chemical Process Principles Chart,* Chart 68.
   Description: Heats of vaporization (60 to 180 Btu/lb) vs. mean average boiling point (100 to 1000°F) with molecular weight and API gravity as parameters.
   Boiling-point basis: Mean average boiling point.

Heat capacity of hydrocarbon liquids[11]

$$C_p = [(0.355 + 0.128 \times 10^{-2}\text{API}) + (0.503 + 0.117 \times 10^{-2}\text{API})(10^{-3}t)] \times (0.05K + 0.41)$$

where API = gravity degrees API and the other units are the same as for vapor heat capacity.

Specific gravity

141.5

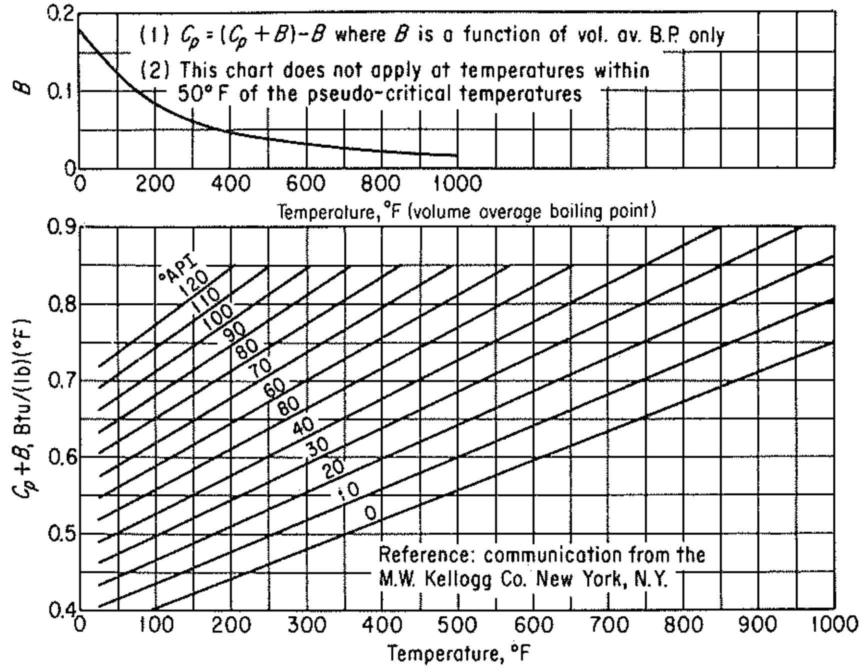


Figure K.1 Specific heats of hydrocarbon liquids.

where sp gr = specific gravity, 60/60°F
°API = degrees API

Pseudo critical temperature[12]

$$t_c' = a_0 + a_1T + a_2T^2 + a_3AT + a_4T^3 + a_5T^2 + a_6A^2T^2$$

where t_c' = pseudo-critical temperature, °R
T = molal average boiling point, °F
A = degrees API
a_i = constants in the correlation (see Table K.3)

Pseudo critical pressure[13]

$$p_c' = b_0 + b_1T + b_2T^2 + b_3AT + b_4T^3 + b_5AT^2 + b_6A^2T + b_7A^2T^2$$

where p_c' = pseudo-critical pressure psia
T = mean average boiling point, °F
A = degrees API
b_i = constants in the correlation (see Table K.3)

[12] R. H. Cavett. "Physical Data for Distillation Calculations-Vapor-Liquid Equilibria," *Proc. Am. Pet. Inst. Div. Refining*. v. 42, p. 351 (1962).

[13] Ibid.

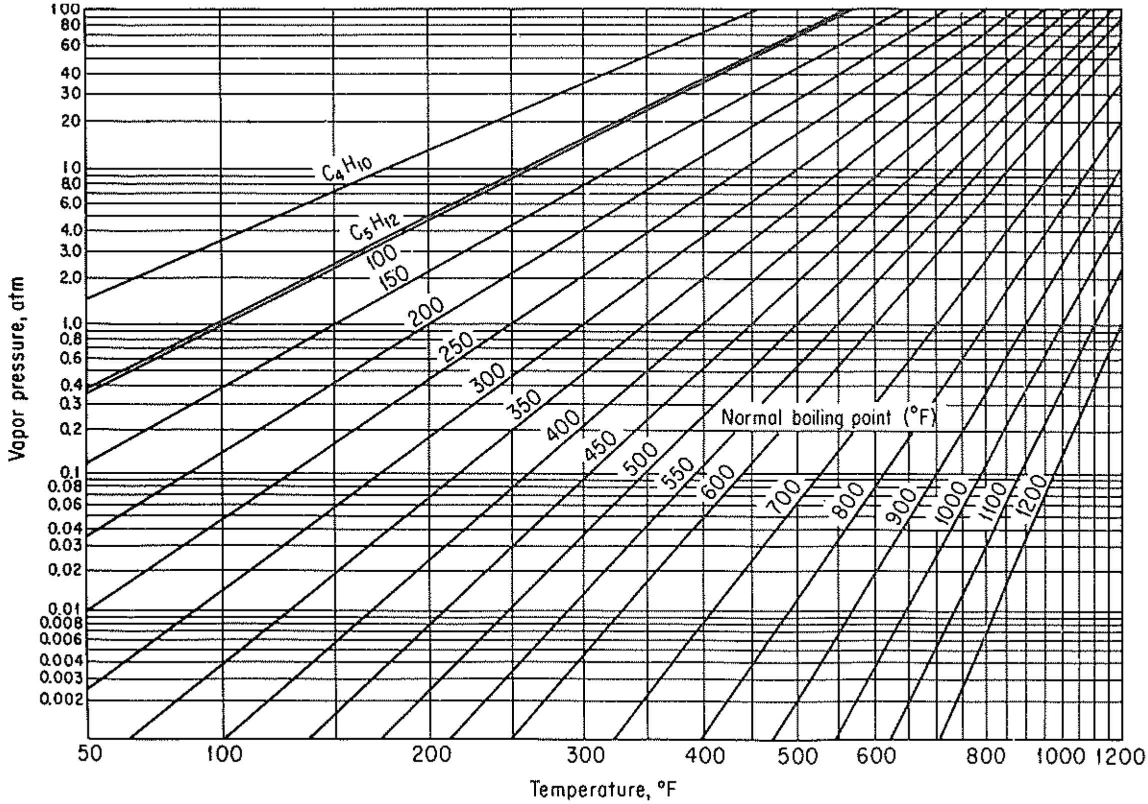


Figure K.2 Vapor pressure of hydrocarbons.

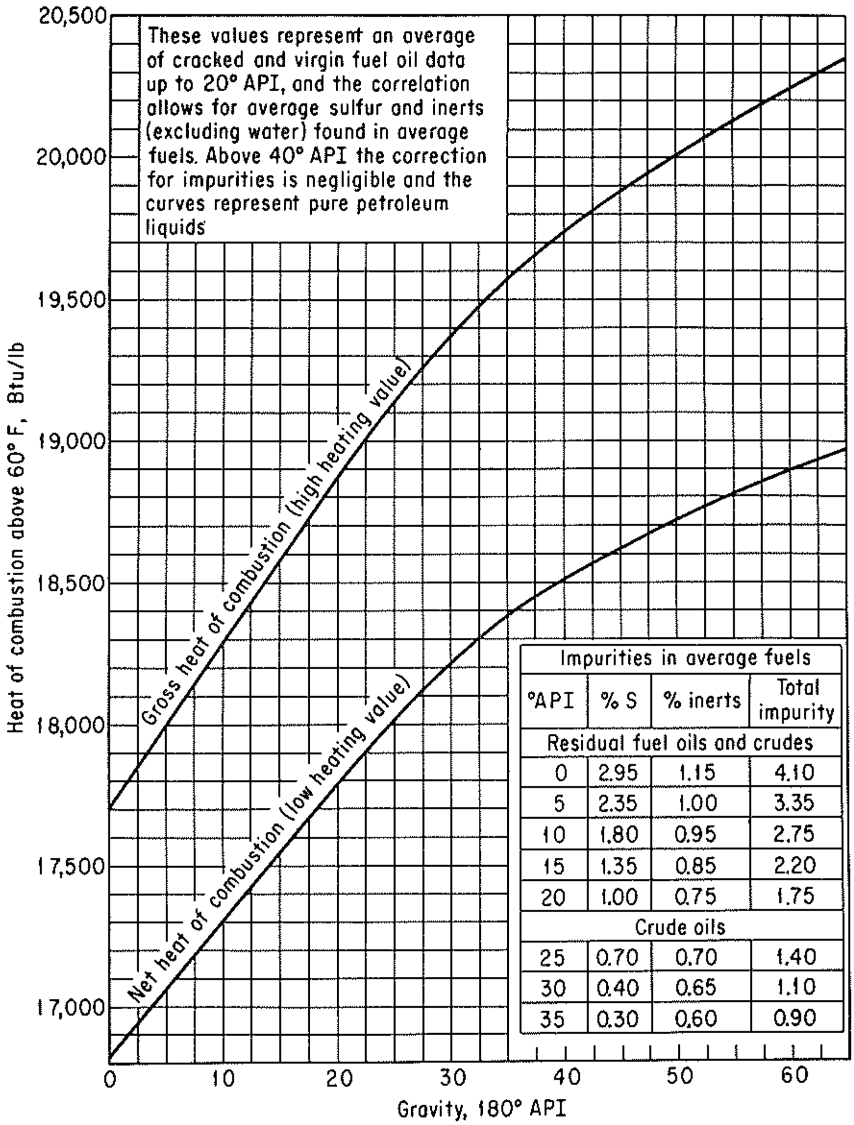
Enthalpy. Calculated as

$$\Delta H = \int_{T_{\text{ref}}}^{\bar{T}_1} C_{p_{\text{liq}}}\, dt + \Delta H_{\text{vaporization at } \bar{T}_1} + \int_{\bar{T}_1}^{T} C_{p_{\text{vap}}}\, dt$$

where $\overline{T_1}$ is the mean average boiling point.

TABLE K.3 Constants for Cavett Correlations

i	a_i	b_i
0	768.07121	2.8290406
1	(0.17133693) (10)	(0.94120109) (10^{-3})
2	(−0.10834003) (10^{-2})	(−0.30474749) (10^{-5})
3	(−0.89212579) (10^{-2})	(−0.20876110) (10^{-4})
4	(0.38890584) (10^{-6})	(0.15184103) (10^{-8})
5	(0.53094920) (10^{-5})	(0.11047899) (10^{-7})



°API	% S	% inerts	Total impurity
Residual fuel oils and crudes			
0	2.95	1.15	4.10
5	2.35	1.00	3.35
10	1.80	0.95	2.75
15	1.35	0.85	2.20
20	1.00	0.75	1.75
Crude oils			
25	0.70	0.70	1.40
30	0.40	0.65	1.10
35	0.30	0.60	0.90

Figure K.3 Heat of combustion of fuel oils and petroleum fractions.

Heat of vaporization[14]

$$\Delta H_{vap} = 2.303\ \{(z_g - z_l)R(T_c)[A + 40T_r^2(T_r - b)e^{-20}(T_r - b)^2]\}$$

where H_{vap} = latent heat of vaporization, Btu/lb_m

$z_g - z_l$ = pressure correction (see Fallon and Watson for table of values)

R = universal gas constant

T_c = critical temperature, °R

[14] Fallon and Watson, "Thermal Properties," p. R-372.

T_r = reduced temperature
A, b = constants (see Fallon and Watson for ways to calculate values)

or (a somewhat less accurate equation)[15]

$$\Delta H_{\text{vap}} = 0.95RT_B\left(\frac{T_B}{T_B - 43}\right)^2\left(\frac{1 - T_r}{1 - T_{rB}}\right)^{0.38}$$

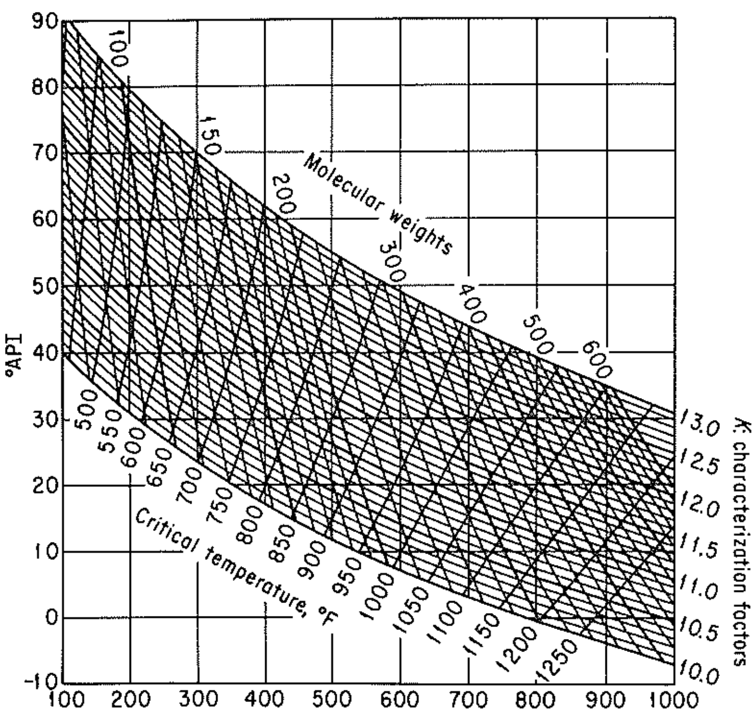
where T_B = molal average boiling point, °R
T_r = reduced temperature
T_{rB} = reduced T_B

Boiling-point relations

(a)
$$\bar{t}_v = \sum_{i=1}^{n} X_{vi} t_{bi}$$

where X_{vi} = volume fraction of the ith component of the petroleum fraction
t_{bi} = normal boiling point of the midpoint of the ith volume fraction, °F or °R
n = number of volume fractions in the distillation curve to characterize the petroleum product

[15] K. M. Watson, "Thermodynamics of the Liquid State," *Ind. Eng. Chem.*, v. 35, p. 398 (1943).



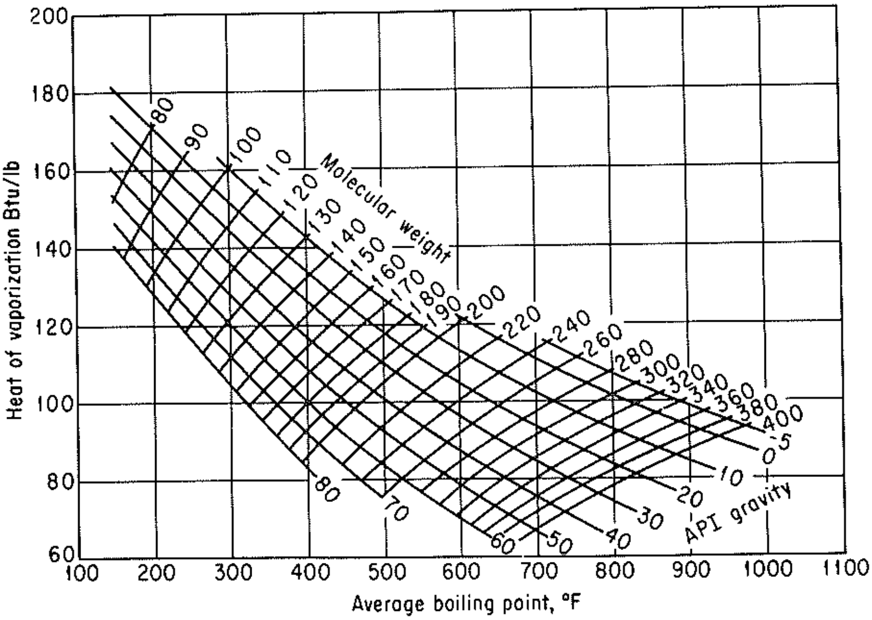


Figure K.5 Heats of vaporization of hydrocarbon and petroleum fractions at 1.0 atmosphere pressure.

(b) $$\bar{t} = (V_1 t_1^{1/3} + V_2 t_2^{1/3} + \cdots + V_n t_n^{1/3})^3$$

where $\bar{t}$ = cubic average boiling point, °F

V_i = volume fraction i of a petroleum product having a normal boiling point of t_i

t_i = temperature, °F

(c) $$\bar{t}_x = x_1 t_1 + x_2 t_2 + \cdots + x_n t_n$$

where $\bar{t}_x$ = molar average boiling point, °F

x_i = the mole fractions of the narrow boiling range with normal boiling points t_i

t_i = temperature, °F

(d) $$\bar{t}_M = \frac{\bar{t}_x + \bar{t}}{2}$$

where $\bar{t}_M$ is the mean average boiling point, °F.

SUPPLEMENTARY REFERENCES

Behar, E., "Natural Gas and Crude Oil Characterization," *Proceedings Use of Computers in Chemical Engineering,* April 26–30, 1987, Taormina, Sicily, Pergamon Press, New York, 1988.

Daubert, T. E., "Property Predictions," *Hydrocarbon Process.,* p. 107 (March 1980).

Appendix L

SOLUTION OF SETS OF EQUATIONS

L.1 INDEPENDENT LINEAR EQUATIONS

This appendix contains a brief summary of methods of solving linear and nonlinear equations. It is only a summary; for details consult one of the numerous texts on numerical analysis that can be found in any library.

If you write several *linear* material balances, say m in number, they will take the form

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\
&\cdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m
\end{aligned}
\tag{L.1}
$$

or in compact matrix notation

$$\mathbf{ax} = \mathbf{b} \tag{L.1a}$$

where $x_1, x_2, \ldots, x_n$ represent the unknown variables, and the a_{ij} and b_i represent the constants and known variables. As an example of Eq. (L.1), we can write the three component mass balances corresponding to Fig. L.1:

$$
\begin{aligned}
0.50(100) &= 0.80(P) + 0.05(W) \\
0.40(100) &= 0.05(P) + 0.925(W) \\
0.10(100) &= 0.15(P) + 0.025(W)
\end{aligned}
\tag{L.2}
$$

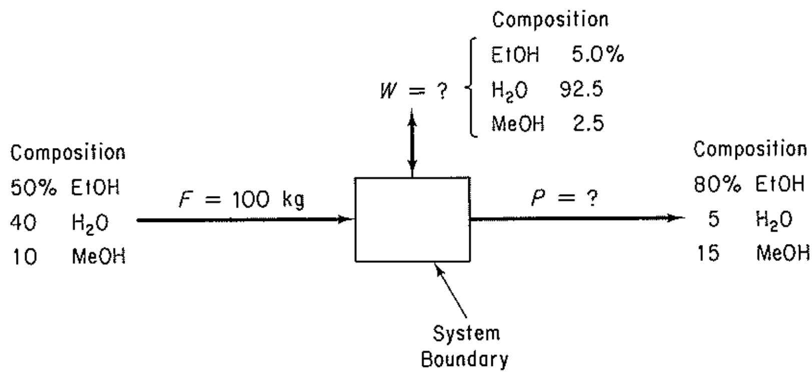


Figure L.1. Data for a material balance.

(a) There is no set of x's that satisfies Eq. (L.1).
(b) There is a unique set of x's that satisfies Eq. (L.1).
(c) There is an infinite number of sets of x's that satisfy Eq. (L.1).

Figure L.2 represents each of the three cases geometrically in two dimensions. Case 1 is usually termed inconsistent, whereas cases 2 and 3 are consistent; but to an engineer who is interested in the solution of practical problems, case 3 is as unsatisfying as case 1. Hence case 2 will be termed *determinate,* and case 3 will be termed *indeterminate*.

To ensure that a system of equations represented by (L.1) has a unique solution, it is necessary to first show that (L.1) is consistent, that is, that the coefficient

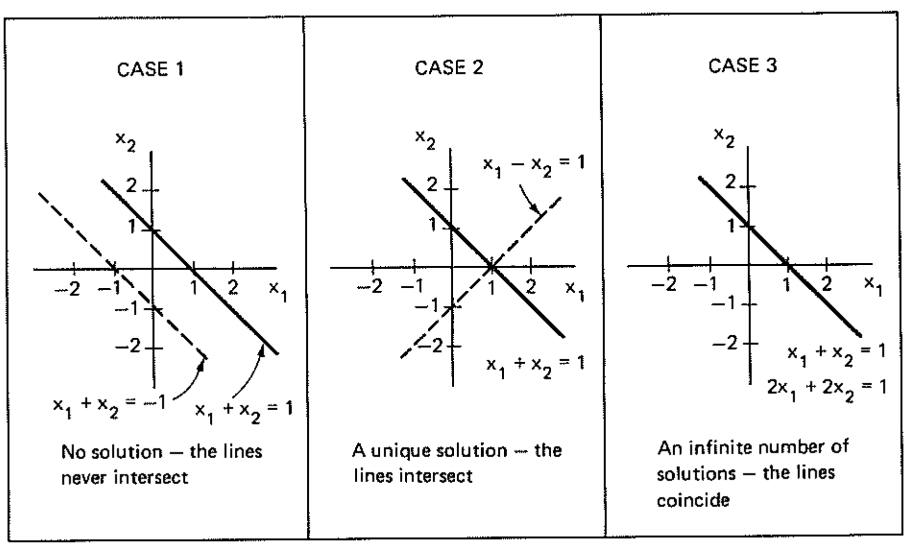


Figure L.2. Types of solutions of linear equations.

matrix **a** and the augmented matrix [**a**, **b**] must have the same rank r. Then, if $n = r$, the system (L.1) is determinate, whereas if $r < n$, as may be the case, then the number $(n - r)$ variables must be specified in some manner or determined by optimization procedures. If the equations are independent, $m = r$.

As an illustration of these ideas, consider the case of the three equations corresponding to the set of Eqs. (L.2) in which we have more equations than unknowns $(m > n)$. The matrix [**a**, **b**] is

$$[\mathbf{a}, \mathbf{b}] = \left[\begin{array}{cc|c} 0.80 & 0.05 & 50 \\ 0.05 & 0.925 & 40 \\ 0.15 & 0.025 & 10 \end{array}\right]$$

Note how the rank of **a** (the rank of a matrix is given by the size of the largest nonzero determinant that can be formed from the matrix), the matrix composed of the first two columns, can at the most be 2 and that the rank of [**a**, **b**] is also 2 because the determinant of [**a**, **b**] is zero. To obtain a consistent set of equations, one of the three material balances must be eliminated, leaving two equations in two unknowns, P and W, that have a unique solution ($m = r = 2$ and $n = 2$). The third equation is a redundant equation. It would probably be best to pick the two equations in which the coefficients were known with the greatest precision.

As another example, consider a set of 10 equations involving 16 unknowns $(m < n)$ so that $(n - m) = 6$. Consequently, six more variables must be specified in some manner before the system of equations becomes determinate.

Next, suppose that you are interested in solving n linear independent equations in n unknown variables:

$$\left.\begin{array}{c} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n \end{array}\right\} \mathbf{ax} = \mathbf{b} \tag{L.3}$$

In general there are two ways to solve Eq. (L.3) for $x_1, \ldots, x_n$: elimination techniques and iterative techniques. Both are easily executed by computer programs. In the pocket in the back cover of this book you will find a disk containing Fortran computer programs that can be used in solving sets of linear equations. We shall illustrate the Gauss–Jordan elimination method. Other techniques can be found in texts on matrices, linear algebra, and numerical analysis.

The essence of the Gauss–Jordan method is to transform Eq. (L.3) into Eq. (L.4) by sequential nonunique elementary operations on Eq. (L.3):

$$\begin{array}{c} x_1 + 0 + \cdots + 0 = b_1' \\ 0 + x_2 + \cdots + 0 = b_2' \\ \cdots \\ 0 + 0 + \cdots + x_n = b_n' \end{array} \tag{L.4}$$

Equation (L.4) has a solution for $x_1, \ldots, x_n$ that can be obtained by inspection.

EXAMPLE L.1

To illustrate the elementary operations that are required to execute the Gauss–Jordan method, consider the following independent set of three equations in three unknowns:

$$4x_1 + 2x_2 + x_3 = 15 \tag{1}$$

$$20x_1 + 5x_2 - 7x_3 = 0 \tag{2}$$

$$8x_1 - 3x_2 + 5x_3 = 24 \tag{3}$$

The augmented matrix is

$$\left[\begin{array}{rrr|r} 4 & 2 & 1 & 15 \\ 20 & 5 & -7 & 0 \\ 8 & -3 & 5 & 24 \end{array}\right]$$

Take the a_{11} element as a pivot. To make it 1 and the other elements in the first column zero, carry out the following elementary operations shown in order for each row:

(a) Subtract $(\frac{20}{4})$ Eq. (1), from Eq. (2).
(b) Subtract $(\frac{8}{4})$ Eq. (1), from Eq. (3).
(c) Multiply Eq. (1) by $\frac{1}{4}$.

to get

$$\begin{array}{cc} & \textit{new eq. no.} \\ \left[\begin{array}{rrr|r} 1 & \frac{1}{2} & \frac{1}{4} & \frac{15}{4} \\ 0 & -5 & -12 & -75 \\ 0 & -7 & 3 & -6 \end{array}\right] & \begin{array}{c} (1a) \\ (2a) \\ (3a) \end{array} \end{array}$$

Carry out the following elementary operations to make the pivot element a_{22} equal to 1 and the other elements in the second column equal to zero:

(d) subtract $[(\frac{1}{2})/-5]$ of Eq. (2a), from Eq. (1a).
(e) Subtract $(-7/-5)$ of Eq. (2a), from Eq. (3a).
(f) Multiply Eq. (2a) by $(1/-5)$.

to obtain

$$\begin{array}{cc} & \textit{new eq. no.} \\ \left[\begin{array}{rrr|r} 1 & 0 & -\frac{19}{20} & -\frac{15}{4} \\ 0 & 1 & \frac{12}{5} & 15 \\ 0 & 0 & \frac{99}{5} & 99 \end{array}\right] & \begin{array}{c} (1b) \\ (2b) \\ (3b) \end{array} \end{array}$$

Another series of elementary operations (left for you to propose) leads to a 1 for the element a_{33} and zeros for the other two elements in the third column:

$$\left[\begin{array}{rrr|r} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

The solution to the original set of equations is

$$x_1 = 1$$

$$x_2 = 3$$

$$x_3 = 5$$

as can be observed from the augmentation column.

To obtain good accuracy and avoid numerical errors, the choice of the pivot should be made by scanning all the eligible coefficients and choosing the one with the greatest magnitude for the next pivot. For example, you might choose a_{21} for the first pivot, and then find that a_{31} would be the next pivot and finally a_{22} the last pivot to give

$$\left[\begin{array}{ccc|c} 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

By exchanging rows 1 and 2, you get exactly the form as (L.4).

EXAMPLE L.2

Determine the number of independent components (which is the same as the number of independent material balances) for a process involving the following two competing reactions:

$$CO + 2H_2 \longrightarrow CH_3OH$$

$$CO + 3H_2 \longrightarrow CH_4 + H_2O$$

Solution

Prepare a matrix in which the rows are the atomic species for which the balances are to be made and the columns are chemical compounds entering and leaving the process. Each element in the matrix is the number of atoms in the chemical compound.

$$\begin{array}{cc} & \begin{array}{ccccc} H_2 & H_2O & CO & CH_4 & CH_3OH \end{array} \\ \begin{array}{c} H \\ O \\ C \end{array} & \begin{bmatrix} 2 & 2 & 0 & 4 & 4 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

Transform the matrix by elementary operations so that there are 1's on the main diagonal starting at the a_{11} position, and only zeros below the main diagonal. The sum of the diagonal elements [starting at the a_{11} position] is the *rank* of the matrix that is equivalent to the number of components. Do you get three for the matrix above?

The number of independent components is not always equal to the number of atomic species, as shown in the next example.

EXAMPLE L.3

Determine the number of independent components for a process involving the following reaction:

$$SO_3 + H_2O \longrightarrow H_2SO_4$$

Solution

Form the species matrix and determine its rank.

$$\begin{array}{c} \\ H \\ S \\ O \end{array}\begin{array}{c} \begin{array}{ccc} H_2O & SO_3 & H_2SO_4 \end{array} \\ \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{bmatrix} \end{array}$$

Are you able to make the transformation to

$$\begin{array}{c} \\ H \\ S \\ O \end{array}\begin{array}{c} \begin{array}{ccc} H_2O & SO_3 & H_2SO_4 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

Note that the rank of the matrix is 2, not 3; hence only two independent components exist for independent material balances.

To summarize, you will find the information flow diagram in Fig. L.3 helpful. Linear equations can also be solved by the iterative methods described in the next section.

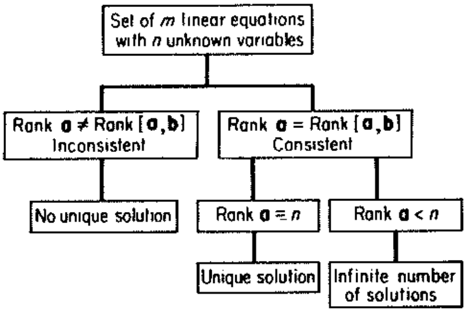


Figure L.3. **a** is the $m \times n$ matrix of the coefficients in the equations, and **[a, b]** is the $m \times (n + 1)$ augmented matrix.

L.2 NONLINEAR INDEPENDENT EQUATIONS

The precise criteria used to ascertain if a linear system of equations is determinate cannot be neatly extended to nonlinear systems of equations. Furthemore, the solution of sets of nonlinear equations requires the use of computer codes that may fail to solve your problem for one or more of a variety of reasons, a few of which are mentioned below. The problem to be solved can be written as

$$\left.\begin{aligned} f_1(x_1, \ldots, x_n) &= 0 \\ f_2(x_1, \ldots, x_n) &= 0 \\ &\cdots \\ f_n(x_1, \ldots, x_n) &= 0 \end{aligned}\right\} \quad \mathbf{f}(\mathbf{x}) = \mathbf{0} \tag{L.5}$$

where each function $f_i(x_1, \ldots, x_n)$ corresponds to a nonlinear function containing one or more of the variable whose values are unknown.

Figure L.4 classifies the major general methods of solving systems of nonlinear equations. Within each category and as combinations of categories you can find innumerable variations and submethods in the literature and available as computer codes. Table L.1 lists several of the computer codes that can be found in most computer center libraries and can be used to solve sets of nonlinear equations. In the disk in the pocket in the back of this book is a simple Fortran code that makes use of the

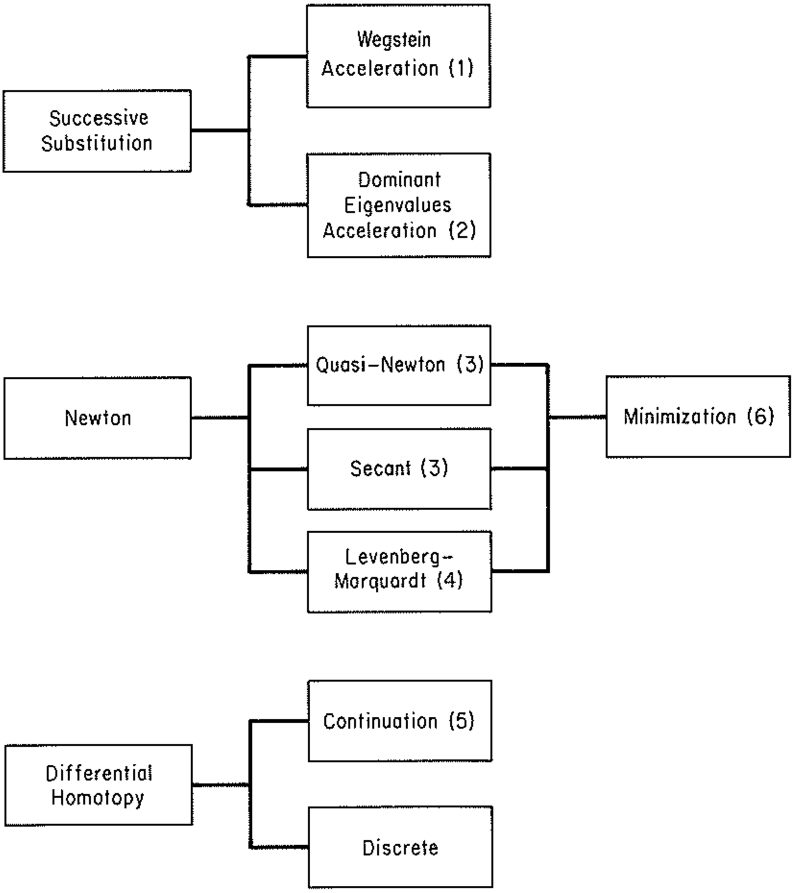


Figure L.4. General categories of techniques to solve nonlinear equations.

References for Figure AL.4

1. Wegstein, J. H., *Commu. ACM*. vol. 1, p. 9 (1958)
2. Orbach, O., and C. M. Crowe, *Can. J. Chem. Eng.*, vol. 49, p. 509 (1971).
3. Dennis, J. E. and R. B. Schnabel, *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, Prentice-Hall, Englewood Cliffs, N.J., 1983.
4. Marquardt, D., *SIAM J Appld. Math.*, vol. 11, p. 431 (1963).
5. Davidenko, D., *Ukrain Math.*, vol. 5, p. 196 (1953).
6. Edgar, T. F. and D. M. Himmelblau, *Optimization of Chemical Processes*, McGraw-Hill, New York, 1988.

TABLE L.1 Computer Codes to Solve Sets of Nonlinear Equations

Code	Method	Source
CO5NAF	Powell's hybrid	NAG, 1101 31st St., Suite 100, Downers Grove, IL 60515
DEPAR	Homotopy plus Newton	Algorithm 502, *ACM Trans. Math. Software,* v. 2, p. 98 (1976)
HYBRD	Powell's hybrid	MINIPACK, Argonne Natl. Lab., Argonne, IL 60439.
	Homotopy	J. D. Seader, Dept. Chem. Eng., University of Utah, Salt Lake City, UT 84112.
SOSNLE	Brown's method	Numerical Math. Div., Sandia Natl. Lab., Albuquerque, NM 87185
Various	Newton, secant	J. E. Dennis, and R. B. Schnabel, *Numerical Methods for Unconstrained Optimization and Nonlinear Equations,* 1983, Appendix A, Prentice-Hall, Englewood Cliffs, NJ 07632
ZSYSTM	Brent's method	IMSL, 7500 Bellaire Blvd., Houston, TX 77036

basic Newton method. However, it is only a simplified code, and for professional work a more robust code should be employed.

Newton's Method

Refer to equations (L.5). For a single equation (and variable), $f(x) = 0$, Newton's method uses the expansion of $f(x)$ in a first-order Taylor series about a reference point (a starting guess for the solution) x_0.

$$f(x) \approx f(x_0) + \frac{df(x_0)}{dx}(x - x_0) \tag{L.6}$$

Note that Eq. (L.6) is a *linear equation* which is tangent to $f(x)$ at x_0. Examine Fig. L.5. The right-hand side of Eq. (L.6) is equated to zero and the resulting equation solved for $(x - x_0)$.

$$x - x_0 = -\frac{f(x_0)}{df(x_0)/dx} \tag{L.7}$$

For example, suppose that $f(x) = 4x^3 - 1 = 0$, hence $df(x)/dx = 12x^2$. The sequence of steps to apply Newton's method using Eq. (L.7) starting at $x_0 = 3$ would be

$$\begin{aligned} x_1 &= x_0 - \frac{4x^3 - 1}{12x^2} \\ &= 3 - \frac{107}{108} = 2.009259 \end{aligned}$$

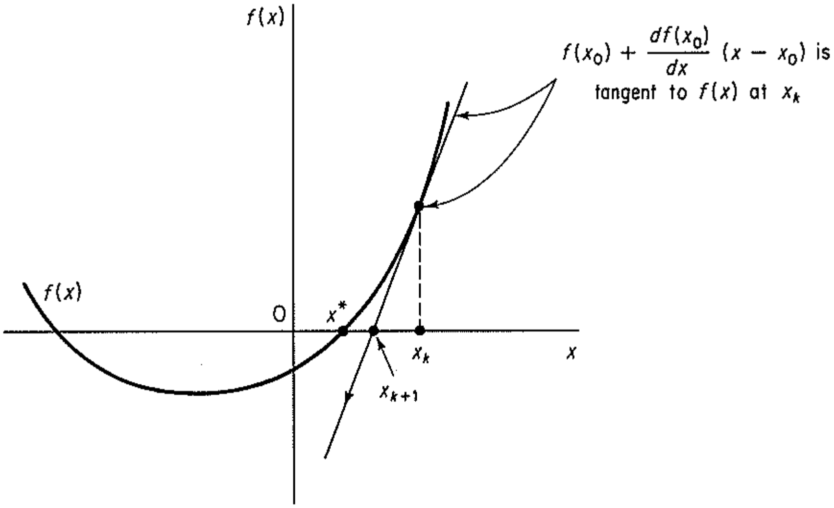


Figure L.5. Newton's Method applied to the solution of $f(x) = 0$ starting at x_k. x_{k+1} is the next reference point for linearization. x^* is the solution.

$$x_2 = 2.00926 - \frac{31.4465}{48.4454} = 1.36015$$

Additional iterations yield the following values for x_k:

k	x_k
0	3.00000
1	2.009259
2	1.3601480
3	0.9518103
4	0.7265254
5	0.6422266
6	0.6301933
7	0.6299606
8	0.6299605
9	0.6299605

so that the solution is given with increasing precision as k increases.

Suppose that two independent equations in two variables whose values are to be determined are

$$\begin{aligned} f_1(x_1, x_2) &= 0 \\ f_2(x_1, x_2) &= 0 \end{aligned} \tag{L.8}$$

To apply Newton's method, expand each equation as a first-order Taylor series to get a set of linear equations at the point (x_{10}, x_{20}).

$$\begin{aligned} f_1(x_1, x_2) &\simeq f_1(x_{10}, x_{20}) + \frac{\partial f_1(x_{10}, x_{20})}{\partial x_1}(x_1 - x_{10}) + \frac{\partial f_1(x_{10}, x_{20})}{\partial x_2}(x_2 - x_{20}) \\ f_2(x_1, x_2) &\simeq f_2(x_{10}, x_{20}) + \frac{\partial f_2(x_{10}, x_{20})}{\partial x_1}(x_1 - x_{10}) + \frac{\partial f_2(x_{10}, x_{20})}{\partial x_2}(x_2 - x_{20}) \end{aligned} \tag{L.9}$$

Let the partial derivatives be designed by the constants $a_{ij} = \partial f_i/\partial x_j$ to simplify the notation, and let $(x_i - x_{i0}) = \Delta x_i$. Then, after equating the right-hand side of Eqs. (L.9) to zero, they become

$$\begin{aligned} a_{11}\,\Delta x_1 + a_{12}\,\Delta x_2 &= -f_1(x_{10}, x_{20}) \\ a_{21}\,\Delta x_1 + a_{22}\,\Delta x_2 &= -f_2(x_{10}, x_{20}) \end{aligned} \tag{L.10}$$

These equations are linear and can be solved by a linear equation solver to get the next reference point (x_{11}, x_{21}). Iteration is continued until a solution of satisfactory precison is reached. Of course, a solution may not be reached, as illustrated in Fig. L.6c, or may not be reached because of round-off or truncation errors. If the Jacobian matrix [see Eq. (L.11) below] is singular, the linearized equations may have no solution or a whole family of solutions, and Newton's method probably will fail to obtain a solution. It is quite common for the Jacobian matrix to become ill-conditioned because if x_0 is far from the solution or the nonlinear equations are badly scaled, the correct solution will not be obtained.

The analog of (L.10) in matrix notation is

$$\mathbf{J}_k(\mathbf{x} - \mathbf{x}_k) = -\mathbf{f}(\mathbf{x}_k) \tag{L.11}$$

where $\mathbf{J}$ is the Jacobian matrix (the matrix whose elements are composed of the first partial derivatives of the equations with respect to the variables). For two equations

$$\mathbf{J} = \begin{bmatrix} \partial f_1(\mathbf{x})/\partial x_1 & \partial f_1(\mathbf{x})/\partial x_2 \\ \partial f_2(\mathbf{x})/\partial x_1 & \partial f_2(\mathbf{x})/\partial x_2 \end{bmatrix} \qquad \mathbf{x} = \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix} \qquad \mathbf{f} = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$$

Quasi-Newton Methods

A quasi-Newton method in general is one that imitates Newton's method. If $f(x)$ is not given by a formula, or the formula is so complicted that analytical derivatives cannot be formulated, you can replace df/dx in Eq. (L.7) with a finite difference approximation

$$x_{k+1} = x_k - \frac{f(x)}{[f(x+h) - f(x-h)]/2h} \tag{L.12}$$

A central difference has been used in Eq. (L.12) but forward differences or any other difference scheme would suffice *as long as the step size h is selected to match the difference formula and the computer (machine) precision* for the computer on which the calculations are to be executed.

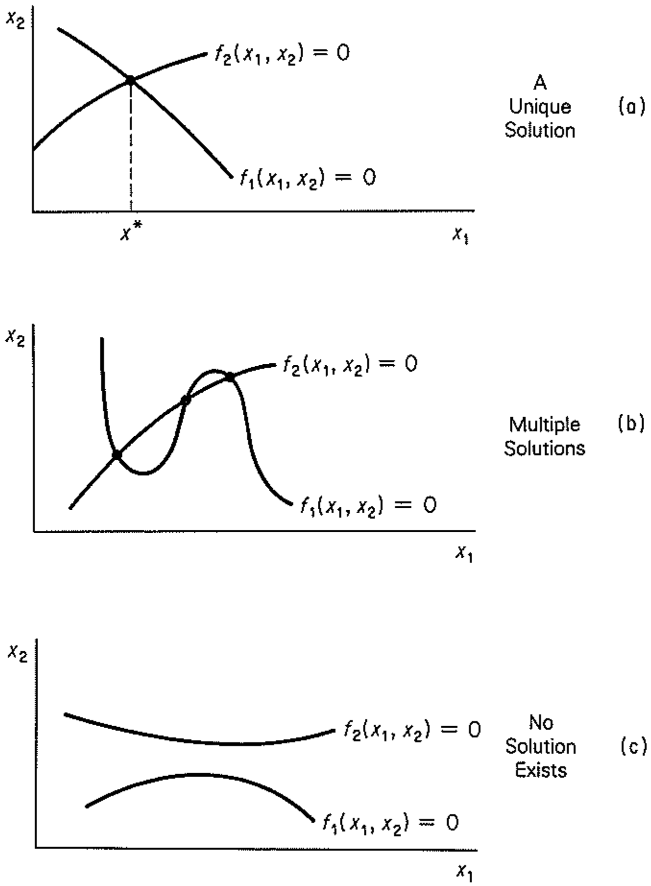


Figure L.6. Possible cases for the solution of two independent nonlinear equations $f_1(x_1, x_2) = 0$ and $f_2(x_1, x_2) = 0$ in two unknown variables.

Other than the problem of the selection of the value of h, the only additional disadvantage of a quasi-Newton method is that additional function evaluations are needed on each iteration k. Eq. (L.12) can be applied to sets of equations if the partial derivatives are replaced by finite difference approximations.

Secant Methods

In the secant method the approximate model analogous to the right hand side of Eq. (4.6) (equated to zero) is

$$f(x_k) + m(x - x_k) = 0 \tag{L.13}$$

where m is the slope of the line connecting the a point x_k and a second point x_q, given by

$$m = \frac{f(x_q) - f(x_k)}{x_q - x_k}$$

Thus the secant method, imitates Newton's method (in this sense the secant method is also a quasi-Newton method (see Figure L.7)

Secant methods start out by using two points x_k and x_q spanning the interval of x, points at which the values of $f(x)$ are of opposite sign. The zero of $f(x)$ is predicted by

$$\tilde{x} = x_q - \frac{f(x_q)}{\dfrac{f(x_q) - f(x_k)}{x_q - x_k}} \tag{L.14}$$

The two points retained for the next step are $\tilde{x}$ and either x_q or x_k, the choice being made so that the pair of values $f(\tilde{x})$, and either $f(x_k)$ or $f(x_q)$, have opposite signs to maintain the bracket on x^*. (This variation is called "regula falsi" or the method of false position.) In Figure L.7, for the $(k + 1)$st stage, $\tilde{x}$ and x_q would be selected as the end points of the secant line. Secant methods may seem crude, but they work well in practice. The details of the computational aspects of a sound algorithm to solve multiple equations by the secant method are too lengthy to outline here (particularly the calculation of a new Jacobian matrix from the former one; instead refer to Dennis and Schnabel[a]).

The application of Eq. (L.14) yields the following results for $f(x) = 4x^3 - 1 = 0$ starting at $x_k = -3$ and $x_q = 3$. Some of the values of $f(x)$ and x during the search are shown below; note that x_q remains unchanged in order to maintain the bracket with $f(x) > 0$.

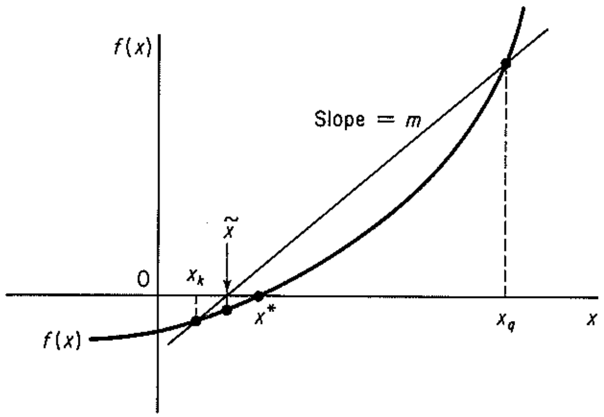


Figure L.7. Secant Method for the solution of $f(x) = 0$. x^* is the solution, $\tilde{x}$ the approximate to x^*, and x_q and x_k the starting points for iteration k of the secant method.

[a] Dennis, J. E. and R. B. Schnabel, *Numerical Methods for Unconstrained Optimization and Nonlinear Equations* (1983, Appendix A) Prentice-Hall, Englewood Cliffs, NJ 07632.

k	x_q	x_k	$f(x_k)$
0	3	-3	-109.0000
1	3	0.0277778	$-$ 0.9991
2	3	0.055296	$-$ 0.9992
3	3	0.0825434	$-$ 0.9977
4	3	0.1094966	$-$ 0.9899
5	3	0.1361213	$-$ 0.9899
20	3	0.4593212	$-$ 0.6124
50	3	0.6223007	$-$ 0.0360
100	3	0.6299311	$-$ 1.399×10^{-4}
132	3	0.6299597	$-$ 3.952×10^{-6}

Brent's and Brown's Methods

Brent's[b] and Brown's[c] methods are variations of Newton's method that improve convergence. The calculation of the elements in $\mathbf{J}_k$ in Eq. (L.11) and the solving of the linear equations are intermingled. Each row of $\mathbf{J}_k$ is obtained as needed using the latest information available. Then one more step in the solution of the linear equations is executed. Brown's method is an extension of Gaussian elimination; Brent's method is an extension of QR factorization. Computer codes are generally implemented by using numerical approximations for the partial derivatives in $\mathbf{J}_k$.

Powell Hybrid and Levenberg-Marquardt Methods

Powell[d] and Levenberg[e]-Marquardt[f] calculated a new point $\mathbf{x}^{(k+1)}$ from the old one $\mathbf{x}^{(k)}$ by (note in the next two equations the superscript (k) is used instead of a subscript k to denote the stage of iteration so that the notation is less confusing)

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \Delta x^{(k)} \tag{L.15}$$

where $\Delta x^{(k)}$ was obtained by solving the set of linear equations

$$\sum_{j=1}^{n} [\mu^{(k)} I_{ij} + \sum_{t=1}^{n} J_{ti}^{(k)} J_{tj}^{(k)}]\Delta x_j^{(k)} = -\sum_{t=1}^{n} J_{ti}^{(k)} f_t(x^{(k)}) \qquad i=1, \ldots, n \quad (L.16)$$

where I_{ij} is an element of the unit matrix $\mathbf{I}$ and $\mu^{(k)}$ is a non-negative parameter whose value is chosen to reduce the sum of squares of the deviations (f_i- 0) on each stage of the calculations. Powell used numerical approximations for the elements of $\mathbf{J}$. In matrix notation Eq. (L.16) can be derived by premultiplying Eq. (L.11) by $\mathbf{J}_k^T$.

$$\mathbf{J}_k^T \mathbf{J}_k \Delta \mathbf{x}_k = -\mathbf{J}_k^T \mathbf{f}(\mathbf{x}_k)$$

and adding a weighting factor $\mu_k \mathbf{I}$ to the left hand side to insure $\mathbf{J}_k^T \mathbf{J}$ is positive definite.

[b] Brent, R. P., *SIAM J. Num. Anal.*, vol. 10, p. 327 (1973).

[c] Brown, K. M., PhD Dissertation, Purdue University, 1966.

[d] Powell, M.J.D. in *Numerical Methods for Nonlinear Algebraic Equations*, ed. P. Rabinowitz, Chapt. 6, Gordon Breach, New York, 1970.

[e] Levenberg, K., *Quart. Appld. Math.*, vol. 2, p. 164 (1944).

[f] Marquardt, D. W., *J. SIAM*, vol. 11, p. 431 (1963).

Minimization Methods

The solution of a set of nonlinear equations can be accomplished by minimizing the sum of squares of the deviations between the function values and zero.

$$\text{Minimize } F = \sum_{i=1}^{n} f_i^2(\mathbf{x}) \qquad i = 1, \ldots, m$$

Edgar and Himmelblau[g] list a number of codes to minimize F including codes that enable you to place constraints on the variables.

Method of Successive Substitutions

Successive substitution (or resubstitution) starts by solving each equation $f_i(x)$ for a single (different) output variable. For example, for three equations, you solve for an output variable (f_2 for x_1, f_3 for x_3, and f_1 for x_2)

$$f_1(x) = 3x_1 + x_2 + 2x_3 - 3 = 0$$

$$f_2(x) = -3x_1 + 5x_2^2 + 2x_1x_3 - 1 = 0$$

$$f_3(x) = 25x_1x_2 + 20x_3 + 12 = 0$$

and rearrange them as follows

$$\left.\begin{aligned} x_1 &= F_1(x) = \frac{5x_2^2}{3} + \frac{2x_3x_1}{3} - \tfrac{1}{3} \\ x_2 &= F_2(x) = -3x_1 - 2x_3 + 3 \\ x_3 &= F_3(x) = -\frac{25x_1x_2}{20} - \frac{12}{20} \end{aligned}\right\} \quad \mathbf{x} = \mathbf{F}(\mathbf{x}) \tag{L.17}$$

An initial vector (x_{10}, x_{20}, x_{30}) is guessed and then introduced into the right hand side of (L.17) to get the next vector (x_{11}, x_{21}, x_{31}), which is in turn introduced into the right hand side and so on. In matrix notation the iteration from k to $k+1$ is

$$\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k) \tag{L.18}$$

For the procedure of successive substitution to be guaranteed to converge, the value of the largest absolute eigenvalue of the Jacobian matrix of $\mathbf{F}(\mathbf{x})$ evaluated at each iteration point must be less than (or equal to) one. If more than one solution exists for Eqs. (L.17), the starting vector and the selection of the variable to solve for in an equation controls the solution located. Also, different arrangements of the equations and different selection of the variable to solve for may yield different convergence results.

The Wegstein and Dominant Eigenvalue methods listed in Figure L.3 are useful techniques to speed up convergence (or avoid non-convergence) of the method of

[g] Edgar, T. F., and D. M. Himmelblau, *Optimization of Chemical Processes*, Chapters 6 and 8, McGraw-Hill, New York, 1988.

successive substitutions. Consult the references cited in Figure L.3 for the specific details.

Wegstein's method, which is used in many flowsheeting codes, accelerates the convergence of the method of successive substitutions on each iteration. In the secant method, the approximate slope is

$$m = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}},$$

where x_k is the value of x on the kth (current) iteration and x_{k-1} is the value of x on the $(k - 1)st$ (previous) iteration. The equation of a line through x_k and $f(x_k)$ with slope m is

$$f(x) - f(x_k) = m(x - x_k).$$

In successive substitutions we solve $x = f(x)$. Introduce $x = f(x)$ into the equation for the line

$$x - f(x_k) = m(x - x_k)$$

and solve for x:

$$x = \frac{1}{1 - m} f(x_k) - \frac{m}{1 - m} x_k.$$

Let $t = \dfrac{1}{1 - m}$. Then for the $(k + 1)st$ (next) iteration,

$$x_{k+1} = (1 - t)x_k + tf(x_k).$$

For the solution of several equations simultaneously, each x is treated independently, a procedure that may possibly cause some instability if the x's interact. In such cases, an upper limit should be placed on t, say $0 \leq t \leq 1$. The Wegstein algorithm is for stage k.

1. Calculate x_k from the previous stage.
2. Evaluate $f(x_k)$.
3. Calculate m and t.
4. Calculate x_{k+1}.
5. Set $x_k \rightarrow x_{k+1}$ and repeat the above, starting with 2.
6. Terminate when $x_{k+1} - x_k <$ tolerance assigned.

Homotopy (Continuation) Methods

Homotopy methods[h,i] can be viewed as methods that widen the domain of conver-

[h] M. Kubicek, "Algorithm 502," *ACM Trans. Math. Software*, v. 2, p. 98 (1976).

[i] W. J. Lin, J. D. Seader, and T. L. Wayburn, *AIChE J.*, v. 33, p. 886 (1987).

gence of any particular method of solving nonlinear equations, or, alternatively, as a method of obtaining starting guesses satisfactorily close to the desired solution. A set of functions $\mathbf{F}(\mathbf{x})$ is modified as follows to be a linear combination of a parameter t:

$$\mathbf{F}(\mathbf{x}, t) = \mathbf{F}(\mathbf{x}) + (1 - t)\mathbf{F}(\mathbf{x}_0) = \mathbf{0} \qquad 0 \le t \le 1 \tag{L.19}$$

where t is a scalar parameter such that when t is a fixed number the trajectory $\mathbf{F}(\mathbf{x}, t) = 0$ occurs [a mapping of $\mathbf{F}(\mathbf{x})$] and when $t = 1$ the trajectory of the set of equations reaches the desired solution $\mathbf{F}(\mathbf{x}^*) = 0$. With this definition $\mathbf{x}$ describes a curve in space (a continuous mapping called a homotopy) with one end point at a known value (the starting guess) for $\mathbf{x}$, namely $\mathbf{x}_0$, and the other end point at a solution of $\mathbf{F}(\mathbf{x}) = \mathbf{0}$, $\mathbf{x}^*$.

Lin et al. outline the procedure, which is first to determine $\mathbf{x}$ and t as functions of the arc length of the homotopy trajectory. Then Eq. (L.19) is differentiated with respect to the arc length to yield an initial value problem in ordinary differential equations. Starting at $\mathbf{x}_0$ and t_0, the initial value problem is transformed by using Euler's method to a set of linear algebraic equations that yield the next step in the trajectory. The trajectory may reach some or all of the solutions of $\mathbf{F}(\mathbf{x}) = \mathbf{0}$; hence several starting points may have to be selected to create paths to all the solutions, and many undesired solutions (from a physical viewpoint) will be obtained. A number of practical matters to make the technique work can be found in the review by Seydel and Hlavacek.[j]

SUPPLEMENTARY REFERENCE

Rheinbolt, W. C., *Numerical Analysis of Parameterized Nonlinear Equations*, Wiley-Interscience, New York, 1986.

[j] R. Seydel, and V. Hlavacek, *Chem. Eng. Sci.*, v. 42, p. 1281 (1987).

Appendix M

FITTING FUNCTIONS TO DATA

Frequently, you would like to estimate the best values of the coefficients in an equation from experimental data. The equation might be a theoretical law or just a polynomial, but the procedure is the same. Let y be the dependent variable in the equation, b_i be the coefficients in the equation, and x_i be the independent variables in the equation so that the model is of the form

$$y = f(b_0, b_1, \ldots; x_0, x_1, \ldots) \tag{M.1}$$

Let ϵ represent the error between the observation of y, Y, and the predicted value of y using the values of x_i and the estimated values of the coefficients b_i:

$$Y = y + \epsilon \tag{M.2}$$

The classical way to get the best estimates of the coefficients is by least squares, that is, by minimizing the sum of the squares of the errors (of the deviations) between Y and y for all the j sets of experimental data:

$$\text{Minimize } F = \sum_{j=1}^{p} (\epsilon_j)^2 = \sum_{j=1}^{p} (Y_j - f_j)^2 \tag{M.3}$$

Let us use a model linear in the coefficients with one independent variable x

$$y = b_0 + b_1 x \tag{M.4}$$

(in which x_0 associated with b_0 always equals 1 in order to have an intercept) to illustrate the principal features of the least-squares method to estimate the model coefficients. The objective function is

$$F = \sum_{j=1}^{p} (Y_j - y_j)^2 = \sum_{j=1}^{p} (Y_j - b_0 - b_1 x_j)^2 \tag{M.5}$$

There are two unknown coefficients, b_0 and b_1, and p known pairs of experimental values of Y_j and x_j. We want to minimize F with respect to b_0 and b_1. Recall from calculus that you take the first partial derivatives of F and equate them to zero to get the necessary conditions for a minimum.

$$\frac{\partial F}{\partial b_0} = 0 = 2\sum_{j=1}^{p}(Y_j - b_0 - b_1x_j)(-1) \tag{M.6a}$$

$$\frac{\partial F_0}{\partial b_1} = 0 = 2\sum_{j=1}^{p}(Y_j - b_0 - b_1x_j)(-x_j) \tag{M.6b}$$

Rearrangement yields a set of linear equations in two unknowns, b_0 and b_1:

$$\sum_{j=1}^{p} b_0 + \sum_{j=1}^{p} b_1x_j = \sum_{j=1}^{p} Y_j$$

$$\sum_{j=1}^{p} b_0x_j + \sum_{j=1}^{p} b_1x_j^2 = \sum_{j=1}^{p} x_jY_j$$

The summation $\Sigma_{j=1}^{p} b_0$ is $(p)(b_0)$ and in the other summations the constants b_0 and b_1 can be removed from within the summation signs so that

$$b_0(p) + b_1\sum_{j=1}^{p} x_j = \sum_{j=1}^{p} Y_j \tag{M.7a}$$

$$b_0\sum_{j=1}^{p} x_j + b_1\sum_{j=1}^{p} x_j^2 = \sum_{j=1}^{p} x_jY_j \tag{M.7b}$$

The two linear equations above in two unknowns, b_0 and b_1, can be solved quite easily for b_0 the intercept and b_1 the slope.

EXAMPLE M.1 Application of Least Squares

Fit the model $y = \beta_0 + \beta_1 x$ to the following data (Y is the measured response and x the independent variable).

x	Y
0	0
1	2
2	4
3	6
4	8
5	10

Solution

The computations needed to solve Eqs. (M.7) are

$$\sum x_j = 15 \qquad \sum x_jY_j = 110$$

$$\sum Y_j = 30 \qquad \sum x_j^2 = 55$$

Then

$$6b_0 + 15b_1 = 30$$
$$15b_0 + 55b_1 = 110$$

Solution of these two equations yields

$$b_0 = 0 \qquad b_1 = 2$$

and the model becomes $\hat{Y} = 2x$, where $\hat{Y}$ is the predicted value for a given x.

The least-squares procedure outlined above can be extended to any number of variables as long as the model is linear in the coefficients. For example, a polynomial

$$y = a + bx + cx^2$$

is linear in the coefficients (but not in x), and would be represented as

$$y = b_0 + b_1 x_1 + b_2 x_2$$

where $a = b_0$, $b = b_1$, $c = b_2$, $x_1 = x$, and $x_2 = x^2$. Linear equations equivalent to Eqs. (M.7) with several independent variables can be solved via a computer. If the equation you want to fit is nonlinear in the coefficients such as

$$y = b_0 e^{b_1 x} + b_2 x^2$$

you can minimize F in Eq. (M.3) directly by the computer program on the disk in the back of this book, or by using a nonlinear least-squares computer code taken from a library of computer codes.

Additional useful information can be extracted from a least-squares analysis if four basic assumptions are made in addition to the presumed linearity of y in x:

1. The x's are deterministic variables (not random).
2. The variance of ϵ_j is constant or varies only with the x's.
3. The observations Y_j are mutually statistically independent.
4. The distribution of Y_j about y_j given x_j is a normal distribution.

For further details, see Box, Hunter, and Hunter[1] or Box and Draper[2].

[1] G. E. P. Box, W. G. Hunter, and J. S. Hunter, *Statistics for Experimenters*, Wiley-Interscience, New York, 1978.

[2] G. E. P. Box and N. R. Draper, *Empirical Model Building and Response Surfaces*, Wiley, New York, 1987.

Appendix N

ANSWERS TO SELECTED PROBLEMS

Chapter 1

1.2 **(a)** 6.707×10^8; **(b)** 116.45; **(c)** 96.6; **(d)** 91.44
1.7 5.86 W; 17.58 W
1.12 **(a)** 40; **(b)** 30; **(c)** 2293; **(d)** 8; **(e)** 400; **(f)** 2
1.18 J/(s)(m^2)(°C)
1.21 No
1.22 1: 1.248×10^6; 2: 2.10×10^7
1.30 **(a)** 9.9×10^3 g; **(b)** 408.3 lb; **(c)** 22.6 lb
1.31 **(a)** 11.45 lb; **(b)** 201 lb; **(c)** 41,100 lb
1.38 0.0311 lb mol H_2SO_4/gal
1.42 59.29 lb/ft^3 and 7.93 lb/gal
1.45 He: 0.167; N_2: 0.389; 0_2: 0.444
1.49 Yes
1.51 **(a)** 42.412 L/min
1.54 **(a)** 34; **(b)** 0.125
1.58 CH_4: 8.5%; C_2H_6 4.0%; CO_2; 87.5%
1.62 Analysis: A: 32.2%; B: 27.2%; C: 40.6% mol
1.66 **(a)** 21.1°C, 294 K, 530°R
1.67 −321.4°F; 138.6°R for N_2; −268.8°C, 4.4 K for He
1.72 **(a)** 6.9×10^4 N/m^2, 69 kN/m^2, 69 kPa; **(b)** 1 atm, 101.3 kN/m^2, 0.1013 MN/m^2
1.74 63.8 psia
1.80 **(a)** 115 ft; **(b)** No
1.84 655 mm Hg
1.94 0.42 kPa

1.97 Yes, yes, yes

1.101 (b) 4.47 g; (d) 3.41 g; (f) 6.08 lb; (h) 8.21 lb

1.104 30.8 lb O_2

1.109 (a) 609 ton S; (b) 913.5 ton O_2; (c) 342.6 ton H_2O

1.115 (a) 33% excess C; (b) 86.0% Fe_2O_3

1.119 (a) C; (b) $Ca_3(PO_4)_2$ and SiO_2; (c) 0.170

Chapter 2

2.2 Yes

2.5 (a) unsteady state, open

2.8 4 balances including total material balance; 3 are independent

2.11 No stream values; 2 compositions; no.

2.13 CO_2(13%), H_2O(14.3%), N_2(67.6%), O_2(5.1%)

2.16 (a) CO_2(14.1%), SO_2(0.32%), H_2O(6.5%), N_2(75.8%), O_2(3.37%)
(b) H_2O(6.40%), SO_2(0.32%), CO_2(11.4%), CO(2.08%), O_2(5.07%), N_2(74.96%)

2.22 CO_2(8.2%), CO(3.5%), O_2(5.4%), N_2(82.3%)

2.25 47.8 kg

2.31 (a) 33%; (b) 8.0

2.36 0.801 ton

2.39 417 spent acid, 176 lb concentrated HNO_3, 407 lb concentrated H_2SO_4

2.45 (a) A(0.600), B(0.350), C(0.05); (b) an infinite number of solutions exist

2.50 7980 gal/hr

2.53 0.236 mol air/mol exhaust gas

2.55 2

2.59 8 unknowns (C, D, E, F, G, H, x_{F,C_2}, x_{F,C_3}; balances: unit 1-3, unit 2-2, unit 3-3 for a total of 8. Note: $1 - x_{F,C_2} - x_{F,C_3}$ can be added by including one more unknown

2.62 $P = 35.9$; $X_{C3} = 0.504$, $X_{C4} = 0.496$

2.67 (a) 53%

2.68 ratio = 0.199 mol R/mol $SiCl_4$ exiting

2.69 R = 7670 kg/hr

2.74 (a) B = 586 lb/hr; (b) D = 414 lb/hr; (c) Fraction = 0.551

2.80 80.8%

2.85 (a) 14.25 kg mol; (b) 29.7 kg mol

Chapter 3

3.1 1.026 L

3.5 1.46×10^{-2} kg H_2O

3.11 (a) 0.107 ft^3 at 125 psia and T; (b) 1.017 ft^3 at 14.7 psia and T

3.16 0.325 ft^3

3.20 3.56 ft^3 at 70° F, 29.4 in. Hg/ft^3 gas at 90°F, 35 in. Hg

3.25 (a) 0.080; (b) 15.47 m^3 at 300 K and 101.4 kPa/kg fuel

3.29 C_3H_6

3.34 (a) 0.0952 kg/m^3; (b) 0.069 = sp.gr.

3.37 pressure CO_2 = 17.7 mm Hg

3.41 **(a)** N_2 (83.35%), CO_2 (11.06%), H_2(5.59%); **(b)** 2.187; **(c)** 28.26

3.44 van der Waals: 10.60 g mol; Redlich-Kwong: 10.61 g mol

3.49 **(a)** 26.5 atm; **(b)** 26.5 atm; **(c)** 25.95 atm

3.51 500°R

3.54 24.6 lb

3.58 435 K

3.63 $(n_2/n_1) = 0.291$ for the real gas

3.67 31.9 kg/m^3 at 393 K and 3500 kPa

3.71 **(a)** 194.6 mm Hg vs. 199.1 from handbook; T = 316.4°K (43.4°C) vs. 42.7 from handbook

3.74 27 atm at 100°C

3.79 **(a)** 47.6 ft^3 at 90°F and 29.80 in. Hg; **(b)** 0.107 lb H_2O

3.83 O_2: 296 ft^3 at 745 mm, 25°C; C_2H_2: 54 ft^3 at 745 mm, 25°C

3.87 T ~ 82°C; mol fr. C_6H_4 ~ 0.73, hence condensation occurs starting at ~ 88°C

3.91 Bubble point: 375.1 K; Dewpoint: 381.5 K

3.96 561.6 total mol liquid and 566.4 total mol vapor

3.100 **(a)** 12.3%; **(b)** 14.9%; **(c)** 75°F

3.104 **(a)** 2.75×10^{-3} mol Bz/mol gas; **(b)** 2.758×10^{-3} mol Bz/mol Bz free gas **(c)** 0.0074; **(d)** 0.023

3.110 717 ft^3 at 30°C and 750 mm Hg

3.115 47.4°C

3.119 **(a)** CO_2 (6.7%), CO(12.4%), O_2(18.1%), N_2(62.8%); **(b)** 35.8 ft^3 air at T and p/lb $H_2C_2O_4$; **(c)** 106.2 ft^3 at T and p/lb $H_2C_2O_4$, **(d)** 111°F

3.122 **(a)** 89 ft^3/min; **(b)** 1180 mm Hg

3.125 37,920 ft^3 at 70°F and 760 mm Hg

3.129 2

3.131 (1):1; (2):1, (3):4

3.133 (1):1

Chapter 4

4.1 **(a)** 2.5×10^4 cal/kg; **(b)** 1.048×10^5 J/kg; **(c)** 2.91×10^{-2} (kW)(hr)/kg; **(d)** 3.5×10^4 (ft)(lb_f)/(lb_m)

4.4 8.56×10^{-4}; 6.20×10^{-2} if the units of D are cm and Gg/(min)(cm^2)

4.12 1.55 (ft)(lb_f)

4.17 4.4 kW

4.21 5×10^8 W

4.24 -2.50×10^6 J/g mol

4.27 All yes except internal energy ($\Delta\hat{U}$ would be yes).

4.29 $C_p = 36.7 + 0.0403T - 0.0000221\ T^2$

4.33 **(a)** $C_p = 6.852 + 1.62 \times 10^{-3}T_{°C} - 0.26 \times 10^{-6}T_{°C}^2$

4.34 **(a)** Kopp's rule: 33.8 cal/(g mol)(°C); 32.8 cal/(g mol)(°C) from Perry; **(b)** 62.4; 65.41 from Perry **(c)** 133.0 cal/(g mol)(°C)

4.37 3759 cal/g mol

4.43 2.563×10^3 J/g mol

4.47 26.8 kJ/kg

4.50 10,590 cal/g mol Cl_2

4.53 117,950 J/g mol (28,190 cal/g mol)

4.59 **(a)** 8640 Btu/lb mol; the data appear valid as they yield a straight line for $\log_{10} p$ vs. $1/T$; **(b)** 9110 Btu/lb mol; **(c)** 8770 Btu/lb mol

4.60 **(a)** 3576 Btu; **(b)** 3576 Btu; **(c)** 1173.35 Btu; **(d)** 182 Btu/lb; **(e)** −964.2 Btu/lb; **(f)** 788.9 Btu/lb

4.64 Use CO_2 chart or tables: −73.4 Btu/lb

4.69 **(a)** $W = 0$; $Q = \Delta U = C_V\Delta T \approx C_p\Delta T = 10$ Btu; **(b)** $Q = 0$, $-W = \Delta U \cong \Delta H = C_p\Delta T = 10$ Btu

4.73 **(a)** 2.120 Btu/lb; **(b)** 2.131 Btu/lb

4.78 2.47 lb steam/lb mol waste gas

4.82 0.1171

4.87 **(a)** 86°C; **(b)** 7,887 kg H_2O

4.91 **(a)** CO_2 is at 800 psia and **(b)** is all liquid; **(c)** $\Delta V \approx 3$ ft^3/lb

4.95 No.

4.96 **(a)** 2.72 kJ/kg **(b)** No. (Q is not known)

4.100 **(a)** Initial volume: 0; **(b)** final volume: 0.218 ft^3

4.105 6.62 hp

4.112 **(a)** \$2.21/10^6 Btu; **(b)** \$6.45/10^6 Btu

4.116 **(a)** −123.94 k cal/g mol; **(b)** 43.200 k cal/g mol; **(c)** −90.434 k cal/g mol

4.123 −199.5 kJ/g mol

4.127 141 Btu/ft^3 at SC of NGI

4.129 a1. −23,600 Btu/lb; b1. −21,500 Btu/lb

4.135 **(a)** 35.84 lb; **(b)** 53,600 Btu output

4.141 -9.92×10^6 kJ/1000 kg

4.144 16.60 kJ/100 g charge or 0.166 kJ/g charge

4.146 2180 K

4.149 C_4H_8 if equilibrium is ignored

4.151 **(a)** 1.70×10^6 Btu; **(b)** 168°F

4.153 **(a)** −33,000 Btu/lb mol; **(b)** −289.65 k cal/g mol $CaCl_2$; **(c)** −1440 Btu/lb mol $CaCl_2$

4.160 28%

4.161 **(a)** 465 lb of 73% NaOH added

4.165 **(a)** 0.079 kg mol H_2O/kg mol air; **(b)** 87.1 kPa; **(c)** 37°C

4.168 **(a)** 0.01813 lb H_2O/lb air; **(b)** 0.031 lb H_2O/lb air; **(c)** 1.14%

Chapter 5

5.1 Degrees of freedom = C + 4

5.5 Degrees of freedom = 2C + 2N + 5; specify p(N), Q(N), F(C + 2), S(C + 2), N(1) for each stage

5.12 Degrees of freedom = 16

5.19 1,129.5 kg/feed

5.21 B = 240,270, C = 95,930, D = 10,270, E = 44,750, F = 4,480, A = 336,200 all lb/hr

5.24 **(a)** 90°F; **(b)** 2,800 lb of 40% NaCl produced/hr.

5.29 **(a)** 3000 lb (NH_3), 11,000 lb $(NH_4)_2SO_4$, 7,500 lb Benzol, 2,500 lb Toluol, 1,500 lb Pyridine; **(c)** 2660 lb 40% NaOH; **(d)** 3720 lb 50% H_2SO_4

5.33 Heat duty for exchangers (in MM kJ/hr): No. 1 = 82.04, No. 2 = 120.33, No. 3 = 165.35 (all temperatures 60°C)

Chapter 6

6.2 230.5 min

6.8 37,700 lb

6.11 121.5 s

6.17 moles $C_6H_{12} = e^{-kt}$

6.21 28 min

6.24 **(b)** 129°F for half-full

NOMENCLATURE*

A = area
A = constant
a = acceleration
$\mathbf{a}$ = coefficient matrix
a = constant in general
a = constant in van der Waals' equation, (Eq. 3.11)
a, b, c = constants in heat capacity equation
$\mathcal{AS}$ = absolute saturation
°API = specific gravity of oil defined in Eq. (1.9)
B = constant
b = constant in van der Waals' equation, Eq. (3.11)
$\dot{B}$ = rate of energy transfer accompanying $\dot{w}$
(c) = crystalline
C = constant in Eq. (1.1)
C = number of chemical components in the phase rule
c^* = rank of the atom matrix
C_p = heat capacity at constant pressure
C_{p_m} = mean heat capacity
C_s = humid heat
C_v = heat capacity at constant volume
D = diameter
D = distillate product
E = total energy in system = $U + K + P$
E_v = irreversible conversion of mechanical energy to internal energy
F = force
F = number of degrees of freedom in the phase rule
F = feed stream
(g) = gas
g = acceleration due to gravity
g_c = conversion factor of $\dfrac{32.174(\text{ft})(\text{lb}_m)}{(\text{sec}^2)(\text{lb}_f)}$
$\hat{H}$ = enthalpy per unit mass or mole
h = distance above reference plane
h = number of stoichiometric constraints
H = enthalpy, with appropriate subscripts, relative to a reference enthalpy
$\mathcal{H}$ = humidity, lb water vapor/lb dry air
(l) = liquid
h_c = heat transfer coefficient
$\Delta\hat{H}$ = enthalpy change per unit mass or mole
ΔH = enthalpy change, with appropriate subscripts
ΔH_0 = constant
ΔH_{rxn} = heat of reaction
ΔH_{soln} = heat of solution
ΔH_c° = standard heat of combustion

*Units are discussed in text.

ΔH_f° = standard heat of formation
K = characterization factor
K = degree Kelvin
K = kinetic energy
k_g' = mass transfer coefficient
K_j = vapor/liquid equilibrium, Eq. (3.39)
L = moles of liquid, Eq. (3.44)
l = distance
lb = pound, as a mass
lb_f = pound, as a force
lb_m = pound, as a mass
m = mass of material
m = number of equations
$\dot{m}$ = rate of mass transport through defined surfaces
mol. wt. = molecular weight
n = number of moles
n = number of unknown variables
N_d = degrees of freedom
N_p = number of equipment parameters
N_Q = number of heat transfer variables
N_{Re} = Reynolds number
N_r = number of independent constraints
N_s = number of streams
N_{sp} = number of chemical species
N_v = total number of variables
N_w = number of work variables
$\mathcal{P}$ = number of phases in the phase rule
p = partial pressure (with a suitable subscript for p)
p = pressure
p_c' = pseudocritical pressure
p^* = vapor pressure
p_c = critical pressure
P = potential energy
P = product
p_r' = pseudoreduced pressure
p_r = reduced pressure = p/p_c
p_t = total pressure in a system
Q = heat transferred
$\dot{Q}$ = rate of heat transferred (per unit time)
q = volumetric flow rate
Q_p = heat evolved in a constant pressure process
Q_v = heat evolved in a constant volume process
(s) = solid
R = recycle stream
R = universal gas constant

r = rank of a matrix in Chap. 2
$\dot{r}_A$ = rate of generation of consumption of component A (by chemical reaction)
$\mathcal{RH}$ = relative humidity
$\mathcal{RS}$ = relative saturation
S = cross-sectional area perpendicular to material flow
s = second
sp gr = specific gravity
T = absolute temperature or temperature in general
t = temperature in °C or °F
t = time
T'_c = pseudocritical temperature
T'_r = pseudoreduced temperature
T_b = normal boiling point (in K or °R)
T_c = critical temperature (absolute)
T_{DB} = dry-bulb temperature
T_f = melting point (in K)
T_r = reduced temperature = T/T_c
T_{WB} = wet-bulb temperature
U = internal energy
$\hat{V}$ = humid volume
$\hat{V}$ = specific volume (volume per unit mass or mole)
V = system volume of fluid volume in general
v = velocity
V_c = critical volume
$\hat{V}_{ci}$ = ideal critical volume = RT_c/p_c
$\hat{V}_{ci}$ = pseudocritical ideal volume
V_g = volume of gas
V_l = volume of liquid
V_r = reduced volume = V/V_c
V_{ri} = ideal reduced volume = V/V_{ci}
$\dot{W}$ = rate of work done by system (per unit time)
W = waste stream
W = work done by the system
$\dot{w}_A$, $\dot{w}$ = rate of mass flow of component A and total mass flow, respectively, through system boundary other than a defined surface
x = mass or mole fraction in general
x = mass or mole fraction in the liquid phase for two-phase systems
x = unknown variable
y = mass or mole fraction in the vapor phase for two-phase systems
z = compressibility factor
z'_c = pseudocritical compressibility factor
z_c = critical compressibility factor
z_j = mole fraction of j
z_m = mean compressibility factor

Greek letters:

α, β, γ = constants in heat capacity equation
γ = constant in Eq. (3.17)
Δ = difference between exit and entering stream; also used for final minus initial times or small time increments
λ = molal heat of vaporization
λ_f = molal latent heat of fusion at the melting point
ρ = density
ρ_A, ρ = mass of component A, or total mass, respectively, per unit volume
ρ_L = liquid density
μ = viscosity
ϵ = error
ρ_V = vapor density
ω = mass fraction

Subscripts:

A, B = components in a mixture
c = critical
i = any component
i = ideal state
r = reduced state
t = at constant temperature
t = total
1, 2 = system boundaries

Superscript:
$\hat{}$ = per unit mass or per mole
$\cdot$ = per unit time (a rate)

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